

Module Catalog

M.Sc. Management and Digital Technology

TUM School of Management

Technische Universität München

www.tum.de/

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Module Catalog: General Information and Notes to the Reader

What is the module catalog?

One of the central components of the Bologna Process consists in the modularization of university curricula, that is, the transition of universities away from earlier seminar/lecture systems to a modular system in which thematically-related courses are bundled together into blocks, or modules.

This module catalog contains descriptions of all modules offered in the course of study.

Serving the goal of transparency in higher education, it provides students, potential students and other internal and external parties with information on the content of individual modules, the goals of academic qualification targeted in each module, as well as their qualitative and quantitative requirements.

Notes to the reader:

Updated Information

An updated module catalog reflecting the current status of module contents and requirements is published every semester. The date on which the module catalog was generated in TUMonline is printed in the footer.

Non-binding Information

Module descriptions serve to increase transparency and improve student orientation with respect to course offerings. They are not legally-binding. Individual modifications of described contents may occur in praxis.

Legally-binding information on all questions concerning the study program and examinations can be found in the subject-specific academic and examination regulations (FPSO) of individual programs, as well as in the general academic and examination regulations of TUM (APSO).

Elective modules

Please note that generally not all elective modules offered within the study program are listed in the module catalog.

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Methods | Methods

Module Description

MGTHN0105: Advanced Multivariate Analysis | Advanced Multivariate Analysis

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Grading will be based on a written exam (120 minutes). Students demonstrate that they can apply advanced multivariate data analysis methods that they have learned about in class. Exam tasks comprise the analysis of given datasets and the application of the methods to solve given problems or questions. Students can also explain the rationale of the statistical approaches. They can discuss which method is appropriate to analyze a dataset and reflect on statistical assumptions of the applied methods.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Undergraduate level courses in research methods, statistics, and marketing

Content:

Students learn to use quantitative marketing research methods. They acquire skills and knowledge that is required to successfully complete research in marketing. The course therefore is designed to provide advanced knowledge of multivariate analysis methods to solve key marketing problems, as well as to give skills in applying these models and interpreting the analysis output. The learning outcome of the course should be that students are able to design, execute and evaluate marketing research projects. Furthermore, students learn to evaluate research carried out and reported by other researchers. Analysis methods covered in the lectures are, for example, conjoint analysis and advanced preference measurement methods, exploratory and confirmatory factor analysis, cluster analysis as well as methods used in social media marketing research.

Intended Learning Outcomes:

Students learn to describe and select relevant uni- and multivariate statistical techniques. They also acquire knowledge about prerequisites for using these techniques. Moreover, students demonstrate their ability to apply the selected uni- and multivariate statistical techniques correctly for the analysis of marketing problems/tasks. Students are able to report the results of an analysis they carry out.

Teaching and Learning Methods:

Lecture-style input, solve exercises in class as a group

Media:**Reading List:****Responsible for Module:**

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0107: Applied Case Study Methodology | Applied Case Study Methodology

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students are expected to write a seminar paper analyzing a specific case study and present and discuss their solution in class. With this the students will demonstrate that they have gained knowledge about the case study method, and have learned to understand, explain, and critically reflect on business problems in the field of strategic management, as well as present corresponding solutions. The final grade is determined by the written submission of the seminar paper and the presentation.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management and organizations. Prior experience with the case-study method is also helpful. Students with consulting experience or in consulting internships are encouraged to participate.

Content:

The case study method is not only employed by top universities (e.g., Harvard) but also common in the field of consulting. Students will study case studies and apply practical methods to analytically discuss and present solutions to the real-world problems posed in their cases. Students with consulting experience or in consulting internships are encouraged to participate.

Intended Learning Outcomes:

After attending this course, students will have gained in-depth knowledge about the case study method often applied in consulting firms and top universities (e.g., Harvard Case Study Method).

Students will be able to analytically examine and solve real cases by applying practical methods used in the consulting business.

Teaching and Learning Methods:

The various consulting methodologies and frameworks are taught in a lecturing format and then interactively applied to real-world problems presented in the cases. Students will also prepare and research their cases in self-study. Afterwards, students discuss and analytically solve their cases in form of discussions.

Media:

PowerPoint, Case Studies

Reading List:

Ellet W. (2018). The case study handbook : a student's guide (Revised). Harvard Business Review Press.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0112: Simulation in Production and Logistics | Simulation in Production and Logistics

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In order to check whether students can perform a simulation study in Python they have to implement a problem and also design an experiment to analyze or optimize the system. Thus, the first part of the exam is an implementation project (programming task, 30%). In the second part of the exam, students must use an implementation to analyze a system. They have to do an output analysis and must be able to present their results (20 minutes, 70%).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Programming skills in Python

Probability calculus and statistics

Content:

Pseudo random numbers and random number generators

Discrete event simulation

Verification and validation

Design of a simulation study

Output analysis

Intended Learning Outcomes:

The student is able to develop a simulation model

The student is able to implement a simulation model in Python

The student knows how to perform verification and validation

The student is able to design a simulation experiment

The student is able to analyze the output of a simulation

The student is able to present the results of a simulation study

Teaching and Learning Methods:

Group work, Programming with Python, Presentations, Exercises

Media:

Power Point Slides, Exercises, Computer Work

Reading List:

A.M. Law. Simulation Modeling and Analysis.

6th Edition. 2024. McGraw-Hill Education

Hands-On Simulation Modeling with Python. 2nd Edition. 2022 Packt publishing

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0150: Applied Regression Analysis | Applied Regression Analysis

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language:	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Essay (50%) and presentation (50%): In this course, students are to independently and individually read foundational and current scientific contributions on strategic management methods and give a presentation on them, as well as actively participate in discussions in class and write a short essay that summarizes the methods and results.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Undergraduate level courses in research methods, statistics, and strategic management

Content:

The course will be taught using a lecture and seminar style. Students will each work on their datasets based on regression methods discussed in class.

Intended Learning Outcomes:

After successful completion of this course, students will have learned important methods used in the regression analysis of data. They will understand basic concepts of the linear regression model and its properties and understand, explain and critically reflect on selected advanced topics such as the analysis of non-linear data, duration data, panel data, and multilevel analysis, and have gained the ability to discuss and elaborate on methods to deal with the problems of endogeneity and sample selection. Students will have built a basic understanding of the theoretical basis of these methods, but the focus of the course is on the application of the methods to data sets.

Teaching and Learning Methods:

Lecture-style input, solve exercises individually in class and as assignments

Media:

Reading List:

Kennedy P. 2008. A Guide to Econometrics; Wooldridge JM. 2015. Introductory Econometrics;
Stock JH, Watson MW. 2015. Introduction to Econometrics.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Applied Regression Analysis (MGTHN0150) (Seminar, 4 SWS)

Meng Y, Schilder J

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0157: Stochastic Models in Production and Logistics | Stochastic Models in Production and Logistics

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students mastery of the content taught in this module is checked with a 60 minutes written exam (electronic). In the exam students have to answer questions about the theory, create stochastic models for small example problems and compute performance measures for stochastic systems. They must also be able to interpret the obtained results. By this the students demonstrate that they have understood and can apply the stochastic models and methods. The overall grade of the module is based on the result obtained in the written exam (electronic).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basics in probability calculus
Programming skills in Python

Content:

The following topics are discussed:

- 1) Random Variables
- 2) Discrete time Markov Chains
- 3) Continuous Time Markov Chains
- 4) Queuing Models
- 5) Stochastic manufacturing systems
- 6) Inventory Systems
- 7) Markov Decision Processes

Intended Learning Outcomes:

In this course you learn

- know how to model uncertainty
- are able to model systems with Markov Chains, queuing models, and Markov Decision Processes
- are able to compute performance measures of stochastic systems
- are able to analyze stochastic manufacturing systems and other logistic systems

Teaching and Learning Methods:

The module consists of a weekly lecture and a weekly exercise course. In the lecture the content is introduced to the student. For the exercises students have to prepare solutions. The solutions are discussed and questions about the content of the lecture and the exercises can be asked.

Media:

Presentation Slides

Reading List:

V.G. Kulkarni. Modeling, Analysis, Design, and Control of Stochastic Systems, 1999, Springer New York

H.C. Tijms. A first Course in Stochastic Models, 2003, John Wiley & Sons

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

Stochastic Models in Production and Logistics - Lecture (MGTHN0157, English) (Vorlesung, 2 SWS)

Kiesmüller G, Lu H

Stochastic Models in Production and Logistics - Exercise (MGTHN0157, English) (Übung, 2 SWS)

Kiesmüller G, Lu H

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0175: Meta-Analysis in Management Research: Techniques and Applications | Meta-Analysis in Management Research: Techniques and Applications

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Essay (about 15 text pages; 80%) and presentation (15 minutes; 20%): In this course (lecture and tutorial), students will study foundational and current scientific contributions on meta-analysis methodology and its application in management research. They will submit an individual short essay in which they apply meta-analysis techniques to a selected management research topic, summarizing their methods, findings, and implications for evidence-based management practice. In addition, students will work in teams to integrate their meta-analytic findings, present their findings in class, and discuss actionable managerial implications. This combination of individual and team assignments ensures that students gain both a theoretical understanding and practical experience in applying meta-analysis to management research.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

No specific skills, abilities, and competencies required

Content:

The course will be taught using a lecture, tutorials, and seminar style. Students will each develop a dataset and work on this datasets based on meta-analytic methods discussed in class. This course provides students with the theoretical foundation and practical skills necessary to conduct meta-analytic research in management. Students will learn how to quantitatively synthesize existing empirical findings to derive insights relevant to both academic research and practical applications. Key topics include:

Introduction to meta-analysis: Understanding the purpose, advantages, and applications of meta-analysis in management research, and its role in evidence-based decision-making.

Systematic literature search: Developing the ability to identify and select suitable studies using advanced literature search strategies and tools (e.g., academic databases and AI-assisted methods).

Bivariate meta-analysis: Gaining proficiency in calculating and interpreting effect sizes to assess the strength and direction of relationships between variables.

Moderator analysis: Exploring how conceptual, contextual, and methodological moderators influence the strength of relationships, and identifying conditions under which associations vary.

Levels of analysis: Applying meta-analytic techniques across individual, team, and organizational levels in management research.

Practical tools for meta-analysis: Hands-on training with open-access tools such as Excel templates, web-based platforms, and AI-assisted resources for efficient literature search and data analysis.

Advanced topics: Addressing challenges such as publication bias, heterogeneity, and multi-level modeling in meta-analytic studies.

The course emphasizes practical application by integrating foundational concepts with opportunities for hands-on learning and project development. Students will develop their own meta-analysis projects, applying learned techniques to a management research topic of their choice, with opportunities to ask questions and clarify challenges along the way.

By the end of the course, students will have the ability to independently conduct and interpret meta-analyses to address meaningful research questions and contribute to evidence-based management practices.

Intended Learning Outcomes:

Upon completion of this course, students...

- ...will gain a comprehensive understanding of meta-analysis as a research tool in the field of management
- ...will learn how to systematically identify, collect, and code existing empirical evidence and use meta-analytic procedures to synthesize previous findings to make managerial decisions based on broad empirical evidence
- ...will gain knowledge of and develop skills related to meta-analytic methodology and will be able to reflect on various management areas through the lense of meta-analytic evidence
- ...will be able to understand psychometric meta-analytic studies and will be able to conduct a meta-analysis themselves

Teaching and Learning Methods:

Type of instruction: Lecture with tutorial
Methods of teaching: Lecture (provides foundational knowledge about meta-analysis concepts, methodologies, and their application in management research), individual in-class exercises (reinforces learning through hands-on application of key concepts, such as identifying keywords for the literature search, calculating effect sizes or interpreting meta-analytic results), in-class group work (facilitates collaboration on complex tasks like designing a meta-analytic study), writing a research paper of about 15 text pages (apply meta-analytic concepts to real-world management research problems), providing a presentation of 15

minutes (allows students to communicate their understanding of meta-analysis methodology and findings as well as to develop critical skills such as clear communication and the ability to defend decisions on critical judgement points)

Media:

Script, Powerpoint slides, exercise sheets, videos, whiteboard

Reading List:

A detailed reading list will be available for students two weeks before the start of the course

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0176: Social Network Analysis | Social Network Analysis

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Written exam (60 min.)

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Undergraduate level courses in research methods, statistics, and strategic management

Content:

The course will be taught using a lecture and seminar style. Students will each work on their datasets based on SNA methods discussed in class.

Intended Learning Outcomes:

After successful completion of this course, students will have learned the most important methods used in social network analysis. They will understand the basic concepts of social networks and relational data. They will further be able to understand, explain, and critically reflect on selected advanced topics such as statistical interdependencies inherent to relational data, how to analyze network data based on descriptive statistical methods as well as applying inferential statistical models for relational network data - both cross-sectional and network panel data.

Teaching and Learning Methods:

Lecture-style input, solve exercises individually in class and as assignments

Media:

Reading List:

- Rawlings/Smith/Moody/McFarland (2023): Network Analysis – Integrating Social Network Theory, Method, and Application with R, Cambridge University Press.
 - Scott (2017): Social Network Analysis, Sage.
 - Borgatti/Everett/Johnson/Agneessens (2022): Analyzing Social Networks Using R, Sage.
 - Yang/Keller/Zheng (2016): Social Network Analysis – Methods and Examples, Sage.
 - Wasserman/Faust (1994): Social Network Analysis: Methods and Applications
- Further literature references will be provided during the course.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0179: Marketing Analytics | Marketing Analytics

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Grading based on a written examination of 90 minutes (100%). The examination evaluates whether participants have gained the competencies stated below (intended learning outcomes). More specifically, the examination consists of two parts. The first part of the examination tests whether the participants properly understand the marketing analytics models introduced in the module. Given a specific marketing problem, participants have to evaluate the suitability of different marketing analytics models and discuss associated data requirements as well as ethical implications. The second part of the examination evaluates whether participants can interpret the output of marketing analytics models and critically reflect on chosen model specifications. In the examination, participants are allowed to use a non-programmable calculator.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Marketing fundamentals, basics of data management, assessment of statistical significance, regression analysis, variance analysis, and basics of statistical analysis with the mainstream statistic software package R.

Content:

This module provides a comprehensive introduction to the field of marketing analytics, equipping participants with both theoretical knowledge and practical skills. It explores a range of analytical models commonly used in marketing practice, enabling participants to conduct data-driven analyses for informed decision-making. Key topics include market segmentation, preference measurement, marketing mix modeling, and the analysis of user-generated data. Throughout the module, participants will develop technical research skills and apply them using R, gaining hands-on experience in leveraging data for strategic marketing insights.

Intended Learning Outcomes:

After successfully completing the module, participants have gained the competence to:

- assess business challenges, determine relevant data requirements, and select appropriate analytical techniques to address marketing problems effectively,
- apply key marketing analytics models to support data-driven decision making, and to
- analyze marketing data, extract meaningful patterns, generate strategic recommendations based on analytical findings, and evaluate ethical implications of the applied decision approach.

Teaching and Learning Methods:

The module consists of an interactive lecture with integrated exercise that combines theoretical instruction with practical application. Participants are engaged in hands-on exercises using R to apply analytical models and techniques in real-time. Participants have to analyze marketing data, generate meaningful insights using the models discussed in the module, and evaluate the ethical implications of the applied decision approach.

Media:

Lecture slides, exercises, codes, and datasets available in TUM Moodle.

Reading List:

- Rao. 2014. Applied conjoint analysis. Springer. (recommended)
- Yildirim and Kübler. 2023. Marketing analytics using R. Sage Publishing. (recommended)
- Further topic-specific readings will be made available via TUM Moodle. (recommended)

Responsible for Module:

Eckert, Christine; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0180: Empirical Consumer Research for Marketing Decisions | Empirical Consumer Research for Marketing Decisions

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language:	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In this seminar, students will actively engage in developing their own research ideas. Throughout the course, they will work in small groups to conceptualize, design, and refine a research project related to the topics covered in the module. Individual feedback sessions will support the development of these projects.

The final examination consists of three components: (1) A scientific paper (15–30 pages) written and submitted as a group project (70% of the final grade), (2) Each student's individual contribution to the research process, assessed through peer evaluation (10%), (3) A written summary and critical evaluation of a scientific paper (approximately 3 pages) (20%). To pass the module, students must receive a grade of 4.0 or higher in each of the three components.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Students should have a solid foundational understanding of marketing principles and should be able to apply basic statistical concepts such as standard deviation, correlation, and linear regression.

Content:

This seminar focuses on a selected psychological construct relevant to marketing decision making. Students will explore its theoretical foundations and measurable impact on consumer behavior, as well as the implications for marketing decision-making. Drawing on current scientific literature, students will identify research gaps related to the construct and formulate research questions with managerial relevance. Lecture sessions will cover topics such as methods of data collection and analysis in marketing research, interpretation of results, academic writing, and the translation of psychological insights into actionable marketing strategies.

Intended Learning Outcomes:

In this seminar, students will learn to:

1. Explain and critically assess key psychological constructs and their relevance to marketing decisions.
2. Identify research gaps related to consumer psychology in marketing decision-making.
3. Design appropriate data collection strategies to gather relevant consumer data.
4. Apply statistical techniques to analyze consumer data and extract actionable insights.
5. Synthesize and interpret findings based on both primary and secondary data to draw evidence-based conclusions and implications for marketing decisions.
6. Critically assess scientific papers, with particular emphasis on their practical implications for marketing and the ethical considerations associated with those implications.

Teaching and Learning Methods:

To support the learning outcomes, the seminar integrates theoretical foundations with practical application, ensuring both academic depth and real-world relevance. Students will actively engage in lectures by analyzing and discussing scientific papers, with particular attention to their managerial implications and ethical dimensions. Working in teams, students will design and carry out empirical research projects addressing real-world consumer behavior challenges. These projects will involve collecting primary data, integrating secondary data where relevant, and applying psychological and statistical methods to derive managerial insights. Through continuous feedback and iterative refinement, students will enhance their research design, analytical skills, and ability to draw practical conclusions from their findings.

Media:

Moodle

Reading List:

Lange, D., & Pfarrer, M. D. (2017). EDITORS' COMMENTS: SENSE AND STRUCTURE—THE CORE BUILDING BLOCKS OF AN "AMR" ARTICLE. *The Academy of Management Review*, 42(3), 407–416. <http://www.jstor.org/stable/26395005>

Responsible for Module:

Eckert, Christine; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Empirical Consumer Research for Marketing Decisions (MGTHN0180) (Seminar, 4 SWS)

Eckert C, Ristau I

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0182: Game Theory | Game Theory

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

This course introduces students to the principles and methods of game theory, the study of strategic interaction among rational decision-makers. Game theory provides a rigorous framework for analyzing situations where outcomes depend not only on one's own choices but also on the actions of others. The course focuses on formal modeling of strategic behavior and develops a deep understanding of equilibrium concepts under complete and incomplete information.

The course is designed for students with strong analytical interest and a willingness to engage with mathematical reasoning. It is particularly suitable for students considering a thesis or academic career that draws on formal modeling, but it is equally valuable for anyone seeking a structured way to think about competition, cooperation, signaling, incentives, or negotiation in business settings.

Students will learn to analyze static and dynamic games, explore repeated interactions and signaling problems, and understand how strategic environments shape decision-making. Weekly sessions will be split into theory lectures and structured exercises. All assessment is based on a final 60 min. written exam (100%).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

a solid quantitative training

Content:

Subject-specific content

- Strategic interaction in economic and managerial contexts
- Conflict, cooperation, and incentives in multi-agent settings

- Decision-making under complete and incomplete information
- Applications to negotiation, pricing, entry deterrence, signaling, reputation, or related concepts

Methodological content

- Formal modeling of static and dynamic games
- Equilibrium concepts (Nash, subgame perfect, Bayesian, sequential)
- Quantitative analysis and logical reasoning in strategic environments

Intended Learning Outcomes:

Upon completion of the module, students can

- formally analyze strategic interactions using foundational models of game theory,
- evaluate the suitability of different equilibrium concepts for modeling competitive and cooperative behavior,
- assess the implications of information structures and incentives in static and dynamic games,
- and critically examine strategic situations in economics and management using formal tools of game-theoretic analysis.

Teaching and Learning Methods:

Lecture

The module is taught as a classical lecture combined with an exercise session, each held weekly for two hours. The lecture introduces core concepts and formal results in game theory, supported by structured examples and analytical reasoning. The accompanying exercise session deepens understanding through problem solving and discussion of selected exercises.

Students are expected to study relevant textbook material in preparation for, or as a follow-up to, each week's sessions. The course is intensive and builds cumulatively—consistent independent study and active engagement with the material are essential for success. The teaching approach is designed to foster deep analytical insight and prepare students to apply game-theoretic reasoning with precision and confidence.

Media:

Slides, whiteboard

Reading List:

Readings for general preparation:

- Osborne, M. J., & Rubinstein, A. (1994). A Course in Game Theory. MIT Press.
- Fudenberg, D., & Tirole, J. (1991). Game Theory. MIT Press.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

Game Theory - Exercise (MGTHN0182) (Übung, 2 SWS)

Golla B

Game Theory - Lecture (MGTHN0182) (Vorlesung, 2 SWS)

Wuttke D

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0186: Research Methods in Business Psychology | Research Methods in Business Psychology

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation and submission of seminar paper and survey (8-10 pages) in groups (70%), 15 min. presentation (30%)

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as a basic understanding of empirical methods

Content:

The course covers key methodological principles in business psychology. Topics include scientific reasoning, hypothesis development and study design (e.g., experiments, field studies). Students will learn how to operationalize constructs such as work motivation and design empirical studies using surveys, interviews, observation and implicit measures (e.g., IAT). Data analysis focuses on descriptive and inferential statistics (e.g., t-tests, ANOVA, regression) with introduction to tools like R or SoSci Survey. Applications in organizational and consumer psychology are discussed, including employee surveys, leadership evaluation, and advertising research. The course also addresses quality criteria (objectivity, reliability, validity) and ethical standards in psychological research.

Intended Learning Outcomes:

After completing this course, students will be able to design and conduct empirical research in business psychology. They will be able to formulate research questions, develop surveys, analyze data and interpret findings in both theoretical and practical contexts. The course equips students with key methodological skills relevant to applied psychological research in business settings.

Teaching and Learning Methods:

Students will complete a group essay (8–10 pages) that outlines the theoretical background, research design, implementation and potential data analysis of a business psychology topic. The essay will be based on academic literature and practical examples, encouraging independent research and critical thinking. Furthermore students will create a survey based on the essay. The course combines weekly theoretical input on topics such as operationalization, study design and data analysis with group work phases, during which students work independently but can receive support in Q&A sessions.

Media:

PowerPoint

Reading List:

Allen, J. A., Bipp, T., Buengeler, C., Dettmers, J., Gerpott, F., Gielnik, M. M., Hirschi, A., Hüffmeier, J., König, C. Organizational, and Business Psychology: An Introductory Textbook. Germany: Kohlhammer Verlag.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Research Methods in Business Psychology (MGTHN0186) (Seminar, 4 SWS)

Khan N

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0188: Customer Relationship Management for Long-Term Profitability | Customer Relationship Management for Long-Term Profitability

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In this seminar, students will actively engage in developing their own research ideas. Throughout the course, they will work in small groups to evaluate current trends in customer relationship management (CRM), and conceptualize, design, and refine a research proposal. Individual feedback sessions will support the development of these proposals.

The final examination consists of three components: (1) A scientific paper proposal (10 pages) written and submitted as a group project (60% of the final grade), (2) Each student's individual contribution to the research process, assessed through peer evaluation (10%), (3) A presentation & discussion of a scientific paper (45min) (30%). To pass the module, students must receive a grade of 4.0 or higher in each of the three components.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Students should have a solid foundational understanding of marketing principles and statistics.

Content:

This seminar explores how organizations can strategically manage customer relationships to drive long-term profitability. Through a series of interactive lectures and case discussions, students will examine key concepts including the fundamentals of CRM, strategic implementation, GDPR compliance, technology platforms, CRM metrics, permission marketing, and loyalty programs. A central component of the seminar is the analysis and application of customer lifetime value to support data-driven marketing decisions. Students will critically engage with current academic literature through presentations. They will develop original research proposals that are grounded in

academic theory, demonstrate managerial relevance, and be informed by in-depth interviews with CRM professionals.

Intended Learning Outcomes:

In this seminar, students will learn to:

1. Understand and apply key theories and strategic frameworks of CRM to real-world business contexts.
2. Evaluate CRM technologies, metrics, and loyalty programs for their effectiveness in driving customer engagement and long-term profitability.
3. Analyze the impact of GDPR and permission marketing on CRM practices and strategic decision-making.
4. Calculate and interpret Customer Lifetime Value (CLV) to inform marketing strategies and resource allocation.
5. Design an academically grounded and managerially relevant CRM research proposal.

Teaching and Learning Methods:

The seminar combines interactive lectures, case-based discussions, and hands-on exercises to foster both theoretical understanding and practical application of CRM concepts. Students will engage in group work, article presentations, and guided discussions to deepen their critical thinking and communication skills. Students will work in teams to design research proposals based on interviews with CRM managers, supported by structured feedback sessions to refine their academic and practical contributions.

Media:

Moodle

Reading List:

Kumar, V., & Reinartz, W. (2018). Customer relationship management: Concept, strategy, and tools (3rd ed.). Springer.

Responsible for Module:

Eckert, Christine; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Customer Relationship Management for Long-Term Profitability (MGTHN0188) (Seminar, 4 SWS)
Eckert C

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0189: Marketing Research | Marketing Research

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Grading is based on a written examination of 90 minutes (100%). In the examination, participants are allowed to use a non-programmable calculator.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Undergraduate courses on statistics and empirical research methods are highly recommended before taking this course (but not a formal requirement).

Content:

The course connects experimental work with data analysis. As such, it is fundamental for doing further marketing lectures as well as for doing a master thesis. In the first block, students learn about experimental design in the consumer research and marketing context (including collecting data using eyetracking and virtual reality). The introductory lectures cover the design of variables and measures, factors influencing experimental designs, the formulation of research hypotheses, how to use the literature for hypotheses formulation as well as moderation and mediation as concepts. In the second block, students learn about how to analyze data collected from experiments using a range of multivariate statistical techniques that allow for the testing of group differences, the classification of observations, the modeling of categorical outcomes, the examination of associations between variables, and the evaluation of complex decision-making processes. Emphasis is placed on understanding the logic behind each method, selecting appropriate techniques based on research questions, and interpreting results in a theoretically grounded and managerially relevant way.

Intended Learning Outcomes:

Key learning goals are:

- Learn about basic principles of experimental design for consumer research and marketing phenomena.
- Learn about important models and methodologies that can be applied to analyze marketing research data.
- Learn about how to apply experimental and analytical methods to real-world marketing problems.
- Be able to explain and design experiments.
- Be able to evaluate the impact that design decisions have on which data analysis can be applied.
- Be able to use marketing research methods, which includes choosing an appropriate method, using statistics programs to do calculations and explain the results.
- Be able to explain limitations of experiments and data analysis.
- Be able to think critically about research questions and methodological choices in marketing research.
- Be able to communicate research findings clearly (experiments as well as data analysis results) to expert audiences and defend the own work.

Teaching and Learning Methods:

The course consists of two building blocks, experimental design and marketing research methods. In both blocks a combination of lectures and corresponding exercises is used. The lectures explain the basics of experimental designs and methodological approaches. In the exercises of the first block, the skills gained are trained by designing and discussing experimental designs based on readings of the marketing literature. In the exercises of the second block, students train their methods skills by analyzing small datasets and interpreting the results. Students will be asked to find and read marketing literature and reflect on the used experimental designs before the lecture. The exercises of the first block also comprise group work in which students discuss the pros and cons of experimental designs.

Media:

Reader, scripts, exercise sheets, PowerPoint, films

Reading List:

Weiber, T., Weiber, R., Gensler, S., Erichson, B., & Backhaus, K. (2021). Multivariate Analysis: An Application-Oriented Introduction. Springer.; further readings will be announced in class

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Marketing Research (Vorlesung mit integrierten Übungen, 4 SWS)

Arslan S, Eckert C

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0194: Advanced Seminar Operations & Technology: Quantum Computing | Advanced Seminar Operations & Technology: Quantum Computing

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination takes the form of exercises to evaluate the understanding of major concepts (20%) and the final problem-solving-oriented term paper (80%). The evaluation of the final project is based on:

- Presentation (~30 min incl. discussion): Oral presentation of the solution approach and results (40%)
- Report (~25 pages): Detailed documentation of the problem, modeling decisions, methodology, and analysis (40%)

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

- solid understanding of mathematics for managers (matrix notation, concepts of proofs, solid basics of linear and nonlinear mathematics modeling, operations with eigen values and eigen vectors, operations with complex numbers)
- solid programming skills in Python, C++, C#, VisualBasic, or a similar programming language

Content:

This course covers basics of quantum computing for managers, outlines its potential for business and current state of technologies, as well as delves into in-depth understanding of current promising research directions and theoretical concepts. Depending on the background of the students and their interest, the course may involve implementation of own projects on quantum simulators or even on a quantum computer.

Intended Learning Outcomes:

After successfully completing the module, participants will:

- Be able to understand in-depth basic concepts of quantum computing
- Understand current developments in quantum computing and quantum technologies and their potential for business, focusing on manufacturing companies
- Analyze real-world case studies and evaluate relevant solution approaches

Teaching and Learning Methods:

The module consists of lectures and dedicated exercise sessions. Theoretical concepts are introduced with illustrative examples and interactive discussions.

Media:

Lecture documents, exercises, and demonstrators available in TUM Moodle and via open-access web-sources.

Reading List:

Books and recent relevant scientific articles published by leading scientific groups, e.g.,
Bochkarev, A., Heese, R., Jäger, S., Schiewe, P., & Schöbel, A. (2024). Quantum Computing for Discrete Optimization: A Highlight of Three Technologies. arXiv:2409.01373

Responsible for Module:

Otto, Alena; Prof. Dr. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology: Quantum Computing (MGTHN0194) (Seminar, 4 SWS)

Otto A

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0196: Advanced Seminar Management & Marketing: Language Analytics and LLMs in Marketing | Advanced Seminar Management & Marketing: Language Analytics and LLMs in Marketing

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 135	Contact Hours: 45

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The assessment consists of a group-based project (70 points) and individual live-quizzes (20 points). In total, students can earn 90 points in the module.

Group project (70 points)

The group project is completed in teams of three to four students. Students independently apply one or more text analytics methods introduced in the seminar to a marketing-related question. The project includes the following steps: Development of a research idea and question; allocation of responsibilities within the group; short literature review; data generation, scraping, or selection; data preprocessing and cleaning, application of text analytics methods, interpretation and discussion of results.

The project outcomes are documented in a scientific paper-style report (10 to 15 pages, excluding references) with the following structure: Theoretical background - Method - Results - Discussion.

The 70 points are distributed as follows: 50 points are given based on the individual contribution to the group project. 20 points are given based on an individual written reflection and a structured peer evaluation of the team collaboration.

Live quizzes (20 points)

Throughout the semester, students complete five short live quizzes based on selected research papers discussed in class (duration per quiz: about 15 minutes, no aids permitted). Each quiz assesses students' understanding of the applied use of text analytics methods in empirical marketing research. Each quiz awards up to 5 points. Students may achieve the full 20 points by participating in any four quizzes. Participation in all five quizzes allows students to earn up to 5 bonus points (maximum of 20 points in total across all quizzes).

Repeat Examination:

(Recommended) Prerequisites:

Basic programming knowledge in R, very good English language skills, undergraduate level courses in research methods, statistics, and marketing. Prior experience with quantitative research projects is also helpful.

Content:

In today's digital economy, vast amounts of unstructured text data are generated by customers, companies, and online platforms. To achieve a strong market orientation, firms and researchers increasingly rely on systematic language analysis to extract insights from customer feedback, online reviews, social media posts, and open-ended survey responses. Modern language analysis enables researchers and practitioners to analyze such data in a scalable, objective, and theory-guided manner.

This seminar provides a both practical and research-oriented introduction to contemporary methods of text analysis in marketing. Topics include:

- String handling, regular expressions, and data structures
- Obtaining text data through web scraping and APIs
- Text preprocessing, TF-IDF, bag-of-words techniques
- Clustering, similarity measures, and grouping
- Text classification through k-Nearest Neighbors and Neural Networks
- Topic Modeling through Latent Dirichlet Allocation (LDA)
- Sentiment Analysis and Machine Learning
- Word Embeddings and Transformer Models (e.g., BERT-style models)

Intended Learning Outcomes:

After successfully completing the module, participants are able to:

- Identify marketing-relevant use cases for the application of text analytics to customer communication, social media, and survey data.
- Select, implement, and apply appropriate empirical text analysis methods to address a given research question.
- Interpret and critically evaluate the results of text-based empirical analyses with respect to their methodological validity, theoretical implications, and strategic recommendations for marketing decisions.
- Design, conduct, and document an independent empirical research project using text analysis methods in the form of a small research paper.

Teaching and Learning Methods:

The module consists of interactive lectures that are immediately followed by hands-on coding exercises. The lectures introduce the theoretical foundations and core methodologies of language analytics, while the exercises enable students to directly apply these methods to real datasets.

In addition, selected sessions focus on the critical discussion of current research papers in text analytics and marketing. Throughout the semester, students work in small groups on an independent project and regularly present intermediate results for feedback and reflection. This combination of lecture, practical exercises, paper discussion, and project-based learning supports the development of methodological, research-oriented, and practical competencies.

Media:

Programming code in R, datasets, PowerPoint slides, scientific articles, Mural boards

Reading List:

Dan, D., & Reutterer, T. (2026). Text analytics in marketing: A practical guide for students and researchers. Springer.

Responsible for Module:

Eckert, Christine; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Specialization in Management | Wirtschaftswissenschaftlichen Schwerpunkt

Specialization in Management: Family Enterprise | Management-Schwerpunkt: Family Enterprise

Advanced Seminar Family Enterprise | Advanced Seminar Family Enterprise

Module Description

MGTHN0096: Advanced Seminar Management & Marketing: Corporate Social Responsibility | Advanced Seminar Management & Marketing: Corporate Social Responsibility

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of handouts and presentation: In this course, students are to independently and individually read foundational and current scientific contributions on CSR and give a presentation on them, as well as actively participate in discussions in class and create a short handout. Additionally, they are supposed to write short essays, applying the gained knowledge to a real-life example.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization, and empirical research methods.

Content:

The course will be taught using a seminar or discussion style. Students will each give a presentation on a CSR topic discussed in class. The course includes for instance different

perspectives, as well as critiques, actors and drivers of CSR. Moreover, managing CSR, a global perspective on the construct, as well as further selected topics with more recent relevance in CSR research will be discussed.

Intended Learning Outcomes:

After successful completion of this course, students will (1) have gained knowledge of different perspectives of corporate social responsibility research in strategic management, (2) have learned to read, understand, explain and critically reflect on scientific literature in the field, (3) have built a basic understanding of research methods in strategic management, (4) have gained the ability to discuss and elaborate on scientific literature on CSR in strategic management, (5) be able to apply this knowledge by analyzing CSR concepts in real-life examples.

Teaching and Learning Methods:

Students should read the assigned literature for the course and give a presentation on a topic that is derived from the literature discussed in class, also creating a short handout. In addition to the presentations students will deepen their knowledge by writing short essays, applying the concepts to real-life examples.

Media:

Powerpoint

Reading List:

Crane, A., McWilliams, A., Matten, D., Moon, J., & Siegel, D. S. (Eds.). (2008). The Oxford handbook of corporate social responsibility. OUP Oxford. Other current literature is selected specifically for each semester and will be distributed in class.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0110: Advanced Seminar Economics & Policy: Management practices in (family) enterprises | Advanced Seminar Economics & Policy: Management practices in (family) enterprises

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on management practices in (family) enterprises.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper, and (ii) oral in-class presentation. Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Content:

The seminar will cover key topics regarding management practices, such as:

1. How can management practices be classified?
2. How do management practices vary across companies?
3. How do management practices differ between family and non-family enterprises?
4. How do management practices relate to economic success?
5. How do management practices relate to work climate?
6. What are the determinants of different management practices employed in practice?

Intended Learning Outcomes:

'Students will learn the key concepts in the research field of management practices, with a special focus on family enterprises. The goal is twofold: First, to understand how management practices relate to economic success of enterprises. Second, to understand what individual and institutional factors determine management practices in enterprises. The course builds on original research on that topic. The course will cover basic theoretical consideration, but will have a clear focus on empirical studies with real-world examples and applications.

Teaching and Learning Methods:

'Students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence. A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups. We plan to complement student presentations by talks from corporate guests who will provide insights into corporate management practices.

Media:

Powerpoint slides

Reading List:

Amongst others (for detailed reading list, please see Moodle):

Bloom, N., & Van Reenen, J. (2010). Why do management practices differ across firms and countries?. *Journal of Economic Perspectives*, 24(1), 203-24.

Bloom, N., Brynjolfsson, E., Foster, L., Jarmin, R., Patnaik, M., Saporta-Eksten, I., & Van Reenen, J. (2019). What drives differences in management practices?. *American Economic Review*, 109(5), 1648-83.

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0171: Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises | Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on a specific topic related to the seminar's overall theme.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper (50%), and (ii) oral in-class presentation (50%). Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Principles of Economics (Master)

Content:

Beyond a lecture-style introduction into the topic, students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence.

A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups.

Intended Learning Outcomes:

Students will explore how technological advancements, ranging from automation to artificial intelligence, transform labor markets and the broader economy. The seminar will delve into the mechanisms behind these changes, examining their impacts on work, skills, and business structures, as well as the role of government policies in managing these transitions. Special attention will be given to comparing family and non-family enterprises.

The course builds on original research on the topic. While it will cover basic theoretical considerations, the focus will be on empirical studies with real-world examples and applications. Students will write a term paper on a selected topic and present their findings, highlighting practical implications for labor markets, policy, and business strategies.

Teaching and Learning Methods:

This seminar will cover key topics within the Economics of Technological Change and Labor Markets, with a focus on the role of family enterprises, such as:

Topic area 1: The impact of skill-biased technological change on wages and employment outcomes.

Topic area 2: Job polarization and the growth of low-skill service jobs in response to automation and routine-biased technological change.

Topic area 3: Labor market adjustments to technological innovations, including the effects of automation, robotics, and artificial intelligence.

Topic area 4: The unique challenges and opportunities for family enterprises in adapting to technological change and leveraging innovations to remain competitive.

Topic area 5: The role of government policies in managing labor market transformations and ensuring equitable outcomes.

Media:

Teaching will be in the form of a lecture and seminar presentations.

Reading List:

Acemoglu, Daron and David Autor (2011). "Skills, tasks and technologies: Implications for employment and earnings", In: Handbook of Labor Economics, Vol. 4, Elsevier, pp. 1043–1171.

Angrist, Joshua and Jörn-Steffen Pischke (2008). Mostly Harmless Econometrics (Princeton: Princeton University Press)

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0028: Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship | Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic related to corporate entrepreneurship. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade, approx. 10 pages) The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade, approx. 10 minutes). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis).

Repeat Examination:

(Recommended) Prerequisites:

Basic knowledge of corporate entrepreneurship and research methodologies is recommended.

Content:

Foundations of Corporate Entrepreneurship (CE): Key concepts, definitions, and the role of CE in modern businesses.

Strategic Thinking in CE: Developing an entrepreneurial mindset to identify and leverage opportunities.

Implementing CE: Steps to integrate entrepreneurial practices into corporate operations.

Building an Entrepreneurial Organization: Creating a culture that fosters innovation and risk-taking.

Sustaining CE: Strategies to maintain entrepreneurial energy and adapt to changing markets.

Leadership and Culture: The impact of leadership styles and organizational culture on entrepreneurship.

Entrepreneurial Traits and Behavior: Key personal characteristics that support entrepreneurial initiatives.

Creativity and Motivation: Harnessing creativity, motivation, and HRM practices to drive innovation.

Incentive Systems and Psychological Ownership: Enhancing engagement through effective incentives and ownership.

Entrepreneurial Intensity: Measuring and increasing the level of entrepreneurial activity within corporations.

Focus Areas of CE: Examining CE at the levels of the phenomenon, organization, individual, and dynamic interactions.

Challenges in CE: Addressing obstacles faced by corporations and strategies to overcome them.

Intended Learning Outcomes:

Understand the current state of research in corporate entrepreneurship.

Develop the ability to critically analyze and synthesize academic literature.

Improve academic writing and presentation skills.

Identify research gaps and formulate theoretical and practical implications.

Teaching and Learning Methods:

In-presence seminar sessions

Literature review and independent research.

Group discussions and collaborative learning (optional group work).

Interactive feedback sessions.

Presentation of results and peer review

Media:

Moodle platform for resources and updates.

Academic databases for literature access.

Presentation tools (e.g., PowerPoint, posters)

Reading List:

Core reading will be provided on Moodle. Students are encouraged to explore additional academic articles and books relevant to their selected topic.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0029: Advanced Seminar Innovation & Entrepreneurship: Managing the Family Enterprise | Advanced Seminar Innovation & Entrepreneurship: Managing the Family Enterprise

Version of module description: Gültig ab summerterm 2020

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work in a group on a specific topic related to managing the family enterprise. The examination consists of two parts:

1) Projekt thesis (75 % of the overall grade) The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to analyze effectively the problem stated in their work and know how to structure and write a project thesis.

2) Presentation of the project thesis (25 % of the overall grade). Students should demonstrate that:

- they have gained the skills to present their project thesis to an academic audience.
- they are able to answer questions related to specific parts of their work.

The final grade will be based on the two parts (75 % project thesis and 25 % presentation and discussion of the seminar thesis).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Good English language skills are required.

Content:

This course deals with the management of family firms. Given that the majority of firms around the world are family businesses (ranging between 75-90 %), it is crucial to understand better the particularities of family businesses. Given that the boundaries between the family and the business

overlap, certain challenges may arise that makes family businesses unique. The objective of this course is to better understand the challenges and opportunities family firms face and how managers, owners, and advisers can best deal with them. By combining various theoretical frameworks with hands-on examples, we will elaborate on how to manage family enterprises successfully. We will cover topics such as:

- strength and weaknesses in family firms,
- succession and governance issues in family enterprises,
- management of key resources in family family firms,
- financial management in family firms and,
- transgenerational entrepreneurship.

Intended Learning Outcomes:

- Understand the family business phenomenon and strengths and weaknesses of family firms.
- Understand the importance of governance in family firms. Evaluate how different types of governance suit the needs of family firms differing in size.
- Reflect on how family firms use the “family” as a strategic asset and how do they manage those resources best. How do different institutional contexts influence the emergence of family firms?
- Reflect on financial management in family firms. Evaluate how does the family background bias financial behaviour, such as investment, financing, and risk taking?
- Know and apply different concepts of transgenerational entrepreneurship and be able to advise family firms how they can sustain in the long run.
- Reflect on governance and succession topics.
- (1) the search, understanding, and analysis of academic literature, (2) methods for data collection and data analysis, (3) the structuring and writing of a paper, and (4) giving presentations including a subsequent discussion

Teaching and Learning Methods:

- Through lectures, supported by a presentation, the instructor will provide the theoretical foundations on managing the family enterprise.
- A series of group discussions and exercises as well as exchanging ideas, questions, and comments will encourage an experiential learning atmosphere.
- Guest speakers will share practical insights and will strengthen the understanding of key concepts and processes, and expose perspectives complementary to the topic of the seminar.
- Within their group work, students identify challenges and develop recommendations. This is supported by feedback sessions with the instructor.
- In a presentation, students present their groupwork.

Media:

Powerpoint slides, Zoom etc.

Reading List:

Amongst others (for detailed reading list, please see Moodle):

- Aldrich, H. E., & Cliff, J. E. (2003). The pervasive effects of family on entrepreneurship: Toward a family embeddedness perspective. *Journal of Business Venturing*, 18(5), 573-596.
- Bird, M. & Zellweger, T. (2018): Relational embeddedness and firm growth: Comparing spousal and sibling entrepreneurs. *Organization Science* 29 (2), 264-283.
- Schulze, W. S., Lubatkin, M. H., Dino, R. N., & Buchholtz, A. K. (2001). Agency relationships in family firms: Theory and evidence. *Organization Science*, 12(2), 99-116.
- Sirmon, D. G., & Hitt, M. A. (2003). Managing resources: Linking unique resources, management, and wealth creation in family firms. *Entrepreneurship Theory and Practice*, 27(4), 339-358.
- Zellweger, T. (2017): *Managing the Family Business*. Edward Elgar Publishing: Massachusetts.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0029): Managing the Family Enterprise
(Seminar, 4 SWS)

Bird M, Kosberg M

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0046: Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises | Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises

Version of module description: Gültig ab winterterm 2020/21

Module Level: Master	Language: English	Duration: one semester	Frequency:
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic in the field of sustainable entrepreneurship / sustainable management in family enterprises. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis). Active engagement in the course will be expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English

Content:

This course addresses various aspects of sustainability available to new ventures and established family enterprises. Sustainable entrepreneurship (SE) emphasizes that entrepreneurs may pursue additional goals such as social and ecological goals that may alleviate social and ecological problems. It explores the opportunities that entrepreneurs create, the challenges entrepreneurs encounter, and the ways in which entrepreneurs exploit opportunities to contribute toward enhancing the society while simultaneously starting and sustaining a profit-oriented business. This course also examines the role of sustainability in family enterprises:

Course content will comprise:

- The importance of sustainable entrepreneurship in solving grand societal challenges of the 21st century
- Motives and barriers of sustainable entrepreneurship
- Overview of theories and concepts in sustainable entrepreneurship
- Stakeholder management for entrepreneurs
- Sustainable management in the context of family enterprises
- Sustainability in entrepreneurship and family enterprises in the context of developing countries
- Recognizing and exploiting social and ecological opportunities
- Practical examples of sustainable ventures

Intended Learning Outcomes:

After completing the seminar students, should understand how entrepreneurs and family enterprises can contribute to solving societal challenges. Therefore, students will be able to:

- understand the concept of sustainable entrepreneurship and sustainability in family enterprises
- understand and evaluate how entrepreneurs can enact on sustainable opportunities
- understand the entrepreneurs' motives and barriers to engage in sustainable entrepreneurship
- reflect on the importance of diversity in achieving sustainability in entrepreneurship and family enterprises
- evaluate strategic actions of sustainable enterprises
- evaluate hands-on examples and cases of how sustainable entrepreneurship can be implemented in practice with specific focus on the Heilbronn region: Those examples may relate to various sectors such as the public governance sector, the food and agriculture, environmental and mobility sector.

In addition, students will be able:

- to search, understand, and analyze academic literature by evaluating key findings and explaining complex research studies in a comprehensible and interesting manner,
- to apply methods for data collection and data analysis, - to structure and write a seminar thesis, and
- to present and subsequently discuss findings of their seminar thesis.

Teaching and Learning Methods:

' -Through lectures, supported by Power-Point presentations, the instructor will provide the theoretical foundations of sustainable entrepreneurship

- The content is discussed in the course by exchanging ideas, questions, and comments to encourage an open learning atmosphere
- Every sessions contains exercises, where the students apply their learnings in a group work (e.g. to present one start-up, enterprise or organization that has already successfully implemented the topic/concept presented in the lecture)
- Guest speakers will share practical insights and will strengthen the understanding of key concepts and processes and expose perspectives complementary to the topic of the seminar
- For their seminar thesis, students will investigate a topic within the subject of this course to develop a seminar thesis. Students will receive feedback from the instructor.
- In a final presentation, students present the results of their project thesis

Media:

Powerpoint, Zoom- & Breakout-Sessions, Online Simulations

Reading List:

Basic literature (for detailed reading list, see Moodle):

- Cruz, C.; Larraza-Kintana, M. Garcés-Galdeano, L. Berrone, P. (2014). Are family firms really more socially responsible? *Entrepreneurship Theory and Practice* 38(6), 1295–1316.
 - Gast, J., Gundolf, K., & Cesinger, B. (2017). Doing business in a green way: A systematic review of the ecological sustainability entrepreneurship literature and future research directions. *Journal of Cleaner Production*, 147, 44-56.
 - Le Breton-Miller, I., & Miller, D. (2016). Family firms and practices of sustainability: A contingency view. *Journal of Family Business Strategy*, 7(1), 26-33.
 - Lumpkin, G. T., Bacq, S., & Pidduck, R. J. (2018). Where change happens: community#level phenomena in social entrepreneurship research. *Journal of Small Business Management*, 56(1), 24-50.
 - Patzelt H., & Shepherd D.A. (2001) Recognizing opportunities for sustainable development. *Entrepreneurship Theory and Practice*. 35(4), 631-652.
 - Sarason, Y., & Dean, T. J. (2019). Lost Battles, Trojan Horses, Open Gates, and Wars Won: How entrepreneurial firms co-create structures to expand and infuse their sustainability missions in the acquisition process. *Academy of Management Perspectives*, 33(4), 469-490.
 - Sarasvathy, S. D., & Ramesh, A. (2019). An effectual model of collective action for addressing sustainability challenges. *Academy of Management Perspectives*, 33(4), 405-424.
 - Shevchenko, A., Levesque, M. & Pagell, M. (2016). Why firms delay reaching true sustainability. *Journal of Management Studies* 53, 911-35.
 - Zahra, S. A., Gedajlovic, E., Neubaum, D. O., & Shulman, J. M. (2009). A typology of social entrepreneurs: Motives, search processes and ethical challenges. *Journal of Business Venturing*, 24(5), 519-532.
- Book(s):
- Lumpkin, T. & Katz, J. (2011). *Social and sustainable entrepreneurship*. Emerald Group Publishing: Bingley (UK).
 - Weidinger, C., Fischler, F. & Schmidpeter, R. (2014). *Sustainable entrepreneurship: Business success through sustainability*. Springer: Berlin (Germany).

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0046): Sustainability in Entrepreneurship and Family Enterprises (Seminar, 4 SWS)

Bird M, Kaufmann L

For further information in this module, please click campus.tum.de or [here](#).

Catalogue of Elective Modules: Family Enterprise | Wahlkatalog: Family Enterprise

Module Description

MGTHN0077: Advanced International Study Trip: Family & non-family firms around the world | Advanced International Study Trip: Family & non-family firms around the world

Version of module description: Gültig ab summerterm 2024

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 9	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each student will individually work on a seminar thesis and participate in the discussion with international firms and universities. The examination consists of two parts:

1) Seminar thesis (70 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics discussed in the course
- they have understood and applied the theoretical frameworks discussed in the course
- they show that they are able to write an academic paper that follows scientific standards
- they are able to structure and write a seminar thesis based on conceptual theories

2) Participation in discussion during excursion (30 % of the overall grade).

The students should demonstrate that:

- they have gained the skills to contribute to conceptual and theoretical discussions with firms and universities
- they have gained an understanding of quantitative-empirical research work that can be applied to companies and their business models
- Active engagement in the course is expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

English oral and written fluency

Content:

Family and non-family firms need to succeed globally in order to survive and thrive in today's business environment. Thus, these firms need managers who are well-versed in international business environments. This course aims to provide students an immersive and intense international experience through a 10-day excursion abroad.

Intended Learning Outcomes:

After completing this course students should be able to:

- Have a scientific understanding on how to do business abroad
- Feel comfortable and gain proficiency in communicating and doing business abroad

Skills Objectives

- Understand theoretical frameworks that allow them to navigate different cultures
- Improve writing and written communication skills
- Gain in-depth understanding of critical thinking, ethnorelative perspectives and interpretation skills

Teaching and Learning Methods:

This module includes the following teaching and learning methods:

- Immersive learning of culture, business, and people in a different country through an excursion
- Key concepts and empirical findings from the field of strategic management of family and non-family firms will be provided through discussions with international firms and universities.
- Students will acquire additional knowledge from academic literature through their own work on their seminar theses.

Media:

PowerPoint, research papers, case scenarios

Reading List:

tba

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced International Study Trip (MGTHN0077): Family & non-family firms around the world (Seminar, 6 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0096: Advanced Seminar Management & Marketing: Corporate Social Responsibility | Advanced Seminar Management & Marketing: Corporate Social Responsibility

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of handouts and presentation: In this course, students are to independently and individually read foundational and current scientific contributions on CSR and give a presentation on them, as well as actively participate in discussions in class and create a short handout. Additionally, they are supposed to write short essays, applying the gained knowledge to a real-life example.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization, and empirical research methods.

Content:

The course will be taught using a seminar or discussion style. Students will each give a presentation on a CSR topic discussed in class. The course includes for instance different perspectives, as well as critiques, actors and drivers of CSR. Moreover, managing CSR, a global perspective on the construct, as well as further selected topics with more recent relevance in CSR research will be discussed.

Intended Learning Outcomes:

After successful completion of this course, students will (1) have gained knowledge of different perspectives of corporate social responsibility research in strategic management, (2) have learned to read, understand, explain and critically reflect on scientific literature in the field, (3) have built a basic understanding of research methods in strategic management, (4) have gained the ability to

discuss and elaborate on scientific literature on CSR in strategic management, (5) be able to apply this knowledge by analyzing CSR concepts in real-life examples.

Teaching and Learning Methods:

Students should read the assigned literature for the course and give a presentation on a topic that is derived from the literature discussed in class, also creating a short handout. In addition to the presentations students will deepen their knowledge by writing short essays, applying the concepts to real-life examples.

Media:

Powerpoint

Reading List:

Crane, A., McWilliams, A., Matten, D., Moon, J., & Siegel, D. S. (Eds.). (2008). The Oxford handbook of corporate social responsibility. OUP Oxford. Other current literature is selected specifically for each semester and will be distributed in class.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0097: CEO Leadership Series | CEO Leadership Series

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The exam is a scientific work that consists of the following two components: 1) Students will individually submit written contributions in the form of questions to the CEOs and voluntarily, in a group of 2 students, an analysis of CEOs and their companies (50%). The individual submission includes 12 questions, which need to be lecture-specific, referring to the concepts presented in the first lecture, such as leadership skills, which include, for instance, decision-making, critical thinking, or employee motivation based on theoretical frameworks that include institutional theory, upper echelon theory, and stakeholder theory. The group work is a report of approximately 500-1,500 words per group, whereby each student adds at least one chapter of analysis. This can include an analysis of leadership style, decision-making processes, organizational culture, succession planning, or the alignment of leadership strategies with overall business objectives. The scientific work examines the extent to which the students can understand leadership skills in theory and practice. It evaluates if students are able to analyze companies and leaders, as well as their behavior, and apply their knowledge of leadership by interacting with CEOs. Students will further participate in a short quiz immediately after each CEO visit to test their understanding of the topics discussed. 2) Oral participation will be graded (50%). In the second half of the lecture, which lasts approximately one hour, students ask questions to the CEOs. They are supposed to actively and individually contribute to the talk on stage. The oral participation checks that the students are able to listen and comprehend the talks and transfer and apply the concepts by asking related questions. Students will also gain practical knowledge and be able to improve their discussion- and soft skills.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization.

Content:

This module is the flagship event of TUM C.HN and will take place on Tuesdays or Fridays at 3 pm (each date will feature one guest). It covers practical insights on firm strategies, whereby lessons can be learned from CEOs themselves. The covered strategy topics may include internationalization, the analysis of the internal and external environment as well as M&A and other types of strategic alliances. This course also provides students with valuable insights for their future work as strategizers in a corporate environment.

Please note: Registration is mandatory. There will be a participants list at the entrance and police presence for each event.

The guests for the winter semester 2023/24 are e.g.: Dr. Roland Busch (CEO, Siemens AG), Christine Grotz (CEO, Weber-Hydraulik), Astrid Teckentrup (CEO, Procter & Gamble (DACH)), René Obermann (CEO, Deutsche Telekom AG (2006-2013)), Gerd Chrzanowski (CEO, Schwarz Gruppe).

Intended Learning Outcomes:

After attending this course, students will have gained knowledge on leadership skills and firm strategies in practice through attending podium discussions and actively participating in Q&A sessions with CEOs. They will be able to analyze companies and leaders, as well as their behavior, and apply their knowledge of leadership by interacting with CEOs. With this approach, students will gain practical knowledge and are additionally able to improve their discussion- and soft skills.

Teaching and Learning Methods:

In this seminar, there will be both Q&As and podium discussions with CEOs. Students will get in contact with several CEOs and managers of real-life companies, analyze their leadership skills and concepts, and interact with them in Q&A sessions. For this, the students should individually prepare and hand in questions for the CEOs as well as analyses of the companies and CEOs themselves. Moreover they should actively participate in all the sessions.

Media:

Powerpoint

Reading List:

Tbd.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0108: Start-up law for founders of family enterprises | Start-up law for founders of family enterprises

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination includes an oral report where the students will present and discuss their work for the seminar paper (30% of the grade, Presentation: 15-20 min, Questions: 15-20 min, Feedback session: 10 min). Moreover, students will prepare a seminar paper. To support students during the writing process, there will be regular discussions about the project's progression (seminar paper and regular discussions = 50% of the grade). The seminar paper contains a case study that addresses the basics of German corporate law and legal challenges associated with the formation of a startup and governance structure. On top of that, students' participation in class discussions of case studies and group work will be evaluated (20% of the grade). The active participation in the case discussions measures the students' ability to use the theoretical knowledge of the seminar to analyze real complex situations. The written examination and the corresponding presentation evaluate the student's ability to work with unknown legal problems and present a sound study incorporating the seminar's topics. In the oral report, students also show that they can summarize, academically present, and discuss the subject. Finally, ongoing discussions within group works and discussion rounds assess the student's capacity to solve a legal issue and elaborate a structured solution within a specified timeframe.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

None

Content:

- Legal structures and industries overview
- Research methods using legal databases
- Handling German legal texts

- Basics of German corporate law
- Overview of GmbH and GbR according to German law
- Startup formation overview
- Contract and agreement requirements for legal entity formation with special attention to the specific situation of family enterprises
- Financing opportunities overview
- Analysis with practical examples (e.g., Campus Founders, local startup, VC Investor)
- Presentation of own legal solutions

Intended Learning Outcomes:

The seminar covers the basics of startup law for founders of family enterprises. The students will gain a basic understanding of German corporate law and the formation process of a startup. Students will delve into a detailed study of German corporate law texts and apply them through practical examples. By the end of the seminar, they will be able to recognize legal challenges that arise during the startup process of a family enterprise. Furthermore, participants will learn how to combine their knowledge with discussing cases. Including practical case studies of newly founded family enterprises will enhance skills in argumentation and analysis, as case studies provide opportunities for debates. The seminar will also encourage the ability to develop legal solutions and justify their decisions through group work and presentations.

Teaching and Learning Methods:

The educational approach is a combination of direct instructions and oral participation of the students as well as reinforcement through interactive group activities. The students present the outcomes of their group work and engage in discussions about the topic with the other students. The practical application of knowledge to real-life cases is demonstrated using illustrative examples. Furthermore, students create detailed checklists and schemes on legal topics. The seminar will feature a presentations by relevant actors (like start-ups or lawyers) who will share their valuable insights and experiences related to legal issues in the startup process. The seminar integrates a case study with the seminar topics so students can apply what they have learned in their analysis. Students will analyze a specific case, demonstrate their understanding, and apply legal principles. Finally, students will present their work in class.

Media:

Case studies, PPT, flipcharts

Reading List:

The German Limited Liability Company, Rosengarten / Burmeister / Klein (9th ed. 2020)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Start-up law for founders of family enterprises (MGTHN0108) (Seminar, 4 SWS)

Back M, Bundrock R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0109: Taxation of Family Enterprises | Taxation of Family Enterprises

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To complete the module, participants have to demonstrate their theoretical/conceptual and practical competences regarding taxation of family business through five written exercise sheets that have to be handed in through the semester.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Basic competences in financial accounting, managerial accounting, and financial management and investing on the Bachelor level.

Content:

The course has the following content elements:• Types of taxes and their collection in the European Union. • Economics of business taxation and tax statistics. • Taxes in company valuation. • Financial accounting under German commercial law. • Income taxation under German tax law. • Tax planning. • International taxation. • Taxation of digital goods and services. • Taxation of family business succession. • Ethical aspects of tax planning and paying. • Methods of empirical tax research.

Intended Learning Outcomes:

After successfully completing the module, the participants have the following four competences:• Participants know the most important concepts and types of corporate taxes and can apply tax planning strategies. • Participants can trade-off taxation effects of different legal forms and governance systems and have particular competences related to the taxation of intergenerational succession of family enterprises. • Participants know the most important challenges of international taxation, in particular aspects related to the taxation of digital goods and services. • Participants

can critically evaluate potentials, incentives, and risks of tax-optimal accounting and managerial policies against the background of ethical principles.

Teaching and Learning Methods:

The module consists of a series of introductory lectures and practitioner-oriented case studies which are offered in blocked two-day courses through the semester.

Media:

Lecture script, case studies, and e-learning (Moodle).

Reading List:

Readings are announced in the module.

Responsible for Module:

Stich, Michael; Prof. Dr. rer. oec.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0110: Advanced Seminar Economics & Policy: Management practices in (family) enterprises | Advanced Seminar Economics & Policy: Management practices in (family) enterprises

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on management practices in (family) enterprises.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper, and (ii) oral in-class presentation. Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Content:

The seminar will cover key topics regarding management practices, such as:

1. How can management practices be classified?
2. How do management practices vary across companies?
3. How do management practices differ between family and non-family enterprises?
4. How do management practices relate to economic success?
5. How do management practices relate to work climate?
6. What are the determinants of different management practices employed in practice?

Intended Learning Outcomes:

'Students will learn the key concepts in the research field of management practices, with a special focus on family enterprises. The goal is twofold: First, to understand how management practices relate to economic success of enterprises. Second, to understand what individual and institutional factors determine management practices in enterprises. The course builds on original research on that topic. The course will cover basic theoretical consideration, but will have a clear focus on empirical studies with real-world examples and applications.

Teaching and Learning Methods:

'Students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence. A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups. We plan to complement student presentations by talks from corporate guests who will provide insights into corporate management practices.

Media:

Powerpoint slides

Reading List:

Amongst others (for detailed reading list, please see Moodle):

Bloom, N., & Van Reenen, J. (2010). Why do management practices differ across firms and countries?. *Journal of Economic Perspectives*, 24(1), 203-24.

Bloom, N., Brynjolfsson, E., Foster, L., Jarmin, R., Patnaik, M., Saporta-Eksten, I., & Van Reenen, J. (2019). What drives differences in management practices?. *American Economic Review*, 109(5), 1648-83.

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0154: Managing Legal Sustainability Requirements in Family Enterprises | Managing Legal Sustainability Requirements in Family Enterprises

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The grade is composed of written and oral performances of the participants. Students will prepare a seminar paper. To support students in the writing process, there will be regular discussions about the progression of the project. The handed in seminar paper and regular discussions, where the progression of the seminar paper is discussed make up 70% of the overall grade. The examination includes an oral presentation where the students will present and discuss the topic of their seminar paper (20% of the grade). Regarding the latter, 10% of the overall course performance is achieved through two separate course activities within two separate sessions of the seminar. 5% of the grade is accounted for the presentation of a poster tailored thematically to the session. The second required activity which accounts for 5 % of the grade is an oral presentation of a case study. The active participation in the two course activities measure the student's ability to use the conveyed legal knowledge on a wide variety of corporate sustainability topics to evaluate the internal and external implications on companies in the European Single Market. The written examination (seminar paper) and the corresponding presentation are means to evaluate the student's ability to present an overview of an unknown topic related to the topics discussed, to analyze the relevant literature in the form of studies and primary literature, and to assess the challenges and opportunities from a corporate perspective. Finally, regular discussions with the instructor measure the student's ability to distinguish subtopics of the chosen topic, locate and evaluate relevant sources and elaborate purposeful analysis from a corporate perspective.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

Introduction (Corporate) Sustainability: ESG-Dimensions

Decarbonisation, human rights in supply chains

circularity, eco-design, right to repair

Greenwashing practices, sustainability reporting, sustainable finance

Intended Learning Outcomes:

The seminar links basic knowledge of the legal framework for corporate sustainability on the European plane with the given organizational and financial structures of family enterprises in the European single market. The students gain the ability to think practical case scenarios through and draw implications for business practices. This is paired with analytical skills for literature review. Handed-out (literature) material is examined, and subsequently, critical opinions are formed and discussed in a team setting. This strengthens the students ability to distinguish the framework of mandatory legal rules from the "business case" for compliance with these rules.

Teaching and Learning Methods:

This seminar comprehensively presents the implementation of the different pillars of the European Green Deal for regulating corporate sustainability. On this basis, the implications for the companies' practices are evaluated critically. The seminar, inter alia, integrates guest lectures and case studies. Students can apply their learnings and implement them in their analysis. Students will study case studies from various industrial settings critical to "Green Transition" in the single market. Based on learned legal and regulatory elements, comparisons to the status quo and future outlook of business practices are drawn. Finally, students will present their work in class.

Media:

Case studies, exercises, PPT, Whiteboard, Videos

Reading List:

E.g.: Ditlev-Simonsen, A Guide to Sustainable Corporate Responsibility (2022), Noeth, The ESG and Sustainability Deskbook for Business (2024), Gaganis et al., Sustainable Finance and ESG (2023), Camara/Morais, The Palgrave Handbook of ESG and Corporate Governance (2022), Lykkesfeldt/KJaergaard, Investor Relations and ESG Reporting in a Regulatory Perspective (2022), Nygaard, Green Marketing and Entrepreneurship (2024), Eisenriegler (ed.), The Circular Economy in the European Union (2020)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Managing Legal Sustainability Requirements in Family Enterprises (MGTHN0154, English)
(Seminar, 4 SWS)

Bundrock R, Dowse M

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0171: Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises | Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on a specific topic related to the seminar's overall theme.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper (50%), and (ii) oral in-class presentation (50%). Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Principles of Economics (Master)

Content:

Beyond a lecture-style introduction into the topic, students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence.

A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups.

Intended Learning Outcomes:

Students will explore how technological advancements, ranging from automation to artificial intelligence, transform labor markets and the broader economy. The seminar will delve into the mechanisms behind these changes, examining their impacts on work, skills, and business structures, as well as the role of government policies in managing these transitions. Special attention will be given to comparing family and non-family enterprises.

The course builds on original research on the topic. While it will cover basic theoretical considerations, the focus will be on empirical studies with real-world examples and applications. Students will write a term paper on a selected topic and present their findings, highlighting practical implications for labor markets, policy, and business strategies.

Teaching and Learning Methods:

This seminar will cover key topics within the Economics of Technological Change and Labor Markets, with a focus on the role of family enterprises, such as:

Topic area 1: The impact of skill-biased technological change on wages and employment outcomes.

Topic area 2: Job polarization and the growth of low-skill service jobs in response to automation and routine-biased technological change.

Topic area 3: Labor market adjustments to technological innovations, including the effects of automation, robotics, and artificial intelligence.

Topic area 4: The unique challenges and opportunities for family enterprises in adapting to technological change and leveraging innovations to remain competitive.

Topic area 5: The role of government policies in managing labor market transformations and ensuring equitable outcomes.

Media:

Teaching will be in the form of a lecture and seminar presentations.

Reading List:

Acemoglu, Daron and David Autor (2011). "Skills, tasks and technologies: Implications for employment and earnings", In: Handbook of Labor Economics, Vol. 4, Elsevier, pp. 1043–1171.
Angrist, Joshua and Jörn-Steffen Pischke (2008). Mostly Harmless Econometrics (Princeton: Princeton University Press)

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0173: Project and Innovation Management: Design thinking methodology | Project and Innovation Management: Design thinking methodology

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The final grade consists of both written and oral components. Students are required to prepare a seminar paper, which accounts for 70% of the final grade. To support them throughout the writing process, there will be regular discussions to monitor the progress of their work. In addition, students will deliver an oral presentation on their seminar paper, which constitutes 20% of the grade. Furthermore, students will complete two class exercises, each accompanied by a discussion. Each exercise and its corresponding discussion will be assessed and will contribute 5% to the final grade, for a total of 10%. Active participation in the discussions and class exercises measures the student's ability to use the theoretical knowledge of the seminar to discuss and analyze real complex situations. The written examination and the corresponding presentation are means to evaluate the student's ability to work with a specific problem and incorporate the methods discussed in class to organize and present a solution to a problem. Finally, regular discussions with the instructor measure the student's ability to develop an idea, shape it, and elaborate a structured project within a given timeframe.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

This seminar presents the methodology of design thinking to generate solutions and tackle identified problems, particularly in connection with innovation initiatives in family enterprises. During the different sessions, the stages of design thinking will be covered, and we will explore how they work in connection with various types of case studies (related to technological

development, sustainability, and internal processes in enterprises). We will also explore how artificial intelligence can contribute to optimizing and enriching the application of the methodology and the generation of innovative solutions.

Intended Learning Outcomes:

1. Participants will be able to articulate the core principles of Design Thinking, including empathy, ideation, prototyping, and iteration.
2. Participants will learn to clearly define and reframe problems from a user-centered perspective, creating problem statements that guide effective solutions.
3. Participants will gain proficiency in various ideation techniques, such as brainstorming, mind mapping, and rapid sketching, to generate innovative ideas.
4. Participants will apply Design Thinking methodologies to real-world challenges, demonstrating the ability to navigate the entire process from problem definition to solution implementation.

Teaching and Learning Methods:

We will have content and application sessions where the methodology will be introduced and case studies of its application will be presented. Students will also receive suggested materials to delve deeper into the methodology after class. In the last part of the session, students will be asked to solve exercises related to the application of the methodology. Paralelly, they will work on the topics while constructing their seminar papers.

Media:

Presentation of the topics, case studies, exercises, PPT, Whiteboard

Reading List:

1. Razzouk, R., & Shute, V. (2012). What Is Design Thinking and Why Is It Important? Review Of Educational Research, 82(3), 330-348. <https://doi.org/10.3102/0034654312457429>
2. Frisendal, T. (2012). Design Thinking for Business Analysis. En Management for professionals (pp. 15-24). https://doi.org/10.1007/978-3-642-32844-2_3
3. You, X. (2022). Applying design thinking for business model innovation. Journal Of Innovation And Entrepreneurship, 11(1). <https://doi.org/10.1186/s13731-022-00251-24>.
4. Martin, R., & Euchner, J. (2012). Design thinking. Research-Technology Management, 55(3), 10-14. <https://doi.org/10.5437/08956308x5503003>
5. Rösch, N., Tiberius, V., & Kraus, S. (2023). Design thinking for innovation: context factors, process, and outcomes. European Journal Of Innovation Management, 26(7), 160-176. <https://doi.org/10.1108/ejim-03-2022-0164>
6. Carella, G., Melazzini, M., Cautela, C., & Zurlo, F. (2024). Boosting Design Thinking adoption in organisations through a game#based toolkit: A gamified approach in building facilitators to overcome Design Thinking adoption barriers. Creativity And Innovation Management. <https://doi.org/10.1111/caim.12627>
7. Docherty, C., Mazzei, M., & Steiner, A. (2023). Exploring the impact of design thinking on social enterprise mission-aligned innovation. Entrepreneurship And Regional Development, 1-17. <https://doi.org/10.1080/08985626.2023.2286506>

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0014: Corporate Campus Challenge | Corporate Campus Challenge

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students will present a prototype and business case in a final presentation. Furthermore, they will write a report which will be examined. The final grade is averaged as follows:

60 %: Final presentation

40 %: Scientific paper (critical reflection on the prototype development process by using relevant scientific literature)

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

- Customer needs
- Research & empathic exploration
- Value proposition
- Design thinking methodology
- Business model and ecosystem modeling
- Rapid prototyping
- Scientific reasoning and writing

Intended Learning Outcomes:

At the end of the module, students are able to

- apply methods of idea generation and need-finding to stimulate creativity and recognize business opportunities
- create design prototypes in order to demonstrate their proposed solutions and gather feedback

- create business plans, presentations, and video prototypes in order to effectively communicate business value
- evaluate their ideas by involving peers, academics, and industry partners
- develop and document their ideas in a scientific seminar paper

Teaching and Learning Methods:

In this course, students will work in interdisciplinary teams to develop innovative solutions for challenges provided by industry partners. Over the course of 8 weeks, students will receive coaching sessions, individual mentoring, tutorials, and practical courses on the use of machines and equipment (3D printer, laser cutter, sensors, etc.). Students will present their first results through a presentation and a live prototype to gather feedback. At the end of the course, a working prototype and the business case will be presented during a final presentation.

Media:

Slides, scientific papers, handbooks, physical events

Reading List:

Lewrick, M., Link, P., & Leifer, L. (2018). The design thinking playbook: Mindful digital transformation of teams, products, services, businesses and ecosystems. John Wiley & Sons.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0028: Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship | Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic related to corporate entrepreneurship. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade, approx. 10 pages) The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade, approx. 10 minutes). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis).

Repeat Examination:

(Recommended) Prerequisites:

Basic knowledge of corporate entrepreneurship and research methodologies is recommended.

Content:

Foundations of Corporate Entrepreneurship (CE): Key concepts, definitions, and the role of CE in modern businesses.

Strategic Thinking in CE: Developing an entrepreneurial mindset to identify and leverage opportunities.

Implementing CE: Steps to integrate entrepreneurial practices into corporate operations.

Building an Entrepreneurial Organization: Creating a culture that fosters innovation and risk-taking.

Sustaining CE: Strategies to maintain entrepreneurial energy and adapt to changing markets.

Leadership and Culture: The impact of leadership styles and organizational culture on entrepreneurship.

Entrepreneurial Traits and Behavior: Key personal characteristics that support entrepreneurial initiatives.

Creativity and Motivation: Harnessing creativity, motivation, and HRM practices to drive innovation.

Incentive Systems and Psychological Ownership: Enhancing engagement through effective incentives and ownership.

Entrepreneurial Intensity: Measuring and increasing the level of entrepreneurial activity within corporations.

Focus Areas of CE: Examining CE at the levels of the phenomenon, organization, individual, and dynamic interactions.

Challenges in CE: Addressing obstacles faced by corporations and strategies to overcome them.

Intended Learning Outcomes:

Understand the current state of research in corporate entrepreneurship.

Develop the ability to critically analyze and synthesize academic literature.

Improve academic writing and presentation skills.

Identify research gaps and formulate theoretical and practical implications.

Teaching and Learning Methods:

In-presence seminar sessions

Literature review and independent research.

Group discussions and collaborative learning (optional group work).

Interactive feedback sessions.

Presentation of results and peer review

Media:

Moodle platform for resources and updates.

Academic databases for literature access.

Presentation tools (e.g., PowerPoint, posters)

Reading List:

Core reading will be provided on Moodle. Students are encouraged to explore additional academic articles and books relevant to their selected topic.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0029: Advanced Seminar Innovation & Entrepreneurship: Managing the Family Enterprise | Advanced Seminar Innovation & Entrepreneurship: Managing the Family Enterprise

Version of module description: Gültig ab summerterm 2020

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work in a group on a specific topic related to managing the family enterprise. The examination consists of two parts:

1) Projekt thesis (75 % of the overall grade) The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to analyze effectively the problem stated in their work and know how to structure and write a project thesis.

2) Presentation of the project thesis (25 % of the overall grade). Students should demonstrate that:

- they have gained the skills to present their project thesis to an academic audience.
- they are able to answer questions related to specific parts of their work.

The final grade will be based on the two parts (75 % project thesis and 25 % presentation and discussion of the seminar thesis).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Good English language skills are required.

Content:

This course deals with the management of family firms. Given that the majority of firms around the world are family businesses (ranging between 75-90 %), it is crucial to understand better the particularities of family businesses. Given that the boundaries between the family and the business

overlap, certain challenges may arise that makes family businesses unique. The objective of this course is to better understand the challenges and opportunities family firms face and how managers, owners, and advisers can best deal with them. By combining various theoretical frameworks with hands-on examples, we will elaborate on how to manage family enterprises successfully. We will cover topics such as:

- strength and weaknesses in family firms,
- succession and governance issues in family enterprises,
- management of key resources in family family firms,
- financial management in family firms and,
- transgenerational entrepreneurship.

Intended Learning Outcomes:

- Understand the family business phenomenon and strengths and weaknesses of family firms.
- Understand the importance of governance in family firms. Evaluate how different types of governance suit the needs of family firms differing in size.
- Reflect on how family firms use the “family” as a strategic asset and how do they manage those resources best. How do different institutional contexts influence the emergence of family firms?
- Reflect on financial management in family firms. Evaluate how does the family background bias financial behaviour, such as investment, financing, and risk taking?
- Know and apply different concepts of transgenerational entrepreneurship and be able to advise family firms how they can sustain in the long run.
- Reflect on governance and succession topics.
- (1) the search, understanding, and analysis of academic literature, (2) methods for data collection and data analysis, (3) the structuring and writing of a paper, and (4) giving presentations including a subsequent discussion

Teaching and Learning Methods:

- Through lectures, supported by a presentation, the instructor will provide the theoretical foundations on managing the family enterprise.
- A series of group discussions and exercises as well as exchanging ideas, questions, and comments will encourage an experiential learning atmosphere.
- Guest speakers will share practical insights and will strengthen the understanding of key concepts and processes, and expose perspectives complementary to the topic of the seminar.
- Within their group work, students identify challenges and develop recommendations. This is supported by feedback sessions with the instructor.
- In a presentation, students present their groupwork.

Media:

Powerpoint slides, Zoom etc.

Reading List:

Amongst others (for detailed reading list, please see Moodle):

- Aldrich, H. E., & Cliff, J. E. (2003). The pervasive effects of family on entrepreneurship: Toward a family embeddedness perspective. *Journal of Business Venturing*, 18(5), 573-596.
- Bird, M. & Zellweger, T. (2018): Relational embeddedness and firm growth: Comparing spousal and sibling entrepreneurs. *Organization Science* 29 (2), 264-283.
- Schulze, W. S., Lubatkin, M. H., Dino, R. N., & Buchholtz, A. K. (2001). Agency relationships in family firms: Theory and evidence. *Organization Science*, 12(2), 99-116.
- Sirmon, D. G., & Hitt, M. A. (2003). Managing resources: Linking unique resources, management, and wealth creation in family firms. *Entrepreneurship Theory and Practice*, 27(4), 339-358.
- Zellweger, T. (2017): *Managing the Family Business*. Edward Elgar Publishing: Massachusetts.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0029): Managing the Family Enterprise
(Seminar, 4 SWS)

Bird M, Kosberg M

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0037: Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations | Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 146	Contact Hours: 34

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The final grade consists of both written and oral components. Students are required to prepare a seminar paper, which accounts for 70% of the final grade. To support them throughout the writing process, there will be regular discussions to monitor the progress of their work. In addition, students will deliver an oral presentation on their seminar paper, which constitutes 20% of the grade. Furthermore, students will complete two class exercises, each accompanied by a discussion. Each exercise and its corresponding discussion will be assessed and will contribute 5% to the final grade, for a total of 10%. The seminar paper and the corresponding presentation are a means to measure the student's ability to understand a scientific subject, evaluate literature, and develop, conduct, and analyze questionnaires/surveys. By doing a presentation, students show that they can summarize the subject, present it to an audience, and to conduct a discussion about the presented subject. Regular discussions with the instructor measure the student's ability to develop an idea from initial concepts to the complete picture within a given timeframe. Participation in class exercises measures the students ability to apply their theoretical knowledge in practice and to reflect on it afterwards.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

Basic terms of contract negotiations (like BATNA and ZOPA etc.);
Negotiation strategies and concepts (Harvard negotiation concept, win-win, win-lose etc.);
Effects of behavioural economics and negotiation tactics based on it (anchoring effect etc.);

Negotiation tactics (based on deception, pressure and defensive tactics etc.);
Principal agent problem in negotiations;
Communication (question techniques, answer techniques, argumentation techniques);
Emotions (core concerns framework);
Gender aspects in contract negotiations.

Intended Learning Outcomes:

Students learn the economic and psychological basics of contract negotiations as well as important communication tools for negotiations. They know the Harvard negotiation concept, including its weaknesses and the most important strategies in contract negotiations. They are familiar with the essential tactics. Students know how to apply these tactics or, respectively, how to react to these tactics if applied by the negotiation partner. They can distinguish different negotiation styles. Students have experienced the functioning of individual tactics during negotiation simulations and are able to understand the dynamics underlying a specific negotiation.

In the area of interdisciplinary competences, students strengthen their communication and argumentative skills (especially through the negotiation simulations) and their ability to work in a team (especially through group work in the context of presentations and negotiation simulations in a team).

Teaching and Learning Methods:

This module is held as an interactive seminar/lecture. Negotiation simulations and games are integrated into the course so that students learn to implement tactics and strategies. The negotiation games are designed to simulate practice. Students are encouraged to actively participate in the negotiation simulations and to get involved in the subsequent discussions. Video recordings are used to reflect on what has been experienced. Moreover, students will do research on a specific research question and write a seminar paper. In this framework, students will have to perform research of reference materials, design a questionnaire, conduct interviews with negotiators and analyse the answers. In order to support the students in their work individual appointments will be offered. Students will present their work in class.

Media:

Simulations, Exercises, Videos, Self-Tests, PPT, Whiteboard

Reading List:

Jung/Krebs, The Essentials of Contract Negotiation (2019)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations (WIHN0037)
(Seminar, 4 SWS)
Gelvez Alvarez L, Jung S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0046: Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises | Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises

Version of module description: Gültig ab winterterm 2020/21

Module Level: Master	Language: English	Duration: one semester	Frequency:
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic in the field of sustainable entrepreneurship / sustainable management in family enterprises. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis). Active engagement in the course will be expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English

Content:

This course addresses various aspects of sustainability available to new ventures and established family enterprises. Sustainable entrepreneurship (SE) emphasizes that entrepreneurs may pursue additional goals such as social and ecological goals that may alleviate social and ecological problems. It explores the opportunities that entrepreneurs create, the challenges entrepreneurs encounter, and the ways in which entrepreneurs exploit opportunities to contribute toward enhancing the society while simultaneously starting and sustaining a profit-oriented business. This course also examines the role of sustainability in family enterprises:

Course content will comprise:

- The importance of sustainable entrepreneurship in solving grand societal challenges of the 21st century
- Motives and barriers of sustainable entrepreneurship
- Overview of theories and concepts in sustainable entrepreneurship
- Stakeholder management for entrepreneurs
- Sustainable management in the context of family enterprises
- Sustainability in entrepreneurship and family enterprises in the context of developing countries
- Recognizing and exploiting social and ecological opportunities
- Practical examples of sustainable ventures

Intended Learning Outcomes:

After completing the seminar students, should understand how entrepreneurs and family enterprises can contribute to solving societal challenges. Therefore, students will be able to:

- understand the concept of sustainable entrepreneurship and sustainability in family enterprises
- understand and evaluate how entrepreneurs can enact on sustainable opportunities
- understand the entrepreneurs' motives and barriers to engage in sustainable entrepreneurship
- reflect on the importance of diversity in achieving sustainability in entrepreneurship and family enterprises
- evaluate strategic actions of sustainable enterprises
- evaluate hands-on examples and cases of how sustainable entrepreneurship can be implemented in practice with specific focus on the Heilbronn region: Those examples may relate to various sectors such as the public governance sector, the food and agriculture, environmental and mobility sector.

In addition, students will be able:

- to search, understand, and analyze academic literature by evaluating key findings and explaining complex research studies in a comprehensible and interesting manner,
- to apply methods for data collection and data analysis, - to structure and write a seminar thesis, and
- to present and subsequently discuss findings of their seminar thesis.

Teaching and Learning Methods:

' -Through lectures, supported by Power-Point presentations, the instructor will provide the theoretical foundations of sustainable entrepreneurship

- The content is discussed in the course by exchanging ideas, questions, and comments to encourage an open learning atmosphere
- Every sessions contains exercises, where the students apply their learnings in a group work (e.g. to present one start-up, enterprise or organization that has already successfully implemented the topic/concept presented in the lecture)
- Guest speakers will share practical insights and will strengthen the understanding of key concepts and processes and expose perspectives complementary to the topic of the seminar
- For their seminar thesis, students will investigate a topic within the subject of this course to develop a seminar thesis. Students will receive feedback from the instructor.
- In a final presentation, students present the results of their project thesis

Media:

Powerpoint, Zoom- & Breakout-Sessions, Online Simulations

Reading List:

Basic literature (for detailed reading list, see Moodle):

- Cruz, C.; Larraza-Kintana, M. Garcés-Galdeano, L. Berrone, P. (2014). Are family firms really more socially responsible? *Entrepreneurship Theory and Practice* 38(6), 1295–1316.
- Gast, J., Gundolf, K., & Cesinger, B. (2017). Doing business in a green way: A systematic review of the ecological sustainability entrepreneurship literature and future research directions. *Journal of Cleaner Production*, 147, 44-56.
- Le Breton-Miller, I., & Miller, D. (2016). Family firms and practices of sustainability: A contingency view. *Journal of Family Business Strategy*, 7(1), 26-33.
- Lumpkin, G. T., Bacq, S., & Pidduck, R. J. (2018). Where change happens: community#level phenomena in social entrepreneurship research. *Journal of Small Business Management*, 56(1), 24-50.
- Patzelt H., & Shepherd D.A. (2001) Recognizing opportunities for sustainable development. *Entrepreneurship Theory and Practice*. 35(4), 631-652.
- Sarason, Y., & Dean, T. J. (2019). Lost Battles, Trojan Horses, Open Gates, and Wars Won: How entrepreneurial firms co-create structures to expand and infuse their sustainability missions in the acquisition process. *Academy of Management Perspectives*, 33(4), 469-490.
- Sarasvathy, S. D., & Ramesh, A. (2019). An effectual model of collective action for addressing sustainability challenges. *Academy of Management Perspectives*, 33(4), 405-424.
- Shevchenko, A., Levesque, M. & Pagell, M. (2016). Why firms delay reaching true sustainability. *Journal of Management Studies* 53, 911-35.
- Zahra, S. A., Gedajlovic, E., Neubaum, D. O., & Shulman, J. M. (2009). A typology of social entrepreneurs: Motives, search processes and ethical challenges. *Journal of Business Venturing*, 24(5), 519-532.
- Lumpkin, T. & Katz, J. (2011). *Social and sustainable entrepreneurship*. Emerald Group Publishing: Bingley (UK).
- Weidinger, C., Fischler, F. & Schmidpeter, R. (2014). *Sustainable entrepreneurship: Business success through sustainability*. Springer: Berlin (Germany).

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0046): Sustainability in Entrepreneurship and Family Enterprises (Seminar, 4 SWS)

Bird M, Kaufmann L

For further information in this module, please click campus.tum.de or [here](#).

Specialization in Management: Digital Enterprises | Management-Schwerpunkt: Digital Enterprises

Advanced Seminar Digital Enterprises | Advanced Seminar Digital Enterprises

Module Description

MGTHN0099: Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy | Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The seminar prepares students to make tactical choices supporting organizational performance and strategy. The seminar enables students to understand the relationships between operations and supply chains and other areas of the organizations and to design reliable, fast, and efficient operations and supply chains.

Students will be guided to learn critically analyzing value chain reports and to identify problems and areas for potential improvements. Later, students will demonstrate their ability to apply these skills in a simulated value chain setting. After completing this seminar, students can identify value chain challenges, identify potential improvements, coordinate critical decisions across functional areas, and forecast results from the solutions implemented. As such, this seminar will prepare students for applying academic knowledge to industry practice.

Central to the seminar is a simulation consisting of several rounds. Students will be assigned to groups of four participants and have to formulate a strategy for their company. During each round, they will be required to make tactical and strategic choices to maximize value and prepare a four-page summary justifying the choices made and the expected results. This summary is part of the overall assessment (75% of the final grade, team grade). This assessment method helps students reason about the options available in each round, make informed decisions, develop predictive capabilities, work together in a team, and condense necessary information formally.

Furthermore, students must submit a four-page report at the end of the course, including a critical reflection about teamwork and individual takeaways (25% of the final grade, individual grade). This assessment method allows students to reflect on their capabilities to work in a team and strengthen their personal learnings from the seminar.

At the beginning of the course, students will be advised on the report structure and submission deadlines.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Desirable but not mandatory: Production and Logistics, Management Science

Content:

Subject-specific content

- Strategic Alignment
- Value Chain
- Supply Chain Tactical Decisions
- Supply Chain Management
- Sales and Operations Planning

Intended Learning Outcomes:

Upon completion of the module, students are able to:

- understand the need to construct operations and supply chains that align with corporate-level strategies and identify appropriate actions
- analyze the relationships between operations and supply chain management and other areas of the organization (e.g., marketing, finance, accounting, engineering, R&D, etc.)
- evaluate the feasibility of functional-level actions with corporate-level strategies and assess the impact of their choices on the organization's performance
- coordinate actions and address competing interests across functional areas
- evaluate alternative tactical decisions and convince with compelling reasoning for their choice of action
- critically reflect upon their ability to work in a team and provide advanced improvement recommendations for effective teamwork

Teaching and Learning Methods:

Experiential learning

The module is designed to enable students to acquire knowledge on current (behavioral) operations and supply chain management topics through hands-on experience and reflection. To this aim, a web-based business simulation game will be employed. The business simulation game will engage participants in turning around a manufacturer of fruit juices by making tactical and strategic decisions on four critical areas of the organization: Purchasing, Operations, Supply Chain, and Sales. Students are required to analyze key performance indicators, define a strategy

for turning around the organization, identify potential improvements, and make tactical decisions in each of the organization's four areas.

Seminar sessions take place physically, and the methods combine mini-lectures, hands-on experience, group reflection, and questions and feedback sessions. Each week, there will be two 90-minutes sessions that will allow for feedback, sparring sessions, and the introduction of new concepts.

Students will get an overview of the course in the first two weeks. They will be informed about the examination and expectations. Furthermore, students will be introduced to highly recognized supply chain strategy and integration papers. Students will also see examples and learn to interpret data from sales, operations, supply chain, and purchasing reports.

After setting the theoretical scene, students will form small groups of size 4 (groups of smaller size might also be formed depending on the total number of participants), and they will start working on the business simulation game. The simulation advances over several rounds and students must present their analysis, evaluation, decisions, and expected results. Further details will be shared during the seminar.

Media:

Research papers and further material to be shared via Moodle, in-class discussions

Reading List:

Readings for general preparation:

Myerson, P. (2012). Lean supply chain and logistics management. McGraw-Hill.

Croson, R., Donohue, K. (2006). Behavioral causes of the bullwhip effect and the observed value of inventory information, *Management Science*, 52(3), pp. 323-336. DOI: 10.1287/mnsc.1050.0436

Flynn, B.B., Huo, B. and Zhao, X. (2010), The impact of supply chain integration on performance: A contingency and configuration approach. *Journal of Operations Management*, 28: 58-71. DOI: 10.1016/j.jom.2009.06.001

Frohlich, M.T., Westbrook, R. (2001). Arcs of integration: an international study of supply chain strategies. *Journal of Operations Management*, 19, pp. 185-200. DOI: 10.1016/S0272-6963(00)00055-3

Further materials will be discussed in the session.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0115: Advanced Seminar: Digital Technologies in Operations Management | Advanced Seminar: Digital Technologies in Operations Management

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The key idea of this seminar is to lead students towards academic work and help prepare them for their master's thesis. In addition, they will learn about recent developments in digital technologies. Accordingly, the coursework and examinations are structured.

During the semesters, students will prepare two short presentations on research methods and skills. They will work in small teams. Each of the two presentations (ca. 45 minutes) accounts for 10% of the final grade and is a team grade.

Upon completion of this module, students will demonstrate their ability to cope with advanced research studies, comprehend and assess study outcomes, compare contributions of several studies, and transfer theoretical concepts to practice through a seminar paper (50% of the final grade; length of 15 pages including references). This method requires students to assess and replicate a study themselves.

In addition, they will prove their ability to communicate even complicated relationships and methods to their peer students through a presentation in a comprehensible fashion. They will further guide and moderate an ensuing discussion throughout which they will demonstrate their ability to criticize and assess innovative approaches and their potential shortcomings. The presentation, including moderation of ensuing discussion, accounts for 30% of the final grade (presentation duration 30 minutes + 30 minutes discussion). Complementing the written seminar paper, the presentation is targeted toward students who have not studied the same paper; this poses the challenge of presenting theoretical work interestingly while breaking down complex relationships into understandable information without losing rigor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Management Science, Production and Logistics

Content:

Subject-specific content

- Digital technologies potentially including but not limited to AR, VR, AI, Blockchain, and IoT.
- Management focus
- Incentive conflicts that arise when introducing new technologies in operations management
- Behavioral operations management

Methodological content (depending on assigned papers and limited to an introductory level)

- Research Design
- Analytical Modelling
- Game Theory
- Event Study Method
- Econometrics

Intended Learning Outcomes:

Upon completion of the module, students can

- understand and analyze state-of-the-art approaches to digital technologies in operations management,
- understand selected research papers and evaluate their key findings,
- present and explain complex research studies in a comprehensible and engaging manner,
- evaluate suggested management solutions in the digital technologies in an operations management context and assess their strengths and weaknesses,
- create management recommendations based on recent academic studies.

Teaching and Learning Methods:

Seminar

The seminar consists of two parts. During the semester, we will meet regularly to discuss select papers on digital technologies in operations management. Students will also explore and present information on select research methods (e.g., econometrics, game theory, and behavioral experiments). This part is complemented by a few interactive lectures.

At the end of the semester, we several sessions take placed blocked where students will present their assigned research studies (final assessment).

Media:

Research papers (to be shared via Moodle), student presentations, further material to be shared via Moodle, online discussions

Reading List:

Readings for general preparation (i.e., what is research? What marks a contribution?):

Wacker, J. G. (1998). A definition of theory: research guidelines for different theory-building research methods in operations management. *Journal of operations management*, 16(4), 361-385.

Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of management review*, 14(4), 490-495.

Examples for typical papers that we will discuss (this list is not meant to be comprehensive but gives some indication of topics to be covered):

Babich, V., & Hilary, G. (2020). OM Forum—Distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management*, 22(2), 223-240.

Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113-122.

Further papers will be provided at the beginning of the semester.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0123: Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals | Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals

Version of module description: Gültig ab summerterm 2023

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To successfully pass the module, the participants have to demonstrate methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations. The participants proof these competences through several presentations with written reports and active participation in discussions.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

No prerequisites required.

Content:

Artificial intelligence (AI) is one of the most profound ongoing technological developments which has the potential to strongly impact various fields of business and society. This module critically analyzes the effects of AI-applications on the sustainable development goals (SDGs) of the United Nations. Units of the module: Political and conceptual background of the SDGs; methods to assess impacts on the micro-, meso-, and macro-level; discussion of supportive and contradictory impacts of specific AI-applications on SDGs.

Intended Learning Outcomes:

After successful completion of the module, the participants have methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations.

Teaching and Learning Methods:

The module consists of an interactive introductory lecture and presentations of the participants with topic-specific written reports.

Media:

Textbook, presentation slides, and e-learning (Moodle).

Reading List:

Dodds, Donoghue und Leiva Roesch. 2016. Negotiating the sustainable development goals: A transformational agenda for an insecure world.

Kanie und Biermann. 2017. Governing through goals: Sustainable development goals as governance innovation.

Saetra. 2022. AI for the sustainable development goals.

Sobti und Sharma. 2018. An introduction to sustainable development goals.

Walker, Pekmezovic und Walker. 2019. Sustainable development goals: Harnessing business to achieve the SDGs through finance, technology, and law reform.

Responsible for Module:

Stich, Michael; Prof. Dr. rer. oec.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0177: Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations | Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1. Group Report (70%)

- **Assessment Details:** In groups, students will collaboratively submit a report (no longer than 5 pages) addressing a specific managerial problem and proposing a solution using VR/AR technologies. Students are allowed to use reference materials and external resources.
- **Scope:**
 - Identify a relevant managerial problem in production, quality inspection, remote support, human resources, or customer experience.
 - Justify the importance of solving this problem.
 - Propose a detailed VR/AR-based solution, including technical implementation steps, management implications, and expected outcomes.
- **Rationale for Assessment:**
 - This assessment evaluates the ability to apply theoretical knowledge to practical problems, foster teamwork, and use technology to prepare solutions with managerial perspectives.
 - The report format develops students' ability to communicate complex ideas concisely and effectively.

2. Individual Critical Reflection (30%)

- **Assessment Details:** Each student will write a critical reflection on the relevance of immersive technologies (VR/AR) in solving managerial challenges. This will be submitted as a separate document, no longer than 2 pages. The reflection must be written independently.
- **Scope:**
 - Reflect on the practical and theoretical insights gained during the module.
 - Critically analyze the advantages and limitations of VR/AR in managerial contexts.

- Discuss how the experience of creating the group report has influenced their understanding of technology's role in management.

- **Rationale for Assessment:**

- This assessment allows students to self-evaluate, critically linking their learning to broader managerial and technological contexts.

- It promotes independent thought, allowing students to articulate their unique perspectives.

Repeat Examination:

(Recommended) Prerequisites:

Basic Programming with C#

Content:

In the module, the students will gain knowledge about the technological and management aspects of VR implementations with hands-on experience and use case examples in

- Production and Assembly
- Quality Inspection and Remote Support
- Training and skill development
- Customer Experience and Prototyping

Technology-specific content

- Introduction to Immersive Technologies (key hardware and software platforms)
- Using Unity for developing VR/AR applications

Management-specific content

- Ethical Considerations
- Behavioral Challenges
- Building a roadmap from prototype to market-ready product

Intended Learning Outcomes:

By the end of this module, students will be able to:

- Understand the fundamental principles of VR and AR technologies.
- Analyze the role of immersive technologies in improving managerial and operational efficiency.
- Design and prototype VR/AR-based solutions for specific management scenarios.
- Critically evaluate the broader implications of VR/AR deployment, including ethical concerns (e.g., data privacy, accessibility).
- Work collaboratively in teams to identify managerial challenges and develop VR/AR solutions.

Teaching and Learning Methods:

The coursework is primarily practical. At the beginning of the module, the students will be introduced to the immersive technologies and different headsets. They will be introduced to Unity Game Development Engine, which they will use to develop application solutions. The students

can choose between VR or AR applications and must program their software solution, keeping a managerial focus in mind.

The students will present the problem statement (in a group) that they want to work on and will have regular feedback sessions during the development phase. The students will present their solutions at the end of the module.

Media:

Material will be shared via Moodle

Reading List:

Readings for general preparation (but not limited to):

Dalton, J. (2021). Reality check: How immersive technologies can transform your business. Kogan Page Publishers.

Jung, T., & Dalton, J. (2021). XR Case Studies. Springer International Publishing.

Wuttke, D., Upadhyay, A., Siemsen, E., & Wuttke-Linnemann, A. (2022). Seeing the bigger picture? Ramping up production with the use of augmented reality. *Manufacturing & Service Operations Management*, 24(4), 2349-2366.

Windhausen, A., Heller, J., Hilken, T., Mahr, D., Di Palma, R., & Quintens, L. (2024). Exploring the impact of augmented reality smart glasses on worker well-being in warehouse order picking. *Computers in Human Behavior*, 155, 108153.

Seeliger, A., Cheng, L., & Netland, T. (2023). Augmented reality for industrial quality inspection: An experiment assessing task performance and human factors. *Computers in Industry*, 151, 103985.

Rejeb, A., Keogh, J. G., Leong, G. K., & Treiblmaier, H. (2021). Potentials and challenges of augmented reality smart glasses in logistics and supply chain management: a systematic literature review. *International Journal of Production Research*, 59(12), 3747-3776.

Eswaran, M., & Bahubalendruni, M. R. (2022). Challenges and opportunities on AR/VR technologies for manufacturing systems in the context of industry 4.0: A state of the art review. *Journal of Manufacturing Systems*, 65, 260-278.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations (MGTHN0177) (Seminar, 4 SWS)

Phadke V

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0181: Advanced Seminar Operations & Technology: Analytics in Manufacturing Management | Advanced Seminar Operations & Technology: Analytics in Manufacturing Management

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours:	Contact Hours:

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

This seminar aims to guide students toward academic research and prepare them for their master's thesis. The topics focus on advanced analytics in manufacturing management, covering AI methods, modeling techniques, optimization algorithms, and rigorous analysis of algorithmic properties, such as performance guarantees. While students are not expected to master all these techniques, they should be willing to develop deep expertise in one and understand its relevance for manufacturing decision-making. Additionally, they should be able to present their findings clearly and didactically to their peers.

Students work in small teams and complete two presentations and a term paper (~15 pages). The first presentation (30 minutes, 20% of the grade) provides a structured overview of a recent research article and formulates a research question for the seminar thesis. The second presentation (30 minutes, 30% of the grade) is the final presentation, showcasing their research findings. The term paper accounts for 50% of the grade.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Foundations in statistics and management science

Content:

This course focuses on advanced analytics in manufacturing, emphasizing algorithms, modeling, simulations, and theoretical analysis. It is open to all advanced analytics methodologies, provided they meaningfully support manufacturing decision-making.

Additionally, the course integrates practical experience by incorporating potential talks from industry practitioners and potential excursions to manufacturing facilities.

Intended Learning Outcomes:

Students will develop a comprehensive understanding of the scientific process, further develop critical thinking skills, and the ability to solve business problems efficiently. They will gain firsthand experience in manufacturing decision-making through practitioner interactions and engagement with cutting-edge research on data-driven manufacturing. Additionally, they will acquire hands-on expertise in advanced analytics methods applied to manufacturing decision-making.

Teaching and Learning Methods:

Teaching methods include presentations, interactive learning, e-learning, group discussions, and (if feasible) discussions and presentations by practitioners. Additionally, if feasible, we aim to include excursions to manufacturing and related facilities, as well as interactions with leading scientists in the field.

Media:

Zoom, literature databases (Ebsco, ScienceDirect etc.), standard solvers (Gurobi, Hexaly etc.), databases and case studies, programming environments and software development platforms, simulation environments etc.

Reading List:

We focus on high-profile and influential ideas in advanced analytics for manufacturing, typically published in leading journals such as Operations Research, Management Science, Manufacturing & Service Operations Management, INFORMS Journal on Optimization, European Journal of Operational Research, and INFORMS Journal on Computing. Below are examples of relevant articles:

[AI] Senoner, J., Netland, T., & Feuerriegel, S. (2022). Using explainable artificial intelligence to improve process quality: Evidence from semiconductor manufacturing. *Management Science*, 68, 5704-5723.

[mathematical analysis] Drozdowski, M., Jaehn, F., & Paszkowski, R. (2017). Scheduling position-dependent maintenance operations. *Operations Research*, 65, 1657-1677.

[modeling] Möhring, R., Schulz, A. S., Stork, F., & Uetz, M. (2003). Solving project scheduling problems by minimum cut computations. *Management Science*, 49, 330-350.

Da Col, G., & Teppan, E. C. (2022). Industrial-size job-shop scheduling with constraint programming. *Operations Research Perspectives*, 9, 100249.

[algorithmics] Koca, Y., Martagan, T. G., Adan, I. J. B. F., Maillart, L., & van Ravenstein, B. (2023). Increasing biomanufacturing yield with bleed-feed: Optimal policies and insights. *Manufacturing & Service Operations Management*, 25, 108-125.

Barnhart, C., Jacquillat, A., & Schmid, A. (2024). Robotic warehousing operations : A learn-then-optimize approach to large-scale neighborhood search. *INFORMS Journal on Optimization*, 1-24.

Responsible for Module:

Otto, Alena; Prof. Dr. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0013: Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance | Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises a scientific coursework (approx. 25-30 pages), which is supposed to address the research questions posed within each given seminar topic. The task is to be conducted as a group work (i.e. 2-3 students per topic). Additional to the written scientific report, candidates are required to showcase main findings within a 30-minute presentation and answer further questions following their presentation.

In doing so, the candidates are required to prove that they have dealt with the given topic in a comprehensive manner, that their selection and evaluation of literature is adequate and that they have in-depth analyzed the topic while simultaneously putting it in a higher context of financial research. Also, it can be assessed whether the students are able to answer further questions in a correct and comprehensive manner.

The written assignment will be weighted 2/3 and the formal presentation will make 1/3 of the final grade. Each student will be assessed separately.

To maintain the seminar atmosphere, the number of participants is generally limited. If the course is offered in a hybrid format, Munich students may also participate in the course. The targeted maximum number of participants is 12. More information will be provided in the introductory session.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;

basic knowledge in scientific writing;
combination with the module "Digital Finance" is recommended

Content:

Candidates are supposed to build groups of 2-3 students and choose one seminar topic out of a set of pre-defined scientific issues. The pre-selection omprises, but is not limited to topics of the following areas:

- Asset Pricing of Cryptocurrencies: Market (In-)Efficiency, Behavioral Finance & Bubbles;
- Financial Innovations: Foundations and Opportunities & Threats to Financial Market Stability;
- Wisdom of the Crowd? Information Asymmetries, Adverse Selection, and Moral Hazard in Peer-to-Peer Lending Markets Around the World;
- Robo-Advisors: Overview of Services, Benefits, and Limitations;
- Social Investing: Overview, Benefits, and Limitations;
- Machine Learning: An Overview of ML Applications for Finance;
- etc.

Candidates are expected to conduct a comprehensive literature review of the most important scientific articles capturing the chosen seminar topic in order to address the given research questions in a systematic and structured manner.

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- describe the effects of current FinTech developments for existing financial products and processes;
- exhibit a comprehensive overview on current research topics in the field of Digital Finance;
- identify chances and opportunities of these developments (especially with regard to the overall macroeconomic stability);
- conduct an independent literature analysis at the highest possible, international level;
- analyze and evaluate references in a systematic manner;
- combine topics in Digital Finance with theories/methods/models of traditional finance in an independent manner;
- deriver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise an initial lecture (overview course) as well as the corresponding main seminar;

Methods of teaching include a group work and the formal presentation of obtained results;

the learning methods of the students primarily comprise the following activities:

- independent literature research (usage of scientific articles published in international top journals);
- Collaborative writing of a scientific report;
- Exercise of a deductive, logic and consistent argumentation to specifically address and answer the posed research questions;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to conduct independent academic work and to elaborate thematically complex contents on their own. It is considered to be a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

Machi, L. A., & McEvoy, B. T. (2012). The literature review: Six steps to success (2nd ed.). Corwin Press.

A separate reading list for each topic will be provided in the introductory lecture.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Finance & Accounting (WIHN0013): Current research Topics in Digital Finance (Seminar, 4 SWS)

Gong Z, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0017: Advanced Seminar Innovation & Entrepreneurship: Digital Innovation | Advanced Seminar Innovation & Entrepreneurship: Digital Innovation

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Seminar paper with presentation: The students independently and individually prepare a scientific seminar paper and present it during the course.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Very good English language skills are required. Students should also have an understanding of basic principles of management, innovation, and organization as well as empirical research methods. Attendance of the course Empirical Research Methods in Management and Economics recommended, but not mandatory.

Content:

The course prepares students for their Master's Thesis by introducing them to (1) the search, understanding, and analysis of academic literature, (2) methods for data collection and data analysis, (3) the structuring and writing of a paper, and (4) giving presentations to an academic audience including a subsequent discussion. The course exposes students to digital platforms and open forms of organizing innovation.

Intended Learning Outcomes:

After the course students will be able to:

- Define, explain and apply selected key concepts in the field of digital innovation
- Read and understand academic literature
- Write and present an academic paper

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via presentations and group discussions
- Build up critical thinking and interpretation skills

Learning Objectives

At the end of this course, students will be able to demonstrate understanding, critical assessment and application of the following:

- Learn how to evaluate academic literature
- Interact with an academic debate
- Prepare, elaborate and defend an academic argument
- Specialist knowledge on their selected topic of interest

Teaching and Learning Methods:

Introductory session will provide introductions into the core topics and scientific writing in lecture style. All other sessions will be focused on the individual student presentations, in which each student presents their seminar paper to the class and all other students are encouraged to discuss the papers.

Media:

Powerpoint

Reading List:

Literature is semester-specific and will be distributed in class

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0031: Advanced Seminar Operations & Supply Chain Management: Digital Operations | Advanced Seminar Operations & Supply Chain Management: Digital Operations

Version of module description: Gültig ab summerterm 2020

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination consists of a written Seminar paper (75%) to check the learning outcomes 1-5, as well as an oral presentation (25%) in order to check learning outcome 6. The seminar paper should cover 15 - 20 pages and is written in the style of current publications of peer-reviewed journal articles. At the end of the module students present their work in a 20 minutes presentation + 15 minutes discussion and have to initialize and moderate the discussion on a selected paper of their fellow participants.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

The courses Management Science and Production and Logistics

Content:

The advanced seminar in logistics and supply chain management focuses on recent research progress on varying topics, e.g. after sales service, procurement.

- In a seminar on After Sales Service the following topics can be discussed: Forecasting Spare parts Demand, Inventory Management of Spare Parts, Reliability Analysis, Maintenance Strategies, Level of Repair Analysis, Repair Kit Problem, Design of Service Supply Chains, Additive Manufacturing of Spare Parts

-In a seminar on procurement the following topics can be discussed: outsourcing, global sourcing, multiple sourcing, order splitting, e-procurement, auctions, procurement on online spot markets,

Intended Learning Outcomes:

The objective of the module is to equip the participants with the necessary skills and tools for a successful master thesis project. Specifically, the aim is to be able to:

- 1 Read and understand recent research contributions
- 2 Pursue interesting research questions
- 3 Conduct a literature study and/or numerical study and/or implementation
- 4 Structure and organize research methods and results
- 5 Write a seminar paper
- 6 Present research finding and defend them in a discussion

Teaching and Learning Methods:

In an introductory session, the current theme of the module is explained by the lecturer and the various available seminar topics are elaborated in detail. Also information on relevant literature for the problem settings is introduced, which forms the basis of the students' seminar paper. After the introductory session, students will work out the topic on their own, by using their abilities of conducting literature research, mathematical modelling, programming and analyses. Throughout the whole time, they receive guidance from a supervisor of the chair. Different milestones are to be achieved at specific dates, such as a preliminary outline of the seminar paper, first research results and the final paper. Following the submission of the final paper, presentations and discussions of all students' seminar papers are conducted, usually spanning one or several days, where amongst others also presentation, moderation and discussion skills are trained.

Media:

Presentation, Various form of literature (Journal Articles, Books, Report, Conference Proceedings, etc.)

Reading List:

'Dependet on seminar focus, e.g.:
R.Dekker, W.D.Rustenburg (2014)

E. vanWingerden, R.J.I.Basten,

More grip on inventory control through improved forecasting:

A comparative study at three companies. IJPE 157, 2020-237

Joeri Poppe, Rob J.I.

Basten, Robert N. Boute, Marc R. Lambrecht. (2017) Numerical study of inventory management under various maintenance policies. Reliability Engineering and System Safety 168, 262-273.

Bram Westerweel, Rob J.I. Basten, Geert-Jan van Houtum (2018). Traditional or Additive Manufacturing? Assessing component design options through lifecycle cost analysis. EJOR 270, 570-585

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Catalogue of Elective Modules: Digital Enterprises | Wahlkatalog: Digital Enterprises

Module Description

MGTHN0074: Contemporary Strategies in the Automotive Industry | Contemporary Strategies in the Automotive Industry

Version of module description: Gültig ab winterterm 2021/22

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The performance of students for this course will be determined by:

- 40% of a comprehensive industry and company analysis in form of PowerPoint presentation
- 40% of continuous monitoring of real-time event, market movement, company announcement
- 20% of processing and consolidating all theory from the seminar and information in self-study into a final project presentation, delivered in PowerPoint deck and verbally

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English, fundamentals in business administration and/or strategic management

Content:

This seminar aims to deliver an overview of recent development of discussion in the automotive industry, and to offer students the chance to consolidate the knowledge from strategic and competitive management to analyse case studies independently.

Automotive is one of the most important pillars of Germany's economy. It is now facing the challenges of transformation: rising conscious for environmental protection from the general public and legislator, pressing need for alternative powertrain technologies, deeper involvement and engagement from the tech powerhouses, ever complicating partnership and competition landscape.

This seminar examines the issues above and the standpoint of major market participants, both established OEMs and new entrants - their reactions, strategic positions, and options. Furthermore, the interpretation of strategy related topics, e.g. core company character, corporate governance, decision-making process, business communication will also be important components to complete the scope of this seminar.

Intended Learning Outcomes:

After the course students will be able to:

- Understand the fundamental theories of automotive management and industry-specific topics
- Apply the knowledge from strategic & international management, as well as competitive strategy in practical oriented course work
- Gain experience in building strategic framework for a market participant and evaluating the impact of influential factors
- Enhance the ability of academic reading, information gathering and research, data analysis
- Further develop communication skills through verbal business presentation and group discussion
- Build up critical thinking and interpretation skills

Teaching and Learning Methods:

This seminar will be aided by PowerPoint slide deck to provide introduction of fundamental theories. Students will be required to self-reading and closely following the real-time market movement.

"War gaming" methods to simulate real-time strategy deployment: each student to prepare keynote to present the strategy analysis of one market participant of their choice - so the students would exercise their ability of information gathering, data processing and business presentation.

Group discussion to ignite the exchange of ideas and provide each other feedbacks.

Industry experts will be invited as guest speakers to provide empirical insights.

Media:

PowerPoint, flipchart, case study materials etc.

Reading List:

Uwe Winkelhake (2022), The Digital Transformation of the Automotive Industry

Jens Diehlmann, Joachim Häcker (2013), Automotive Management. Navigating the next decade of auto industry transformation

Léa Platel (2010), Rethinking communication strategies in the automotive industry

Andreas Blum (2009), Co-opetition in the Automotive Industry

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0084: Advanced International Excursion: Global Strategy | Advanced International Excursion: Global Strategy

Version of module description: Gültig ab summerterm 2022

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 9	Total Hours: 270	Self-study Hours: 180	Contact Hours: 90

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each student will work on a seminar thesis and participate in the discussion with international firms and universities. The examination consists of two parts:

1) Seminar thesis (70 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics discussed in the course
- they have applied the frameworks introduced in the course
- they show that they are able to write a paper that follows scientific standards
- they are able to structure and write a seminar thesis

2) Participation in discussion during excursion (30 % of the overall grade).

The students should demonstrate that:

- they have gained the skills to contribute to discussions with firms and universities
- Active engagement in the course is expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

English oral and written fluency

Content:

Multinational firms need to succeed globally to thrive in today's business environment. Thus, these firms need managers who are well-versed in international business environments. This course aims to provide students an immersive and intense international experience through a 9-day excursion abroad.

For 2022, the destination will be Paris.

Intended Learning Outcomes:

After completing this course students should be able to:

- Feel comfortable and gain proficiency in communicating and doing business abroad

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via in-class discussions
- Build up critical thinking, ethnorelative perspectives and interpretation skills

Teaching and Learning Methods:

This module includes the following teaching and learning methods:

- Immersive learning of strategy, localization, and value chain activities in a different country through an excursion
- Key concepts and empirical findings from the field of strategic management will be provided through discussions with multinational firms.
- Students will acquire additional knowledge from practical literature through their own work on their seminar theses.

Media:

PowerPoint, research papers, case scenarios

Reading List:**Responsible for Module:**

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced International Excursion: Global Strategy (MGTHN0084) (Seminar, 6 SWS)

Li C, Schilder J

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0095: Social Media Marketing | Social Media Marketing

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Grading based on a written multiple-choice examination of 90 minutes (100%) with 35 multiple-choice assignments, designed to assess students' ability to outline and explain core concepts of social media marketing in a structured manner. The exam includes both knowledge-based questions and applied tasks that require students to demonstrate their understanding of electronic word-of-mouth marketing, influencer marketing, and digital consumer feedback. Additionally, case studies or practical scenarios will be used to evaluate students' ability to critically reflect on emerging topics such as meme marketing or social media firestorms.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Completed course on fundamentals in Marketing (mandatory)

Content:

Social media have gained strong importance in both practice and academia. Corresponding digital technologies changed the way how consumers communicate, how they gather information, and how they spend large portions of their life. In this spirit, social media also opened up new opportunities for marketers to target and reach potential customers online. The lecture will cover the basics of social media (marketing), (electronic) word-of-mouth marketing, endorser marketing, cancelling, meme marketing, decoding digital consumer feedback, and new social media formats.

Intended Learning Outcomes:

Upon successful participation, students will be able to outline core concepts of social media and social media marketing. Students will understand and be able to explain the importance of electronic word of mouth marketing. Moreover, students will be able to outline differences between endorser and influencer marketing. They will be capable to explain key mechanism of influencer

marketing and persuasion processes. Students will have an understanding of novel topics in social media marketing as well, these include meme marketing, the threat of cancelling or social media firestorms. Students will have an overview of approaches to analyze social media data for marketing purposes. Students will be aware of new directions in social media and will be able to discuss what this implies for stakeholders. Finally, students will be capable to read and understand current research in social media marketing and to discuss their findings.

Teaching and Learning Methods:

The lecture will consist of input by the lecturer with interactive class discussions. Lectures will include critical discussions of scientific papers.

Media:

PowerPoint, videos, AI image generation software, research papers, virtual reality devices, exercise sheets

Reading List:

Ismagilova, Elvira; Dwivedi, Yogesh K.; Slade, Emma; Williams, Michael (2017): Electronic Word of Mouth (eWOM) in the Marketing Context. A State of the Art Analysis and Future Directions: Springer.

Appel, G., Grewal, L., Hadi, R., & Stephen, A. T. (2020). The future of social media in marketing. Journal of the Academy of Marketing Science, 48(1), 79-95.

Further reading material will be provided in the course.

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0099: Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy | Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The seminar prepares students to make tactical choices supporting organizational performance and strategy. The seminar enables students to understand the relationships between operations and supply chains and other areas of the organizations and to design reliable, fast, and efficient operations and supply chains.

Students will be guided to learn critically analyzing value chain reports and to identify problems and areas for potential improvements. Later, students will demonstrate their ability to apply these skills in a simulated value chain setting. After completing this seminar, students can identify value chain challenges, identify potential improvements, coordinate critical decisions across functional areas, and forecast results from the solutions implemented. As such, this seminar will prepare students for applying academic knowledge to industry practice.

Central to the seminar is a simulation consisting of several rounds. Students will be assigned to groups of four participants and have to formulate a strategy for their company. During each round, they will be required to make tactical and strategic choices to maximize value and prepare a four-page summary justifying the choices made and the expected results. This summary is part of the overall assessment (75% of the final grade, team grade). This assessment method helps students reason about the options available in each round, make informed decisions, develop predictive capabilities, work together in a team, and condense necessary information formally. Furthermore, students must submit a four-page report at the end of the course, including a critical reflection about teamwork and individual takeaways (25% of the final grade, individual grade). This assessment method allows students to reflect on their capabilities to work in a team and strengthen their personal learnings from the seminar.

At the beginning of the course, students will be advised on the report structure and submission deadlines.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Desirable but not mandatory: Production and Logistics, Management Science

Content:

Subject-specific content

- Strategic Alignment
- Value Chain
- Supply Chain Tactical Decisions
- Supply Chain Management
- Sales and Operations Planning

Intended Learning Outcomes:

Upon completion of the module, students are able to:

- understand the need to construct operations and supply chains that align with corporate-level strategies and identify appropriate actions
- analyze the relationships between operations and supply chain management and other areas of the organization (e.g., marketing, finance, accounting, engineering, R&D, etc.)
- evaluate the feasibility of functional-level actions with corporate-level strategies and assess the impact of their choices on the organization's performance
- coordinate actions and address competing interests across functional areas
- evaluate alternative tactical decisions and convince with compelling reasoning for their choice of action
- critically reflect upon their ability to work in a team and provide advanced improvement recommendations for effective teamwork

Teaching and Learning Methods:

Experiential learning

The module is designed to enable students to acquire knowledge on current (behavioral) operations and supply chain management topics through hands-on experience and reflection. To this aim, a web-based business simulation game will be employed. The business simulation game will engage participants in turning around a manufacturer of fruit juices by making tactical and strategic decisions on four critical areas of the organization: Purchasing, Operations, Supply Chain, and Sales. Students are required to analyze key performance indicators, define a strategy for turning around the organization, identify potential improvements, and make tactical decisions in each of the organization's four areas.

Seminar sessions take place physically, and the methods combine mini-lectures, hands-on experience, group reflection, and questions and feedback sessions. Each week, there will be two

90-minutes sessions that will allow for feedback, sparring sessions, and the introduction of new concepts.

Students will get an overview of the course in the first two weeks. They will be informed about the examination and expectations. Furthermore, students will be introduced to highly recognized supply chain strategy and integration papers. Students will also see examples and learn to interpret data from sales, operations, supply chain, and purchasing reports.

After setting the theoretical scene, students will form small groups of size 4 (groups of smaller size might also be formed depending on the total number of participants), and they will start working on the business simulation game. The simulation advances over several rounds and students must present their analysis, evaluation, decisions, and expected results. Further details will be shared during the seminar.

Media:

Research papers and further material to be shared via Moodle, in-class discussions

Reading List:

Readings for general preparation:

Myerson, P. (2012). Lean supply chain and logistics management. McGraw-Hill.

Croson, R., Donohue, K. (2006). Behavioral causes of the bullwhip effect and the observed value of inventory information, *Management Science*, 52(3), pp. 323-336. DOI: 10.1287/mnsc.1050.0436

Flynn, B.B., Huo, B. and Zhao, X. (2010), The impact of supply chain integration on performance: A contingency and configuration approach. *Journal of Operations Management*, 28: 58-71. DOI: 10.1016/j.jom.2009.06.001

Frohlich, M.T., Westbrook, R. (2001). Arcs of integration: an international study of supply chain strategies. *Journal of Operations Management*, 19, pp. 185-200. DOI: 10.1016/S0272-6963(00)00055-3

Further materials will be discussed in the session.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0111: Cases in industry 4.0 | Cases in industry 4.0

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The idea of this course is to train students in managerial decision-making to strengthen their positive impact on their next job. Accordingly, the coursework and examinations are structured.

Each week during this semester, there will be sessions of 2 times 90 minutes where we will discuss teaching case studies revolving around management issues in Industry 4.0. Students must prepare each case and answer preparatory questions (60% of the final grade, individual grade). Class participation and active contributions are graded as 40% of the final grade (individual grade). This course will allow students to learn firsthand about management issues in Industry 4.0. During those sessions, we will cover a variety of relevant technologies and their interplay.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Production and Logistics, Strategic Management

Content:

Subject-specific content

- Industry 4.0
- Management issues/conflicts of interests
- Implementation of digital technologies
- Blockchain, IoT, Cloud, AI, AR, VR, etc.

Methodological content

- Quantitative assessment
- Modelling

Intended Learning Outcomes:

Upon completion of the module, students can

- understand and analyze state-of-the-art approaches to industry 4.0,
- evaluate the advantages and disadvantages of the most recent technologies
- critically analyze firms and supply chains within their context and make superior recommendations,
- and criticize innovative approaches by assessing potential shortcomings.

Teaching and Learning Methods:

Lecture

Each week, students will be given one case study (approx. 10-20 pages), read those, answer a few graded case-related questions (approx. 2-4), and attend the classroom session. In those sessions, students will actively contribute and engage in the discussion. While correctness and plausibility are essential, this class also seeks diverse views and appreciates reasonably different perspectives.

Media:

Case studies (to be shared via Moodle)

Reading List:

Readings for general preparation:

Babich, V., & Hilary, G. (2020). OM Forum—Distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management*, 22(2), 223-240.

Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113-122.

Wuttke, D., Upadhyay, A., Siemsen, E., & Wuttke-Linnemann, A. (2022). Seeing the Bigger Picture? Ramping up Production with the Use of Augmented Reality. *Manufacturing & Service Operations Management*.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0113: How to launch a platform start-up | How to launch a platform start-up

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1) A final written exam determines 60% of the grade. The final written exam (60 minutes) is a means to assess students' understanding of the basic and advanced theories of the economics and management of digital platforms. Students have to show that they understand the core economic concepts underlying platform business models. They have to prove that they are familiar with the strategies and tactics to start a digital platform business. They have to show that they can critically evaluate the different options for designing a platform business model. Students may use a non-programmable calculator and a non-electronic dictionary for the exam.

2) Case study group work with presentation determines the remaining 40% of the grade. Students will be assigned into groups to work on a case study about a platform company. Students have to show that they can analyze the case based on the knowledge acquired in the course and suggest recommendations for the platform company. The grade for the case study group work is based on an in-class presentation by the group.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

none

Content:

In the 21st century, we see a number of companies like Uber or Airbnb that provide value to their customers by operating a platform and not by selling products. This rather new business model allows users and producers to interact with each other while the platform provider sets up the infrastructure for doing so.

This course gives an introduction to digital platforms to understand the underlying concepts of those business models and highlights the importance of the critical mass and network effects. Important principles for launching a digital platform are explained and different challenges that can appear in the process are outlined. By the end of the course, students will be able to understand digital platforms as a business model in general and be able to develop promising strategies in the launching phase. For a deeper understanding of the theoretical concepts, participants will apply the learnings on a case study which provides them with the necessary skills for analyzing cases and using previously acquired knowledge to assess them.

Intended Learning Outcomes:

After the course students will be able to:

- understand the concepts of digital platforms as a business model.
- develop strategies for launching a platform and understand the importance of the critical mass and network effects.
- apply instruments that allow to overcome the challenges in the launch process in a market with and without competition.

Teaching and Learning Methods:

Lecture: the lecture serves to build a thorough theoretical understanding of the concepts and methods introduced in the course.

Case studies: students will be assigned into groups to work on a case study of a real-world platform start-up with the goal to identify solutions for their challenges.

Media:

Slides

Case studies

Reading List:

Förderer, Jens: Erfolgsmodell Digitale Plattformen: Geschäftsmodelle – Netzwerkeffekte – Community-Management. 2022. Schäffer-Poeschel Verlag, Stuttgart.

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0115: Advanced Seminar: Digital Technologies in Operations Management | Advanced Seminar: Digital Technologies in Operations Management

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The key idea of this seminar is to lead students towards academic work and help prepare them for their master's thesis. In addition, they will learn about recent developments in digital technologies. Accordingly, the coursework and examinations are structured.

During the semesters, students will prepare two short presentations on research methods and skills. They will work in small teams. Each of the two presentations (ca. 45 minutes) accounts for 10% of the final grade and is a team grade.

Upon completion of this module, students will demonstrate their ability to cope with advanced research studies, comprehend and assess study outcomes, compare contributions of several studies, and transfer theoretical concepts to practice through a seminar paper (50% of the final grade; length of 15 pages including references). This method requires students to assess and replicate a study themselves.

In addition, they will prove their ability to communicate even complicated relationships and methods to their peer students through a presentation in a comprehensible fashion. They will further guide and moderate an ensuing discussion throughout which they will demonstrate their ability to criticize and assess innovative approaches and their potential shortcomings. The presentation, including moderation of ensuing discussion, accounts for 30% of the final grade (presentation duration 30 minutes + 30 minutes discussion). Complementing the written seminar paper, the presentation is targeted toward students who have not studied the same paper; this poses the challenge of presenting theoretical work interestingly while breaking down complex relationships into understandable information without losing rigor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Management Science, Production and Logistics

Content:

Subject-specific content

- Digital technologies potentially including but not limited to AR, VR, AI, Blockchain, and IoT.
- Management focus
- Incentive conflicts that arise when introducing new technologies in operations management
- Behavioral operations management

Methodological content (depending on assigned papers and limited to an introductory level)

- Research Design
- Analytical Modelling
- Game Theory
- Event Study Method
- Econometrics

Intended Learning Outcomes:

Upon completion of the module, students can

- understand and analyze state-of-the-art approaches to digital technologies in operations management,
- understand selected research papers and evaluate their key findings,
- present and explain complex research studies in a comprehensible and engaging manner,
- evaluate suggested management solutions in the digital technologies in an operations management context and assess their strengths and weaknesses,
- create management recommendations based on recent academic studies.

Teaching and Learning Methods:

Seminar

The seminar consists of two parts. During the semester, we will meet regularly to discuss select papers on digital technologies in operations management. Students will also explore and present information on select research methods (e.g., econometrics, game theory, and behavioral experiments). This part is complemented by a few interactive lectures.

At the end of the semester, we several sessions take placed blocked where students will present their assigned research studies (final assessment).

Media:

Research papers (to be shared via Moodle), student presentations, further material to be shared via Moodle, online discussions

Reading List:

Readings for general preparation (i.e., what is research? What marks a contribution?):

Wacker, J. G. (1998). A definition of theory: research guidelines for different theory-building research methods in operations management. *Journal of operations management*, 16(4), 361-385.

Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of management review*, 14(4), 490-495.

Examples for typical papers that we will discuss (this list is not meant to be comprehensive but gives some indication of topics to be covered):

Babich, V., & Hilary, G. (2020). OM Forum—Distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management*, 22(2), 223-240.

Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113-122.

Further papers will be provided at the beginning of the semester.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0122: Campus Challenge: Successful Investing with AI and Large Language Models | Campus Challenge: Successful Investing with AI and Large Language Models

Version of module description: Gültig ab summerterm 2023

Module Level:	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises group-based scientific coursework (approx. 20-30 pages) about the group's investment strategy.

The investment strategy should incorporate language models systematically and creatively.

Students will receive more information on the available tools during the kick-off meeting.

The coursework should comprehensively document the investment strategy (investment idea, tools used, prompt engineering, data preparation, data analysis, performance evaluation and interpretation, and suggestions for further improvement). In addition to the written scientific report, candidates must present their strategy within an approximately 20-minute presentation and answer questions from their audience.

To maintain the seminar atmosphere, the number of participating groups is generally limited. While Heilbronn students are prioritized, students from all TUM locations may also participate if the course is offered in a hybrid/online format. The targeted maximum number of participants is 24. More information will be provided in the introductory session.

The written assignment will be weighted 80%, and the formal presentation will make up 20% of the final grade. The group whose investment strategy performs best may receive an additional grade improvement in the live-test. The assessment is generally based on group work. If there are significant differences across the quality of the report or presentation, grades may be adapted individually. The students should therefore highlight their contributions to the report.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;
Basic knowledge in scientific writing;
Interest in investing in financial markets and the application of AI tools

Content:

The students will motivate, create and implement an investment strategy for the equity market using AI and/or large language models and/or sophisticated chatbots. The investment strategy will be developed based on group-work.

Candidates are expected to motivate their investment strategies with respective literature to ensure a systematic and structured manner. Candidates are expected to learn how to work with sophisticated chatbots, language models, and Application Programming Interfaces (APIs).

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- motivate, create, and implement an investment strategy;
- work with language models / sophisticated chatbots;
- conduct an empirical project in a team;
- deliver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise regular meetings with the team members about the current status of the stock portfolio;

Methods of teaching comprise group-based coursework and the formal presentation of obtained results;

The learning methods of the students primarily comprise the following activities:

- Collaborative Implementation of the investment strategy (theory, tools used, prompt engineering, data preparation, data analysis, results, and interpretation, room for improvements);
- The theory may be developed based on an independent literature review (usage of scientific articles published in top international journals);
- Collaborative writing of a scientific report;
- Exercise a deductive, logical, and consistent argumentation to derive an investment strategy;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction/teaching methods are adequate to foster/extend the student's ability to conduct independent academic work and elaborate thematically complex contents on their own. It is considered a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

The students will receive a list of relevant literature during the kick-off meeting at the beginning of the semester.

De Prado, Marcos Lopez. Advances in financial machine learning. John Wiley & Sons, 2018.

Lopez-Lira, Alejandro and Tang, Yuehua, Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models, 2023

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Campus Challenge: Successful Investing with AI and Large Language Models (MGTHN0122)
(Seminar, 4 SWS)

Breitung C, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0123: Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals | Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals

Version of module description: Gültig ab summerterm 2023

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To successfully pass the module, the participants have to demonstrate methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations. The participants proof these competences through several presentations with written reports and active participation in discussions.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

No prerequisites required.

Content:

Artificial intelligence (AI) is one of the most profound ongoing technological developments which has the potential to strongly impact various fields of business and society. This module critically analyzes the effects of AI-applications on the sustainable development goals (SDGs) of the United Nations. Units of the module: Political and conceptual background of the SDGs; methods to assess impacts on the micro-, meso-, and macro-level; discussion of supportive and contradictory impacts of specific AI-applications on SDGs.

Intended Learning Outcomes:

After successful completion of the module, the participants have methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations.

Teaching and Learning Methods:

The module consists of an interactive introductory lecture and presentations of the participants with topic-specific written reports.

Media:

Textbook, presentation slides, and e-learning (Moodle).

Reading List:

Dodds, Donoghue und Leiva Roesch. 2016. Negotiating the sustainable development goals: A transformational agenda for an insecure world.

Kanie und Biermann. 2017. Governing through goals: Sustainable development goals as governance innovation.

Saetra. 2022. AI for the sustainable development goals.

Sobti und Sharma. 2018. An introduction to sustainable development goals.

Walker, Pekmezovic und Walker. 2019. Sustainable development goals: Harnessing business to achieve the SDGs through finance, technology, and law reform.

Responsible for Module:

Stich, Michael; Prof. Dr. rer. oec.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0130: CEO Strategy Series | CEO Strategy Series

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The assessment of this course is based on short quizzes administered at the end of each session. These quizzes both test listening comprehension and encourage students to critically reflect on the content in light of the theoretical foundations introduced at the beginning of the semester. To successfully complete the course, students must submit at least 70% of the quizzes. Each quiz contains 7 questions. The final grade is calculated from the aggregated quiz results.

In addition, students may earn a grade bonus of up to 30% through active oral participation. This requires asking relevant and well-prepared questions during the sessions, which should be linked to the visiting CEOs and their companies as well as to current developments, global trends, or issues related to digital strategy.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization.

Content:

This module is the flagship event of TUM C.HN and will take place on Tuesdays or Fridays at 3 pm (each date will feature one guest). It covers practical insights on firm strategies, whereby lessons can be learned directly from CEOs themselves. The covered strategy topics may include internationalization, the analysis of the internal and external environment, M&A and other types of strategic alliances, as well as digital transformation, the role of digital enterprises, and the strategic challenges arising from new technologies, online platforms, and data-driven business models. This course therefore combines classical perspectives of strategic management with current

developments in the digital economy, providing students with valuable insights for their future work as strategizers in a corporate environment.

Please note: Registration is mandatory. There will be a participants list at the entrance and police presence for each event.

Intended Learning Outcomes:

After attending this course, students will have learned about different perspectives on developing and executing firm strategies in practice. They will be able to analyze companies and CEOs in their function as strategizers, and apply the knowledge gained in an in-depth study of relevant strategy literature by interacting with CEOs. With this approach, students will gain theoretical and practical knowledge, strengthen their understanding of digital enterprises and the role of digital transformation in strategy, and are additionally able to improve their discussion- and soft skills.

Teaching and Learning Methods:

In this seminar, there will be both Q&As and podium discussions with CEOs. Students will meet several CEOs and managers of real-life companies, gain insights into their strategies and related concepts, and interact with them in Q&A sessions. To prepare, students should develop questions that build on the concepts introduced in the first session and reflect the relevance of digital enterprises and digital transformation in corporate strategy. Active participation in all sessions is expected.

Media:

Powerpoint

Reading List:

- 1) Hanson, D., Hitt, M. A., Ireland, R. D., & Hoskisson, R. E. (2016). Strategic management: Competitiveness and globalisation. Cengage AU.
- 2) Barney, J. B. (2002). Strategic management: From informed conversation to academic discipline. Academy of Management Perspectives, 16(2), 53-57.
- 3) Nag, R., Hambrick, D. C., & Chen, M. J. (2007). What is strategic management, really? Inductive derivation of a consensus definition of the field. Strategic management journal, 28(9), 935-955
- 4) Smith, K. G., & Hitt, M. A. (Eds.). (2005). Great minds in management: The process of theory development. OUP Oxford.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

CEO Strategy Series (MGTHN0130) (Seminar, 4 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0165: Sustainable Management - Bridging Economy and Ecology through Technology | Sustainable Management - Bridging Economy and Ecology through Technology

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 160	Contact Hours: 20

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of seminar papers (60%) and presentation (40%): The seminar begins with block lectures, after which students independently write a seminar paper, present their findings, and actively engage in discussions about their work. Possible seminar paper topics include the development of energy-efficient building technologies, advancements in CO2 capture and storage, the implementation and potential of smart grids, modern heat recovery systems, innovations in sustainable waste management, the decarbonization of industrial processes, and urban planning.

Repeat Examination:

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management and organizations.

Content:

This course introduces the fundamentals of sustainability, covering ecological principles, the significance of ecosystems, and the science behind climate change, including its global impacts. It examines historical and modern responses to climate-related disasters, focusing on technological solutions like clean air regulations and sulfur dioxide controls, and highlights the importance of a circular economy.

Intended Learning Outcomes:

After attending this course, students will have gained a comprehensive understanding of sustainability, including ecological principles, climate change, and its impacts. They will be able to explain, analyze, and critically reflect on historical and current climate-related challenges and

solutions driven by technological advancements. Students will have built an understanding of political, legal, and economic frameworks related to sustainability, with practical insights into corporate sustainability practices, exemplified by Knauf. Moreover, they will acquire knowledge on energy efficiency, sustainable energy sources, and technological innovations. They will prepare a written assignment, hold a short presentation and lead a discussion on a sustainability related topic, which enhances their ability to apply these concepts practically and improves their discussion skills. Furthermore, students will have the option to discuss actual activities at the multinational company Knauf, where they can observe and learn firsthand how a leading company integrates sustainability into its operations and practices.

Teaching and Learning Methods:

Students will write a seminar paper on various sustainability topics and present their findings to the class. After each presentation, there will be discussions to elaborate on the topics and reflect on the concepts. Additionally, students can see how sustainability practices are implemented in a real-world setting at Knauf. These methods aim to deepen the understanding, enhance analytical and presentation skills, and apply theoretical knowledge practically.

Media:

Powerpoint

Reading List:

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Sustainable Management - Bridging Economy and Ecology through Technology (MGTHN0165)
(Seminar, 4 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0177: Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations | Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1. Group Report (70%)

- **Assessment Details:** In groups, students will collaboratively submit a report (no longer than 5 pages) addressing a specific managerial problem and proposing a solution using VR/AR technologies. Students are allowed to use reference materials and external resources.
- **Scope:**
 - Identify a relevant managerial problem in production, quality inspection, remote support, human resources, or customer experience.
 - Justify the importance of solving this problem.
 - Propose a detailed VR/AR-based solution, including technical implementation steps, management implications, and expected outcomes.
- **Rationale for Assessment:**
 - This assessment evaluates the ability to apply theoretical knowledge to practical problems, foster teamwork, and use technology to prepare solutions with managerial perspectives.
 - The report format develops students' ability to communicate complex ideas concisely and effectively.

2. Individual Critical Reflection (30%)

- **Assessment Details:** Each student will write a critical reflection on the relevance of immersive technologies (VR/AR) in solving managerial challenges. This will be submitted as a separate document, no longer than 2 pages. The reflection must be written independently.
- **Scope:**
 - Reflect on the practical and theoretical insights gained during the module.
 - Critically analyze the advantages and limitations of VR/AR in managerial contexts.

- Discuss how the experience of creating the group report has influenced their understanding of technology's role in management.

- **Rationale for Assessment:**

- This assessment allows students to self-evaluate, critically linking their learning to broader managerial and technological contexts.

- It promotes independent thought, allowing students to articulate their unique perspectives.

Repeat Examination:

(Recommended) Prerequisites:

Basic Programming with C#

Content:

In the module, the students will gain knowledge about the technological and management aspects of VR implementations with hands-on experience and use case examples in

- Production and Assembly
- Quality Inspection and Remote Support
- Training and skill development
- Customer Experience and Prototyping

Technology-specific content

- Introduction to Immersive Technologies (key hardware and software platforms)
- Using Unity for developing VR/AR applications

Management-specific content

- Ethical Considerations
- Behavioral Challenges
- Building a roadmap from prototype to market-ready product

Intended Learning Outcomes:

By the end of this module, students will be able to:

- Understand the fundamental principles of VR and AR technologies.
- Analyze the role of immersive technologies in improving managerial and operational efficiency.
- Design and prototype VR/AR-based solutions for specific management scenarios.
- Critically evaluate the broader implications of VR/AR deployment, including ethical concerns (e.g., data privacy, accessibility).
- Work collaboratively in teams to identify managerial challenges and develop VR/AR solutions.

Teaching and Learning Methods:

The coursework is primarily practical. At the beginning of the module, the students will be introduced to the immersive technologies and different headsets. They will be introduced to Unity Game Development Engine, which they will use to develop application solutions. The students

can choose between VR or AR applications and must program their software solution, keeping a managerial focus in mind.

The students will present the problem statement (in a group) that they want to work on and will have regular feedback sessions during the development phase. The students will present their solutions at the end of the module.

Media:

Material will be shared via Moodle

Reading List:

Readings for general preparation (but not limited to):

Dalton, J. (2021). Reality check: How immersive technologies can transform your business. Kogan Page Publishers.

Jung, T., & Dalton, J. (2021). XR Case Studies. Springer International Publishing.

Wuttke, D., Upadhyay, A., Siemsen, E., & Wuttke-Linnemann, A. (2022). Seeing the bigger picture? Ramping up production with the use of augmented reality. *Manufacturing & Service Operations Management*, 24(4), 2349-2366.

Windhausen, A., Heller, J., Hilken, T., Mahr, D., Di Palma, R., & Quintens, L. (2024). Exploring the impact of augmented reality smart glasses on worker well-being in warehouse order picking. *Computers in Human Behavior*, 155, 108153.

Seeliger, A., Cheng, L., & Netland, T. (2023). Augmented reality for industrial quality inspection: An experiment assessing task performance and human factors. *Computers in Industry*, 151, 103985.

Rejeb, A., Keogh, J. G., Leong, G. K., & Treiblmaier, H. (2021). Potentials and challenges of augmented reality smart glasses in logistics and supply chain management: a systematic literature review. *International Journal of Production Research*, 59(12), 3747-3776.

Eswaran, M., & Bahubalendruni, M. R. (2022). Challenges and opportunities on AR/VR technologies for manufacturing systems in the context of industry 4.0: A state of the art review. *Journal of Manufacturing Systems*, 65, 260-278.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations (MGTHN0177) (Seminar, 4 SWS)

Phadke V

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0181: Advanced Seminar Operations & Technology: Analytics in Manufacturing Management | Advanced Seminar Operations & Technology: Analytics in Manufacturing Management

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours:	Contact Hours:

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

This seminar aims to guide students toward academic research and prepare them for their master's thesis. The topics focus on advanced analytics in manufacturing management, covering AI methods, modeling techniques, optimization algorithms, and rigorous analysis of algorithmic properties, such as performance guarantees. While students are not expected to master all these techniques, they should be willing to develop deep expertise in one and understand its relevance for manufacturing decision-making. Additionally, they should be able to present their findings clearly and didactically to their peers.

Students work in small teams and complete two presentations and a term paper (~15 pages). The first presentation (30 minutes, 20% of the grade) provides a structured overview of a recent research article and formulates a research question for the seminar thesis. The second presentation (30 minutes, 30% of the grade) is the final presentation, showcasing their research findings. The term paper accounts for 50% of the grade.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Foundations in statistics and management science

Content:

This course focuses on advanced analytics in manufacturing, emphasizing algorithms, modeling, simulations, and theoretical analysis. It is open to all advanced analytics methodologies, provided they meaningfully support manufacturing decision-making.

Additionally, the course integrates practical experience by incorporating potential talks from industry practitioners and potential excursions to manufacturing facilities.

Intended Learning Outcomes:

Students will develop a comprehensive understanding of the scientific process, further develop critical thinking skills, and the ability to solve business problems efficiently. They will gain firsthand experience in manufacturing decision-making through practitioner interactions and engagement with cutting-edge research on data-driven manufacturing. Additionally, they will acquire hands-on expertise in advanced analytics methods applied to manufacturing decision-making.

Teaching and Learning Methods:

Teaching methods include presentations, interactive learning, e-learning, group discussions, and (if feasible) discussions and presentations by practitioners. Additionally, if feasible, we aim to include excursions to manufacturing and related facilities, as well as interactions with leading scientists in the field.

Media:

Zoom, literature databases (Ebsco, ScienceDirect etc.), standard solvers (Gurobi, Hexaly etc.), databases and case studies, programming environments and software development platforms, simulation environments etc.

Reading List:

We focus on high-profile and influential ideas in advanced analytics for manufacturing, typically published in leading journals such as Operations Research, Management Science, Manufacturing & Service Operations Management, INFORMS Journal on Optimization, European Journal of Operational Research, and INFORMS Journal on Computing. Below are examples of relevant articles:

[AI] Senoner, J., Netland, T., & Feuerriegel, S. (2022). Using explainable artificial intelligence to improve process quality: Evidence from semiconductor manufacturing. *Management Science*, 68, 5704-5723.

[mathematical analysis] Drozdowski, M., Jaehn, F., & Paszkowski, R. (2017). Scheduling position-dependent maintenance operations. *Operations Research*, 65, 1657-1677.

[modeling] Möhring, R., Schulz, A. S., Stork, F., & Uetz, M. (2003). Solving project scheduling problems by minimum cut computations. *Management Science*, 49, 330-350.

Da Col, G., & Teppan, E. C. (2022). Industrial-size job-shop scheduling with constraint programming. *Operations Research Perspectives*, 9, 100249.

[algorithmics] Koca, Y., Martagan, T. G., Adan, I. J. B. F., Maillart, L., & van Ravenstein, B. (2023). Increasing biomanufacturing yield with bleed-feed: Optimal policies and insights. *Manufacturing & Service Operations Management*, 25, 108-125.

Barnhart, C., Jacquillat, A., & Schmid, A. (2024). Robotic warehousing operations : A learn-then-optimize approach to large-scale neighborhood search. *INFORMS Journal on Optimization*, 1-24.

Responsible for Module:

Otto, Alena; Prof. Dr. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0012: Digital Finance | Digital Finance [DF]

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The module examination consists of a 90-minute written exam, in which the candidates should reproduce the knowledge imparted in the lectures. The students are also required to provide transfer knowledge. It is tested whether students have understood how Digital Finance relates to Traditional Finance and other financial research disciplines; which interdependencies exist among the research disciplines, etc. Dangers and Opportunities of FinTech applications should be understood in detail. Further, students are required to conduct and assess mathematical calculations. The written exam also comprises open questions as well as multiple-choice-tasks.

Students may have the opportunity to improve their final grade by up to 0.3 points by voluntarily participating in group work and presentations on selected topics in Digital Finance. This opportunity depends on factors such as the number of participants and available capacity, and is decided on a semester-by-semester basis by the instructor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge of corporate finance/financial mathematics is required (e.g. material covered in the course Corporate Finance); joint election with the module "Advanced Seminar in Finance/Accounting: Current Research Topics in Digital Finance" highly recommended

Content:

The aim of the course is to give a comprehensive overview on Digital Finance, not covered elsewhere in courses of Corporate Finance within the Master in Management. It provides a coverage of the most recent relevant topics in the field of Digital Finance. Basic knowledge in the field of financial mathematics is a pre-requisite.

The module comprises the most important and most recent developments in Digital Finance. Specifically, it comprises (but is not limited to) the following areas:

- History of Digital Finance and how it relates to other financial research disciplines;
- Robo-Advisors;
- Crowd Funding;
- Peer-2-Peer-Lending;
- Social Media in Finance;
- Digital Payments;
- Cryptocurrencies;
- Digital Technologies (Big Data and Machine Learning; Textual Analysis; Blockchain)
- Cybersecurity;
- Digital Finance and the Market Efficiency Debate;
- The Future of Digital Finance;
- Regulation;
- etc.

Intended Learning Outcomes:

The students acquire detailed knowledge on how technological innovations are changing existing financial services and products, their implications and possible applications within the financial industry. Specifically, upon completion of the module, students will be able to identify changes and opportunities of Digital Finance and its implications for the overall macroeconomic stability.

Also, students will be able to fully understand limitations of the digital transformation within the financial industry and its underlying causes. Finally, the students will be able to assess how and why digital finance applications influence the market (in-)efficiency of existing financial markets. In particular, the students are able to analyze complex financial market relations while taking into consideration various influencing factors (as for instance irrational investor behavior).

Teaching and Learning Methods:

Types of instruction comprise lecture courses as well as practical courses;

Methods of teaching include lectures, presentations as well as guest lectures;

the learning methods of the students primarily comprise the following activities:

- Follow-up of course contents;
- Exercise and execution of financial calculations;
- Preparation and execution of presentations;
- Ability to answer advanced thematic issues;
- etc.

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to fully understand and elaborate the in-depth the thematic content.

Media:

Lecture slides; whiteboard; exercise sheets; exercise portfolio; flipchart; powerpoint; films

Reading List:

Due to the topicality of this lecture, the usage of reference books is only possible in a limited manner. Still, the below-listed references are considered as a solid starting point for the contents of this lecture.

Goldstein, I., Jiang, W., & Karolyi, G. A. (2019). To FinTech and Beyond. The Review of Financial Studies, 32(5), 1647-1661.

Chuen, D. L. K., & Deng, R. H. (2017). Handbook of blockchain, digital finance, and inclusion: Cryptocurrency, FinTech, InsurTech, regulation, ChinaTech, mobile security, and distributed ledger. Academic Press.

Fatas, A. (Ed.; 2019). The Economics of Fintech and Digital Currencies. CEPR Press.

If applicable, further literature will be announced at the first session.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Digital Finance (WIHN0012) (Vorlesung mit integrierten Übungen, 4 SWS)

Müller S, Osterrieder D

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0013: Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance | Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises a scientific coursework (approx. 25-30 pages), which is supposed to address the research questions posed within each given seminar topic. The task is to be conducted as a group work (i.e. 2-3 students per topic). Additional to the written scientific report, candidates are required to showcase main findings within a 30-minute presentation and answer further questions following their presentation.

In doing so, the candidates are required to prove that they have dealt with the given topic in a comprehensive manner, that their selection and evaluation of literature is adequate and that they have in-depth analyzed the topic while simultaneously putting it in a higher context of financial research. Also, it can be assessed whether the students are able to answer further questions in a correct and comprehensive manner.

The written assignment will be weighted 2/3 and the formal presentation will make 1/3 of the final grade. Each student will be assessed separately.

To maintain the seminar atmosphere, the number of participants is generally limited. If the course is offered in a hybrid format, Munich students may also participate in the course. The targeted maximum number of participants is 12. More information will be provided in the introductory session.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;

basic knowledge in scientific writing;
combination with the module "Digital Finance" is recommended

Content:

Candidates are supposed to build groups of 2-3 students and choose one seminar topic out of a set of pre-defined scientific issues. The pre-selection omprises, but is not limited to topics of the following areas:

- Asset Pricing of Cryptocurrencies: Market (In-)Efficiency, Behavioral Finance & Bubbles;
- Financial Innovations: Foundations and Opportunities & Threats to Financial Market Stability;
- Wisdom of the Crowd? Information Asymmetries, Adverse Selection, and Moral Hazard in Peer-to-Peer Lending Markets Around the World;
- Robo-Advisors: Overview of Services, Benefits, and Limitations;
- Social Investing: Overview, Benefits, and Limitations;
- Machine Learning: An Overview of ML Applications for Finance;
- etc.

Candidates are expected to conduct a comprehensive literature review of the most important scientific articles capturing the chosen seminar topic in order to address the given research questions in a systematic and structured manner.

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- describe the effects of current FinTech developments for existing financial products and processes;
- exhibit a comprehensive overview on current research topics in the field of Digital Finance;
- identify chances and opportunities of these developments (especially with regard to the overall macroeconomic stability);
- conduct an independent literature analysis at the highest possible, international level;
- analyze and evaluate references in a systematic manner;
- combine topics in Digital Finance with theories/methods/models of traditional finance in an independent manner;
- deriver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise an initial lecture (overview course) as well as the corresponding main seminar;

Methods of teaching include a group work and the formal presentation of obtained results;

the learning methods of the students primarily comprise the following activities:

- independent literature research (usage of scientific articles published in international top journals);
- Collaborative writing of a scientific report;
- Exercise of a deductive, logic and consistent argumentation to specifically address and answer the posed research questions;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to conduct independent academic work and to elaborate thematically complex contents on their own. It is considered to be a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

Machi, L. A., & McEvoy, B. T. (2012). The literature review: Six steps to success (2nd ed.). Corwin Press.

A separate reading list for each topic will be provided in the introductory lecture.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Finance & Accounting (WIHN0013): Current research Topics in Digital Finance (Seminar, 4 SWS)

Gong Z, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0017: Advanced Seminar Innovation & Entrepreneurship: Digital Innovation | Advanced Seminar Innovation & Entrepreneurship: Digital Innovation

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Seminar paper with presentation: The students independently and individually prepare a scientific seminar paper and present it during the course.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Very good English language skills are required. Students should also have an understanding of basic principles of management, innovation, and organization as well as empirical research methods. Attendance of the course Empirical Research Methods in Management and Economics recommended, but not mandatory.

Content:

The course prepares students for their Master's Thesis by introducing them to (1) the search, understanding, and analysis of academic literature, (2) methods for data collection and data analysis, (3) the structuring and writing of a paper, and (4) giving presentations to an academic audience including a subsequent discussion. The course exposes students to digital platforms and open forms of organizing innovation.

Intended Learning Outcomes:

After the course students will be able to:

- Define, explain and apply selected key concepts in the field of digital innovation
- Read and understand academic literature
- Write and present an academic paper

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via presentations and group discussions
- Build up critical thinking and interpretation skills

Learning Objectives

At the end of this course, students will be able to demonstrate understanding, critical assessment and application of the following:

- Learn how to evaluate academic literature
- Interact with an academic debate
- Prepare, elaborate and defend an academic argument
- Specialist knowledge on their selected topic of interest

Teaching and Learning Methods:

Introductory session will provide introductions into the core topics and scientific writing in lecture style. All other sessions will be focused on the individual student presentations, in which each student presents their seminar paper to the class and all other students are encouraged to discuss the papers.

Media:

Powerpoint

Reading List:

Literature is semester-specific and will be distributed in class

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0019: Inventory Management | Inventory Management

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In order to check the learning outcomes an oral exam (20 minutes) is performed.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

- the content of the course Production and Logistics
- Basics in probability calculus
- Programming skills in Python

Content:

The following topics are discussed

Demand Data for Inventory Control,

3) Performance measures for inventory control
control policies

approaches

7) Joint replenishments

safety stock

10) Multi echelon inventory systems

11) Inventory Management in the digital Age

1) Analysis of

2) ABC Analysis, XYZ Analysis

4) Inventory

5) Service and cost oriented

6) Optimization of cycle stock

8) Optimization of

9) Managing Spare Parts Inventories

Intended Learning Outcomes:

The students

- know fundamental problems and trade-offs while managing inventories in supply chains,
- are able to model inventory problems,
- know common inventory policies and are able to analyse them,

- can determine optimal policy parameters for common inventory policies,
- are able to use stochastic inventory models to optimize safety stock.

Teaching and Learning Methods:

Case studies, presentation, discussions, computer work

Media:

Reading List:

- Kiesmüller, G.P. (2024) Inventory Management. Books on demand
- Tempelmeier, H. (2006): Inventory Management in Supply Networks. 2nd edition. Books on demand
- Axsäter, S. (2004): Inventory Control. Kluwer's International Series.
- Montgomery D.C.; Runger, G.C. (2003) Applied Statistics and Probability for Engineers. 3rd edition. Wiley

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

Inventory Management (WIHN0019) (Vorlesung mit integrierten Übungen, 4 SWS)

Kiesmüller G, Sahin K

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0032: Business Forecasting | Business Forecasting

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In order to check whether students can calculate forecasts using the Python/nixtla, they receive data that needs to be analysed. They have to select a suitable method and determine the forecasts. Thus, the first part of the exam is a practice performance (programming task, 30%). In the second part of the exam, students must present the results of a small project that focuses on forecasting in a presentation (20 minutes, 70%) and be able to defend their approach in the subsequent discussion.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

- Basics in statistics
- Basics in probability calculus
- Programming skills in Python
- Basics in Forecasting

Content:

The following topics are discussed
analysis

1) Basics of time series

2) Arima

3) Hierarchical Forecasting

4) Global models

5) Random Forrest

6) Multivariate Forecasting

7) LGBM, XGBoost

8) Neural Networks

Intended Learning Outcomes:

The students

- understand the importance of data analysis for business forecasting
- know different forecasting techniques
- are able to select the right forecasting method and apply it
- can use Python and the forecasting libraries from nixtla to compute forecasts
- are able to present forecasts and explain their derivation

Teaching and Learning Methods:

Group work, Programming with Python, Presentations, Exercises

Media:

PowerPoint Slides, Screencasts, Video conferences, Exercises

Reading List:

- Hyndman, R.J., Athanasopoulos, G. (2012) Forecasting: principles and practice. Otexts
- Ord, K., Fildes, R. (2013) Principles of Business Forecasting. Cengage Learning.
- Joseph, J., Tackes, J., (2024) Modern Time Series Forecasting with Python. 2nd edition.

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

No management specialization chosen | No management specialization chosen

Advanced Seminar | Advanced Seminar

Module Description

MGTHN0096: Advanced Seminar Management & Marketing: Corporate Social Responsibility | Advanced Seminar Management & Marketing: Corporate Social Responsibility

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of handouts and presentation: In this course, students are to independently and individually read foundational and current scientific contributions on CSR and give a presentation on them, as well as actively participate in discussions in class and create a short handout. Additionally, they are supposed to write short essays, applying the gained knowledge to a real-life example.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization, and empirical research methods.

Content:

The course will be taught using a seminar or discussion style. Students will each give a presentation on a CSR topic discussed in class. The course includes for instance different perspectives, as well as critiques, actors and drivers of CSR. Moreover, managing CSR, a global perspective on the construct, as well as further selected topics with more recent relevance in CSR research will be discussed.

Intended Learning Outcomes:

After successful completion of this course, students will (1) have gained knowledge of different perspectives of corporate social responsibility research in strategic management, (2) have learned to read, understand, explain and critically reflect on scientific literature in the field, (3) have built a basic understanding of research methods in strategic management, (4) have gained the ability to discuss and elaborate on scientific literature on CSR in strategic management, (5) be able to apply this knowledge by analyzing CSR concepts in real-life examples.

Teaching and Learning Methods:

Students should read the assigned literature for the course and give a presentation on a topic that is derived from the literature discussed in class, also creating a short handout. In addition to the presentations students will deepen their knowledge by writing short essays, applying the concepts to real-life examples.

Media:

Powerpoint

Reading List:

Crane, A., McWilliams, A., Matten, D., Moon, J., & Siegel, D. S. (Eds.). (2008). The Oxford handbook of corporate social responsibility. OUP Oxford. Other current literature is selected specifically for each semester and will be distributed in class.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0099: Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy | Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The seminar prepares students to make tactical choices supporting organizational performance and strategy. The seminar enables students to understand the relationships between operations and supply chains and other areas of the organizations and to design reliable, fast, and efficient operations and supply chains.

Students will be guided to learn critically analyzing value chain reports and to identify problems and areas for potential improvements. Later, students will demonstrate their ability to apply these skills in a simulated value chain setting. After completing this seminar, students can identify value chain challenges, identify potential improvements, coordinate critical decisions across functional areas, and forecast results from the solutions implemented. As such, this seminar will prepare students for applying academic knowledge to industry practice.

Central to the seminar is a simulation consisting of several rounds. Students will be assigned to groups of four participants and have to formulate a strategy for their company. During each round, they will be required to make tactical and strategic choices to maximize value and prepare a four-page summary justifying the choices made and the expected results. This summary is part of the overall assessment (75% of the final grade, team grade). This assessment method helps students reason about the options available in each round, make informed decisions, develop predictive capabilities, work together in a team, and condense necessary information formally. Furthermore, students must submit a four-page report at the end of the course, including a critical reflection about teamwork and individual takeaways (25% of the final grade, individual grade). This assessment method allows students to reflect on their capabilities to work in a team and strengthen their personal learnings from the seminar.

At the beginning of the course, students will be advised on the report structure and submission deadlines.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Desirable but not mandatory: Production and Logistics, Management Science

Content:

Subject-specific content

- Strategic Alignment
- Value Chain
- Supply Chain Tactical Decisions
- Supply Chain Management
- Sales and Operations Planning

Intended Learning Outcomes:

Upon completion of the module, students are able to:

- understand the need to construct operations and supply chains that align with corporate-level strategies and identify appropriate actions
- analyze the relationships between operations and supply chain management and other areas of the organization (e.g., marketing, finance, accounting, engineering, R&D, etc.)
- evaluate the feasibility of functional-level actions with corporate-level strategies and assess the impact of their choices on the organization's performance
- coordinate actions and address competing interests across functional areas
- evaluate alternative tactical decisions and convince with compelling reasoning for their choice of action
- critically reflect upon their ability to work in a team and provide advanced improvement recommendations for effective teamwork

Teaching and Learning Methods:

Experiential learning

The module is designed to enable students to acquire knowledge on current (behavioral) operations and supply chain management topics through hands-on experience and reflection. To this aim, a web-based business simulation game will be employed. The business simulation game will engage participants in turning around a manufacturer of fruit juices by making tactical and strategic decisions on four critical areas of the organization: Purchasing, Operations, Supply Chain, and Sales. Students are required to analyze key performance indicators, define a strategy for turning around the organization, identify potential improvements, and make tactical decisions in each of the organization's four areas.

Seminar sessions take place physically, and the methods combine mini-lectures, hands-on experience, group reflection, and questions and feedback sessions. Each week, there will be two

90-minutes sessions that will allow for feedback, sparring sessions, and the introduction of new concepts.

Students will get an overview of the course in the first two weeks. They will be informed about the examination and expectations. Furthermore, students will be introduced to highly recognized supply chain strategy and integration papers. Students will also see examples and learn to interpret data from sales, operations, supply chain, and purchasing reports.

After setting the theoretical scene, students will form small groups of size 4 (groups of smaller size might also be formed depending on the total number of participants), and they will start working on the business simulation game. The simulation advances over several rounds and students must present their analysis, evaluation, decisions, and expected results. Further details will be shared during the seminar.

Media:

Research papers and further material to be shared via Moodle, in-class discussions

Reading List:

Readings for general preparation:

Myerson, P. (2012). Lean supply chain and logistics management. McGraw-Hill.

Croson, R., Donohue, K. (2006). Behavioral causes of the bullwhip effect and the observed value of inventory information, *Management Science*, 52(3), pp. 323-336. DOI: 10.1287/mnsc.1050.0436

Flynn, B.B., Huo, B. and Zhao, X. (2010), The impact of supply chain integration on performance: A contingency and configuration approach. *Journal of Operations Management*, 28: 58-71. DOI: 10.1016/j.jom.2009.06.001

Frohlich, M.T., Westbrook, R. (2001). Arcs of integration: an international study of supply chain strategies. *Journal of Operations Management*, 19, pp. 185-200. DOI: 10.1016/S0272-6963(00)00055-3

Further materials will be discussed in the session.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0110: Advanced Seminar Economics & Policy: Management practices in (family) enterprises | Advanced Seminar Economics & Policy: Management practices in (family) enterprises

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on management practices in (family) enterprises.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper, and (ii) oral in-class presentation. Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Content:

The seminar will cover key topics regarding management practices, such as:

1. How can management practices be classified?
2. How do management practices vary across companies?
3. How do management practices differ between family and non-family enterprises?
4. How do management practices relate to economic success?
5. How do management practices relate to work climate?
6. What are the determinants of different management practices employed in practice?

Intended Learning Outcomes:

'Students will learn the key concepts in the research field of management practices, with a special focus on family enterprises. The goal is twofold: First, to understand how management practices relate to economic success of enterprises. Second, to understand what individual and institutional factors determine management practices in enterprises. The course builds on original research on that topic. The course will cover basic theoretical consideration, but will have a clear focus on empirical studies with real-world examples and applications.

Teaching and Learning Methods:

'Students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence. A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups. We plan to complement student presentations by talks from corporate guests who will provide insights into corporate management practices.

Media:

Powerpoint slides

Reading List:

Amongst others (for detailed reading list, please see Moodle):

Bloom, N., & Van Reenen, J. (2010). Why do management practices differ across firms and countries?. *Journal of Economic Perspectives*, 24(1), 203-24.

Bloom, N., Brynjolfsson, E., Foster, L., Jarmin, R., Patnaik, M., Saporta-Eksten, I., & Van Reenen, J. (2019). What drives differences in management practices?. *American Economic Review*, 109(5), 1648-83.

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0115: Advanced Seminar: Digital Technologies in Operations Management | Advanced Seminar: Digital Technologies in Operations Management

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The key idea of this seminar is to lead students towards academic work and help prepare them for their master's thesis. In addition, they will learn about recent developments in digital technologies. Accordingly, the coursework and examinations are structured.

During the semesters, students will prepare two short presentations on research methods and skills. They will work in small teams. Each of the two presentations (ca. 45 minutes) accounts for 10% of the final grade and is a team grade.

Upon completion of this module, students will demonstrate their ability to cope with advanced research studies, comprehend and assess study outcomes, compare contributions of several studies, and transfer theoretical concepts to practice through a seminar paper (50% of the final grade; length of 15 pages including references). This method requires students to assess and replicate a study themselves.

In addition, they will prove their ability to communicate even complicated relationships and methods to their peer students through a presentation in a comprehensible fashion. They will further guide and moderate an ensuing discussion throughout which they will demonstrate their ability to criticize and assess innovative approaches and their potential shortcomings. The presentation, including moderation of ensuing discussion, accounts for 30% of the final grade (presentation duration 30 minutes + 30 minutes discussion). Complementing the written seminar paper, the presentation is targeted toward students who have not studied the same paper; this poses the challenge of presenting theoretical work interestingly while breaking down complex relationships into understandable information without losing rigor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Management Science, Production and Logistics

Content:

Subject-specific content

- Digital technologies potentially including but not limited to AR, VR, AI, Blockchain, and IoT.
- Management focus
- Incentive conflicts that arise when introducing new technologies in operations management
- Behavioral operations management

Methodological content (depending on assigned papers and limited to an introductory level)

- Research Design
- Analytical Modelling
- Game Theory
- Event Study Method
- Econometrics

Intended Learning Outcomes:

Upon completion of the module, students can

- understand and analyze state-of-the-art approaches to digital technologies in operations management,
- understand selected research papers and evaluate their key findings,
- present and explain complex research studies in a comprehensible and engaging manner,
- evaluate suggested management solutions in the digital technologies in an operations management context and assess their strengths and weaknesses,
- create management recommendations based on recent academic studies.

Teaching and Learning Methods:

Seminar

The seminar consists of two parts. During the semester, we will meet regularly to discuss select papers on digital technologies in operations management. Students will also explore and present information on select research methods (e.g., econometrics, game theory, and behavioral experiments). This part is complemented by a few interactive lectures.

At the end of the semester, we several sessions take placed blocked where students will present their assigned research studies (final assessment).

Media:

Research papers (to be shared via Moodle), student presentations, further material to be shared via Moodle, online discussions

Reading List:

Readings for general preparation (i.e., what is research? What marks a contribution?):

Wacker, J. G. (1998). A definition of theory: research guidelines for different theory-building research methods in operations management. *Journal of operations management*, 16(4), 361-385.

Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of management review*, 14(4), 490-495.

Examples for typical papers that we will discuss (this list is not meant to be comprehensive but gives some indication of topics to be covered):

Babich, V., & Hilary, G. (2020). OM Forum—Distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management*, 22(2), 223-240.

Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113-122.

Further papers will be provided at the beginning of the semester.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0123: Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals | Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals

Version of module description: Gültig ab summerterm 2023

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To successfully pass the module, the participants have to demonstrate methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations. The participants proof these competences through several presentations with written reports and active participation in discussions.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

No prerequisites required.

Content:

Artificial intelligence (AI) is one of the most profound ongoing technological developments which has the potential to strongly impact various fields of business and society. This module critically analyzes the effects of AI-applications on the sustainable development goals (SDGs) of the United Nations. Units of the module: Political and conceptual background of the SDGs; methods to assess impacts on the micro-, meso-, and macro-level; discussion of supportive and contradictory impacts of specific AI-applications on SDGs.

Intended Learning Outcomes:

After successful completion of the module, the participants have methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations.

Teaching and Learning Methods:

The module consists of an interactive introductory lecture and presentations of the participants with topic-specific written reports.

Media:

Textbook, presentation slides, and e-learning (Moodle).

Reading List:

Dodds, Donoghue und Leiva Roesch. 2016. Negotiating the sustainable development goals: A transformational agenda for an insecure world.

Kanie und Biermann. 2017. Governing through goals: Sustainable development goals as governance innovation.

Saetra. 2022. AI for the sustainable development goals.

Sobti und Sharma. 2018. An introduction to sustainable development goals.

Walker, Pekmezovic und Walker. 2019. Sustainable development goals: Harnessing business to achieve the SDGs through finance, technology, and law reform.

Responsible for Module:

Stich, Michael; Prof. Dr. rer. oec.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0171: Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises | Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on a specific topic related to the seminar's overall theme.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper (50%), and (ii) oral in-class presentation (50%). Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Principles of Economics (Master)

Content:

Beyond a lecture-style introduction into the topic, students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence.

A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups.

Intended Learning Outcomes:

Students will explore how technological advancements, ranging from automation to artificial intelligence, transform labor markets and the broader economy. The seminar will delve into the mechanisms behind these changes, examining their impacts on work, skills, and business structures, as well as the role of government policies in managing these transitions. Special attention will be given to comparing family and non-family enterprises.

The course builds on original research on the topic. While it will cover basic theoretical considerations, the focus will be on empirical studies with real-world examples and applications. Students will write a term paper on a selected topic and present their findings, highlighting practical implications for labor markets, policy, and business strategies.

Teaching and Learning Methods:

This seminar will cover key topics within the Economics of Technological Change and Labor Markets, with a focus on the role of family enterprises, such as:

Topic area 1: The impact of skill-biased technological change on wages and employment outcomes.

Topic area 2: Job polarization and the growth of low-skill service jobs in response to automation and routine-biased technological change.

Topic area 3: Labor market adjustments to technological innovations, including the effects of automation, robotics, and artificial intelligence.

Topic area 4: The unique challenges and opportunities for family enterprises in adapting to technological change and leveraging innovations to remain competitive.

Topic area 5: The role of government policies in managing labor market transformations and ensuring equitable outcomes.

Media:

Teaching will be in the form of a lecture and seminar presentations.

Reading List:

Acemoglu, Daron and David Autor (2011). "Skills, tasks and technologies: Implications for employment and earnings", In: Handbook of Labor Economics, Vol. 4, Elsevier, pp. 1043–1171.

Angrist, Joshua and Jörn-Steffen Pischke (2008). Mostly Harmless Econometrics (Princeton: Princeton University Press)

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0177: Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations | Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1. Group Report (70%)

- **Assessment Details:** In groups, students will collaboratively submit a report (no longer than 5 pages) addressing a specific managerial problem and proposing a solution using VR/AR technologies. Students are allowed to use reference materials and external resources.
- **Scope:**
 - Identify a relevant managerial problem in production, quality inspection, remote support, human resources, or customer experience.
 - Justify the importance of solving this problem.
 - Propose a detailed VR/AR-based solution, including technical implementation steps, management implications, and expected outcomes.
- **Rationale for Assessment:**
 - This assessment evaluates the ability to apply theoretical knowledge to practical problems, foster teamwork, and use technology to prepare solutions with managerial perspectives.
 - The report format develops students' ability to communicate complex ideas concisely and effectively.

2. Individual Critical Reflection (30%)

- **Assessment Details:** Each student will write a critical reflection on the relevance of immersive technologies (VR/AR) in solving managerial challenges. This will be submitted as a separate document, no longer than 2 pages. The reflection must be written independently.
- **Scope:**
 - Reflect on the practical and theoretical insights gained during the module.
 - Critically analyze the advantages and limitations of VR/AR in managerial contexts.

- Discuss how the experience of creating the group report has influenced their understanding of technology's role in management.

- **Rationale for Assessment:**

- This assessment allows students to self-evaluate, critically linking their learning to broader managerial and technological contexts.

- It promotes independent thought, allowing students to articulate their unique perspectives.

Repeat Examination:

(Recommended) Prerequisites:

Basic Programming with C#

Content:

In the module, the students will gain knowledge about the technological and management aspects of VR implementations with hands-on experience and use case examples in

- Production and Assembly
- Quality Inspection and Remote Support
- Training and skill development
- Customer Experience and Prototyping

Technology-specific content

- Introduction to Immersive Technologies (key hardware and software platforms)
- Using Unity for developing VR/AR applications

Management-specific content

- Ethical Considerations
- Behavioral Challenges
- Building a roadmap from prototype to market-ready product

Intended Learning Outcomes:

By the end of this module, students will be able to:

- Understand the fundamental principles of VR and AR technologies.
- Analyze the role of immersive technologies in improving managerial and operational efficiency.
- Design and prototype VR/AR-based solutions for specific management scenarios.
- Critically evaluate the broader implications of VR/AR deployment, including ethical concerns (e.g., data privacy, accessibility).
- Work collaboratively in teams to identify managerial challenges and develop VR/AR solutions.

Teaching and Learning Methods:

The coursework is primarily practical. At the beginning of the module, the students will be introduced to the immersive technologies and different headsets. They will be introduced to Unity Game Development Engine, which they will use to develop application solutions. The students

can choose between VR or AR applications and must program their software solution, keeping a managerial focus in mind.

The students will present the problem statement (in a group) that they want to work on and will have regular feedback sessions during the development phase. The students will present their solutions at the end of the module.

Media:

Material will be shared via Moodle

Reading List:

Readings for general preparation (but not limited to):

Dalton, J. (2021). Reality check: How immersive technologies can transform your business. Kogan Page Publishers.

Jung, T., & Dalton, J. (2021). XR Case Studies. Springer International Publishing.

Wuttke, D., Upadhyay, A., Siemsen, E., & Wuttke-Linnemann, A. (2022). Seeing the bigger picture? Ramping up production with the use of augmented reality. *Manufacturing & Service Operations Management*, 24(4), 2349-2366.

Windhausen, A., Heller, J., Hilken, T., Mahr, D., Di Palma, R., & Quintens, L. (2024). Exploring the impact of augmented reality smart glasses on worker well-being in warehouse order picking. *Computers in Human Behavior*, 155, 108153.

Seeliger, A., Cheng, L., & Netland, T. (2023). Augmented reality for industrial quality inspection: An experiment assessing task performance and human factors. *Computers in Industry*, 151, 103985.

Rejeb, A., Keogh, J. G., Leong, G. K., & Treiblmaier, H. (2021). Potentials and challenges of augmented reality smart glasses in logistics and supply chain management: a systematic literature review. *International Journal of Production Research*, 59(12), 3747-3776.

Eswaran, M., & Bahubalendruni, M. R. (2022). Challenges and opportunities on AR/VR technologies for manufacturing systems in the context of industry 4.0: A state of the art review. *Journal of Manufacturing Systems*, 65, 260-278.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations (MGTHN0177) (Seminar, 4 SWS)

Phadke V

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0181: Advanced Seminar Operations & Technology: Analytics in Manufacturing Management | Advanced Seminar Operations & Technology: Analytics in Manufacturing Management

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours:	Contact Hours:

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

This seminar aims to guide students toward academic research and prepare them for their master's thesis. The topics focus on advanced analytics in manufacturing management, covering AI methods, modeling techniques, optimization algorithms, and rigorous analysis of algorithmic properties, such as performance guarantees. While students are not expected to master all these techniques, they should be willing to develop deep expertise in one and understand its relevance for manufacturing decision-making. Additionally, they should be able to present their findings clearly and didactically to their peers.

Students work in small teams and complete two presentations and a term paper (~15 pages). The first presentation (30 minutes, 20% of the grade) provides a structured overview of a recent research article and formulates a research question for the seminar thesis. The second presentation (30 minutes, 30% of the grade) is the final presentation, showcasing their research findings. The term paper accounts for 50% of the grade.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Foundations in statistics and management science

Content:

This course focuses on advanced analytics in manufacturing, emphasizing algorithms, modeling, simulations, and theoretical analysis. It is open to all advanced analytics methodologies, provided they meaningfully support manufacturing decision-making.

Additionally, the course integrates practical experience by incorporating potential talks from industry practitioners and potential excursions to manufacturing facilities.

Intended Learning Outcomes:

Students will develop a comprehensive understanding of the scientific process, further develop critical thinking skills, and the ability to solve business problems efficiently. They will gain firsthand experience in manufacturing decision-making through practitioner interactions and engagement with cutting-edge research on data-driven manufacturing. Additionally, they will acquire hands-on expertise in advanced analytics methods applied to manufacturing decision-making.

Teaching and Learning Methods:

Teaching methods include presentations, interactive learning, e-learning, group discussions, and (if feasible) discussions and presentations by practitioners. Additionally, if feasible, we aim to include excursions to manufacturing and related facilities, as well as interactions with leading scientists in the field.

Media:

Zoom, literature databases (Ebsco, ScienceDirect etc.), standard solvers (Gurobi, Hexaly etc.), databases and case studies, programming environments and software development platforms, simulation environments etc.

Reading List:

We focus on high-profile and influential ideas in advanced analytics for manufacturing, typically published in leading journals such as Operations Research, Management Science, Manufacturing & Service Operations Management, INFORMS Journal on Optimization, European Journal of Operational Research, and INFORMS Journal on Computing. Below are examples of relevant articles:

[AI] Senoner, J., Netland, T., & Feuerriegel, S. (2022). Using explainable artificial intelligence to improve process quality: Evidence from semiconductor manufacturing. *Management Science*, 68, 5704-5723.

[mathematical analysis] Drozdowski, M., Jaehn, F., & Paszkowski, R. (2017). Scheduling position-dependent maintenance operations. *Operations Research*, 65, 1657-1677.

[modeling] Möhring, R., Schulz, A. S., Stork, F., & Uetz, M. (2003). Solving project scheduling problems by minimum cut computations. *Management Science*, 49, 330-350.

Da Col, G., & Teppan, E. C. (2022). Industrial-size job-shop scheduling with constraint programming. *Operations Research Perspectives*, 9, 100249.

[algorithmics] Koca, Y., Martagan, T. G., Adan, I. J. B. F., Maillart, L., & van Ravenstein, B. (2023). Increasing biomanufacturing yield with bleed-feed: Optimal policies and insights. *Manufacturing & Service Operations Management*, 25, 108-125.

Barnhart, C., Jacquillat, A., & Schmid, A. (2024). Robotic warehousing operations : A learn-then-optimize approach to large-scale neighborhood search. *INFORMS Journal on Optimization*, 1-24.

Responsible for Module:

Otto, Alena; Prof. Dr. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0183: Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research | Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students work in groups of 3-4 to design and implement a VR-based marketing research experiment. Following experimental design, students will create VR scenarios using industry-standard software and hardware. Final deliverables include: (1) an in-class presentation demonstrating the functional VR experiment and discussing findings (30%), and (2) a comprehensive written report (15-20 pages) documenting the research question, methodology, technical implementation with code, results, data analysis, and reflection on the research process (70%).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic programming experience, preferably C# or similar object-oriented language (highly-recommended)

Content:

In an introductory block, participants will acquire key methodological skills in experimental design, virtual reality implementation, and programming for marketing research in immersive environments. Students will critically reflect on how to use VR for marketing research based on relevant scientific literature. Based on literature review and supervisor consultations, students will conceptualize and develop original VR implementations.

The seminar will conclude with in-class project presentations and written documentation of all VR implementation code and assets.

Intended Learning Outcomes:

Students will acquire skills in the following areas:

- Software development: Using a game engine (Unity) and programming (C#) to implement virtual reality studies in a marketing context
- Experimental design: Conceptualizing and planning VR study implementations
- Literature analysis: Reviewing, evaluating, and synthesizing relevant scientific literature
- Academic communication: Presenting research results and defending methodological choices
- Technical documentation: Preparing a comprehensive seminar report that includes software architecture, code documentation, and programming implementation

Teaching and Learning Methods:

The seminar starts with introductory lectures providing examples of virtual reality as a research environment for investigating marketing research questions. Lectures introduce the Extended Reality Lab equipment and guide the implementation of VR experimental designs. Subsequently, student groups develop their experimental implementations while receiving ongoing supervision. Students are also expected to pilot-test their VR implementations and collect data with actual participants. Finally, all groups present their results in class and document their work in a seminar report.

Media:

PowerPoint, videos, AI image generation software, research papers, virtual reality devices, exercise sheets

Reading List:

Will be published at the start of the semester

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research (MGTHN0183) (Seminar, 4 SWS)

Meißner M, Siqueira Rodrigues L

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0196: Advanced Seminar Management & Marketing: Language Analytics and LLMs in Marketing | Advanced Seminar Management & Marketing: Language Analytics and LLMs in Marketing

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 135	Contact Hours: 45

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The assessment consists of a group-based project (70 points) and individual live-quizzes (20 points). In total, students can earn 90 points in the module.

Group project (70 points)

The group project is completed in teams of three to four students. Students independently apply one or more text analytics methods introduced in the seminar to a marketing-related question. The project includes the following steps: Development of a research idea and question; allocation of responsibilities within the group; short literature review; data generation, scraping, or selection; data preprocessing and cleaning, application of text analytics methods, interpretation and discussion of results.

The project outcomes are documented in a scientific paper-style report (10 to 15 pages, excluding references) with the following structure: Theoretical background - Method - Results - Discussion.

The 70 points are distributed as follows: 50 points are given based on the individual contribution to the group project. 20 points are given based on an individual written reflection and a structured peer evaluation of the team collaboration.

Live quizzes (20 points)

Throughout the semester, students complete five short live quizzes based on selected research papers discussed in class (duration per quiz: about 15 minutes, no aids permitted). Each quiz assesses students' understanding of the applied use of text analytics methods in empirical marketing research. Each quiz awards up to 5 points. Students may achieve the full 20 points by participating in any four quizzes. Participation in all five quizzes allows students to earn up to 5 bonus points (maximum of 20 points in total across all quizzes).

Repeat Examination:

(Recommended) Prerequisites:

Basic programming knowledge in R, very good English language skills, undergraduate level courses in research methods, statistics, and marketing. Prior experience with quantitative research projects is also helpful.

Content:

In today's digital economy, vast amounts of unstructured text data are generated by customers, companies, and online platforms. To achieve a strong market orientation, firms and researchers increasingly rely on systematic language analysis to extract insights from customer feedback, online reviews, social media posts, and open-ended survey responses. Modern language analysis enables researchers and practitioners to analyze such data in a scalable, objective, and theory-guided manner.

This seminar provides a both practical and research-oriented introduction to contemporary methods of text analysis in marketing. Topics include:

- String handling, regular expressions, and data structures
- Obtaining text data through web scraping and APIs
- Text preprocessing, TF-IDF, bag-of-words techniques
- Clustering, similarity measures, and grouping
- Text classification through k-Nearest Neighbors and Neural Networks
- Topic Modeling through Latent Dirichlet Allocation (LDA)
- Sentiment Analysis and Machine Learning
- Word Embeddings and Transformer Models (e.g., BERT-style models)

Intended Learning Outcomes:

After successfully completing the module, participants are able to:

- Identify marketing-relevant use cases for the application of text analytics to customer communication, social media, and survey data.
- Select, implement, and apply appropriate empirical text analysis methods to address a given research question.
- Interpret and critically evaluate the results of text-based empirical analyses with respect to their methodological validity, theoretical implications, and strategic recommendations for marketing decisions.
- Design, conduct, and document an independent empirical research project using text analysis methods in the form of a small research paper.

Teaching and Learning Methods:

The module consists of interactive lectures that are immediately followed by hands-on coding exercises. The lectures introduce the theoretical foundations and core methodologies of language analytics, while the exercises enable students to directly apply these methods to real datasets.

In addition, selected sessions focus on the critical discussion of current research papers in text analytics and marketing. Throughout the semester, students work in small groups on an independent project and regularly present intermediate results for feedback and reflection. This combination of lecture, practical exercises, paper discussion, and project-based learning supports the development of methodological, research-oriented, and practical competencies.

Media:

Programming code in R, datasets, PowerPoint slides, scientific articles, Mural boards

Reading List:

Dan, D., & Reutterer, T. (2026). Text analytics in marketing: A practical guide for students and researchers. Springer.

Responsible for Module:

Eckert, Christine; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0013: Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance | Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises a scientific coursework (approx. 25-30 pages), which is supposed to address the research questions posed within each given seminar topic. The task is to be conducted as a group work (i.e. 2-3 students per topic). Additional to the written scientific report, candidates are required to showcase main findings within a 30-minute presentation and answer further questions following their presentation.

In doing so, the candidates are required to prove that they have dealt with the given topic in a comprehensive manner, that their selection and evaluation of literature is adequate and that they have in-depth analyzed the topic while simultaneously putting it in a higher context of financial research. Also, it can be assessed whether the students are able to answer further questions in a correct and comprehensive manner.

The written assignment will be weighted 2/3 and the formal presentation will make 1/3 of the final grade. Each student will be assessed separately.

To maintain the seminar atmosphere, the number of participants is generally limited. If the course is offered in a hybrid format, Munich students may also participate in the course. The targeted maximum number of participants is 12. More information will be provided in the introductory session.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;

basic knowledge in scientific writing;
combination with the module "Digital Finance" is recommended

Content:

Candidates are supposed to build groups of 2-3 students and choose one seminar topic out of a set of pre-defined scientific issues. The pre-selection omprises, but is not limited to topics of the following areas:

- Asset Pricing of Cryptocurrencies: Market (In-)Efficiency, Behavioral Finance & Bubbles;
- Financial Innovations: Foundations and Opportunities & Threats to Financial Market Stability;
- Wisdom of the Crowd? Information Asymmetries, Adverse Selection, and Moral Hazard in Peer-to-Peer Lending Markets Around the World;
- Robo-Advisors: Overview of Services, Benefits, and Limitations;
- Social Investing: Overview, Benefits, and Limitations;
- Machine Learning: An Overview of ML Applications for Finance;
- etc.

Candidates are expected to conduct a comprehensive literature review of the most important scientific articles capturing the chosen seminar topic in order to address the given research questions in a systematic and structured manner.

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- describe the effects of current FinTech developments for existing financial products and processes;
- exhibit a comprehensive overview on current research topics in the field of Digital Finance;
- identify chances and opportunities of these developments (especially with regard to the overall macroeconomic stability);
- conduct an independent literature analysis at the highest possible, international level;
- analyze and evaluate references in a systematic manner;
- combine topics in Digital Finance with theories/methods/models of traditional finance in an independent manner;
- deriver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise an initial lecture (overview course) as well as the corresponding main seminar;

Methods of teaching include a group work and the formal presentation of obtained results;

the learning methods of the students primarily comprise the following activities:

- independent literature research (usage of scientific articles published in international top journals);
- Collaborative writing of a scientific report;
- Exercise of a deductive, logic and consistent argumentation to specifically address and answer the posed research questions;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to conduct independent academic work and to elaborate thematically complex contents on their own. It is considered to be a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

Machi, L. A., & McEvoy, B. T. (2012). The literature review: Six steps to success (2nd ed.). Corwin Press.

A separate reading list for each topic will be provided in the introductory lecture.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Finance & Accounting (WIHN0013): Current research Topics in Digital Finance (Seminar, 4 SWS)

Gong Z, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0017: Advanced Seminar Innovation & Entrepreneurship: Digital Innovation | Advanced Seminar Innovation & Entrepreneurship: Digital Innovation

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Seminar paper with presentation: The students independently and individually prepare a scientific seminar paper and present it during the course.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Very good English language skills are required. Students should also have an understanding of basic principles of management, innovation, and organization as well as empirical research methods. Attendance of the course Empirical Research Methods in Management and Economics recommended, but not mandatory.

Content:

The course prepares students for their Master's Thesis by introducing them to (1) the search, understanding, and analysis of academic literature, (2) methods for data collection and data analysis, (3) the structuring and writing of a paper, and (4) giving presentations to an academic audience including a subsequent discussion. The course exposes students to digital platforms and open forms of organizing innovation.

Intended Learning Outcomes:

After the course students will be able to:

- Define, explain and apply selected key concepts in the field of digital innovation
- Read and understand academic literature
- Write and present an academic paper

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via presentations and group discussions
- Build up critical thinking and interpretation skills

Learning Objectives

At the end of this course, students will be able to demonstrate understanding, critical assessment and application of the following:

- Learn how to evaluate academic literature
- Interact with an academic debate
- Prepare, elaborate and defend an academic argument
- Specialist knowledge on their selected topic of interest

Teaching and Learning Methods:

Introductory session will provide introductions into the core topics and scientific writing in lecture style. All other sessions will be focused on the individual student presentations, in which each student presents their seminar paper to the class and all other students are encouraged to discuss the papers.

Media:

Powerpoint

Reading List:

Literature is semester-specific and will be distributed in class

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0028: Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship | Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic related to corporate entrepreneurship. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade, approx. 10 pages) The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade, approx. 10 minutes). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis).

Repeat Examination:

(Recommended) Prerequisites:

Basic knowledge of corporate entrepreneurship and research methodologies is recommended.

Content:

Foundations of Corporate Entrepreneurship (CE): Key concepts, definitions, and the role of CE in modern businesses.

Strategic Thinking in CE: Developing an entrepreneurial mindset to identify and leverage opportunities.

Implementing CE: Steps to integrate entrepreneurial practices into corporate operations.

Building an Entrepreneurial Organization: Creating a culture that fosters innovation and risk-taking.

Sustaining CE: Strategies to maintain entrepreneurial energy and adapt to changing markets.

Leadership and Culture: The impact of leadership styles and organizational culture on entrepreneurship.

Entrepreneurial Traits and Behavior: Key personal characteristics that support entrepreneurial initiatives.

Creativity and Motivation: Harnessing creativity, motivation, and HRM practices to drive innovation.

Incentive Systems and Psychological Ownership: Enhancing engagement through effective incentives and ownership.

Entrepreneurial Intensity: Measuring and increasing the level of entrepreneurial activity within corporations.

Focus Areas of CE: Examining CE at the levels of the phenomenon, organization, individual, and dynamic interactions.

Challenges in CE: Addressing obstacles faced by corporations and strategies to overcome them.

Intended Learning Outcomes:

Understand the current state of research in corporate entrepreneurship.

Develop the ability to critically analyze and synthesize academic literature.

Improve academic writing and presentation skills.

Identify research gaps and formulate theoretical and practical implications.

Teaching and Learning Methods:

In-presence seminar sessions

Literature review and independent research.

Group discussions and collaborative learning (optional group work).

Interactive feedback sessions.

Presentation of results and peer review

Media:

Moodle platform for resources and updates.

Academic databases for literature access.

Presentation tools (e.g., PowerPoint, posters)

Reading List:

Core reading will be provided on Moodle. Students are encouraged to explore additional academic articles and books relevant to their selected topic.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0031: Advanced Seminar Operations & Supply Chain Management: Digital Operations | Advanced Seminar Operations & Supply Chain Management: Digital Operations

Version of module description: Gültig ab summerterm 2020

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination consists of a written Seminar paper (75%) to check the learning outcomes 1-5, as well as an oral presentation (25%) in order to check learning outcome 6. The seminar paper should cover 15 - 20 pages and is written in the style of current publications of peer-reviewed journal articles. At the end of the module students present their work in a 20 minutes presentation + 15 minutes discussion and have to initialize and moderate the discussion on a selected paper of their fellow participants.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

The courses Management Science and Production and Logistics

Content:

The advanced seminar in logistics and supply chain management focuses on recent research progress on varying topics, e.g. after sales service, procurement.

- In a seminar on After Sales Service the following topics can be discussed: Forecasting Spare parts Demand, Inventory Management of Spare Parts, Reliability Analysis, Maintenance Strategies, Level of Repair Analysis, Repair Kit Problem, Design of Service Supply Chains, Additive Manufacturing of Spare Parts

-In a seminar on procurement the following topics can be discussed: outsourcing, global sourcing, multiple sourcing, order splitting, e-procurement, auctions, procurement on online spot markets,

Intended Learning Outcomes:

The objective of the module is to equip the participants with the necessary skills and tools for a successful master thesis project. Specifically, the aim is to be able to:

- 1 Read and understand recent research contributions
- 2 Pursue interesting research questions
- 3 Conduct a literature study and/or numerical study and/or implementation
- 4 Structure and organize research methods and results
- 5 Write a seminar paper
- 6 Present research finding and defend them in a discussion

Teaching and Learning Methods:

In an introductory session, the current theme of the module is explained by the lecturer and the various available seminar topics are elaborated in detail. Also information on relevant literature for the problem settings is introduced, which forms the basis of the students' seminar paper. After the introductory session, students will work out the topic on their own, by using their abilities of conducting literature research, mathematical modelling, programming and analyses. Throughout the whole time, they receive guidance from a supervisor of the chair. Different milestones are to be achieved at specific dates, such as a preliminary outline of the seminar paper, first research results and the final paper. Following the submission of the final paper, presentations and discussions of all students' seminar papers are conducted, usually spanning one or several days, where amongst others also presentation, moderation and discussion skills are trained.

Media:

Presentation, Various form of literature (Journal Articles, Books, Report, Conference Proceedings, etc.)

Reading List:

'Dependet on seminar focus, e.g.:
R.Dekker, W.D.Rustenburg (2014)

E. vanWingerden, R.J.I.Basten,

More grip on inventory control through improved forecasting:

A comparative study at three companies. IJPE 157, 2020-237

Joeri Poppe, Rob J.I.

Basten, Robert N. Boute, Marc R. Lambrecht. (2017) Numerical study of inventory management under various maintenance policies. Reliability Engineering and System Safety 168, 262-273.

Bram Westerweel, Rob J.I. Basten, Geert-Jan van Houtum (2018). Traditional or Additive Manufacturing? Assessing component design options through lifecycle cost analysis. EJOR 270, 570-585

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0046: Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises | Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises

Version of module description: Gültig ab winterterm 2020/21

Module Level: Master	Language: English	Duration: one semester	Frequency:
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic in the field of sustainable entrepreneurship / sustainable management in family enterprises. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis). Active engagement in the course will be expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English

Content:

This course addresses various aspects of sustainability available to new ventures and established family enterprises. Sustainable entrepreneurship (SE) emphasizes that entrepreneurs may pursue additional goals such as social and ecological goals that may alleviate social and ecological problems. It explores the opportunities that entrepreneurs create, the challenges entrepreneurs encounter, and the ways in which entrepreneurs exploit opportunities to contribute toward enhancing the society while simultaneously starting and sustaining a profit-oriented business. This course also examines the role of sustainability in family enterprises:

Course content will comprise:

- The importance of sustainable entrepreneurship in solving grand societal challenges of the 21st century
- Motives and barriers of sustainable entrepreneurship
- Overview of theories and concepts in sustainable entrepreneurship
- Stakeholder management for entrepreneurs
- Sustainable management in the context of family enterprises
- Sustainability in entrepreneurship and family enterprises in the context of developing countries
- Recognizing and exploiting social and ecological opportunities
- Practical examples of sustainable ventures

Intended Learning Outcomes:

After completing the seminar students, should understand how entrepreneurs and family enterprises can contribute to solving societal challenges. Therefore, students will be able to:

- understand the concept of sustainable entrepreneurship and sustainability in family enterprises
- understand and evaluate how entrepreneurs can enact on sustainable opportunities
- understand the entrepreneurs' motives and barriers to engage in sustainable entrepreneurship
- reflect on the importance of diversity in achieving sustainability in entrepreneurship and family enterprises
- evaluate strategic actions of sustainable enterprises
- evaluate hands-on examples and cases of how sustainable entrepreneurship can be implemented in practice with specific focus on the Heilbronn region: Those examples may relate to various sectors such as the public governance sector, the food and agriculture, environmental and mobility sector.

In addition, students will be able:

- to search, understand, and analyze academic literature by evaluating key findings and explaining complex research studies in a comprehensible and interesting manner,
- to apply methods for data collection and data analysis, - to structure and write a seminar thesis, and
- to present and subsequently discuss findings of their seminar thesis.

Teaching and Learning Methods:

' -Through lectures, supported by Power-Point presentations, the instructor will provide the theoretical foundations of sustainable entrepreneurship

- The content is discussed in the course by exchanging ideas, questions, and comments to encourage an open learning atmosphere
- Every sessions contains exercises, where the students apply their learnings in a group work (e.g. to present one start-up, enterprise or organization that has already successfully implemented the topic/concept presented in the lecture)
- Guest speakers will share practical insights and will strengthen the understanding of key concepts and processes and expose perspectives complementary to the topic of the seminar
- For their seminar thesis, students will investigate a topic within the subject of this course to develop a seminar thesis. Students will receive feedback from the instructor.
- In a final presentation, students present the results of their project thesis

Media:

Powerpoint, Zoom- & Breakout-Sessions, Online Simulations

Reading List:

Basic literature (for detailed reading list, see Moodle):

- Cruz, C.; Larrazza-Kintana, M. Garcés-Galdeano, L. Berrone, P. (2014). Are family firms really more socially responsible? *Entrepreneurship Theory and Practice* 38(6), 1295–1316.
 - Gast, J., Gundolf, K., & Cesinger, B. (2017). Doing business in a green way: A systematic review of the ecological sustainability entrepreneurship literature and future research directions. *Journal of Cleaner Production*, 147, 44-56.
 - Le Breton-Miller, I., & Miller, D. (2016). Family firms and practices of sustainability: A contingency view. *Journal of Family Business Strategy*, 7(1), 26-33.
 - Lumpkin, G. T., Bacq, S., & Pidduck, R. J. (2018). Where change happens: community#level phenomena in social entrepreneurship research. *Journal of Small Business Management*, 56(1), 24-50.
 - Patzelt H., & Shepherd D.A. (2001) Recognizing opportunities for sustainable development. *Entrepreneurship Theory and Practice*. 35(4), 631-652.
 - Sarason, Y., & Dean, T. J. (2019). Lost Battles, Trojan Horses, Open Gates, and Wars Won: How entrepreneurial firms co-create structures to expand and infuse their sustainability missions in the acquisition process. *Academy of Management Perspectives*, 33(4), 469-490.
 - Sarasvathy, S. D., & Ramesh, A. (2019). An effectual model of collective action for addressing sustainability challenges. *Academy of Management Perspectives*, 33(4), 405-424.
 - Shevchenko, A., Levesque, M. & Pagell, M. (2016). Why firms delay reaching true sustainability. *Journal of Management Studies* 53, 911-35.
 - Zahra, S. A., Gedajlovic, E., Neubaum, D. O., & Shulman, J. M. (2009). A typology of social entrepreneurs: Motives, search processes and ethical challenges. *Journal of Business Venturing*, 24(5), 519-532.
- Book(s):
- Lumpkin, T. & Katz, J. (2011). *Social and sustainable entrepreneurship*. Emerald Group Publishing: Bingley (UK).
 - Weidinger, C., Fischler, F. & Schmidpeter, R. (2014). *Sustainable entrepreneurship: Business success through sustainability*. Springer: Berlin (Germany).

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0046): Sustainability in Entrepreneurship and Family Enterprises (Seminar, 4 SWS)

Bird M, Kaufmann L

For further information in this module, please click campus.tum.de or [here](#).

Catalogue of Elective Modules | Wahlkatalog

Module Description

MGTHN0074: Contemporary Strategies in the Automotive Industry | Contemporary Strategies in the Automotive Industry

Version of module description: Gültig ab winterterm 2021/22

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The performance of students for this course will be determined by:

- 40% of a comprehensive industry and company analysis in form of PowerPoint presentation
- 40% of continuous monitoring of real-time event, market movement, company announcement
- 20% of processing and consolidating all theory from the seminar and information in self-study into a final project presentation, delivered in PowerPoint deck and verbally

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English, fundamentals in business administration and/or strategic management

Content:

This seminar aims to deliver an overview of recent development of discussion in the automotive industry, and to offer students the chance to consolidate the knowledge from strategic and competitive management to analyse case studys independently.

Automotive is one of the most important pillars of Germany's economy. It is now facing the challenges of transformation: rising conscious for environmental protection from the general public and legislator, pressing need for alternative powertrain technologies, deeper involvement and engagement from the tech powerhouses, ever complicating partnership and competition landscape.

This seminar examines the issues above and the standpoint of major market participants, both established OEMs and new entrants - their reactions, strategic positions, and options.

Furthermore, the interpretation of strategy related topics, e.g. core company character, corporate governance, decision-making process, business communication will also be important components to complete the scope of this seminar.

Intended Learning Outcomes:

After the course students will be able to:

- Understand the fundamental theories of automotive management and industry-specific topics
- Apply the knowledge from strategic & international management, as well as competitive strategy in practical oriented course work
- Gain experience in building strategic framework for a market participant and evaluating the impact of influential factors
- Enhance the ability of academic reading, information gathering and research, data analysis
- Further develop communication skills through verbal business presentation and group discussion
- Build up critical thinking and interpretation skills

Teaching and Learning Methods:

This seminar will be aided by PowerPoint slide deck to provide introduction of fundamental theories. Students will be required to self-reading and closely following the real-time market movement.

"War gaming" methods to simulate real-time strategy deployment: each student to prepare keynote to present the strategy analysis of one market participant of their choice - so the students would exercise their ability of information gathering, data processing and business presentation.

Group discussion to ignite the exchange of ideas and provide each other feedbacks.

Industry experts will be invited as guest speakers to provide empirical insights.

Media:

PowerPoint, flipchart, case study materials etc.

Reading List:

Uwe Winkelhake (2022), The Digital Transformation of the Automotive Industry

Jens Diehlmann, Joachim Häcker (2013), Automotive Management. Navigating the next decade of auto industry transformation

Léa Platel (2010), Rethinking communication strategies in the automotive industry

Andreas Blum (2009), Co-opetition in the Automotive Industry

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0077: Advanced International Study Trip: Family & non-family firms around the world | Advanced International Study Trip: Family & non-family firms around the world

Version of module description: Gültig ab summerterm 2024

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 9	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each student will individually work on a seminar thesis and participate in the discussion with international firms and universities. The examination consists of two parts:

1) Seminar thesis (70 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics discussed in the course
- they have understood and applied the theoretical frameworks discussed in the course
- they show that they are able to write an academic paper that follows scientific standards
- they are able to structure and write a seminar thesis based on conceptual theories

2) Participation in discussion during excursion (30 % of the overall grade).

The students should demonstrate that:

- they have gained the skills to contribute to conceptual and theoretical discussions with firms and universities
- they have gained an understanding of quantitative-empirical research work that can be applied to companies and their business models
- Active engagement in the course is expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

English oral and written fluency

Content:

Family and non-family firms need to succeed globally in order to survive and thrive in today's business environment. Thus, these firms need managers who are well-versed in international

business environments. This course aims to provide students an immersive and intense international experience through a 10-day excursion abroad.

Intended Learning Outcomes:

After completing this course students should be able to:

- Have a scientific understanding on how to do business abroad
- Feel comfortable and gain proficiency in communicating and doing business abroad

Skills Objectives

- Understand theoretical frameworks that allow them to navigate different cultures
- Improve writing and written communication skills
- Gain in-depth understanding of critical thinking, ethnorelative perspectives and interpretation skills

Teaching and Learning Methods:

This module includes the following teaching and learning methods:

- Immersive learning of culture, business, and people in a different country through an excursion
- Key concepts and empirical findings from the field of strategic management of family and non-family firms will be provided through discussions with international firms and universities.
- Students will acquire additional knowledge from academic literature through their own work on their seminar theses.

Media:

PowerPoint, research papers, case scenarios

Reading List:

tba

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced International Study Trip (MGTHN0077): Family & non-family firms around the world (Seminar, 6 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0084: Advanced International Excursion: Global Strategy | Advanced International Excursion: Global Strategy

Version of module description: Gültig ab summerterm 2022

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 9	Total Hours: 270	Self-study Hours: 180	Contact Hours: 90

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each student will work on a seminar thesis and participate in the discussion with international firms and universities. The examination consists of two parts:

1) Seminar thesis (70 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics discussed in the course
- they have applied the frameworks introduced in the course
- they show that they are able to write a paper that follows scientific standards
- they are able to structure and write a seminar thesis

2) Participation in discussion during excursion (30 % of the overall grade).

The students should demonstrate that:

- they have gained the skills to contribute to discussions with firms and universities
- Active engagement in the course is expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

English oral and written fluency

Content:

Multinational firms need to succeed globally to thrive in today's business environment. Thus, these firms need managers who are well-versed in international business environments. This course aims to provide students an immersive and intense international experience through a 9-day excursion abroad.

For 2022, the destination will be Paris.

Intended Learning Outcomes:

After completing this course students should be able to:

- Feel comfortable and gain proficiency in communicating and doing business abroad

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via in-class discussions
- Build up critical thinking, ethnorelative perspectives and interpretation skills

Teaching and Learning Methods:

This module includes the following teaching and learning methods:

- Immersive learning of strategy, localization, and value chain activities in a different country through an excursion
- Key concepts and empirical findings from the field of strategic management will be provided through discussions with multinational firms.
- Students will acquire additional knowledge from practical literature through their own work on their seminar theses.

Media:

PowerPoint, research papers, case scenarios

Reading List:**Responsible for Module:**

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced International Excursion: Global Strategy (MGTHN0084) (Seminar, 6 SWS)

Li C, Schilder J

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0095: Social Media Marketing | Social Media Marketing

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Grading based on a written multiple-choice examination of 90 minutes (100%) with 35 multiple-choice assignments, designed to assess students' ability to outline and explain core concepts of social media marketing in a structured manner. The exam includes both knowledge-based questions and applied tasks that require students to demonstrate their understanding of electronic word-of-mouth marketing, influencer marketing, and digital consumer feedback. Additionally, case studies or practical scenarios will be used to evaluate students' ability to critically reflect on emerging topics such as meme marketing or social media firestorms.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Completed course on fundamentals in Marketing (mandatory)

Content:

Social media have gained strong importance in both practice and academia. Corresponding digital technologies changed the way how consumers communicate, how they gather information, and how they spend large portions of their life. In this spirit, social media also opened up new opportunities for marketers to target and reach potential customers online. The lecture will cover the basics of social media (marketing), (electronic) word-of-mouth marketing, endorser marketing, cancelling, meme marketing, decoding digital consumer feedback, and new social media formats.

Intended Learning Outcomes:

Upon successful participation, students will be able to outline core concepts of social media and social media marketing. Students will understand and be able to explain the importance of electronic word of mouth marketing. Moreover, students will be able to outline differences between endorser and influencer marketing. They will be capable to explain key mechanism of influencer

marketing and persuasion processes. Students will have an understanding of novel topics in social media marketing as well, these include meme marketing, the threat of cancelling or social media firestorms. Students will have an overview of approaches to analyze social media data for marketing purposes. Students will be aware of new directions in social media and will be able to discuss what this implies for stakeholders. Finally, students will be capable to read and understand current research in social media marketing and to discuss their findings.

Teaching and Learning Methods:

The lecture will consist of input by the lecturer with interactive class discussions. Lectures will include critical discussions of scientific papers.

Media:

PowerPoint, videos, AI image generation software, research papers, virtual reality devices, exercise sheets

Reading List:

Ismagilova, Elvira; Dwivedi, Yogesh K.; Slade, Emma; Williams, Michael (2017): Electronic Word of Mouth (eWOM) in the Marketing Context. A State of the Art Analysis and Future Directions: Springer.

Appel, G., Grewal, L., Hadi, R., & Stephen, A. T. (2020). The future of social media in marketing. Journal of the Academy of Marketing Science, 48(1), 79-95.

Further reading material will be provided in the course.

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0096: Advanced Seminar Management & Marketing: Corporate Social Responsibility | Advanced Seminar Management & Marketing: Corporate Social Responsibility

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of handouts and presentation: In this course, students are to independently and individually read foundational and current scientific contributions on CSR and give a presentation on them, as well as actively participate in discussions in class and create a short handout. Additionally, they are supposed to write short essays, applying the gained knowledge to a real-life example.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization, and empirical research methods.

Content:

The course will be taught using a seminar or discussion style. Students will each give a presentation on a CSR topic discussed in class. The course includes for instance different perspectives, as well as critiques, actors and drivers of CSR. Moreover, managing CSR, a global perspective on the construct, as well as further selected topics with more recent relevance in CSR research will be discussed.

Intended Learning Outcomes:

After successful completion of this course, students will (1) have gained knowledge of different perspectives of corporate social responsibility research in strategic management, (2) have learned to read, understand, explain and critically reflect on scientific literature in the field, (3) have built a basic understanding of research methods in strategic management, (4) have gained the ability to

discuss and elaborate on scientific literature on CSR in strategic management, (5) be able to apply this knowledge by analyzing CSR concepts in real-life examples.

Teaching and Learning Methods:

Students should read the assigned literature for the course and give a presentation on a topic that is derived from the literature discussed in class, also creating a short handout. In addition to the presentations students will deepen their knowledge by writing short essays, applying the concepts to real-life examples.

Media:

Powerpoint

Reading List:

Crane, A., McWilliams, A., Matten, D., Moon, J., & Siegel, D. S. (Eds.). (2008). The Oxford handbook of corporate social responsibility. OUP Oxford. Other current literature is selected specifically for each semester and will be distributed in class.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0097: CEO Leadership Series | CEO Leadership Series

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The exam is a scientific work that consists of the following two components: 1) Students will individually submit written contributions in the form of questions to the CEOs and voluntarily, in a group of 2 students, an analysis of CEOs and their companies (50%). The individual submission includes 12 questions, which need to be lecture-specific, referring to the concepts presented in the first lecture, such as leadership skills, which include, for instance, decision-making, critical thinking, or employee motivation based on theoretical frameworks that include institutional theory, upper echelon theory, and stakeholder theory. The group work is a report of approximately 500-1,500 words per group, whereby each student adds at least one chapter of analysis. This can include an analysis of leadership style, decision-making processes, organizational culture, succession planning, or the alignment of leadership strategies with overall business objectives. The scientific work examines the extent to which the students can understand leadership skills in theory and practice. It evaluates if students are able to analyze companies and leaders, as well as their behavior, and apply their knowledge of leadership by interacting with CEOs. Students will further participate in a short quiz immediately after each CEO visit to test their understanding of the topics discussed. 2) Oral participation will be graded (50%). In the second half of the lecture, which lasts approximately one hour, students ask questions to the CEOs. They are supposed to actively and individually contribute to the talk on stage. The oral participation checks that the students are able to listen and comprehend the talks and transfer and apply the concepts by asking related questions. Students will also gain practical knowledge and be able to improve their discussion- and soft skills.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization.

Content:

This module is the flagship event of TUM C.HN and will take place on Tuesdays or Fridays at 3 pm (each date will feature one guest). It covers practical insights on firm strategies, whereby lessons can be learned from CEOs themselves. The covered strategy topics may include internationalization, the analysis of the internal and external environment as well as M&A and other types of strategic alliances. This course also provides students with valuable insights for their future work as strategizers in a corporate environment.

Please note: Registration is mandatory. There will be a participants list at the entrance and police presence for each event.

The guests for the winter semester 2023/24 are e.g.: Dr. Roland Busch (CEO, Siemens AG), Christine Grotz (CEO, Weber-Hydraulik), Astrid Teckentrup (CEO, Procter & Gamble (DACH)), René Obermann (CEO, Deutsche Telekom AG (2006-2013)), Gerd Chrzanowski (CEO, Schwarz Gruppe).

Intended Learning Outcomes:

After attending this course, students will have gained knowledge on leadership skills and firm strategies in practice through attending podium discussions and actively participating in Q&A sessions with CEOs. They will be able to analyze companies and leaders, as well as their behavior, and apply their knowledge of leadership by interacting with CEOs. With this approach, students will gain practical knowledge and are additionally able to improve their discussion- and soft skills.

Teaching and Learning Methods:

In this seminar, there will be both Q&As and podium discussions with CEOs. Students will get in contact with several CEOs and managers of real-life companies, analyze their leadership skills and concepts, and interact with them in Q&A sessions. For this, the students should individually prepare and hand in questions for the CEOs as well as analyses of the companies and CEOs themselves. Moreover they should actively participate in all the sessions.

Media:

Powerpoint

Reading List:

Tbd.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0099: Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy | Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The seminar prepares students to make tactical choices supporting organizational performance and strategy. The seminar enables students to understand the relationships between operations and supply chains and other areas of the organizations and to design reliable, fast, and efficient operations and supply chains.

Students will be guided to learn critically analyzing value chain reports and to identify problems and areas for potential improvements. Later, students will demonstrate their ability to apply these skills in a simulated value chain setting. After completing this seminar, students can identify value chain challenges, identify potential improvements, coordinate critical decisions across functional areas, and forecast results from the solutions implemented. As such, this seminar will prepare students for applying academic knowledge to industry practice.

Central to the seminar is a simulation consisting of several rounds. Students will be assigned to groups of four participants and have to formulate a strategy for their company. During each round, they will be required to make tactical and strategic choices to maximize value and prepare a four-page summary justifying the choices made and the expected results. This summary is part of the overall assessment (75% of the final grade, team grade). This assessment method helps students reason about the options available in each round, make informed decisions, develop predictive capabilities, work together in a team, and condense necessary information formally. Furthermore, students must submit a four-page report at the end of the course, including a critical reflection about teamwork and individual takeaways (25% of the final grade, individual grade). This assessment method allows students to reflect on their capabilities to work in a team and strengthen their personal learnings from the seminar.

At the beginning of the course, students will be advised on the report structure and submission deadlines.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Desirable but not mandatory: Production and Logistics, Management Science

Content:

Subject-specific content

- Strategic Alignment
- Value Chain
- Supply Chain Tactical Decisions
- Supply Chain Management
- Sales and Operations Planning

Intended Learning Outcomes:

Upon completion of the module, students are able to:

- understand the need to construct operations and supply chains that align with corporate-level strategies and identify appropriate actions
- analyze the relationships between operations and supply chain management and other areas of the organization (e.g., marketing, finance, accounting, engineering, R&D, etc.)
- evaluate the feasibility of functional-level actions with corporate-level strategies and assess the impact of their choices on the organization's performance
- coordinate actions and address competing interests across functional areas
- evaluate alternative tactical decisions and convince with compelling reasoning for their choice of action
- critically reflect upon their ability to work in a team and provide advanced improvement recommendations for effective teamwork

Teaching and Learning Methods:

Experiential learning

The module is designed to enable students to acquire knowledge on current (behavioral) operations and supply chain management topics through hands-on experience and reflection. To this aim, a web-based business simulation game will be employed. The business simulation game will engage participants in turning around a manufacturer of fruit juices by making tactical and strategic decisions on four critical areas of the organization: Purchasing, Operations, Supply Chain, and Sales. Students are required to analyze key performance indicators, define a strategy for turning around the organization, identify potential improvements, and make tactical decisions in each of the organization's four areas.

Seminar sessions take place physically, and the methods combine mini-lectures, hands-on experience, group reflection, and questions and feedback sessions. Each week, there will be two

90-minutes sessions that will allow for feedback, sparring sessions, and the introduction of new concepts.

Students will get an overview of the course in the first two weeks. They will be informed about the examination and expectations. Furthermore, students will be introduced to highly recognized supply chain strategy and integration papers. Students will also see examples and learn to interpret data from sales, operations, supply chain, and purchasing reports.

After setting the theoretical scene, students will form small groups of size 4 (groups of smaller size might also be formed depending on the total number of participants), and they will start working on the business simulation game. The simulation advances over several rounds and students must present their analysis, evaluation, decisions, and expected results. Further details will be shared during the seminar.

Media:

Research papers and further material to be shared via Moodle, in-class discussions

Reading List:

Readings for general preparation:

Myerson, P. (2012). Lean supply chain and logistics management. McGraw-Hill.

Croson, R., Donohue, K. (2006). Behavioral causes of the bullwhip effect and the observed value of inventory information, *Management Science*, 52(3), pp. 323-336. DOI: 10.1287/mnsc.1050.0436

Flynn, B.B., Huo, B. and Zhao, X. (2010), The impact of supply chain integration on performance: A contingency and configuration approach. *Journal of Operations Management*, 28: 58-71. DOI: 10.1016/j.jom.2009.06.001

Frohlich, M.T., Westbrook, R. (2001). Arcs of integration: an international study of supply chain strategies. *Journal of Operations Management*, 19, pp. 185-200. DOI: 10.1016/S0272-6963(00)00055-3

Further materials will be discussed in the session.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0108: Start-up law for founders of family enterprises | Start-up law for founders of family enterprises

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination includes an oral report where the students will present and discuss their work for the seminar paper (30% of the grade, Presentation: 15-20 min, Questions: 15-20 min, Feedback session: 10 min). Moreover, students will prepare a seminar paper. To support students during the writing process, there will be regular discussions about the project's progression (seminar paper and regular discussions = 50% of the grade). The seminar paper contains a case study that addresses the basics of German corporate law and legal challenges associated with the formation of a startup and governance structure. On top of that, students' participation in class discussions of case studies and group work will be evaluated (20% of the grade). The active participation in the case discussions measures the students' ability to use the theoretical knowledge of the seminar to analyze real complex situations. The written examination and the corresponding presentation evaluate the student's ability to work with unknown legal problems and present a sound study incorporating the seminar's topics. In the oral report, students also show that they can summarize, academically present, and discuss the subject. Finally, ongoing discussions within group works and discussion rounds assess the student's capacity to solve a legal issue and elaborate a structured solution within a specified timeframe.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

None

Content:

- Legal structures and industries overview
- Research methods using legal databases
- Handling German legal texts

- Basics of German corporate law
- Overview of GmbH and GbR according to German law
- Startup formation overview
- Contract and agreement requirements for legal entity formation with special attention to the specific situation of family enterprises
- Financing opportunities overview
- Analysis with practical examples (e.g., Campus Founders, local startup, VC Investor)
- Presentation of own legal solutions

Intended Learning Outcomes:

The seminar covers the basics of startup law for founders of family enterprises. The students will gain a basic understanding of German corporate law and the formation process of a startup. Students will delve into a detailed study of German corporate law texts and apply them through practical examples. By the end of the seminar, they will be able to recognize legal challenges that arise during the startup process of a family enterprise. Furthermore, participants will learn how to combine their knowledge with discussing cases. Including practical case studies of newly founded family enterprises will enhance skills in argumentation and analysis, as case studies provide opportunities for debates. The seminar will also encourage the ability to develop legal solutions and justify their decisions through group work and presentations.

Teaching and Learning Methods:

The educational approach is a combination of direct instructions and oral participation of the students as well as reinforcement through interactive group activities. The students present the outcomes of their group work and engage in discussions about the topic with the other students. The practical application of knowledge to real-life cases is demonstrated using illustrative examples. Furthermore, students create detailed checklists and schemes on legal topics. The seminar will feature a presentations by relevant actors (like start-ups or lawyers) who will share their valuable insights and experiences related to legal issues in the startup process. The seminar integrates a case study with the seminar topics so students can apply what they have learned in their analysis. Students will analyze a specific case, demonstrate their understanding, and apply legal principles. Finally, students will present their work in class.

Media:

Case studies, PPT, flipcharts

Reading List:

The German Limited Liability Company, Rosengarten / Burmeister / Klein (9th ed. 2020)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Start-up law for founders of family enterprises (MGTHN0108) (Seminar, 4 SWS)

Back M, Bundrock R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0109: Taxation of Family Enterprises | Taxation of Family Enterprises

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To complete the module, participants have to demonstrate their theoretical/conceptual and practical competences regarding taxation of family business through five written exercise sheets that have to be handed in through the semester.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Basic competences in financial accounting, managerial accounting, and financial management and investing on the Bachelor level.

Content:

The course has the following content elements:• Types of taxes and their collection in the European Union. • Economics of business taxation and tax statistics. • Taxes in company valuation. • Financial accounting under German commercial law. • Income taxation under German tax law. • Tax planning. • International taxation. • Taxation of digital goods and services. • Taxation of family business succession. • Ethical aspects of tax planning and paying. • Methods of empirical tax research.

Intended Learning Outcomes:

After successfully completing the module, the participants have the following four competences:• Participants know the most important concepts and types of corporate taxes and can apply tax planning strategies. • Participants can trade-off taxation effects of different legal forms and governance systems and have particular competences related to the taxation of intergenerational succession of family enterprises. • Participants know the most important challenges of international taxation, in particular aspects related to the taxation of digital goods and services. • Participants

can critically evaluate potentials, incentives, and risks of tax-optimal accounting and managerial policies against the background of ethical principles.

Teaching and Learning Methods:

The module consists of a series of introductory lectures and practitioner-oriented case studies which are offered in blocked two-day courses through the semester.

Media:

Lecture script, case studies, and e-learning (Moodle).

Reading List:

Readings are announced in the module.

Responsible for Module:

Stich, Michael; Prof. Dr. rer. oec.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0110: Advanced Seminar Economics & Policy: Management practices in (family) enterprises | Advanced Seminar Economics & Policy: Management practices in (family) enterprises

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on management practices in (family) enterprises.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper, and (ii) oral in-class presentation. Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Content:

The seminar will cover key topics regarding management practices, such as:

1. How can management practices be classified?
2. How do management practices vary across companies?
3. How do management practices differ between family and non-family enterprises?
4. How do management practices relate to economic success?
5. How do management practices relate to work climate?
6. What are the determinants of different management practices employed in practice?

Intended Learning Outcomes:

'Students will learn the key concepts in the research field of management practices, with a special focus on family enterprises. The goal is twofold: First, to understand how management practices relate to economic success of enterprises. Second, to understand what individual and institutional factors determine management practices in enterprises. The course builds on original research on that topic. The course will cover basic theoretical consideration, but will have a clear focus on empirical studies with real-world examples and applications.

Teaching and Learning Methods:

'Students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence. A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups. We plan to complement student presentations by talks from corporate guests who will provide insights into corporate management practices.

Media:

Powerpoint slides

Reading List:

Amongst others (for detailed reading list, please see Moodle):

Bloom, N., & Van Reenen, J. (2010). Why do management practices differ across firms and countries?. *Journal of Economic Perspectives*, 24(1), 203-24.

Bloom, N., Brynjolfsson, E., Foster, L., Jarmin, R., Patnaik, M., Saporta-Eksten, I., & Van Reenen, J. (2019). What drives differences in management practices?. *American Economic Review*, 109(5), 1648-83.

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0111: Cases in industry 4.0 | Cases in industry 4.0

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The idea of this course is to train students in managerial decision-making to strengthen their positive impact on their next job. Accordingly, the coursework and examinations are structured.

Each week during this semester, there will be sessions of 2 times 90 minutes where we will discuss teaching case studies revolving around management issues in Industry 4.0. Students must prepare each case and answer preparatory questions (60% of the final grade, individual grade). Class participation and active contributions are graded as 40% of the final grade (individual grade). This course will allow students to learn firsthand about management issues in Industry 4.0. During those sessions, we will cover a variety of relevant technologies and their interplay.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Production and Logistics, Strategic Management

Content:

Subject-specific content

- Industry 4.0
- Management issues/conflicts of interests
- Implementation of digital technologies
- Blockchain, IoT, Cloud, AI, AR, VR, etc.

Methodological content

- Quantitative assessment
- Modelling

Intended Learning Outcomes:

Upon completion of the module, students can

- understand and analyze state-of-the-art approaches to industry 4.0,
- evaluate the advantages and disadvantages of the most recent technologies
- critically analyze firms and supply chains within their context and make superior recommendations,
- and criticize innovative approaches by assessing potential shortcomings.

Teaching and Learning Methods:

Lecture

Each week, students will be given one case study (approx. 10-20 pages), read those, answer a few graded case-related questions (approx. 2-4), and attend the classroom session. In those sessions, students will actively contribute and engage in the discussion. While correctness and plausibility are essential, this class also seeks diverse views and appreciates reasonably different perspectives.

Media:

Case studies (to be shared via Moodle)

Reading List:

Readings for general preparation:

Babich, V., & Hilary, G. (2020). OM Forum—Distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management*, 22(2), 223-240.

Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113-122.

Wuttke, D., Upadhyay, A., Siemsen, E., & Wuttke-Linnemann, A. (2022). Seeing the Bigger Picture? Ramping up Production with the Use of Augmented Reality. *Manufacturing & Service Operations Management*.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0113: How to launch a platform start-up | How to launch a platform start-up

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1) A final written exam determines 60% of the grade. The final written exam (60 minutes) is a means to assess students' understanding of the basic and advanced theories of the economics and management of digital platforms. Students have to show that they understand the core economic concepts underlying platform business models. They have to prove that they are familiar with the strategies and tactics to start a digital platform business. They have to show that they can critically evaluate the different options for designing a platform business model. Students may use a non-programmable calculator and a non-electronic dictionary for the exam.

2) Case study group work with presentation determines the remaining 40% of the grade. Students will be assigned into groups to work on a case study about a platform company. Students have to show that they can analyze the case based on the knowledge acquired in the course and suggest recommendations for the platform company. The grade for the case study group work is based on an in-class presentation by the group.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

none

Content:

In the 21st century, we see a number of companies like Uber or Airbnb that provide value to their customers by operating a platform and not by selling products. This rather new business model allows users and producers to interact with each other while the platform provider sets up the infrastructure for doing so.

This course gives an introduction to digital platforms to understand the underlying concepts of those business models and highlights the importance of the critical mass and network effects. Important principles for launching a digital platform are explained and different challenges that can appear in the process are outlined. By the end of the course, students will be able to understand digital platforms as a business model in general and be able to develop promising strategies in the launching phase. For a deeper understanding of the theoretical concepts, participants will apply the learnings on a case study which provides them with the necessary skills for analyzing cases and using previously acquired knowledge to assess them.

Intended Learning Outcomes:

After the course students will be able to:

- understand the concepts of digital platforms as a business model.
- develop strategies for launching a platform and understand the importance of the critical mass and network effects.
- apply instruments that allow to overcome the challenges in the launch process in a market with and without competition.

Teaching and Learning Methods:

Lecture: the lecture serves to build a thorough theoretical understanding of the concepts and methods introduced in the course.

Case studies: students will be assigned into groups to work on a case study of a real-world platform start-up with the goal to identify solutions for their challenges.

Media:

Slides

Case studies

Reading List:

Förderer, Jens: Erfolgsmodell Digitale Plattformen: Geschäftsmodelle – Netzwerkeffekte – Community-Management. 2022. Schäffer-Poeschel Verlag, Stuttgart.

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0115: Advanced Seminar: Digital Technologies in Operations Management | Advanced Seminar: Digital Technologies in Operations Management

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The key idea of this seminar is to lead students towards academic work and help prepare them for their master's thesis. In addition, they will learn about recent developments in digital technologies. Accordingly, the coursework and examinations are structured.

During the semesters, students will prepare two short presentations on research methods and skills. They will work in small teams. Each of the two presentations (ca. 45 minutes) accounts for 10% of the final grade and is a team grade.

Upon completion of this module, students will demonstrate their ability to cope with advanced research studies, comprehend and assess study outcomes, compare contributions of several studies, and transfer theoretical concepts to practice through a seminar paper (50% of the final grade; length of 15 pages including references). This method requires students to assess and replicate a study themselves.

In addition, they will prove their ability to communicate even complicated relationships and methods to their peer students through a presentation in a comprehensible fashion. They will further guide and moderate an ensuing discussion throughout which they will demonstrate their ability to criticize and assess innovative approaches and their potential shortcomings. The presentation, including moderation of ensuing discussion, accounts for 30% of the final grade (presentation duration 30 minutes + 30 minutes discussion). Complementing the written seminar paper, the presentation is targeted toward students who have not studied the same paper; this poses the challenge of presenting theoretical work interestingly while breaking down complex relationships into understandable information without losing rigor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Management Science, Production and Logistics

Content:

Subject-specific content

- Digital technologies potentially including but not limited to AR, VR, AI, Blockchain, and IoT.
- Management focus
- Incentive conflicts that arise when introducing new technologies in operations management
- Behavioral operations management

Methodological content (depending on assigned papers and limited to an introductory level)

- Research Design
- Analytical Modelling
- Game Theory
- Event Study Method
- Econometrics

Intended Learning Outcomes:

Upon completion of the module, students can

- understand and analyze state-of-the-art approaches to digital technologies in operations management,
- understand selected research papers and evaluate their key findings,
- present and explain complex research studies in a comprehensible and engaging manner,
- evaluate suggested management solutions in the digital technologies in an operations management context and assess their strengths and weaknesses,
- create management recommendations based on recent academic studies.

Teaching and Learning Methods:

Seminar

The seminar consists of two parts. During the semester, we will meet regularly to discuss select papers on digital technologies in operations management. Students will also explore and present information on select research methods (e.g., econometrics, game theory, and behavioral experiments). This part is complemented by a few interactive lectures.

At the end of the semester, we several sessions take placed blocked where students will present their assigned research studies (final assessment).

Media:

Research papers (to be shared via Moodle), student presentations, further material to be shared via Moodle, online discussions

Reading List:

Readings for general preparation (i.e., what is research? What marks a contribution?):

Wacker, J. G. (1998). A definition of theory: research guidelines for different theory-building research methods in operations management. *Journal of operations management*, 16(4), 361-385.

Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of management review*, 14(4), 490-495.

Examples for typical papers that we will discuss (this list is not meant to be comprehensive but gives some indication of topics to be covered):

Babich, V., & Hilary, G. (2020). OM Forum—Distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management*, 22(2), 223-240.

Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113-122.

Further papers will be provided at the beginning of the semester.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0122: Campus Challenge: Successful Investing with AI and Large Language Models | Campus Challenge: Successful Investing with AI and Large Language Models

Version of module description: Gültig ab summerterm 2023

Module Level:	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises group-based scientific coursework (approx. 20-30 pages) about the group's investment strategy.

The investment strategy should incorporate language models systematically and creatively.

Students will receive more information on the available tools during the kick-off meeting.

The coursework should comprehensively document the investment strategy (investment idea, tools used, prompt engineering, data preparation, data analysis, performance evaluation and interpretation, and suggestions for further improvement). In addition to the written scientific report, candidates must present their strategy within an approximately 20-minute presentation and answer questions from their audience.

To maintain the seminar atmosphere, the number of participating groups is generally limited. While Heilbronn students are prioritized, students from all TUM locations may also participate if the course is offered in a hybrid/online format. The targeted maximum number of participants is 24. More information will be provided in the introductory session.

The written assignment will be weighted 80%, and the formal presentation will make up 20% of the final grade. The group whose investment strategy performs best may receive an additional grade improvement in the live-test. The assessment is generally based on group work. If there are significant differences across the quality of the report or presentation, grades may be adapted individually. The students should therefore highlight their contributions to the report.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;
Basic knowledge in scientific writing;
Interest in investing in financial markets and the application of AI tools

Content:

The students will motivate, create and implement an investment strategy for the equity market using AI and/or large language models and/or sophisticated chatbots. The investment strategy will be developed based on group-work.

Candidates are expected to motivate their investment strategies with respective literature to ensure a systematic and structured manner. Candidates are expected to learn how to work with sophisticated chatbots, language models, and Application Programming Interfaces (APIs).

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- motivate, create, and implement an investment strategy;
- work with language models / sophisticated chatbots;
- conduct an empirical project in a team;
- deliver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise regular meetings with the team members about the current status of the stock portfolio;

Methods of teaching comprise group-based coursework and the formal presentation of obtained results;

The learning methods of the students primarily comprise the following activities:

- Collaborative Implementation of the investment strategy (theory, tools used, prompt engineering, data preparation, data analysis, results, and interpretation, room for improvements);
- The theory may be developed based on an independent literature review (usage of scientific articles published in top international journals);
- Collaborative writing of a scientific report;
- Exercise a deductive, logical, and consistent argumentation to derive an investment strategy;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction/teaching methods are adequate to foster/extend the student's ability to conduct independent academic work and elaborate thematically complex contents on their own. It is considered a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

The students will receive a list of relevant literature during the kick-off meeting at the beginning of the semester.

De Prado, Marcos Lopez. Advances in financial machine learning. John Wiley & Sons, 2018.

Lopez-Lira, Alejandro and Tang, Yuehua, Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models, 2023

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Campus Challenge: Successful Investing with AI and Large Language Models (MGTHN0122)
(Seminar, 4 SWS)

Breitung C, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0123: Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals | Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals

Version of module description: Gültig ab summerterm 2023

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To successfully pass the module, the participants have to demonstrate methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations. The participants proof these competences through several presentations with written reports and active participation in discussions.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

No prerequisites required.

Content:

Artificial intelligence (AI) is one of the most profound ongoing technological developments which has the potential to strongly impact various fields of business and society. This module critically analyzes the effects of AI-applications on the sustainable development goals (SDGs) of the United Nations. Units of the module: Political and conceptual background of the SDGs; methods to assess impacts on the micro-, meso-, and macro-level; discussion of supportive and contradictory impacts of specific AI-applications on SDGs.

Intended Learning Outcomes:

After successful completion of the module, the participants have methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations.

Teaching and Learning Methods:

The module consists of an interactive introductory lecture and presentations of the participants with topic-specific written reports.

Media:

Textbook, presentation slides, and e-learning (Moodle).

Reading List:

Dodds, Donoghue und Leiva Roesch. 2016. Negotiating the sustainable development goals: A transformational agenda for an insecure world.

Kanie und Biermann. 2017. Governing through goals: Sustainable development goals as governance innovation.

Saetra. 2022. AI for the sustainable development goals.

Sobti und Sharma. 2018. An introduction to sustainable development goals.

Walker, Pekmezovic und Walker. 2019. Sustainable development goals: Harnessing business to achieve the SDGs through finance, technology, and law reform.

Responsible for Module:

Stich, Michael; Prof. Dr. rer. oec.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0130: CEO Strategy Series | CEO Strategy Series

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The assessment of this course is based on short quizzes administered at the end of each session. These quizzes both test listening comprehension and encourage students to critically reflect on the content in light of the theoretical foundations introduced at the beginning of the semester. To successfully complete the course, students must submit at least 70% of the quizzes. Each quiz contains 7 questions. The final grade is calculated from the aggregated quiz results.

In addition, students may earn a grade bonus of up to 30% through active oral participation. This requires asking relevant and well-prepared questions during the sessions, which should be linked to the visiting CEOs and their companies as well as to current developments, global trends, or issues related to digital strategy.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization.

Content:

This module is the flagship event of TUM C.HN and will take place on Tuesdays or Fridays at 3 pm (each date will feature one guest). It covers practical insights on firm strategies, whereby lessons can be learned directly from CEOs themselves. The covered strategy topics may include internationalization, the analysis of the internal and external environment, M&A and other types of strategic alliances, as well as digital transformation, the role of digital enterprises, and the strategic challenges arising from new technologies, online platforms, and data-driven business models. This course therefore combines classical perspectives of strategic management with current

developments in the digital economy, providing students with valuable insights for their future work as strategizers in a corporate environment.

Please note: Registration is mandatory. There will be a participants list at the entrance and police presence for each event.

Intended Learning Outcomes:

After attending this course, students will have learned about different perspectives on developing and executing firm strategies in practice. They will be able to analyze companies and CEOs in their function as strategizers, and apply the knowledge gained in an in-depth study of relevant strategy literature by interacting with CEOs. With this approach, students will gain theoretical and practical knowledge, strengthen their understanding of digital enterprises and the role of digital transformation in strategy, and are additionally able to improve their discussion- and soft skills.

Teaching and Learning Methods:

In this seminar, there will be both Q&As and podium discussions with CEOs. Students will meet several CEOs and managers of real-life companies, gain insights into their strategies and related concepts, and interact with them in Q&A sessions. To prepare, students should develop questions that build on the concepts introduced in the first session and reflect the relevance of digital enterprises and digital transformation in corporate strategy. Active participation in all sessions is expected.

Media:

Powerpoint

Reading List:

- 1) Hanson, D., Hitt, M. A., Ireland, R. D., & Hoskisson, R. E. (2016). Strategic management: Competitiveness and globalisation. Cengage AU.
- 2) Barney, J. B. (2002). Strategic management: From informed conversation to academic discipline. Academy of Management Perspectives, 16(2), 53-57.
- 3) Nag, R., Hambrick, D. C., & Chen, M. J. (2007). What is strategic management, really? Inductive derivation of a consensus definition of the field. Strategic management journal, 28(9), 935-955
- 4) Smith, K. G., & Hitt, M. A. (Eds.). (2005). Great minds in management: The process of theory development. OUP Oxford.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

CEO Strategy Series (MGTHN0130) (Seminar, 4 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0154: Managing Legal Sustainability Requirements in Family Enterprises | Managing Legal Sustainability Requirements in Family Enterprises

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The grade is composed of written and oral performances of the participants. Students will prepare a seminar paper. To support students in the writing process, there will be regular discussions about the progression of the project. The handed in seminar paper and regular discussions, where the progression of the seminar paper is discussed make up 70% of the overall grade. The examination includes an oral presentation where the students will present and discuss the topic of their seminar paper (20% of the grade). Regarding the latter, 10% of the overall course performance is achieved through two separate course activities within two separate sessions of the seminar. 5% of the grade is accounted for the presentation of a poster tailored thematically to the session. The second required activity which accounts for 5 % of the grade is an oral presentation of a case study. The active participation in the two course activities measure the student's ability to use the conveyed legal knowledge on a wide variety of corporate sustainability topics to evaluate the internal and external implications on companies in the European Single Market. The written examination (seminar paper) and the corresponding presentation are means to evaluate the student's ability to present an overview of an unknown topic related to the topics discussed, to analyze the relevant literature in the form of studies and primary literature, and to assess the challenges and opportunities from a corporate perspective. Finally, regular discussions with the instructor measure the student's ability to distinguish subtopics of the chosen topic, locate and evaluate relevant sources and elaborate purposeful analysis from a corporate perspective.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

Introduction (Corporate) Sustainability: ESG-Dimensions

Decarbonisation, human rights in supply chains

circularity, eco-design, right to repair

Greenwashing practices, sustainability reporting, sustainable finance

Intended Learning Outcomes:

The seminar links basic knowledge of the legal framework for corporate sustainability on the European plane with the given organizational and financial structures of family enterprises in the European single market. The students gain the ability to think practical case scenarios through and draw implications for business practices. This is paired with analytical skills for literature review. Handed-out (literature) material is examined, and subsequently, critical opinions are formed and discussed in a team setting. This strengthens the students ability to distinguish the framework of mandatory legal rules from the "business case" for compliance with these rules.

Teaching and Learning Methods:

This seminar comprehensively presents the implementation of the different pillars of the European Green Deal for regulating corporate sustainability. On this basis, the implications for the companies' practices are evaluated critically. The seminar, inter alia, integrates guest lectures and case studies. Students can apply their learnings and implement them in their analysis. Students will study case studies from various industrial settings critical to "Green Transition" in the single market. Based on learned legal and regulatory elements, comparisons to the status quo and future outlook of business practices are drawn. Finally, students will present their work in class.

Media:

Case studies, exercises, PPT, Whiteboard, Videos

Reading List:

E.g.: Ditlev-Simonsen, A Guide to Sustainable Corporate Responsibility (2022), Noeth, The ESG and Sustainability Deskbook for Business (2024), Gaganis et al., Sustainable Finance and ESG (2023), Camara/Morais, The Palgrave Handbook of ESG and Corporate Governance (2022), Lykkesfeldt/KJaergaard, Investor Relations and ESG Reporting in a Regulatory Perspective (2022), Nygaard, Green Marketing and Entrepreneurship (2024), Eisenriegler (ed.), The Circular Economy in the European Union (2020)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Managing Legal Sustainability Requirements in Family Enterprises (MGTHN0154, English)
(Seminar, 4 SWS)

Bundrock R, Dowse M

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0165: Sustainable Management - Bridging Economy and Ecology through Technology | Sustainable Management - Bridging Economy and Ecology through Technology

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 160	Contact Hours: 20

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of seminar papers (60%) and presentation (40%): The seminar begins with block lectures, after which students independently write a seminar paper, present their findings, and actively engage in discussions about their work. Possible seminar paper topics include the development of energy-efficient building technologies, advancements in CO2 capture and storage, the implementation and potential of smart grids, modern heat recovery systems, innovations in sustainable waste management, the decarbonization of industrial processes, and urban planning.

Repeat Examination:

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management and organizations.

Content:

This course introduces the fundamentals of sustainability, covering ecological principles, the significance of ecosystems, and the science behind climate change, including its global impacts. It examines historical and modern responses to climate-related disasters, focusing on technological solutions like clean air regulations and sulfur dioxide controls, and highlights the importance of a circular economy.

Intended Learning Outcomes:

After attending this course, students will have gained a comprehensive understanding of sustainability, including ecological principles, climate change, and its impacts. They will be able to explain, analyze, and critically reflect on historical and current climate-related challenges and

solutions driven by technological advancements. Students will have built an understanding of political, legal, and economic frameworks related to sustainability, with practical insights into corporate sustainability practices, exemplified by Knauf. Moreover, they will acquire knowledge on energy efficiency, sustainable energy sources, and technological innovations. They will prepare a written assignment, hold a short presentation and lead a discussion on a sustainability related topic, which enhances their ability to apply these concepts practically and improves their discussion skills. Furthermore, students will have the option to discuss actual activities at the multinational company Knauf, where they can observe and learn firsthand how a leading company integrates sustainability into its operations and practices.

Teaching and Learning Methods:

Students will write a seminar paper on various sustainability topics and present their findings to the class. After each presentation, there will be discussions to elaborate on the topics and reflect on the concepts. Additionally, students can see how sustainability practices are implemented in a real-world setting at Knauf. These methods aim to deepen the understanding, enhance analytical and presentation skills, and apply theoretical knowledge practically.

Media:

Powerpoint

Reading List:**Responsible for Module:**

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Sustainable Management - Bridging Economy and Ecology through Technology (MGTHN0165)
(Seminar, 4 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0171: Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises | Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on a specific topic related to the seminar's overall theme.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper (50%), and (ii) oral in-class presentation (50%). Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Principles of Economics (Master)

Content:

Beyond a lecture-style introduction into the topic, students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence.

A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups.

Intended Learning Outcomes:

Students will explore how technological advancements, ranging from automation to artificial intelligence, transform labor markets and the broader economy. The seminar will delve into the mechanisms behind these changes, examining their impacts on work, skills, and business structures, as well as the role of government policies in managing these transitions. Special attention will be given to comparing family and non-family enterprises.

The course builds on original research on the topic. While it will cover basic theoretical considerations, the focus will be on empirical studies with real-world examples and applications. Students will write a term paper on a selected topic and present their findings, highlighting practical implications for labor markets, policy, and business strategies.

Teaching and Learning Methods:

This seminar will cover key topics within the Economics of Technological Change and Labor Markets, with a focus on the role of family enterprises, such as:

Topic area 1: The impact of skill-biased technological change on wages and employment outcomes.

Topic area 2: Job polarization and the growth of low-skill service jobs in response to automation and routine-biased technological change.

Topic area 3: Labor market adjustments to technological innovations, including the effects of automation, robotics, and artificial intelligence.

Topic area 4: The unique challenges and opportunities for family enterprises in adapting to technological change and leveraging innovations to remain competitive.

Topic area 5: The role of government policies in managing labor market transformations and ensuring equitable outcomes.

Media:

Teaching will be in the form of a lecture and seminar presentations.

Reading List:

Acemoglu, Daron and David Autor (2011). "Skills, tasks and technologies: Implications for employment and earnings", In: Handbook of Labor Economics, Vol. 4, Elsevier, pp. 1043–1171.

Angrist, Joshua and Jörn-Steffen Pischke (2008). Mostly Harmless Econometrics (Princeton: Princeton University Press)

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0173: Project and Innovation Management: Design thinking methodology | Project and Innovation Management: Design thinking methodology

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The final grade consists of both written and oral components. Students are required to prepare a seminar paper, which accounts for 70% of the final grade. To support them throughout the writing process, there will be regular discussions to monitor the progress of their work. In addition, students will deliver an oral presentation on their seminar paper, which constitutes 20% of the grade. Furthermore, students will complete two class exercises, each accompanied by a discussion. Each exercise and its corresponding discussion will be assessed and will contribute 5% to the final grade, for a total of 10%. Active participation in the discussions and class exercises measures the student's ability to use the theoretical knowledge of the seminar to discuss and analyze real complex situations. The written examination and the corresponding presentation are means to evaluate the student's ability to work with a specific problem and incorporate the methods discussed in class to organize and present a solution to a problem. Finally, regular discussions with the instructor measure the student's ability to develop an idea, shape it, and elaborate a structured project within a given timeframe.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

This seminar presents the methodology of design thinking to generate solutions and tackle identified problems, particularly in connection with innovation initiatives in family enterprises. During the different sessions, the stages of design thinking will be covered, and we will explore how they work in connection with various types of case studies (related to technological

development, sustainability, and internal processes in enterprises). We will also explore how artificial intelligence can contribute to optimizing and enriching the application of the methodology and the generation of innovative solutions.

Intended Learning Outcomes:

1. Participants will be able to articulate the core principles of Design Thinking, including empathy, ideation, prototyping, and iteration.
2. Participants will learn to clearly define and reframe problems from a user-centered perspective, creating problem statements that guide effective solutions.
3. Participants will gain proficiency in various ideation techniques, such as brainstorming, mind mapping, and rapid sketching, to generate innovative ideas.
4. Participants will apply Design Thinking methodologies to real-world challenges, demonstrating the ability to navigate the entire process from problem definition to solution implementation.

Teaching and Learning Methods:

We will have content and application sessions where the methodology will be introduced and case studies of its application will be presented. Students will also receive suggested materials to delve deeper into the methodology after class. In the last part of the session, students will be asked to solve exercises related to the application of the methodology. Paralelly, they will work on the topics while constructing their seminar papers.

Media:

Presentation of the topics, case studies, exercises, PPT, Whiteboard

Reading List:

1. Razzouk, R., & Shute, V. (2012). What Is Design Thinking and Why Is It Important? Review Of Educational Research, 82(3), 330-348. <https://doi.org/10.3102/0034654312457429>
2. Frisendal, T. (2012). Design Thinking for Business Analysis. En Management for professionals (pp. 15-24). https://doi.org/10.1007/978-3-642-32844-2_3
3. You, X. (2022). Applying design thinking for business model innovation. Journal Of Innovation And Entrepreneurship, 11(1). <https://doi.org/10.1186/s13731-022-00251-24>.
4. Martin, R., & Euchner, J. (2012). Design thinking. Research-Technology Management, 55(3), 10-14. <https://doi.org/10.5437/08956308x5503003>
5. Rösch, N., Tiberius, V., & Kraus, S. (2023). Design thinking for innovation: context factors, process, and outcomes. European Journal Of Innovation Management, 26(7), 160-176. <https://doi.org/10.1108/ejim-03-2022-0164>
6. Carella, G., Melazzini, M., Cautela, C., & Zurlo, F. (2024). Boosting Design Thinking adoption in organisations through a game#based toolkit: A gamified approach in building facilitators to overcome Design Thinking adoption barriers. Creativity And Innovation Management. <https://doi.org/10.1111/caim.12627>
7. Docherty, C., Mazzei, M., & Steiner, A. (2023). Exploring the impact of design thinking on social enterprise mission-aligned innovation. Entrepreneurship And Regional Development, 1-17. <https://doi.org/10.1080/08985626.2023.2286506>

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0177: Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations | Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1. Group Report (70%)

- **Assessment Details:** In groups, students will collaboratively submit a report (no longer than 5 pages) addressing a specific managerial problem and proposing a solution using VR/AR technologies. Students are allowed to use reference materials and external resources.
- **Scope:**
 - Identify a relevant managerial problem in production, quality inspection, remote support, human resources, or customer experience.
 - Justify the importance of solving this problem.
 - Propose a detailed VR/AR-based solution, including technical implementation steps, management implications, and expected outcomes.
- **Rationale for Assessment:**
 - This assessment evaluates the ability to apply theoretical knowledge to practical problems, foster teamwork, and use technology to prepare solutions with managerial perspectives.
 - The report format develops students' ability to communicate complex ideas concisely and effectively.

2. Individual Critical Reflection (30%)

- **Assessment Details:** Each student will write a critical reflection on the relevance of immersive technologies (VR/AR) in solving managerial challenges. This will be submitted as a separate document, no longer than 2 pages. The reflection must be written independently.
- **Scope:**
 - Reflect on the practical and theoretical insights gained during the module.
 - Critically analyze the advantages and limitations of VR/AR in managerial contexts.

- Discuss how the experience of creating the group report has influenced their understanding of technology's role in management.

- **Rationale for Assessment:**

- This assessment allows students to self-evaluate, critically linking their learning to broader managerial and technological contexts.

- It promotes independent thought, allowing students to articulate their unique perspectives.

Repeat Examination:

(Recommended) Prerequisites:

Basic Programming with C#

Content:

In the module, the students will gain knowledge about the technological and management aspects of VR implementations with hands-on experience and use case examples in

- Production and Assembly
- Quality Inspection and Remote Support
- Training and skill development
- Customer Experience and Prototyping

Technology-specific content

- Introduction to Immersive Technologies (key hardware and software platforms)
- Using Unity for developing VR/AR applications

Management-specific content

- Ethical Considerations
- Behavioral Challenges
- Building a roadmap from prototype to market-ready product

Intended Learning Outcomes:

By the end of this module, students will be able to:

- Understand the fundamental principles of VR and AR technologies.
- Analyze the role of immersive technologies in improving managerial and operational efficiency.
- Design and prototype VR/AR-based solutions for specific management scenarios.
- Critically evaluate the broader implications of VR/AR deployment, including ethical concerns (e.g., data privacy, accessibility).
- Work collaboratively in teams to identify managerial challenges and develop VR/AR solutions.

Teaching and Learning Methods:

The coursework is primarily practical. At the beginning of the module, the students will be introduced to the immersive technologies and different headsets. They will be introduced to Unity Game Development Engine, which they will use to develop application solutions. The students

can choose between VR or AR applications and must program their software solution, keeping a managerial focus in mind.

The students will present the problem statement (in a group) that they want to work on and will have regular feedback sessions during the development phase. The students will present their solutions at the end of the module.

Media:

Material will be shared via Moodle

Reading List:

Readings for general preparation (but not limited to):

Dalton, J. (2021). Reality check: How immersive technologies can transform your business. Kogan Page Publishers.

Jung, T., & Dalton, J. (2021). XR Case Studies. Springer International Publishing.

Wuttke, D., Upadhyay, A., Siemsen, E., & Wuttke-Linnemann, A. (2022). Seeing the bigger picture? Ramping up production with the use of augmented reality. *Manufacturing & Service Operations Management*, 24(4), 2349-2366.

Windhausen, A., Heller, J., Hilken, T., Mahr, D., Di Palma, R., & Quintens, L. (2024). Exploring the impact of augmented reality smart glasses on worker well-being in warehouse order picking. *Computers in Human Behavior*, 155, 108153.

Seeliger, A., Cheng, L., & Netland, T. (2023). Augmented reality for industrial quality inspection: An experiment assessing task performance and human factors. *Computers in Industry*, 151, 103985.

Rejeb, A., Keogh, J. G., Leong, G. K., & Treiblmaier, H. (2021). Potentials and challenges of augmented reality smart glasses in logistics and supply chain management: a systematic literature review. *International Journal of Production Research*, 59(12), 3747-3776.

Eswaran, M., & Bahubalendruni, M. R. (2022). Challenges and opportunities on AR/VR technologies for manufacturing systems in the context of industry 4.0: A state of the art review. *Journal of Manufacturing Systems*, 65, 260-278.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations (MGTHN0177) (Seminar, 4 SWS)

Phadke V

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0181: Advanced Seminar Operations & Technology: Analytics in Manufacturing Management | Advanced Seminar Operations & Technology: Analytics in Manufacturing Management

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours:	Contact Hours:

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

This seminar aims to guide students toward academic research and prepare them for their master's thesis. The topics focus on advanced analytics in manufacturing management, covering AI methods, modeling techniques, optimization algorithms, and rigorous analysis of algorithmic properties, such as performance guarantees. While students are not expected to master all these techniques, they should be willing to develop deep expertise in one and understand its relevance for manufacturing decision-making. Additionally, they should be able to present their findings clearly and didactically to their peers.

Students work in small teams and complete two presentations and a term paper (~15 pages). The first presentation (30 minutes, 20% of the grade) provides a structured overview of a recent research article and formulates a research question for the seminar thesis. The second presentation (30 minutes, 30% of the grade) is the final presentation, showcasing their research findings. The term paper accounts for 50% of the grade.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Foundations in statistics and management science

Content:

This course focuses on advanced analytics in manufacturing, emphasizing algorithms, modeling, simulations, and theoretical analysis. It is open to all advanced analytics methodologies, provided they meaningfully support manufacturing decision-making.

Additionally, the course integrates practical experience by incorporating potential talks from industry practitioners and potential excursions to manufacturing facilities.

Intended Learning Outcomes:

Students will develop a comprehensive understanding of the scientific process, further develop critical thinking skills, and the ability to solve business problems efficiently. They will gain firsthand experience in manufacturing decision-making through practitioner interactions and engagement with cutting-edge research on data-driven manufacturing. Additionally, they will acquire hands-on expertise in advanced analytics methods applied to manufacturing decision-making.

Teaching and Learning Methods:

Teaching methods include presentations, interactive learning, e-learning, group discussions, and (if feasible) discussions and presentations by practitioners. Additionally, if feasible, we aim to include excursions to manufacturing and related facilities, as well as interactions with leading scientists in the field.

Media:

Zoom, literature databases (Ebsco, ScienceDirect etc.), standard solvers (Gurobi, Hexaly etc.), databases and case studies, programming environments and software development platforms, simulation environments etc.

Reading List:

We focus on high-profile and influential ideas in advanced analytics for manufacturing, typically published in leading journals such as Operations Research, Management Science, Manufacturing & Service Operations Management, INFORMS Journal on Optimization, European Journal of Operational Research, and INFORMS Journal on Computing. Below are examples of relevant articles:

[AI] Senoner, J., Netland, T., & Feuerriegel, S. (2022). Using explainable artificial intelligence to improve process quality: Evidence from semiconductor manufacturing. *Management Science*, 68, 5704-5723.

[mathematical analysis] Drozdowski, M., Jaehn, F., & Paszkowski, R. (2017). Scheduling position-dependent maintenance operations. *Operations Research*, 65, 1657-1677.

[modeling] Möhring, R., Schulz, A. S., Stork, F., & Uetz, M. (2003). Solving project scheduling problems by minimum cut computations. *Management Science*, 49, 330-350.

Da Col, G., & Teppan, E. C. (2022). Industrial-size job-shop scheduling with constraint programming. *Operations Research Perspectives*, 9, 100249.

[algorithmics] Koca, Y., Martagan, T. G., Adan, I. J. B. F., Maillart, L., & van Ravenstein, B. (2023). Increasing biomanufacturing yield with bleed-feed: Optimal policies and insights. *Manufacturing & Service Operations Management*, 25, 108-125.

Barnhart, C., Jacquillat, A., & Schmid, A. (2024). Robotic warehousing operations : A learn-then-optimize approach to large-scale neighborhood search. *INFORMS Journal on Optimization*, 1-24.

Responsible for Module:

Otto, Alena; Prof. Dr. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0012: Digital Finance | Digital Finance [DF]

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The module examination consists of a 90-minute written exam, in which the candidates should reproduce the knowledge imparted in the lectures. The students are also required to provide transfer knowledge. It is tested whether students have understood how Digital Finance relates to Traditional Finance and other financial research disciplines; which interdependencies exist among the research disciplines, etc. Dangers and Opportunities of FinTech applications should be understood in detail. Further, students are required to conduct and assess mathematical calculations. The written exam also comprises open questions as well as multiple-choice-tasks.

Students may have the opportunity to improve their final grade by up to 0.3 points by voluntarily participating in group work and presentations on selected topics in Digital Finance. This opportunity depends on factors such as the number of participants and available capacity, and is decided on a semester-by-semester basis by the instructor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge of corporate finance/financial mathematics is required (e.g. material covered in the course Corporate Finance); joint election with the module "Advanced Seminar in Finance/Accounting: Current Research Topics in Digital Finance" highly recommended

Content:

The aim of the course is to give a comprehensive overview on Digital Finance, not covered elsewhere in courses of Corporate Finance within the Master in Management. It provides a coverage of the most recent relevant topics in the field of Digital Finance. Basic knowledge in the field of financial mathematics is a pre-requisite.

The module comprises the most important and most recent developments in Digital Finance. Specifically, it comprises (but is not limited to) the following areas:

- History of Digital Finance and how it relates to other financial research disciplines;
- Robo-Advisors;
- Crowd Funding;
- Peer-2-Peer-Lending;
- Social Media in Finance;
- Digital Payments;
- Cryptocurrencies;
- Digital Technologies (Big Data and Machine Learning; Textual Analysis; Blockchain)
- Cybersecurity;
- Digital Finance and the Market Efficiency Debate;
- The Future of Digital Finance;
- Regulation;
- etc.

Intended Learning Outcomes:

The students acquire detailed knowledge on how technological innovations are changing existing financial services and products, their implications and possible applications within the financial industry. Specifically, upon completion of the module, students will be able to identify changes and opportunities of Digital Finance and its implications for the overall macroeconomic stability.

Also, students will be able to fully understand limitations of the digital transformation within the financial industry and its underlying causes. Finally, the students will be able to assess how and why digital finance applications influence the market (in-)efficiency of existing financial markets. In particular, the students are able to analyze complex financial market relations while taking into consideration various influencing factors (as for instance irrational investor behavior).

Teaching and Learning Methods:

Types of instruction comprise lecture courses as well as practical courses;

Methods of teaching include lectures, presentations as well as guest lectures;

the learning methods of the students primarily comprise the following activities:

- Follow-up of course contents;
- Exercise and execution of financial calculations;
- Preparation and execution of presentations;
- Ability to answer advanced thematic issues;
- etc.

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to fully understand and elaborate the in-depth the thematic content.

Media:

Lecture slides; whiteboard; exercise sheets; exercise portfolio; flipchart; powerpoint; films

Reading List:

Due to the topicality of this lecture, the usage of reference books is only possible in a limited manner. Still, the below-listed references are considered as a solid starting point for the contents of this lecture.

Goldstein, I., Jiang, W., & Karolyi, G. A. (2019). To FinTech and Beyond. The Review of Financial Studies, 32(5), 1647-1661.

Chuen, D. L. K., & Deng, R. H. (2017). Handbook of blockchain, digital finance, and inclusion: Cryptocurrency, FinTech, InsurTech, regulation, ChinaTech, mobile security, and distributed ledger. Academic Press.

Fatas, A. (Ed.; 2019). The Economics of Fintech and Digital Currencies. CEPR Press.

If applicable, further literature will be announced at the first session.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Digital Finance (WIHN0012) (Vorlesung mit integrierten Übungen, 4 SWS)

Müller S, Osterrieder D

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0013: Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance | Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises a scientific coursework (approx. 25-30 pages), which is supposed to address the research questions posed within each given seminar topic. The task is to be conducted as a group work (i.e. 2-3 students per topic). Additional to the written scientific report, candidates are required to showcase main findings within a 30-minute presentation and answer further questions following their presentation.

In doing so, the candidates are required to prove that they have dealt with the given topic in a comprehensive manner, that their selection and evaluation of literature is adequate and that they have in-depth analyzed the topic while simultaneously putting it in a higher context of financial research. Also, it can be assessed whether the students are able to answer further questions in a correct and comprehensive manner.

The written assignment will be weighted 2/3 and the formal presentation will make 1/3 of the final grade. Each student will be assessed separately.

To maintain the seminar atmosphere, the number of participants is generally limited. If the course is offered in a hybrid format, Munich students may also participate in the course. The targeted maximum number of participants is 12. More information will be provided in the introductory session.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;

basic knowledge in scientific writing;
combination with the module "Digital Finance" is recommended

Content:

Candidates are supposed to build groups of 2-3 students and choose one seminar topic out of a set of pre-defined scientific issues. The pre-selection omprises, but is not limited to topics of the following areas:

- Asset Pricing of Cryptocurrencies: Market (In-)Efficiency, Behavioral Finance & Bubbles;
- Financial Innovations: Foundations and Opportunities & Threats to Financial Market Stability;
- Wisdom of the Crowd? Information Asymmetries, Adverse Selection, and Moral Hazard in Peer-to-Peer Lending Markets Around the World;
- Robo-Advisors: Overview of Services, Benefits, and Limitations;
- Social Investing: Overview, Benefits, and Limitations;
- Machine Learning: An Overview of ML Applications for Finance;
- etc.

Candidates are expected to conduct a comprehensive literature review of the most important scientific articles capturing the chosen seminar topic in order to address the given research questions in a systematic and structured manner.

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- describe the effects of current FinTech developments for existing financial products and processes;
- exhibit a comprehensive overview on current research topics in the field of Digital Finance;
- identify chances and opportunities of these developments (especially with regard to the overall macroeconomic stability);
- conduct an independent literature analysis at the highest possible, international level;
- analyze and evaluate references in a systematic manner;
- combine topics in Digital Finance with theories/methods/models of traditional finance in an independent manner;
- deriver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise an initial lecture (overview course) as well as the corresponding main seminar;

Methods of teaching include a group work and the formal presentation of obtained results;

the learning methods of the students primarily comprise the following activities:

- independent literature research (usage of scientific articles published in international top journals);
- Collaborative writing of a scientific report;
- Exercise of a deductive, logic and consistent argumentation to specifically address and answer the posed research questions;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to conduct independent academic work and to elaborate thematically complex contents on their own. It is considered to be a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

Machi, L. A., & McEvoy, B. T. (2012). The literature review: Six steps to success (2nd ed.). Corwin Press.

A separate reading list for each topic will be provided in the introductory lecture.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Finance & Accounting (WIHN0013): Current research Topics in Digital Finance (Seminar, 4 SWS)

Gong Z, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0014: Corporate Campus Challenge | Corporate Campus Challenge

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students will present a prototype and business case in a final presentation. Furthermore, they will write a report which will be examined. The final grade is averaged as follows:

60 %: Final presentation

40 %: Scientific paper (critical reflection on the prototype development process by using relevant scientific literature)

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

- Customer needs
- Research & empathic exploration
- Value proposition
- Design thinking methodology
- Business model and ecosystem modeling
- Rapid prototyping
- Scientific reasoning and writing

Intended Learning Outcomes:

At the end of the module, students are able to

- apply methods of idea generation and need-finding to stimulate creativity and recognize business opportunities
- create design prototypes in order to demonstrate their proposed solutions and gather feedback

- create business plans, presentations, and video prototypes in order to effectively communicate business value
- evaluate their ideas by involving peers, academics, and industry partners
- develop and document their ideas in a scientific seminar paper

Teaching and Learning Methods:

In this course, students will work in interdisciplinary teams to develop innovative solutions for challenges provided by industry partners. Over the course of 8 weeks, students will receive coaching sessions, individual mentoring, tutorials, and practical courses on the use of machines and equipment (3D printer, laser cutter, sensors, etc.). Students will present their first results through a presentation and a live prototype to gather feedback. At the end of the course, a working prototype and the business case will be presented during a final presentation.

Media:

Slides, scientific papers, handbooks, physical events

Reading List:

Lewrick, M., Link, P., & Leifer, L. (2018). The design thinking playbook: Mindful digital transformation of teams, products, services, businesses and ecosystems. John Wiley & Sons.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0017: Advanced Seminar Innovation & Entrepreneurship: Digital Innovation | Advanced Seminar Innovation & Entrepreneurship: Digital Innovation

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Seminar paper with presentation: The students independently and individually prepare a scientific seminar paper and present it during the course.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Very good English language skills are required. Students should also have an understanding of basic principles of management, innovation, and organization as well as empirical research methods. Attendance of the course Empirical Research Methods in Management and Economics recommended, but not mandatory.

Content:

The course prepares students for their Master's Thesis by introducing them to (1) the search, understanding, and analysis of academic literature, (2) methods for data collection and data analysis, (3) the structuring and writing of a paper, and (4) giving presentations to an academic audience including a subsequent discussion. The course exposes students to digital platforms and open forms of organizing innovation.

Intended Learning Outcomes:

After the course students will be able to:

- Define, explain and apply selected key concepts in the field of digital innovation
- Read and understand academic literature
- Write and present an academic paper

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via presentations and group discussions
- Build up critical thinking and interpretation skills

Learning Objectives

At the end of this course, students will be able to demonstrate understanding, critical assessment and application of the following:

- Learn how to evaluate academic literature
- Interact with an academic debate
- Prepare, elaborate and defend an academic argument
- Specialist knowledge on their selected topic of interest

Teaching and Learning Methods:

Introductory session will provide introductions into the core topics and scientific writing in lecture style. All other sessions will be focused on the individual student presentations, in which each student presents their seminar paper to the class and all other students are encouraged to discuss the papers.

Media:

Powerpoint

Reading List:

Literature is semester-specific and will be distributed in class

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0019: Inventory Management | Inventory Management

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In order to check the learning outcomes an oral exam (20 minutes) is performed.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

- the content of the course Production and Logistics
- Basics in probability calculus
- Programming skills in Python

Content:

The following topics are discussed

Demand Data for Inventory Control,

3) Performance measures for inventory control
control policies

approaches

7) Joint replenishments

safety stock

10) Multi echelon inventory systems

11) Inventory Management in the digital Age

1) Analysis of

2) ABC Analysis, XYZ Analysis

4) Inventory

5) Service and cost oriented

6) Optimization of cycle stock

8) Optimization of

9) Managing Spare Parts Inventories

Intended Learning Outcomes:

The students

- know fundamental problems and trade-offs while managing inventories in supply chains,
- are able to model inventory problems,
- know common inventory policies and are able to analyse them,

- can determine optimal policy parameters for common inventory policies,
- are able to use stochastic inventory models to optimize safety stock.

Teaching and Learning Methods:

Case studies, presentation, discussions, computer work

Media:

Reading List:

- Kiesmüller, G.P. (2024) Inventory Management. Books on demand
- Tempelmeier, H. (2006): Inventory Management in Supply Networks. 2nd edition. Books on demand
- Axsäter, S. (2004): Inventory Control. Kluwer's International Series.
- Montgomery D.C.; Runger, G.C. (2003) Applied Statistics and Probability for Engineers. 3rd edition. Wiley

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

Inventory Management (WIHN0019) (Vorlesung mit integrierten Übungen, 4 SWS)

Kiesmüller G, Sahin K

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0028: Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship | Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic related to corporate entrepreneurship. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade, approx. 10 pages) The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade, approx. 10 minutes). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis).

Repeat Examination:

(Recommended) Prerequisites:

Basic knowledge of corporate entrepreneurship and research methodologies is recommended.

Content:

Foundations of Corporate Entrepreneurship (CE): Key concepts, definitions, and the role of CE in modern businesses.

Strategic Thinking in CE: Developing an entrepreneurial mindset to identify and leverage opportunities.

Implementing CE: Steps to integrate entrepreneurial practices into corporate operations.

Building an Entrepreneurial Organization: Creating a culture that fosters innovation and risk-taking.

Sustaining CE: Strategies to maintain entrepreneurial energy and adapt to changing markets.

Leadership and Culture: The impact of leadership styles and organizational culture on entrepreneurship.

Entrepreneurial Traits and Behavior: Key personal characteristics that support entrepreneurial initiatives.

Creativity and Motivation: Harnessing creativity, motivation, and HRM practices to drive innovation.

Incentive Systems and Psychological Ownership: Enhancing engagement through effective incentives and ownership.

Entrepreneurial Intensity: Measuring and increasing the level of entrepreneurial activity within corporations.

Focus Areas of CE: Examining CE at the levels of the phenomenon, organization, individual, and dynamic interactions.

Challenges in CE: Addressing obstacles faced by corporations and strategies to overcome them.

Intended Learning Outcomes:

Understand the current state of research in corporate entrepreneurship.

Develop the ability to critically analyze and synthesize academic literature.

Improve academic writing and presentation skills.

Identify research gaps and formulate theoretical and practical implications.

Teaching and Learning Methods:

In-presence seminar sessions

Literature review and independent research.

Group discussions and collaborative learning (optional group work).

Interactive feedback sessions.

Presentation of results and peer review

Media:

Moodle platform for resources and updates.

Academic databases for literature access.

Presentation tools (e.g., PowerPoint, posters)

Reading List:

Core reading will be provided on Moodle. Students are encouraged to explore additional academic articles and books relevant to their selected topic.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0032: Business Forecasting | Business Forecasting

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In order to check whether students can calculate forecasts using the Python/nixtla, they receive data that needs to be analysed. They have to select a suitable method and determine the forecasts. Thus, the first part of the exam is a practice performance (programming task, 30%). In the second part of the exam, students must present the results of a small project that focuses on forecasting in a presentation (20 minutes, 70%) and be able to defend their approach in the subsequent discussion.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

- Basics in statistics
- Basics in probability calculus
- Programming skills in Python
- Basics in Forecasting

Content:

The following topics are discussed
analysis

1) Basics of time series

2) Arima

3) Hierarchical Forecasting

4) Global models

5) Random Forrest

6) Multivariate Forecasting

7) LGBM, XGBoost

8) Neural Networks

Intended Learning Outcomes:

The students

- understand the importance of data analysis for business forecasting
- know different forecasting techniques
- are able to select the right forecasting method and apply it
- can use Python and the forecasting libraries from nixtla to compute forecasts
- are able to present forecasts and explain their derivation

Teaching and Learning Methods:

Group work, Programming with Python, Presentations, Exercises

Media:

PowerPoint Slides, Screencasts, Video conferences, Exercises

Reading List:

- Hyndman, R.J., Athanasopoulos, G. (2012) Forecasting: principles and practice. Otexts
- Ord, K., Fildes, R. (2013) Principles of Business Forecasting. Cengage Learning.
- Joseph, J., Tackes, J., (2024) Modern Time Series Forecasting with Python. 2nd edition.

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0037: Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations | Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 146	Contact Hours: 34

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The final grade consists of both written and oral components. Students are required to prepare a seminar paper, which accounts for 70% of the final grade. To support them throughout the writing process, there will be regular discussions to monitor the progress of their work. In addition, students will deliver an oral presentation on their seminar paper, which constitutes 20% of the grade. Furthermore, students will complete two class exercises, each accompanied by a discussion. Each exercise and its corresponding discussion will be assessed and will contribute 5% to the final grade, for a total of 10%. The seminar paper and the corresponding presentation are a means to measure the student's ability to understand a scientific subject, evaluate literature, and develop, conduct, and analyze questionnaires/surveys. By doing a presentation, students show that they can summarize the subject, present it to an audience, and to conduct a discussion about the presented subject. Regular discussions with the instructor measure the student's ability to develop an idea from initial concepts to the complete picture within a given timeframe. Participation in class exercises measures the students ability to apply their theoretical knowledge in practice and to reflect on it afterwards.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

Basic terms of contract negotiations (like BATNA and ZOPA etc.);
Negotiation strategies and concepts (Harvard negotiation concept, win-win, win-lose etc.);
Effects of behavioural economics and negotiation tactics based on it (anchoring effect etc.);

Negotiation tactics (based on deception, pressure and defensive tactics etc.);
Principal agent problem in negotiations;
Communication (question techniques, answer techniques, argumentation techniques);
Emotions (core concerns framework);
Gender aspects in contract negotiations.

Intended Learning Outcomes:

Students learn the economic and psychological basics of contract negotiations as well as important communication tools for negotiations. They know the Harvard negotiation concept, including its weaknesses and the most important strategies in contract negotiations. They are familiar with the essential tactics. Students know how to apply these tactics or, respectively, how to react to these tactics if applied by the negotiation partner. They can distinguish different negotiation styles. Students have experienced the functioning of individual tactics during negotiation simulations and are able to understand the dynamics underlying a specific negotiation.

In the area of interdisciplinary competences, students strengthen their communication and argumentative skills (especially through the negotiation simulations) and their ability to work in a team (especially through group work in the context of presentations and negotiation simulations in a team).

Teaching and Learning Methods:

This module is held as an interactive seminar/lecture. Negotiation simulations and games are integrated into the course so that students learn to implement tactics and strategies. The negotiation games are designed to simulate practice. Students are encouraged to actively participate in the negotiation simulations and to get involved in the subsequent discussions. Video recordings are used to reflect on what has been experienced. Moreover, students will do research on a specific research question and write a seminar paper. In this framework, students will have to perform research of reference materials, design a questionnaire, conduct interviews with negotiators and analyse the answers. In order to support the students in their work individual appointments will be offered. Students will present their work in class.

Media:

Simulations, Exercises, Videos, Self-Tests, PPT, Whiteboard

Reading List:

Jung/Krebs, The Essentials of Contract Negotiation (2019)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations (WIHN0037)
(Seminar, 4 SWS)
Gelvez Alvarez L, Jung S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0046: Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises | Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises

Version of module description: Gültig ab winterterm 2020/21

Module Level: Master	Language: English	Duration: one semester	Frequency:
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic in the field of sustainable entrepreneurship / sustainable management in family enterprises. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis). Active engagement in the course will be expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English

Content:

This course addresses various aspects of sustainability available to new ventures and established family enterprises. Sustainable entrepreneurship (SE) emphasizes that entrepreneurs may pursue additional goals such as social and ecological goals that may alleviate social and ecological problems. It explores the opportunities that entrepreneurs create, the challenges entrepreneurs encounter, and the ways in which entrepreneurs exploit opportunities to contribute toward enhancing the society while simultaneously starting and sustaining a profit-oriented business. This course also examines the role of sustainability in family enterprises:

Course content will comprise:

- The importance of sustainable entrepreneurship in solving grand societal challenges of the 21st century
- Motives and barriers of sustainable entrepreneurship
- Overview of theories and concepts in sustainable entrepreneurship
- Stakeholder management for entrepreneurs
- Sustainable management in the context of family enterprises
- Sustainability in entrepreneurship and family enterprises in the context of developing countries
- Recognizing and exploiting social and ecological opportunities
- Practical examples of sustainable ventures

Intended Learning Outcomes:

After completing the seminar students, should understand how entrepreneurs and family enterprises can contribute to solving societal challenges. Therefore, students will be able to:

- understand the concept of sustainable entrepreneurship and sustainability in family enterprises
- understand and evaluate how entrepreneurs can enact on sustainable opportunities
- understand the entrepreneurs' motives and barriers to engage in sustainable entrepreneurship
- reflect on the importance of diversity in achieving sustainability in entrepreneurship and family enterprises
- evaluate strategic actions of sustainable enterprises
- evaluate hands-on examples and cases of how sustainable entrepreneurship can be implemented in practice with specific focus on the Heilbronn region: Those examples may relate to various sectors such as the public governance sector, the food and agriculture, environmental and mobility sector.

In addition, students will be able:

- to search, understand, and analyze academic literature by evaluating key findings and explaining complex research studies in a comprehensible and interesting manner,
- to apply methods for data collection and data analysis, - to structure and write a seminar thesis, and
- to present and subsequently discuss findings of their seminar thesis.

Teaching and Learning Methods:

' -Through lectures, supported by Power-Point presentations, the instructor will provide the theoretical foundations of sustainable entrepreneurship

- The content is discussed in the course by exchanging ideas, questions, and comments to encourage an open learning atmosphere
- Every sessions contains exercises, where the students apply their learnings in a group work (e.g. to present one start-up, enterprise or organization that has already successfully implemented the topic/concept presented in the lecture)
- Guest speakers will share practical insights and will strengthen the understanding of key concepts and processes and expose perspectives complementary to the topic of the seminar
- For their seminar thesis, students will investigate a topic within the subject of this course to develop a seminar thesis. Students will receive feedback from the instructor.
- In a final presentation, students present the results of their project thesis

Media:

Powerpoint, Zoom- & Breakout-Sessions, Online Simulations

Reading List:

Basic literature (for detailed reading list, see Moodle):

- Cruz, C.; Larraza-Kintana, M. Garcés-Galdeano, L. Berrone, P. (2014). Are family firms really more socially responsible? *Entrepreneurship Theory and Practice* 38(6), 1295–1316.
 - Gast, J., Gundolf, K., & Cesinger, B. (2017). Doing business in a green way: A systematic review of the ecological sustainability entrepreneurship literature and future research directions. *Journal of Cleaner Production*, 147, 44-56.
 - Le Breton-Miller, I., & Miller, D. (2016). Family firms and practices of sustainability: A contingency view. *Journal of Family Business Strategy*, 7(1), 26-33.
 - Lumpkin, G. T., Bacq, S., & Pidduck, R. J. (2018). Where change happens: community#level phenomena in social entrepreneurship research. *Journal of Small Business Management*, 56(1), 24-50.
 - Patzelt H., & Shepherd D.A. (2001) Recognizing opportunities for sustainable development. *Entrepreneurship Theory and Practice*. 35(4), 631-652.
 - Sarason, Y., & Dean, T. J. (2019). Lost Battles, Trojan Horses, Open Gates, and Wars Won: How entrepreneurial firms co-create structures to expand and infuse their sustainability missions in the acquisition process. *Academy of Management Perspectives*, 33(4), 469-490.
 - Sarasvathy, S. D., & Ramesh, A. (2019). An effectual model of collective action for addressing sustainability challenges. *Academy of Management Perspectives*, 33(4), 405-424.
 - Shevchenko, A., Levesque, M. & Pagell, M. (2016). Why firms delay reaching true sustainability. *Journal of Management Studies* 53, 911-35.
 - Zahra, S. A., Gedajlovic, E., Neubaum, D. O., & Shulman, J. M. (2009). A typology of social entrepreneurs: Motives, search processes and ethical challenges. *Journal of Business Venturing*, 24(5), 519-532.
- Book(s):
- Lumpkin, T. & Katz, J. (2011). *Social and sustainable entrepreneurship*. Emerald Group Publishing: Bingley (UK).
 - Weidinger, C., Fischler, F. & Schmidpeter, R. (2014). *Sustainable entrepreneurship: Business success through sustainability*. Springer: Berlin (Germany).

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0046): Sustainability in Entrepreneurship and Family Enterprises (Seminar, 4 SWS)

Bird M, Kaufmann L

For further information in this module, please click campus.tum.de or [here](#).

Digital Technology | Digital Technology

Module Description

CITHN2001: Advanced Software Testing and Analysis | Advanced Software Testing and Analysis [ASTA]

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

There is a written examination (60 minutes) on lecture and exercise contents. Only a written exam can cover the breadth and depth of the covered topics and methods. The final grade is the grade from this exam. In addition, students get a 0.3 bonus if they receive more than 80 % of the marks on their voluntary homework project. In the homework project, the students can demonstrate that they can apply the methods taught practically. This bonus cannot decide on passing/ failing the course.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

At least basic knowledge of software testing as from the module "Introduction to Software Engineering" (INHN0006)

Content:

Participants will work using recent research papers and corresponding tools to explore, apply, and evaluate cutting-edge software engineering research.

- Foundations of software quality and continuous quality control
- Repetition of the basics of software testing and analysis
- Specific static program analysis techniques, e.g., clone detection
- Specific techniques for test case prioritization and test suite optimization, e.g., mutation testing
- Specific test case generation techniques, e.g., fuzzing
- Specific debugging techniques, e.g., slicing
- Theoretical and empirical evaluation of all techniques

- Running example project with homework for all topics

Intended Learning Outcomes:

Participants will understand a selection of state-of-the-art methods and research tools for software testing and analysis and will be able to evaluate and apply them.

Teaching and Learning Methods:

The module combines lectures and tutorials as the main teaching methods. The lecture sessions contain a mix of traditional presentations by the lecturer and flipped classroom concepts where the students work through text and video material beforehand. This is especially important as examples from current research are used. The lecture sessions are then used for peer instruction, detailed examples by the lecturer, and discussing questions by the students. The tutorials follow the content of the lecture sessions. For each tutorial session, there is a worksheet that the students can work on during the sessions. There are tutors present to support the students. Finally, the students have to work on an example project as homework with feedback from the tutors.

Media:

- Slides for lectures and input in the tutorials
- Short instructional videos and scientific papers for lecture preparation
- Exercise sheets
- Moodle learning forum

Reading List:

Wagner. Software Product Quality Control. Springer, 2013.

Zeller et al. Generating Software Tests, fuzzingbook.org

Zeller. The Debugging Book, debuggingbook.org

Responsible for Module:

Wagner, Stefan; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

CITHN2004: Process Mining | Process Mining

Version of module description: Gültig ab winterterm 2024/25

Module Level: Bachelor/Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 5	Total Hours: 150	Self-study Hours: 90	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination will be in the form of a written exam (max. 120 minutes) in which the student's ability to explain discussed process mining concepts and apply it to small event logs or given use cases.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Recommended: Bachelor INHN0019 - Business Process Management

Content:

Process mining is a family of new data analysis methods that aims to discover, monitor, and improve business processes and is increasingly adopted by industry. It analyses so-called event logs, which capture the real execution data of business processes and are usually collected from one or several information systems supporting the execution of business processes. The main difference to traditional data analysis techniques is that process mining explicitly focuses on the process perspective. That is, it aims to reveal the complex order relations among the activities captured in the event log.

This lecture gives an introduction to the field of process mining. After introducing basic formalisms, the lecture provides a detailed and algorithmic perspective on the three key process mining perspectives: process discovery, conformance checking, and enhancement. Furthermore, it is discussed how typically process mining projects are executed. Finally, the lecture shows how process mining is currently used in industrial environments and advanced in process mining research (e.g., application to blockchain data).

Intended Learning Outcomes:

After successfully completing this course, students will be able to:

- Explain the principles of process mining and its application area
- Understand the structure of event logs (the basic input for process mining) and how it is built
- Explain and apply basic process mining algorithms for the different perspectives: discovery, conformance, and enhancement, and explain their strength and limitations
- Understand and apply different process mining tools/libraries and apply them to event logs
- Understand how to run a process mining project and explain the different phases, and how it is applied in industrial settings
- Know advances in current process mining research

Teaching and Learning Methods:

The teaching format consists of a lecture and a content-aligned exercise. The lecturer presents the content in the lecture. Some videos of the lecture content are also made available on the lecture website as an option for "flipped classroom" and for repetition. The lecture content consists of theory on process mining and their illustration based on practical examples. Students will be encouraged to study the literature and engage with the topics' content.

The exercise part consists of in-depth tasks on the theoretical part of the lecture. The exercise part serves to understand and apply the concepts taught. It is also intended to give students an overview of the existing tool landscape and apply process mining on real-world data.

Media:

Slides, Videos, Moodle, Exercises

Reading List:

- Wil van der Aalst. Process Mining: Data Science in Action. Springer 2016
- Josep Carmona et al. Conformance Checking: Relating Processes and Models. Springer 2018
- Marlon Dumas et al. Fundamentals of Business Process Management, 2nd edition. Springer 2018

Responsible for Module:

Pufahl, Luise; Prof. Dr. rer. nat.

Courses (Type of course, Weekly hours per semester), Instructor:

Exercise Process Mining (CITHN2004) (Übung, 1 SWS)

Oliveira da Nobrega Costa A, Pufahl L

Process Mining (CITHN2004) (Vorlesung, 3 SWS)

Oliveira da Nobrega Costa A, Pufahl L

For further information in this module, please click campus.tum.de or [here](#).

Module Description

CITHN2013: Process Mining Project | Process Mining Project

Version of module description: Gültig ab winterterm 2024/25

Module Level: Bachelor/Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 3	Total Hours: 90	Self-study Hours: 60	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To be able to apply and evaluate process mining techniques in the real world, students are asked to solve a close-to-the-real world process mining project in a team. The examination is a project work consisting of a written part (70 %, max. 15 pages for the team) and an oral presentation (30%, max. 30 min.). The written part is used to demonstrate the extent to which students can apply the methods they have learned to a specific real-world data set with analysis questions, the extent to which students can transparently document what analysis steps they have applied, and the extent to which they can communicate and critically evaluate their analysis results. Each student has to provide in this project report a bullet point list with their project role, their list of activities in the project, and the part that they have been involved in writing. The presentation demonstrates that students can present and communicate their analysis approach and results to an audience. Each student of a team has their own part in the team presentation.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Master Process Mining

Content:

Process mining is a family of new data analysis methods that aims to discover, monitor, and improve business processes and is increasingly adopted by industry. It analyses so-called event logs, which capture the real execution data of business processes and are usually collected from one or several information systems supporting the execution of business processes. In this project, students are working in a team on a process mining project with real-world data and with different goals and questions. Thereby, they can work with different academic process mining tools/libraries and can also develop own algorithms.

Intended Learning Outcomes:

After successfully completing this course, students will be able to:

- Run a process mining project and understand the different phases in detail
- Apply different process mining and data mining techniques in combination to solve complex questions in a process mining project
- Communicate and evaluate the results of a data science project in an oral and written format and document their applied approach transparently
- Be sensible in handling ethical concerns in their oral and written documentation.

Teaching and Learning Methods:

Participants work in groups on one specific process mining project close to the real-world. The data and the goals of the project are introduced in the beginning and different tutorials for different process mining tools/libraries might be provided. Project-specific mentoring sessions are provided.

Media:

Slides, Videos, Moodle

Reading List:

- Wil van der Aalst. Process Mining: Data Science in Action. Springer 2016
- Josep Carmona et al. Conformance Checking: Relating Processes and Models. Springer 2018
- Marlon Dumas et al. Fundamentals of Business Process Management, 2nd edition. Springer 2018

Responsible for Module:

Pufahl, Luise; Prof. Dr. rer. nat.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

CITHN2014: Foundations and Application of Generative AI | Foundations and Application of Generative AI

Version of module description: Gültig ab winterterm 2025/26

Module Level: Bachelor/Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 105	Contact Hours: 75

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The evaluation is based on a combination of preparatory assignments completed during exercise sessions (15%), the project implementation and report (60%), and a group presentation (25%).

Skills to be tested:

- Assignment: theoretical thinking, and hands-on capability using LLM
- Implementation: team-working, problem solving, integration of knowledge, project management
- Presentation: summarization, communication and collaboration

The assignments are completed individually and each student contributes 3-4 pages to the final group report and 2-3 minutes to the final group presentation.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Master: no prerequisite

Bachelor: INHN0001 Introduction to Informatics

INHN0002 Fundamentals of Programming

Content:

Basic principles of generative AI:

- Transformer architecture
- Training and inference of generative AI
- Large language model
- Stable diffusion

Prompt engineering:

- In-context learning
- Chain of thoughts
- Other advanced techniques like role-playing...

Applications of generative AI:

- LLM use cases and tasks
- GPT API call and plugin development

Open-source generative AI:

- Open-source models e.g., Llama 7B
- Loading & running
- Instruction tuning + fine tuning

Risk and security of generative AI:

- Limitations of generative AI
- Ethics issues
- Attack and defence of Generative AI (e.g., Jailbreak)

Intended Learning Outcomes:

Upon completion of the module, students are able to

- Understand generative AI deeply, including its historical development
- Comprehend and implement prompt engineering to enhance productivity
- Apply generative AI into different areas
- Navigate the ethical, compliance, and risk aspects associated with Generative AI

Teaching and Learning Methods:

The lecture is supplemented by integrated exercises, demonstrations and practical components.

The exercises serve to consolidate the theoretical content and should be solved in small groups.

Occasionally, individual exercises can also be set as voluntary homework. Students should be encouraged to repeat and experiment independently. The evaluation of your learning will be based on project report and the final presentation.

Media:

Slides, exercise sheets, board etc.

Reading List:

- Generative AI: Perspectives from Stanford HAI:

https://hai.stanford.edu/sites/default/files/2023-03/Generative_AI_HAI_Perspectives

- Generative AI at Work:

https://www.nber.org/system/files/working_papers/w31161/w31161.pdf

- Proactive Risk Management in Generative AI:

<https://www2.deloitte.com/us/en/pages/consulting/articles/responsible-use-of-generative-ai.html>

- Attention is All You Need:

<https://research.google/pubs/pub46201/>

Responsible for Module:

Chen, Chunyang, Prof: chun-yang.chen@tum.de

Courses (Type of course, Weekly hours per semester), Instructor:

Übungen zu: Foundations and Application of Generative AI (CITHN2014) (Übung, 2 SWS)
Chen C

Foundations and Application of Generative AI (CITHN2014) (Vorlesung, 3 SWS)
Chen C

For further information in this module, please click campus.tum.de or [here](#).

Module Description

CITHN7501: Master Seminar Information Engineering | Master Seminar Information Engineering

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 5	Total Hours: 150	Self-study Hours: 120	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Type of examination: Scientific paper (consisting of a written part (7-14 pages), as well as a presentation with discussion (30-60 minutes), weighting: 50/50)

Before the start of the seminar (usually at the end of the previous semester), all participants will be given a challenging scientific topic on which they will prepare a written paper using scientific literature they have researched themselves. The results of the individual papers are presented to the other participants of the seminar in a presentation and supported by visual media such as beamers or slides. The presentation also includes the extent to which the students can respond to questions, suggestions and discussion points about their own work and presentation and critically reflect on results. The extent to which they participate in the scientific discussion on the work and presentations of the other participants is also evaluated.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge of the specific subject area may be necessary (elective course).

Content:

Teaching of the following competencies:

- Independent assessment of a scientifically demanding theme
- Dealing with scientific literature
- Preparation of a seminar paper with an overview of sources
- Presentation and discussion of scientific results

The chairs (professors) choose suitable topics from their subject area that are related to information engineering, usually a sub-aspect of one of their research projects. They support the students in learning the technical and scientific skills. Topics may come from the following areas of information engineering, among others: databases, compiler construction, information systems, networks, groupware, graphics, robotics, image processing.

Intended Learning Outcomes:

After completing the module the participants possess the necessary methodological skills to familiarize themselves with a challenging scientific topic in information engineering, to independently prepare a scientific seminar paper, to present it and to analyze and comparatively evaluate the results. They are proficient in working with scientific literature (researching, evaluating, citing) and can present essential aspects of their topic in an elaboration and in a presentation.

Teaching and Learning Methods:

Participants independently assess a demanding scientific topic, prepare a presentation and discuss their findings.

The accompanying seminar paper summarizes the main concepts of the subject and provides an overview over related work.

Media:

Beamer, slides, whiteboard, report, possibly tool presentation and/or animations

Reading List:

The scientific literature and scientific publications in the respective area used in the module vary depending on the area and are announced at the beginning of the module.

TUM guidelines for good scientific practice:

<https://www.gs.tum.de/en/gs/during-the-doctorate/good-scientific-practice/>

Responsible for Module:

Matthes, Florian; Prof. Dr. rer. nat.

Courses (Type of course, Weekly hours per semester), Instructor:

Masterseminar Representation Learning (CITHN7501, INHN4512) (Seminar, 2 SWS)

Acosta Deibe M [L], Acosta Deibe M, Schwabe T

MSc Seminar High-Performance Computing (CITHN7501, INHN4501) (Seminar, 2 SWS)

Anzt H [L], Anzt H

Masterseminar Responsible AI and Language Technologies (CITHN7501, INHN4510) (Seminar, 2 SWS)

Dementieva D [L], Dementieva D

Masterseminar on Multimodal Models (CITHN7501, INHN4509) (Seminar, 2 SWS)
Edman J

Master Seminar Digital Transformation (CITHN7501, INHN4507) (Seminar, 2 SWS)
Fantino E, Neß N, König D

Masterseminar Generative Models on Text (CITHN7501, INHN4504) (Seminar, 2 SWS)
Fraser A [L], Di Marco M, Fraser A

Masterseminar Python under the Hood: Implementation and System Design of Machine Learning
Packages (CITHN7501, INHN4508) (Seminar, 2 SWS)
Hook B [L], Hook B

Master Seminar Gamified Information Systems (CITHN7501, INHN4503) (Seminar, 2 SWS)
Krohmann S, Hu S, Schmidt-Kraepelin M, Fantino E

Masterseminar Natural Language Processing for Low-Resource Languages (CITHN7501,
INHN4511) (Seminar, 2 SWS)
Okabe S [L], Okabe S

Master Seminar Digital Health (CITHN7501, INHN4505) (Seminar, 2 SWS)
Schmidt-Kraepelin M, Hu S, Thiebes S

Masterseminar Responsible AI (CITHN7501, INHN4506) (Seminar, 2 SWS)
Toussaint P, Du G, Fantino E
For further information in this module, please click campus.tum.de or [here](#).

Module Description

INHN0012: Computer Networking and IT Security | Rechnernetze und IT-Sicherheit

Version of module description: Gültig ab winterterm 2022/23

Module Level: Bachelor	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 105	Contact Hours: 75

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The exam takes the form of a 90 minutes written test. Comprehension questions and calculation tasks allow to assess acquaintance with the technologies and methods of computer networking and security, and the understanding obtained by implementation of protocol mechanisms. Calculation tasks also allow to assess the ability to determine the performance of selected computer networks and distributed applications.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

INHN0001 Introduction to Informatics

INHN0002 Fundamentals of Programming (Exercises & Laboratory)

Content:

Computer Networking

- Physical Layer: line coding and digital modulation schemes
- Data Link Layer: media access / multiplexing, framing, addressing, error detection, switching, wireless networks
- Network Layer: IPv4/v6 including address resolution and assignment, subnetting, multicast, static and dynamic routing
- Transport Layer: connection-less vs. connection-oriented protocols and sliding window protocols (UDP/TCP), NAT, code demos and live programming (implementation of UDP/TCP-based relay chats)
- Session / Presentation Layer: connection-less vs. connection-oriented protocols from the view of the Session Layer, examples (e.g. HTTP, TLS), encoding, data compression (Huffman)

- Application Layer: name resolution (DNS), case studies of various protocols such as Telnet, HTTP, FTP, SMTP, SSH, etc.

Security:

- Firewalls
- Authentication, authorization privacy, integrity
- Basics of cryptography (symmetric vs. asymmetric encryption)
- Case study of basic encryption algorithms (e.g. RC4, AES)
- Protocols with security mechanisms, e.g. SSH, TLS, IPSec

Intended Learning Outcomes:

After successful completion of the module, participants understand the key concepts of technologies and methods of computer networks and security and are able to use key layered network architecture protocols to explain what protocol mechanisms are used in each layer and how they work. They understand the architecture and implementation of basic encryption algorithms and protocols and their application in computer networking.

Participants can determine the performance of selected networks and security implementations, and can implement specific protocol mechanisms.

Teaching and Learning Methods:

The interactive lecture with slide presentations, animations, demonstrations and live programming presents the basic knowledge of computer networks and security and explains them using examples. Quizzes help students to recognize whether they have understood the basic concepts and essential contexts. Accompanying tutor exercises deepen the understanding of the contents of the lecture by means of suitable tasks and show the application of the various methods on the basis of manageable problems. Programming tasks allow computeraided deepening and application of conceptual knowledge to practical problems.

A midterm examination motivates students to continuously follow the lecture.

Media:

Lecture slides, exercise sheets, demonstrations, code demos, past exams

Reading List:

Literature

There is no (single) literature covering the lecture. Taking the lecture is therefore recommended. A standard book about computer networking could be:

1. James F. Kurose, Keith W. Ross Computernetzwerke Pearson Studium; 5. aktualisierte Auflage, 2012
2. Andrew S. Tanenbaum / Prof. David J. Wetherall Computernetzwerke Pearson Studium, 5. aktualisierte Auflage, 2012

Responsible for Module:

Günther, Stephan; Prof. Dr.-Ing.

Courses (Type of course, Weekly hours per semester), Instructor:

Übungen zu Rechnernetze und IT-Sicherheit (INHN0012) Campus Heilbronn (Übung, 2 SWS)

Günther S [L], Carle G, Günther S, Lehle L

Rechnernetze und IT-Sicherheit (INHN0012) Campus Heilbronn (Vorlesung, 3 SWS)

Günther S [L], Carle G, Günther S, Lehle L

For further information in this module, please click campus.tum.de or [here](#).

Module Description

INHN0017: Enterprise Architecture Management and Reference Models | Enterprise Architecture Management und Referenzmodelle

Version of module description: Gültig ab summerterm 2024

Module Level: Bachelor	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 60	Contact Hours: 120

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination will be in the form of a written exam (120 minutes) in which the students' ability to describe, evaluate and apply models, methods, visualizations and tools of EAM and reference modeling for manageable problems in a context-dependent manner will be tested.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

None

Content:

- Motivation: Use of Enterprise Architecture Models, Reference models and Patterns to support Strategic IT Management and Digital Transformation
- Foundations of Modeling
- Enterprise Architecture Models
- Reference Models and Modeling Patterns
- Capability-based Planning
- Agile Enterprise Architecture Management
- EAM Standards, Frameworks and Notations
- Large-Scale Scaling Agile Frameworks
- Agile (IT) Program Management

Intended Learning Outcomes:

Students will have a basic knowledge of enterprise architecture

management and reference modeling, including models, methods, metrics, visualizations, and tools after taking the module. In addition, students will be able to:

- describe the business architecture, application architecture and IT infrastructure of an organization holistically and formally using standardized notations and models.
- describe (digital) transformation processes in organizations using actual, plan, and target models and roadmaps.
- use reference models and design patterns in the process.

Teaching and Learning Methods:

lecture, tutorial, problems for individual study

Media:

Slides, Moodle, modeling, exercises

Reading List:

Hanschke, I. (2013). Strategisches Management der IT-Landschaft: Ein praktischer Leitfaden für das Enterprise Architecture Management. Carl Hanser Verlag GmbH Co KG.

Kaplan, J. D. (2005). Strategic IT portfolio management: governing enterprise transformation. PRTM.

Buckl, S., Ernst, A. M., Matthes, F., Ramacher, R., & Schweda, C. M. (2009, September). Using enterprise architecture management patterns to complement TOGAF. In 2009 IEEE International Enterprise Distributed Object Computing Conference (pp. 34-41). IEEE.

Buckl, S., Ernst, J., Lankes, A. M., Matthes (2008). Enterprise architecture management pattern catalog (version 1.0, february 2008)

Sandkuhl, K., Fill, H. G., Hoppenbrouwers, S., Krogstie, J., Matthes, F., Opdahl, A., ... & Winter, R. (2018). From expert discipline to common practice: a vision and research agenda for extending the reach of enterprise modeling. Business & Information Systems Engineering, 60(1), 69-80. Chicago

Uludag, Ö., Kleehaus, M., Caprano, C., & Matthes, F. (2018, October). Identifying and structuring challenges in large-scale agile development based on a structured literature review. In 2018 IEEE

22nd International Enterprise Distributed Object Computing Conference (EDOC) (pp. 191-197). IEEE.

Dikert, K., Paasivaara, M., & Lassenius, C. (2016). Challenges and success factors for large-scale agile transformations: A systematic literature review. Journal of Systems and Software, 119, 87-108.

Paasivaara, M., Behm, B., Lassenius, C., & Hallikainen, M. (2018). Large-scale agile transformation at Ericsson: a case study. Empirical Software Engineering, 23(5), 2550-2596. Chicago

Responsible for Module:

Pufahl, Luise; Prof. Dr. rer. nat.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

INHN0019: Business Process Management | Business Process Management

Version of module description: Gültig ab winterterm 2022/23

Module Level: Bachelor	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 5	Total Hours: 150	Self-study Hours: 60	Contact Hours: 90

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination takes the form of a written test with the duration of 90 minutes. It is intended to demonstrate that the fundamentals and modeling methods of business process management have been understood and can be applied. This includes the implementation of selected scenarios in process models using the studied process modeling languages such as BPMN. Furthermore, the ability to transform process models into other languages like Petri nets will be demonstrated by example and presented process verification methods like reachability analysis will be applied. Furthermore, findings from the exercises can be reported and corresponding questions can be answered.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

None

Content:

- Basic terms of business process management
- Business process management in companies
- Modeling of business processes and data (BPMN, EPK, petri nets)
- Verification of selected properties with process models
- Basics of the development of process-oriented applications
- Use of process-oriented applications in practice
- Contribution of business process management to digital transformation
- Basics of process mining

Intended Learning Outcomes:

After participating in the module, students are able to understand terms and methods of business process management and classify the use of processes and process models with their respective benefits in different industries. They will also be able to create correct process models as a basis for designing information systems. They will be able to prove correctness of process models via accessibility analysis. Students understand how process models contribute to the development of process-oriented applications and are familiar with examples of process engines and process-aware information systems. Students recognize the importance of process-oriented applications based on real-world processes and can thus classify their contribution to business informatics and digital transformation in companies. Finally, students understand the basics of process mining for the analysis of processes and can apply selected process mining techniques.

Teaching and Learning Methods:

Lecture, tutorial, self-study assignments

Media:

Slides, videos, moodle, exercises

Reading List:

Weske, M. (2019) Business Process Management - Concepts, Languages, Architectures, Third Edition. Springer 2019, ISBN 978-3-662-59431-5, pp. 1-417 2019

Journal and conference articles for the individual topics will be announced separately.

Responsible for Module:

Pufahl, Luise; Prof. Dr. rer. nat.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0094: Python and Advanced Data Science | Python and Advanced Data Science

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination consists of one project work on the programming language Python. Students work in groups of two or three. A project work consists of several cases that are given to them during the semester. Groups write a project report (70% of final grade) and present the results to their peers (30% of final grade).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

It is recommended that students have basic knowledge how to use Python. Ideally, students have visited an introductory course to Python.

Content:

Data Science and Machine Learning are the major technologies and have a huge impact for large private businesses, public institutions, and research. With technological developments, globalization and dynamic environment in every industry, the amount of generated data is growing exponentially. Data science therefore is crucial and deals with the methods and tools needed to analyze (large amounts of) data and draw actionable conclusions from the results gained in the process.

The lecture covers the Python programming and applying advanced Data Science concepts with Python implementations. The module covers the following topics:

1. Introduction and Organization
2. Python Data Structures
 - a. Lists
 - b. Dictionaries

- c. Tuples
- d. Sets
- e. Numpy & Pandas
- 3. Working with files and databases, SQL
 - a. Working with different file types
 - b. Database Management System (DBMS)
 - c. SQL
- 4. Data Analysis & Visualization
 - a. Data understanding and analysis
 - b. Data visualization
- 5. Data Wrangling & Manipulation
 - a. Data Cleaning & Preparation
 - b. Exploratory Data Analysis (EDA) techniques
 - c. Hands-on exercises
- 6. Building Machine Learning Models & advanced techniques-I
 - a. Bias-variance Trade-off
 - b. Regression and regularizations
 - c. Multicollinearity, VIF, Heteroscedasticity
 - d. Ensemble models, Random Forest, Bagging, Boosting, tuning parameters
- 7. Building Machine Learning Models & advanced techniques-II
 - a. Logistic Regression, Sigmoid, AUC-ROC, precision-recall
 - b. Unsupervised learning
 - c. Hands-on exercises
- 8. New Technologies & ML Models
 - a. AutoML, H2O
 - b. More new ML technologies
 - c. Hands-on exercises
- 9. Deep Learning-1
 - a. Neural Networks
 - c. Recurrent Neural Networks (RNN)
- 10. Deep Learning-2
 - a. CNN
- 11. Natural Language Processing-1
 - a. NLP pipelines
 - b. TF-IDF
- 12. Natural Language Processing-2
 - a. Sentiment Analysis
 - b. Hands-on Exercises
- 13. Project

Intended Learning Outcomes:

Upon the completion of this course, students will acquire knowledge of advanced Data Science concepts and learn how Python can be used in this context.

Students will also learn about data structures in Python and how to write programs/scripts using Python data structures. They will learn how relational databases and SQL can be used for data analysis and storage. They also develop skills on using Python libraries for data analysis, dealing with raw data, data pre-processing and visualization methods. Through exercises and projects, they will gain hands-on experience and solve data analysis problems.

They will gain theoretical knowledge of advanced techniques for Machine Learning (ML) models and their applications. Students will also learn how to choose ML models and then build them with hands-on examples. In the projects, they will be able to build advanced models through interactive exercises and real-world examples through projects, which use a variety of data sources, project scenarios, data analysis tools and advanced ML techniques. They will also learn how to evaluate their models and improve them by tuning.

Students will gain knowledge about new technologies and applications in ML and Data Science fields. Moreover, Deep Learning topics will be introduced. Moreover, students will learn about Natural Languages Processing and its pipelines with some hands-on exercises.

Finally, students will be able to communicate their findings behind the data to relevant audiences by visualizing and presenting their idea through presentations and reports.

Teaching and Learning Methods:

The course comprises lectures, exercises, group work and student presentations. Lectures familiarize students with Python and advanced techniques for data science. Exercises allow students to practice using Python and applying advanced theories. Group work is selected as it allows students to have a joint learning experience. Students will have the opportunity to reflect on their work by discussing their project progress with the supervisor. Presenting their results to others, encourages students to reflect on their learning experience.

Media:

Lecture slides; whiteboard; programming assignment; powerpoint.

Reading List:

Deitel, P., & Deitel, H. (2020). Intro to Python for Computer Science and Data Science, Global Edition. Pearson Education

McKinney W. (2017). Python for Data Analysis: Data Wrangling with pandas, numpy and ipython

Jake VanderPlas (2022) Python Data Science Handbook: Essential Tools for Working with Data

Tshepo Chris Nokeri (2022) Data Science Solutions with Python: Fast and Scalable Models Using Keras, PySpark MLlib, H2O, XGBoost, and Scikit-Learn

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Python and Advanced Data Science (MGTHN0094) (Vorlesung, 4 SWS)

Can B

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0116: Advanced Seminar Management & Marketing: Eyetracking and Decision Making | Advanced Seminar Management & Marketing: Eyetracking and Decision Making

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students work in groups. The task is to design an eyetracking experiment based on a self-chosen research question and topic. Students are encouraged to replicate and/or add constructs to existing experimental designs. Students are required to collect valid eye movement data from at least 30 respondents (after data cleaning). Students analyze the data and test the formulated research hypotheses. The empirical results are presented in class (20%) and documented in a written report (80%, 20 pages).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

It is required that participants have successfully completed undergraduate courses on statistics and empirical research methods before taking this course.

Content:

In an introductory block, seminar participants will be familiarized with key methodological know-how regarding eye movement recording. Students will critically reflect on how to design and implement eyetracking experiments using examples from the literature. Based on their assessment of prior research as well as discussions with the supervisors, students learn to formulate a research question and research hypotheses. Students are then required to apply the methodological know-how they have learned by implementing an experiment and collecting eye-tracking data in the laboratory. Furthermore, students are requested to analyze the empirical data, to present their results in class and defend their work, as well as to document the research process (including the results of the experimental data) in a written report. eers.

Intended Learning Outcomes:

Acquire methodological know-how (learn how to collect eyetracking data and set up small scale experiment) / Formulate a research question and research hypotheses / Work with scientific literature / Analyze empirical (eyetracking) data, including statistical tests and visualizations of key results / Present research results to others and defend own work / Structure and write a seminar report that includes statistical analyses

Teaching and Learning Methods:

The seminar starts with introductory lectures explaining the basics of eyetracking. Afterwards, students start working in groups and discuss the progress of their research projects with supervisors. Students are familiarized with the eyetracking equipment in a practical lecture in the laboratory. Students continue to work in groups and collect and analyze the empirical data. Finally, all groups present the results of their empirical work in class and document their work in a seminar report.

Media:

PowerPoint, videos, AI image generation software, research papers, virtual reality devices, exercise sheets

Reading List:

Will be published at the start of the semester

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Management & Marketing: Eyetracking and Decision Making (MGTHN0116)
(Seminar, 4 SWS)

Arslan S, Kleffke K, Meißner M

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0125: Data Science in Business and Economics with R | Data Science in Business and Economics with R

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves academic elaboration, research and statistical programming on a business or economics topic with the software R. Data sets for statistical examination will be provided by the lecturer.

The students will be asked to

- (i) apply statistical programming in R to gain empirical insights on the seminar topic,
- (ii) summarize the key insights and empirical results on the seminar topic, and to
- (iii) present their findings in class in a presentation of about 15 to 20 minutes.

The examination will consist of an oral in-class presentation (no group work). Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Principles of Economics (for Master in Management)

Content:

The seminar is divided into three parts: a theory part, a methodology part on conducting data analysis with R and a seminar part with own data analysis under guidance.

Topic area 1: Socio-economic characteristics and working from home

Topic area 2: Individual preferences and working from home

Topic area 3: Stress overload and socio-economic characteristics

Topic area 4: Individual preferences and stress overload

Topic area 5: Socio-economic characteristics and mental health

Topic area 6: Individual preferences and mental health

Topic area 7: Healthy eating and socio-economic characteristics

Topic area 8: Individual preferences and healthy eating

Topic area 9: Food consumption and waste behavior and socio-economic characteristics

Topic area 10: Individual preferences and food consumption and waste decisions

Intended Learning Outcomes:

Students deepen their knowledge and understanding about microeconomic theories. They develop detailed and reflected views on potential biases in (consumption) choices. They are further able to understand and critically judge the challenge to scientifically testing (consumer) theories. Students acquire knowledge of empirical studies (on both measurement and effect identification) and integrate the existing and new knowledge to conduct their own data analysis with a provided data set under guidance.

Teaching and Learning Methods:

After being introduced to basic economic concepts in consumer theory, students will be guided to do own steps in data analysis with R. They will be coached throughout the semester by an instructor. A key learning objective is the ability to perform an own data analysis with R and to integrate the results of this analysis into a literature- and theory-backed simple framework to write a term paper. Students will perform their own literature research. A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups.

Media:

Lecture and seminar presentations, and interactive coding in R.

Reading List:

Seminar refers to content covered in the following books:

Hal Varian: Intermediate Microeconomics, chapters 2-5

James Stock und Mark Watson: Introduction to Econometrics

Joshua Angrist and Jörn-Steffen Pischke: Mostly harmless econometrics

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

Data Science in Business and Economics with R (MGTHN0125) (Seminar, 4 SWS)

Zeidler H

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0177: Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations | Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1. Group Report (70%)

- **Assessment Details:** In groups, students will collaboratively submit a report (no longer than 5 pages) addressing a specific managerial problem and proposing a solution using VR/AR technologies. Students are allowed to use reference materials and external resources.
- **Scope:**
 - Identify a relevant managerial problem in production, quality inspection, remote support, human resources, or customer experience.
 - Justify the importance of solving this problem.
 - Propose a detailed VR/AR-based solution, including technical implementation steps, management implications, and expected outcomes.
- **Rationale for Assessment:**
 - This assessment evaluates the ability to apply theoretical knowledge to practical problems, foster teamwork, and use technology to prepare solutions with managerial perspectives.
 - The report format develops students' ability to communicate complex ideas concisely and effectively.

2. Individual Critical Reflection (30%)

- **Assessment Details:** Each student will write a critical reflection on the relevance of immersive technologies (VR/AR) in solving managerial challenges. This will be submitted as a separate document, no longer than 2 pages. The reflection must be written independently.
- **Scope:**
 - Reflect on the practical and theoretical insights gained during the module.
 - Critically analyze the advantages and limitations of VR/AR in managerial contexts.

- Discuss how the experience of creating the group report has influenced their understanding of technology's role in management.

- **Rationale for Assessment:**

- This assessment allows students to self-evaluate, critically linking their learning to broader managerial and technological contexts.

- It promotes independent thought, allowing students to articulate their unique perspectives.

Repeat Examination:

(Recommended) Prerequisites:

Basic Programming with C#

Content:

In the module, the students will gain knowledge about the technological and management aspects of VR implementations with hands-on experience and use case examples in

- Production and Assembly
- Quality Inspection and Remote Support
- Training and skill development
- Customer Experience and Prototyping

Technology-specific content

- Introduction to Immersive Technologies (key hardware and software platforms)
- Using Unity for developing VR/AR applications

Management-specific content

- Ethical Considerations
- Behavioral Challenges
- Building a roadmap from prototype to market-ready product

Intended Learning Outcomes:

By the end of this module, students will be able to:

- Understand the fundamental principles of VR and AR technologies.
- Analyze the role of immersive technologies in improving managerial and operational efficiency.
- Design and prototype VR/AR-based solutions for specific management scenarios.
- Critically evaluate the broader implications of VR/AR deployment, including ethical concerns (e.g., data privacy, accessibility).
- Work collaboratively in teams to identify managerial challenges and develop VR/AR solutions.

Teaching and Learning Methods:

The coursework is primarily practical. At the beginning of the module, the students will be introduced to the immersive technologies and different headsets. They will be introduced to Unity Game Development Engine, which they will use to develop application solutions. The students

can choose between VR or AR applications and must program their software solution, keeping a managerial focus in mind.

The students will present the problem statement (in a group) that they want to work on and will have regular feedback sessions during the development phase. The students will present their solutions at the end of the module.

Media:

Material will be shared via Moodle

Reading List:

Readings for general preparation (but not limited to):

Dalton, J. (2021). Reality check: How immersive technologies can transform your business. Kogan Page Publishers.

Jung, T., & Dalton, J. (2021). XR Case Studies. Springer International Publishing.

Wuttke, D., Upadhyay, A., Siemsen, E., & Wuttke-Linnemann, A. (2022). Seeing the bigger picture? Ramping up production with the use of augmented reality. *Manufacturing & Service Operations Management*, 24(4), 2349-2366.

Windhausen, A., Heller, J., Hilken, T., Mahr, D., Di Palma, R., & Quintens, L. (2024). Exploring the impact of augmented reality smart glasses on worker well-being in warehouse order picking. *Computers in Human Behavior*, 155, 108153.

Seeliger, A., Cheng, L., & Netland, T. (2023). Augmented reality for industrial quality inspection: An experiment assessing task performance and human factors. *Computers in Industry*, 151, 103985.

Rejeb, A., Keogh, J. G., Leong, G. K., & Treiblmaier, H. (2021). Potentials and challenges of augmented reality smart glasses in logistics and supply chain management: a systematic literature review. *International Journal of Production Research*, 59(12), 3747-3776.

Eswaran, M., & Bahubalendruni, M. R. (2022). Challenges and opportunities on AR/VR technologies for manufacturing systems in the context of industry 4.0: A state of the art review. *Journal of Manufacturing Systems*, 65, 260-278.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology: Immersive Technologies for Business and Operations (MGTHN0177) (Seminar, 4 SWS)

Phadke V

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0178: Data Science in Finance | Data Science in Finance

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The module examination consists of a 90-minute written exam assessing students’ understanding of key concepts in AI and data science in finance, including data manipulation, statistical and machine learning models, NLP techniques (e.g., LLMs), and ethical considerations. The exam includes open-ended questions, multiple-choice tasks, and problem-solving components.

Students may have the opportunity to improve their final grade by up to 0.3 points by voluntarily participating in group work and presentations on selected topics in AI and Data Science. This opportunity depends on factors such as the number of participants and available capacity, and is decided on a semester-by-semester basis by the instructor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

none

Content:

This course offers a comprehensive exploration of data science techniques and their applications to address challenges in the financial sector. It combines foundational principles with advanced methodologies to equip students with the skills needed to analyze financial data, build predictive models, and critically interpret results. The course emphasizes practical applications of machine learning, artificial intelligence (AI), and natural language processing (NLP) while also addressing ethical considerations and future trends in financial technology.

Key Topics Covered

- 1. Introduction to AI and Data Science in Finance

- Overview of artificial intelligence and data science concepts for finance
- Understanding financial data structures and their unique characteristics
- 2. Machine Learning Basics
 - Supervised regression and classification techniques tailored to financial data
- 3. Performance Evaluation and Advanced ML Techniques
 - Evaluation metrics, random forests, and deep learning methods for financial applications
- 4. Unsupervised Learning
 - Clustering methods and principal component analysis (PCA) for financial data analysis
- 5. Generative AI in Finance
 - Exploring the innovative use of generative AI techniques in financial applications
- 6. Natural Language Processing (NLP)
 - Fundamentals of NLP
 - Understanding and applying large language models (LLMs) in finance
 - Fine-tuning and evaluating LLMs for financial use cases
- 6. AI Ethics, Privacy, and Emerging Trends
 - Ethical considerations, privacy, and compliance in AI-driven finance
- 7. Emerging trends in AI, machine learning, and NLP for the financial sector

Intended Learning Outcomes:

Dieser Kurs bietet eine umfassende Einführung in die Anwendung von Data-Science-Methoden zur Bewältigung von Herausforderungen im Finanzsektor. Die Studierenden entwickeln grundlegende und fortgeschrittene Fähigkeiten in Datenanalyse, statistischer Modellierung und maschinellem Lernen, wobei der Schwerpunkt auf spezifischen Anwendungen im Finanzbereich liegt. Am Ende des Kurses verfügen die Studierenden über das Wissen und die Werkzeuge, um Finanzdaten zu analysieren, prädiktive Modelle zu erstellen und die Ergebnisse im Kontext der Finanzwelt zu interpretieren.

1. Einführung in KI und Datenwissenschaft im Finanzwesen
 - Überblick über Konzepte der künstlichen Intelligenz und der Datenwissenschaft im Finanzbereich
 - Verstehen von Finanzdatenstrukturen und ihren einzigartigen Eigenschaften
2. Grundlagen des maschinellen Lernens
 - Supervised Regressions- und Klassifizierungstechniken, zugeschnitten auf Finanzdaten
3. Leistungsbewertung und fortgeschrittene ML-Techniken
 - Bewertungsmetriken, Random Forests und Deep Learning-Methoden für Finanzanwendungen
4. Unsupervised Lernen
 - Clustering-Methoden und Hauptkomponentenanalyse (PCA) für die Finanzdatenanalyse
5. Generative KI im Finanzwesen
 - Erforschung des innovativen Einsatzes von generativen KI-Techniken in Finanzanwendungen
6. Natürliche Sprachverarbeitung (NLP)
 - Grundlagen des NLP
 - Verständnis und Anwendung großer Sprachmodelle (LLMs) im Finanzbereich
 - Feinabstimmung und Bewertung von LLMs für finanzielle Anwendungsfälle
6. KI-Ethik, Datenschutz und aufkommende Trends
 - Ethische Überlegungen, Datenschutz und Compliance im KI-gesteuerten Finanzwesen

7. Aufkommende Trends in KI, maschinellem Lernen und NLP für den Finanzsektor

Teaching and Learning Methods:

The course "Data Science in Finance" is designed to provide students with a comprehensive understanding of integrating data science methods in the financial sector. To achieve this goal, various teaching methods are used:

- Lectures: Provide foundational knowledge on topics such as applications of artificial intelligence in finance, statistical modeling, and machine learning techniques.
- Practical programming exercises: Offer hands-on experience in implementing data science methods using programming languages like Python.

These diverse learning methods aim to equip students with both theoretical knowledge and practical skills essential for roles in the field of data science in finance.

Media:

Exercise sheets, PowerPoint, and further material to be shared via Moodle.

Reading List:

To become familiar with the topic, the following literature might be useful:

1. Alammari, Jay and Maarten Grootendorst (2024). Hands-On Large Language Models: Language Understanding and Generation. 1st. O'Reilly Media. isbn: 978-1492099206.
2. Bartram, Söhnke M., Jürgen Branke, and Mehrshad Motahari (2020). Artificial Intelligence in Asset Management. Charlottesville, VA: CFA Institute Research Foundation. isbn: 978-1-952927-02-7. url: <https://www.cfainstitute.org/-/media/documents/book/rf-lit-review/2020/rflr-artificial-intelligence-in-asset-management.pdf>.
3. Bothmann, Ludwig et al. (19–23 Sep 2023). "Developing Open Source Educational Resources for Machine Learning and Data Science". In: Proceedings of the Third Teaching Machine Learning and Artificial Intelligence Workshop. Ed. by Katherine M. Kinnaird, Peter Steinbach, and Oliver Guhr. Vol. 207. Proceedings of Machine Learning Research. PMLR, pp. 1–6. url: <https://proceedings.mlr.press/v207/bothmann23a.html>.
4. Bruce, Peter C., Grant Fleming, and Andrew Bruce (2021). Responsible Data Science. Hoboken, NJ: Wiley. isbn: 978-1-119-74175-6. url: <https://www.wiley.com/en-us/Responsible+Data+Science-p-9781119741640>.
5. Hugging Face (2022). The Hugging Face Course, 2022. <https://huggingface.co/course>. [Online; accessed].
6. James, Gareth et al. (2017). An Introduction to Statistical Learning. 7th ed. Springer Texts in Statistics. New York City: Springer.
7. Prado, Marcos López de (2018). Advances in Financial Machine Learning. Hoboken, NJ: Wiley. isbn: 978-1-119-48208-6.
8. — (2020). Machine Learning for Asset Managers. Cambridge: Cambridge University Press. isbn: 978-1-108-49905-6.

9. Tatsat, Hilary, Sahil Puri, and Brad Lookabaugh (2020). Machine Learning and Data Science Blueprints for Finance. Sebastopol, CA: O'Reilly Media. isbn: 978-1-492-05743-3.
10. Wooldridge, Jeffrey M. (2019). Introductory Econometrics: A Modern Approach. 7th. Boston, MA: Cengage Learning. isbn: 978-1-337-55886-0.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0183: Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research | Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students work in groups of 3-4 to design and implement a VR-based marketing research experiment. Following experimental design, students will create VR scenarios using industry-standard software and hardware. Final deliverables include: (1) an in-class presentation demonstrating the functional VR experiment and discussing findings (30%), and (2) a comprehensive written report (15-20 pages) documenting the research question, methodology, technical implementation with code, results, data analysis, and reflection on the research process (70%).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic programming experience, preferably C# or similar object-oriented language (highly-recommended)

Content:

In an introductory block, participants will acquire key methodological skills in experimental design, virtual reality implementation, and programming for marketing research in immersive environments. Students will critically reflect on how to use VR for marketing research based on relevant scientific literature. Based on literature review and supervisor consultations, students will conceptualize and develop original VR implementations.

The seminar will conclude with in-class project presentations and written documentation of all VR implementation code and assets.

Intended Learning Outcomes:

Students will acquire skills in the following areas:

- Software development: Using a game engine (Unity) and programming (C#) to implement virtual reality studies in a marketing context
- Experimental design: Conceptualizing and planning VR study implementations
- Literature analysis: Reviewing, evaluating, and synthesizing relevant scientific literature
- Academic communication: Presenting research results and defending methodological choices
- Technical documentation: Preparing a comprehensive seminar report that includes software architecture, code documentation, and programming implementation

Teaching and Learning Methods:

The seminar starts with introductory lectures providing examples of virtual reality as a research environment for investigating marketing research questions. Lectures introduce the Extended Reality Lab equipment and guide the implementation of VR experimental designs. Subsequently, student groups develop their experimental implementations while receiving ongoing supervision. Students are also expected to pilot-test their VR implementations and collect data with actual participants. Finally, all groups present their results in class and document their work in a seminar report.

Media:

PowerPoint, videos, AI image generation software, research papers, virtual reality devices, exercise sheets

Reading List:

Will be published at the start of the semester

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research (MGTHN0183) (Seminar, 4 SWS)

Meißner M, Siqueira Rodrigues L

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0185: Methods and Databases in Empirical Finance | Methods and Databases in Empirical Finance

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The module examination consists of a 90-minute written exam that tests the understanding of core theoretical concepts related to databases and empirical methods in finance. This includes data analysis, statistical and machine learning models, as well as generative AI (e.g., LLMs). The exam comprises open-ended questions, multiple-choice tasks, and problem-solving exercises.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;

Basic knowledge in mathematical and statistical analysis methods;

Content:

The topic of this course is the introduction to core theoretical and methodological concepts in empirical finance. In addition, various databases (e.g., Bloomberg, Thomson Reuters, WRDS) available at TUM—particularly at the Heilbronn campus—as well as the use of software packages relevant to empirical capital market research are introduced.

The lecture is structured into three main components: an introductory session, a presentation of theoretical concepts, and an overview of empirical methods in finance.

The databases and their functionalities are presented in the accompanying exercise sessions. In subsequent sections, students learn to work with these databases and appropriate programming languages through practical finance-related sub-projects. Key topics include data acquisition, data processing, and the application of analytical methods, all of which are covered in depth.

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- work with various capital market databases available at TUM, particularly on the Heilbronn campus;
- use software-based analysis tools to import, process, and analyze data from different systems;
- apply a range of analytical methods.

The module provides students with the foundational skills for empirical research projects, particularly in the field of finance.

Teaching and Learning Methods:

Types of Instruction:

The course content is delivered in weekly sessions comprising lectures and practical exercises.

Teaching Methods:

Theoretical content is first presented in lectures. In accompanying exercises, students complete smaller tasks that help them independently develop skills in working with various databases and software tools. In case of difficulties, they are encouraged to seek guidance from the instructors.

Learning Methods:

- Collaborative development of theoretical concepts and introduction to empirical methods during lectures;
- Independent work on a variety of problem-based questions;
- Application of theoretical knowledge—from data collection and preparation to data analysis—within the context of hands-on exercises.

The selected teaching formats and methods are designed to ensure that students acquire or strengthen their ability to carry out scientific work. The module can serve as a preparatory course for a master's thesis or other empirical research projects, particularly in the field of finance.

Media:

Exercise sheets, PowerPoint, Hands-on coding

Reading List:

Ang and Clifford (2015): Analyzing Financial Data and Implementing Financial Models Using R

Cochrane (2005) : Asset Pricing

Bali et al (2016): Empirical Asset Pricing: The Cross Section of Stock Returns

More Literature will be announced during the first session.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Methods and Databases in Empirical Finance (MGTHN0185) - Lecture (Vorlesung, 2 SWS)

Gabelmann Y, Osterrieder D

Methods and Databases in Empirical Finance (MGTHN0185) - Exercise (Übung, 2 SWS)

Gabelmann Y, Osterrieder D

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0192: Intelligent Scheduling | Intelligent Scheduling

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination takes the form of exercise tasks to understand major concepts (20%) and the final problem-solving-oriented project. The evaluation of the final project is based on:

- Presentation (~30 min incl. discussion): Oral presentation of the solution approach and results (40%)
- Report (~25 pages): Detailed documentation of the problem, modeling decisions, methodology, and analysis (40%)

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

- solid understanding of mathematics for managers (matrix notation, concepts of proofs, solid basics of mathematics modeling)
- solid programming skills in Python, C++, C#, VisualBasic, or a similar programming language

Content:

If we talk about digitization in the context of smart manufacturing, Industry 4.0, and Industry 5.0, one of the central challenges is the development of predominantly automated, intelligent scheduling systems. Imagine networked machines, products, robots, and workers that can react on-the-fly to changes in the operational environment (e.g., machine breakdowns, shifts in customer orders) in a way that is both intelligent and business-oriented. Achieving this is far from trivial, and in practice we are still some distance away from such fully adaptive systems.

In this course, we begin with an introduction to scheduling and then explore the theory behind selected promising algorithmic approaches, such as constraint propagation, AI-guided heuristics,

and methods for dynamic scheduling. By the end of the course, students will implement their own project, designing, applying, and evaluating an algorithm based on the methods covered in class.

Intended Learning Outcomes:

After successfully completing the module, participants will:

- understand the fundamentals of scheduling complexity
- apply exact and approximate solution approaches
- leverage AI and advanced heuristics for scheduling
- analyze and interpret results in a managerial and operational context
- develop hands-on experience through projects

Teaching and Learning Methods:

The module consists of lectures and dedicated exercise sessions. Theoretical concepts are introduced with illustrative examples and interactive discussions, encouraging participants to understand and program covered algorithmic concepts. Through exercise sessions and AI-assisted coding support, students independently apply techniques to deepen their competencies.

Media:

Lecture documents, exercises, and demonstrators available in TUM Moodle and via open-access web-sources.

Reading List:

Blazewicz, J., Ecker, K. H., Pesch, E., Schmidt, G., Sterna, M., & Weglarz, J. (2019). Handbook on Scheduling: From Theory to Practice. Springer.

Influential relevant scientific articles published in top scientific journals, e.g.:

Dorndorf, U., Pesch, E., Phan-Huy, T. (2000). Constraint propagation techniques for the disjunctive scheduling problem. Artificial Intelligence, 122, 189--240.

Hottung, A., Kwon, Y.-D., & Tierney, K. (2022). Efficient active search for combinatorial problems. arXiv:2106.05126.

Responsible for Module:

Otto, Alena; Prof. Dr. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0195: Advanced Seminar Operations & Technology: Machine Learning | Advanced Seminar Operations & Technology: Machine Learning

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1. Four-to-Five in-Class Presentation in Total (Group Work) (50%)
2. Contribution to Discussion (Group Work) (10%)
3. One Written Six-page Review Essay (Individual Work) (40%)

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Statistics, Mathematics

Content:

This seminar happens bi-weekly to give students enough time to prepare their paper and understand the topics

1. Supervised, Unsupervised, and Semi-Supervised Learning
2. High-dimensional Data, Tensor, Clustering, Principal Component Analysis
3. Neural Network, Convolutional Neural Network, Graph Convolutional Network
4. Natural Language Process and Large Language Models
5. Transfer Learning, Meta Learning, and Knowledge Distillation
6. Self-supervised Learning
7. Reinforcement Learning

Intended Learning Outcomes:

After this seminar course, the students will be able to understand the most front-tier machine learning papers that shaped today's AI reign. The students will also get an entry level

understanding to machine learning models and machine learning research and how to apply them into industries. The students will also get full training for scientific presentation and writing.

Teaching and Learning Methods:

The teaching and learning method will be a reverse classroom: the student will finish the basic knowledge ad lecture off the class, and then read the paper and prepare the presentation as a group; during the class, the student will present their paper as a group and discuss with other groups. At the end of each class, the lecture will also give a in-class task to be solved by the groups.

Media:

Slides, Videos, Whiteboards

Reading List:

tba

Responsible for Module:

Li, Ziyue; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology: Machine Learning (MGTHN0195) (Seminar, 4 SWS)

Li Z

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0018: Economics and Management of Platforms | Economics and Management of Platforms

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Seminar paper with presentation: Each student will be assigned a reader of research papers, book chapters, and case studies to work on. To pass the course, students have to write a seminar paper and present the seminar paper in class.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

very strongly recommend you previously attend an introductory course on economics (e.g., "Principles of Economics") and management (e.g., "International Management"). Also, the attendance of a course on empirical research methods (e.g., "Empirical Research Methods in Management and Economics") is recommended.

Content:

Digital platforms (e.g., Uber, iOS, AirBnB) are transforming entire industries. The drivers of platform markets' success are complex, and the successful setup and management of a platform requires a sound theoretical understanding of the concepts of network effects, two-sided markets, and complementarity.

The seminar seeks to provide an advanced understanding of the following questions:

- What are digital platforms and how do they impact markets?
- Which economic principles underlie platforms?
- What strategies enable firms to successfully establish platforms?

Intended Learning Outcomes:

Knowledge Objectives

After the course, students will be able to:

- Understand the concept of digital platforms from a managerial and economic perspective
- Understand the economic impact of platforms
- Outline the economic principles underlying platforms
- Craft and evaluate strategies for the effective management of digital platforms

Skills Objectives

- Improve diagnostic and analytical skills (i.e., structured problem-solving)
- Build up critical thinking and interpretation skills
- Enhance verbal and argumentation skills via presentations and group discussions

Teaching and Learning Methods:

The introductory session will provide an introduction into the key concepts and academic writing in a lecture-style. The largest share of this course will be based on interactive discussions among course participants and based on the presentations of the course participants. A large share of learning will occur through you preparing for the in-class session.

Media:

Reading List:

- Shapiro, Carl, and Hal R. Varian. Information rules: a strategic guide to the network economy. Harvard Business Press, 1998.
- Parker, Geoffrey G., Marshall W. Van Alstyne, and Sangeet Paul Choudary. Platform revolution: how networked markets are transforming the economy and how to make them work for you. WW Norton & Company, 2016.

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0033: Deep Reinforcement Learning | Deep Reinforcement Learning [IRL]

Version of module description: Gültig ab summerterm 2020

Module Level:	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The module is evaluated by laboratory assignments. Students are paired into groups of two and work on homeworks and a project throughout the semester. Each group should submit the homeworks (30%), a written report (weighs 50%) and present in the last session of the course (weighs 20%).

In the report, the students show the understanding of the theories and methods in the fields of reinforcement learning, and their ability to apply them to model real world problems, and to implement the solution with a programming language (Python). The presentation takes 20 minutes with 20 minutes discussion.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

The module requires a solid knowledge in advanced mathematics. It is better to know machine learning and deep learning. The experience of programming is helpful, because the students will use Python to finish the project.

Content:

The module covers different state of the art methods for reinforcement learning. Specifically the topics cover:

- Review of supervised/unsupervised learning and deep learning
- Definition of reinforcement learning
- Markov decision process
- Dynamic programming
- Monte Carlo methods
- Temporal difference learning

- Q-learning
- SARSA
- Policy gradient
- Proximal policy optimization (PPO)
- Actor critic

Intended Learning Outcomes:

After successful completion of this module, the students will (1) have a deep understanding of the concepts of reinforcement learning, (2) can explain the classical algorithms, such as Q-learning, SARAS, DQN, policy gradient and so on, based on the theoretical backgrounds, the students are able to (3) model the real world problem by reinforcement learning, and (4) implement the solution approaches by Python.

Teaching and Learning Methods:

The module consists of a series of lectures that introduce the theory and illustrate the examples and applications in practical.

The project for the students is aim to practice the classical algorithms learned in classes. In the written report, the students should learn to model a real world problem and implement by programming.

Media:

Presentation slides, technical papers

Reading List:

Bach F., Sutton R., Barton A. Reinforcement learning: An introduction, The MIT Press (2018)

Responsible for Module:

Xie, Jingui; Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

Deep Reinforcement Learning (WIHN0033) (Seminar, 4 SWS)

Sun M, Xie J

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0034: Advanced Seminar Operations & Technology Management: Business Analytics and its Application in Healthcare | Advanced Seminar Operations & Technology Management: Business Analytics and its Application in Healthcare

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 135	Contact Hours: 45

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination consists of two parts: 1. The presentation (25% of the overall grade) by which it is proved that the students: have a solid overview over business and data analytics, and have modelling and data analytics skills in solving healthcare problems. 2. The research proposal (75% of the overall grade) by which it is proved that the students: understand the literature on a subtopic in the field of business analytics with application in healthcare management, have learned the most important research methods in business analytics, are able to detect promising research questions and finding innovative ways to investigate them using data. The overall grade is calculated by averaging written work (75%) and presentation (25%).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Foundations in statistics and management science

Content:

In this seminar, we will discuss how to use big data to solve business problems, and have a good understanding of business analytics. In particular, we will focus in the area of healthcare management. The seminar generally has four parts. Part 1: Review the literature on business analytics, review the literature on healthcare data analytics. Part 2: Examine and identify various public databases, for example, open source data (google mobility data), data from organizations such as WHO. Part 3: Propose a research question, build the correct model and find the appropriate dataset to support the analysis. Part 4. Present research ideas and write an academic

report. Thus, this course will be especially valuable to inspire research ideas and prepare for scientific work on the subsequent master thesis or PhD study.

Intended Learning Outcomes:

Students who have participated in this seminar: have a comprehensive understanding of the scientific process; know about the most important issues of current business and management research; have a solid overview over diverse and innovative ways of collecting data and analyzing data; be able to think critically; solve business problems efficiently and innovatively using big data.

Teaching and Learning Methods:

Presentation, interactive teaching, e-learning, group discussions

Media:

Zoom, literature databases (Ebsco, ScienceDirect etc.), Databases, software development platform (e.g. GitHub)

Reading List:

Top 24 leading business journals - see <https://jindal.utdallas.edu/the-utd-top-100-business-school-research-rankings/>. In particular, we focus on the following five journals: Management Science, Operations Research, Journal of Operations Management, Manufacturing and Service Operations Management, Production and Operations Management. For example: Liu, J., Xie, J., Yang, K.K. and Zheng, Z., 2019. Effects of rescheduling on patient no-show behavior in outpatient clinics. Manufacturing & Service Operations Management, 21(4), pp.780-797. Xie, J., Zhuang, W., Ang, M., Chou, M.C., Luo, L. and Yao, D.D., Analytics for Hospital Resource Planning—Two Case Studies. Production and Operations Management.

Responsible for Module:

Xie, Jingui; Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology Management (WIHN0034) : Business Analytics and its Application in Healthcare (Seminar, 4 SWS)

Lou Z, Xie J

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0039: Business Analytics with Python and R | Business Analytics with Python and R

Version of module description: Gültig ab winterterm 2020/21

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The module is evaluated by assignments and projects. Each student should finish the assignments during the semester (30%) and work on a final project, which includes a written report (weighs 50%) and present in the last session of the course (weighs 20%).

In the report, the students show the understanding of the theories and methods in the fields of business analytics, and their ability to apply them to analyze real world data, and to implement the solution with a programming language (Python and R). The presentation takes 20 minutes with 20 minutes discussion.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Prerequisites: Statistics, Machine Learning, Reinforcement Learning, Econometrics, Management Science

Content:

The module covers different state of the art methods for business analytics. Specifically the topics cover:

1. Descriptive Analytics

Review of statistics

Introduction to R

Introduction to Python

2. Predictive Analytics

Review of machine learning (deep learning)

Review of econometrics

Regression models

Time series models

Simulation

3. Prescriptive Analytics

Review of Linear programming (Management Science)

Review of Dynamic programming (Reinforcement Learning)

Nonlinear optimization

Convex optimization

Robust optimization

Intended Learning Outcomes:

After successful completion of this module, the students will (1) have a deep understanding of the key concepts and the most important issues of business analytics, (2) have a solid overview over diverse and innovative ways of collecting and analyzing data. Furthermore, by the software exercise, the students (3) are required to implement the data analysis approaches by Python and R.

Teaching and Learning Methods:

The module consists of a series of lectures and software exercise. The lecture introduce the theory and illustrate the examples and applications in practical.

The software seminar offers instructions of Python and R, by which the business analytics methods could be implemented.

The final project for the students is aim to practice the classical algorithms learned in classes.

In the written report, the students should learn to model a real world problem and implement by programming. .

Media:

Presentation slides, software exercise, technical papers

Reading List:

Business Analytics: Data Analysis & Decision Making, 7th Edition, S.Christian Albright, Wayne L. Winston

Business Analytics, 4th Edition, Jeffrey D. Camm, James J. Cochran, Michael J Fry, Jeffrey W. Ohlmann

An Introduction to Statistical Learning with Applications in R, Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani.

Responsible for Module:

Xie, Jingui; Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0043: Advanced Seminar Operations & Technology Management: Real-World Applications of Deep Learning and Reinforcement Learning | Advanced Seminar Operations & Technology Management: Real-World Applications of Deep Learning and Reinforcement Learning

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 135	Contact Hours: 45

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The module is evaluated by two parts. The students should finish a written report (weighs 75%) and present in the last session of the course (weighs 25%).

In the report, the students show the understanding of the theories, methods and literature in the fields of deep reinforcement learning, and the application in the healthcare management area, and their ability to model real world problems in a innovative way, and to implement the solution with a programming language (Python). The presentation takes 20 minutes with 20 minutes discussion.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

The module requires a solid knowledge in advanced mathematics. It is better to know machine learning, deep learning and programming.

Content:

In this seminar, we will discuss how to apply reinforcement learning to solve business problems. In particular, we will focus in the area of healthcare management. The seminar generally has three parts. Part 1: Review the literature on deep reinforcement learning, review the literature on healthcare decision making. Part 2: Propose a research question, build the correct model and find the appropriate algorithm. Part 3. Present research ideas and write an academic report. Thus, this course will be especially valuable to inspire research ideas and prepare for scientific work on the subsequent master thesis or PhD study.

Intended Learning Outcomes:

After successful completion of this module, the students will (1) have a comprehensive understanding of the scientific process; (2) get deep understanding of the concepts and algorithms of deep learning and reinforcement learning, (3) have a solid overview over diverse and innovatives of the application of deep learning and reinforcement learning , based on the theoretical backgrounds, the students are able to (4) think critically, (5) model problems efficiently and innovatively using deep learning and reinforcement learning, and (6) find optimal solution using R and Python.

Teaching and Learning Methods:

Presentation, interactive teaching, e-learning, group discussions

Media:

Zoom, technical papers

Reading List:

Top 24 leading business journals - see <https://jindal.utdallas.edu/the-utd-top-100-business-school-research-rankings/>. In particular, we focus on the following five journals: Management Science, Operations Research, Journal of Operations Management, Manufacturing and Service Operations Management, Production and Operations Management.

Responsible for Module:

Xie, Jingui; Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Electives in Management | Wirtschaftswissenschaftlichen Wahlmodule

Catalogue of Elective Modules: Family Enterprise | Wahlkatalog: Family Enterprise

Module Description

MGTHN0077: Advanced International Study Trip: Family & non-family firms around the world | Advanced International Study Trip: Family & non-family firms around the world

Version of module description: Gültig ab summerterm 2024

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 9	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each student will individually work on a seminar thesis and participate in the discussion with international firms and universities. The examination consists of two parts:

1) Seminar thesis (70 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics discussed in the course
- they have understood and applied the theoretical frameworks discussed in the course
- they show that they are able to write an academic paper that follows scientific standards
- they are able to structure and write a seminar thesis based on conceptual theories

2) Participation in discussion during excursion (30 % of the overall grade).

The students should demonstrate that:

- they have gained the skills to contribute to conceptual and theoretical discussions with firms and universities
- they have gained an understanding of quantitative-empirical research work that can be applied to companies and their business models
- Active engagement in the course is expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

English oral and written fluency

Content:

Family and non-family firms need to succeed globally in order to survive and thrive in today's business environment. Thus, these firms need managers who are well-versed in international business environments. This course aims to provide students an immersive and intense international experience through a 10-day excursion abroad.

Intended Learning Outcomes:

After completing this course students should be able to:

- Have a scientific understanding on how to do business abroad
- Feel comfortable and gain proficiency in communicating and doing business abroad

Skills Objectives

- Understand theoretical frameworks that allow them to navigate different cultures
- Improve writing and written communication skills
- Gain in-depth understanding of critical thinking, ethnorelative perspectives and interpretation skills

Teaching and Learning Methods:

This module includes the following teaching and learning methods:

- Immersive learning of culture, business, and people in a different country through an excursion
- Key concepts and empirical findings from the field of strategic management of family and non-family firms will be provided through discussions with international firms and universities.
- Students will acquire additional knowledge from academic literature through their own work on their seminar theses.

Media:

PowerPoint, research papers, case scenarios

Reading List:

tba

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced International Study Trip (MGTHN0077): Family & non-family firms around the world (Seminar, 6 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0096: Advanced Seminar Management & Marketing: Corporate Social Responsibility | Advanced Seminar Management & Marketing: Corporate Social Responsibility

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of handouts and presentation: In this course, students are to independently and individually read foundational and current scientific contributions on CSR and give a presentation on them, as well as actively participate in discussions in class and create a short handout. Additionally, they are supposed to write short essays, applying the gained knowledge to a real-life example.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization, and empirical research methods.

Content:

The course will be taught using a seminar or discussion style. Students will each give a presentation on a CSR topic discussed in class. The course includes for instance different perspectives, as well as critiques, actors and drivers of CSR. Moreover, managing CSR, a global perspective on the construct, as well as further selected topics with more recent relevance in CSR research will be discussed.

Intended Learning Outcomes:

After successful completion of this course, students will (1) have gained knowledge of different perspectives of corporate social responsibility research in strategic management, (2) have learned to read, understand, explain and critically reflect on scientific literature in the field, (3) have built a basic understanding of research methods in strategic management, (4) have gained the ability to

discuss and elaborate on scientific literature on CSR in strategic management, (5) be able to apply this knowledge by analyzing CSR concepts in real-life examples.

Teaching and Learning Methods:

Students should read the assigned literature for the course and give a presentation on a topic that is derived from the literature discussed in class, also creating a short handout. In addition to the presentations students will deepen their knowledge by writing short essays, applying the concepts to real-life examples.

Media:

Powerpoint

Reading List:

Crane, A., McWilliams, A., Matten, D., Moon, J., & Siegel, D. S. (Eds.). (2008). The Oxford handbook of corporate social responsibility. OUP Oxford. Other current literature is selected specifically for each semester and will be distributed in class.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0097: CEO Leadership Series | CEO Leadership Series

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The exam is a scientific work that consists of the following two components: 1) Students will individually submit written contributions in the form of questions to the CEOs and voluntarily, in a group of 2 students, an analysis of CEOs and their companies (50%). The individual submission includes 12 questions, which need to be lecture-specific, referring to the concepts presented in the first lecture, such as leadership skills, which include, for instance, decision-making, critical thinking, or employee motivation based on theoretical frameworks that include institutional theory, upper echelon theory, and stakeholder theory. The group work is a report of approximately 500-1,500 words per group, whereby each student adds at least one chapter of analysis. This can include an analysis of leadership style, decision-making processes, organizational culture, succession planning, or the alignment of leadership strategies with overall business objectives. The scientific work examines the extent to which the students can understand leadership skills in theory and practice. It evaluates if students are able to analyze companies and leaders, as well as their behavior, and apply their knowledge of leadership by interacting with CEOs. Students will further participate in a short quiz immediately after each CEO visit to test their understanding of the topics discussed. 2) Oral participation will be graded (50%). In the second half of the lecture, which lasts approximately one hour, students ask questions to the CEOs. They are supposed to actively and individually contribute to the talk on stage. The oral participation checks that the students are able to listen and comprehend the talks and transfer and apply the concepts by asking related questions. Students will also gain practical knowledge and be able to improve their discussion- and soft skills.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization.

Content:

This module is the flagship event of TUM C.HN and will take place on Tuesdays or Fridays at 3 pm (each date will feature one guest). It covers practical insights on firm strategies, whereby lessons can be learned from CEOs themselves. The covered strategy topics may include internationalization, the analysis of the internal and external environment as well as M&A and other types of strategic alliances. This course also provides students with valuable insights for their future work as strategizers in a corporate environment.

Please note: Registration is mandatory. There will be a participants list at the entrance and police presence for each event.

The guests for the winter semester 2023/24 are e.g.: Dr. Roland Busch (CEO, Siemens AG), Christine Grotz (CEO, Weber-Hydraulik), Astrid Teckentrup (CEO, Procter & Gamble (DACH)), René Obermann (CEO, Deutsche Telekom AG (2006-2013)), Gerd Chrzanowski (CEO, Schwarz Gruppe).

Intended Learning Outcomes:

After attending this course, students will have gained knowledge on leadership skills and firm strategies in practice through attending podium discussions and actively participating in Q&A sessions with CEOs. They will be able to analyze companies and leaders, as well as their behavior, and apply their knowledge of leadership by interacting with CEOs. With this approach, students will gain practical knowledge and are additionally able to improve their discussion- and soft skills.

Teaching and Learning Methods:

In this seminar, there will be both Q&As and podium discussions with CEOs. Students will get in contact with several CEOs and managers of real-life companies, analyze their leadership skills and concepts, and interact with them in Q&A sessions. For this, the students should individually prepare and hand in questions for the CEOs as well as analyses of the companies and CEOs themselves. Moreover they should actively participate in all the sessions.

Media:

Powerpoint

Reading List:

Tbd.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0108: Start-up law for founders of family enterprises | Start-up law for founders of family enterprises

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination includes an oral report where the students will present and discuss their work for the seminar paper (30% of the grade, Presentation: 15-20 min, Questions: 15-20 min, Feedback session: 10 min). Moreover, students will prepare a seminar paper. To support students during the writing process, there will be regular discussions about the project's progression (seminar paper and regular discussions = 50% of the grade). The seminar paper contains a case study that addresses the basics of German corporate law and legal challenges associated with the formation of a startup and governance structure. On top of that, students' participation in class discussions of case studies and group work will be evaluated (20% of the grade). The active participation in the case discussions measures the students' ability to use the theoretical knowledge of the seminar to analyze real complex situations. The written examination and the corresponding presentation evaluate the student's ability to work with unknown legal problems and present a sound study incorporating the seminar's topics. In the oral report, students also show that they can summarize, academically present, and discuss the subject. Finally, ongoing discussions within group works and discussion rounds assess the student's capacity to solve a legal issue and elaborate a structured solution within a specified timeframe.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

None

Content:

- Legal structures and industries overview
- Research methods using legal databases
- Handling German legal texts

- Basics of German corporate law
- Overview of GmbH and GbR according to German law
- Startup formation overview
- Contract and agreement requirements for legal entity formation with special attention to the specific situation of family enterprises
- Financing opportunities overview
- Analysis with practical examples (e.g., Campus Founders, local startup, VC Investor)
- Presentation of own legal solutions

Intended Learning Outcomes:

The seminar covers the basics of startup law for founders of family enterprises. The students will gain a basic understanding of German corporate law and the formation process of a startup. Students will delve into a detailed study of German corporate law texts and apply them through practical examples. By the end of the seminar, they will be able to recognize legal challenges that arise during the startup process of a family enterprise. Furthermore, participants will learn how to combine their knowledge with discussing cases. Including practical case studies of newly founded family enterprises will enhance skills in argumentation and analysis, as case studies provide opportunities for debates. The seminar will also encourage the ability to develop legal solutions and justify their decisions through group work and presentations.

Teaching and Learning Methods:

The educational approach is a combination of direct instructions and oral participation of the students as well as reinforcement through interactive group activities. The students present the outcomes of their group work and engage in discussions about the topic with the other students. The practical application of knowledge to real-life cases is demonstrated using illustrative examples. Furthermore, students create detailed checklists and schemes on legal topics. The seminar will feature a presentations by relevant actors (like start-ups or lawyers) who will share their valuable insights and experiences related to legal issues in the startup process. The seminar integrates a case study with the seminar topics so students can apply what they have learned in their analysis. Students will analyze a specific case, demonstrate their understanding, and apply legal principles. Finally, students will present their work in class.

Media:

Case studies, PPT, flipcharts

Reading List:

The German Limited Liability Company, Rosengarten / Burmeister / Klein (9th ed. 2020)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Start-up law for founders of family enterprises (MGTHN0108) (Seminar, 4 SWS)

Back M, Bundrock R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0109: Taxation of Family Enterprises | Taxation of Family Enterprises

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To complete the module, participants have to demonstrate their theoretical/conceptual and practical competences regarding taxation of family business through five written exercise sheets that have to be handed in through the semester.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Basic competences in financial accounting, managerial accounting, and financial management and investing on the Bachelor level.

Content:

The course has the following content elements:• Types of taxes and their collection in the European Union. • Economics of business taxation and tax statistics. • Taxes in company valuation. • Financial accounting under German commercial law. • Income taxation under German tax law. • Tax planning. • International taxation. • Taxation of digital goods and services. • Taxation of family business succession. • Ethical aspects of tax planning and paying. • Methods of empirical tax research.

Intended Learning Outcomes:

After successfully completing the module, the participants have the following four competences:• Participants know the most important concepts and types of corporate taxes and can apply tax planning strategies. • Participants can trade-off taxation effects of different legal forms and governance systems and have particular competences related to the taxation of intergenerational succession of family enterprises. • Participants know the most important challenges of international taxation, in particular aspects related to the taxation of digital goods and services. • Participants

can critically evaluate potentials, incentives, and risks of tax-optimal accounting and managerial policies against the background of ethical principles.

Teaching and Learning Methods:

The module consists of a series of introductory lectures and practitioner-oriented case studies which are offered in blocked two-day courses through the semester.

Media:

Lecture script, case studies, and e-learning (Moodle).

Reading List:

Readings are announced in the module.

Responsible for Module:

Stich, Michael; Prof. Dr. rer. oec.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0110: Advanced Seminar Economics & Policy: Management practices in (family) enterprises | Advanced Seminar Economics & Policy: Management practices in (family) enterprises

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on management practices in (family) enterprises.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper, and (ii) oral in-class presentation. Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Content:

The seminar will cover key topics regarding management practices, such as:

1. How can management practices be classified?
2. How do management practices vary across companies?
3. How do management practices differ between family and non-family enterprises?
4. How do management practices relate to economic success?
5. How do management practices relate to work climate?
6. What are the determinants of different management practices employed in practice?

Intended Learning Outcomes:

'Students will learn the key concepts in the research field of management practices, with a special focus on family enterprises. The goal is twofold: First, to understand how management practices relate to economic success of enterprises. Second, to understand what individual and institutional factors determine management practices in enterprises. The course builds on original research on that topic. The course will cover basic theoretical consideration, but will have a clear focus on empirical studies with real-world examples and applications.

Teaching and Learning Methods:

'Students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence. A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups. We plan to complement student presentations by talks from corporate guests who will provide insights into corporate management practices.

Media:

Powerpoint slides

Reading List:

Amongst others (for detailed reading list, please see Moodle):

Bloom, N., & Van Reenen, J. (2010). Why do management practices differ across firms and countries?. *Journal of Economic Perspectives*, 24(1), 203-24.

Bloom, N., Brynjolfsson, E., Foster, L., Jarmin, R., Patnaik, M., Saporta-Eksten, I., & Van Reenen, J. (2019). What drives differences in management practices?. *American Economic Review*, 109(5), 1648-83.

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0154: Managing Legal Sustainability Requirements in Family Enterprises | Managing Legal Sustainability Requirements in Family Enterprises

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The grade is composed of written and oral performances of the participants. Students will prepare a seminar paper. To support students in the writing process, there will be regular discussions about the progression of the project. The handed in seminar paper and regular discussions, where the progression of the seminar paper is discussed make up 70% of the overall grade. The examination includes an oral presentation where the students will present and discuss the topic of their seminar paper (20% of the grade). Regarding the latter, 10% of the overall course performance is achieved through two separate course activities within two separate sessions of the seminar. 5% of the grade is accounted for the presentation of a poster tailored thematically to the session. The second required activity which accounts for 5 % of the grade is an oral presentation of a case study. The active participation in the two course activities measure the student's ability to use the conveyed legal knowledge on a wide variety of corporate sustainability topics to evaluate the internal and external implications on companies in the European Single Market. The written examination (seminar paper) and the corresponding presentation are means to evaluate the student's ability to present an overview of an unknown topic related to the topics discussed, to analyze the relevant literature in the form of studies and primary literature, and to assess the challenges and opportunities from a corporate perspective. Finally, regular discussions with the instructor measure the student's ability to distinguish subtopics of the chosen topic, locate and evaluate relevant sources and elaborate purposeful analysis from a corporate perspective.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

Introduction (Corporate) Sustainability: ESG-Dimensions

Decarbonisation, human rights in supply chains

circularity, eco-design, right to repair

Greenwashing practices, sustainability reporting, sustainable finance

Intended Learning Outcomes:

The seminar links basic knowledge of the legal framework for corporate sustainability on the European plane with the given organizational and financial structures of family enterprises in the European single market. The students gain the ability to think practical case scenarios through and draw implications for business practices. This is paired with analytical skills for literature review. Handed-out (literature) material is examined, and subsequently, critical opinions are formed and discussed in a team setting. This strengthens the students ability to distinguish the framework of mandatory legal rules from the "business case" for compliance with these rules.

Teaching and Learning Methods:

This seminar comprehensively presents the implementation of the different pillars of the European Green Deal for regulating corporate sustainability. On this basis, the implications for the companies' practices are evaluated critically. The seminar, inter alia, integrates guest lectures and case studies. Students can apply their learnings and implement them in their analysis. Students will study case studies from various industrial settings critical to "Green Transition" in the single market. Based on learned legal and regulatory elements, comparisons to the status quo and future outlook of business practices are drawn. Finally, students will present their work in class.

Media:

Case studies, exercises, PPT, Whiteboard, Videos

Reading List:

E.g.: Ditlev-Simonsen, A Guide to Sustainable Corporate Responsibility (2022), Noeth, The ESG and Sustainability Deskbook for Business (2024), Gaganis et al., Sustainable Finance and ESG (2023), Camara/Morais, The Palgrave Handbook of ESG and Corporate Governance (2022), Lykkesfeldt/KJaergaard, Investor Relations and ESG Reporting in a Regulatory Perspective (2022), Nygaard, Green Marketing and Entrepreneurship (2024), Eisenriegler (ed.), The Circular Economy in the European Union (2020)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Managing Legal Sustainability Requirements in Family Enterprises (MGTHN0154, English)
(Seminar, 4 SWS)

Bundrock R, Dowse M

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0171: Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises | Advanced Seminar Economics & Policy: Technological Change, AI, and Labor Market Dynamics in Family and Non-Family Enterprises

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on a specific topic related to the seminar's overall theme.

The students will be asked to

- (i) summarize the key insights on one of the seminar topics in a written essay (seminar paper), and to
- (ii) present their finding in class in a 20 minutes presentation.

The examination will consist of these two parts: (i) written seminar paper (50%), and (ii) oral in-class presentation (50%). Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Principles of Economics (Master)

Content:

Beyond a lecture-style introduction into the topic, students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence.

A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups.

Intended Learning Outcomes:

Students will explore how technological advancements, ranging from automation to artificial intelligence, transform labor markets and the broader economy. The seminar will delve into the mechanisms behind these changes, examining their impacts on work, skills, and business structures, as well as the role of government policies in managing these transitions. Special attention will be given to comparing family and non-family enterprises.

The course builds on original research on the topic. While it will cover basic theoretical considerations, the focus will be on empirical studies with real-world examples and applications. Students will write a term paper on a selected topic and present their findings, highlighting practical implications for labor markets, policy, and business strategies.

Teaching and Learning Methods:

This seminar will cover key topics within the Economics of Technological Change and Labor Markets, with a focus on the role of family enterprises, such as:

Topic area 1: The impact of skill-biased technological change on wages and employment outcomes.

Topic area 2: Job polarization and the growth of low-skill service jobs in response to automation and routine-biased technological change.

Topic area 3: Labor market adjustments to technological innovations, including the effects of automation, robotics, and artificial intelligence.

Topic area 4: The unique challenges and opportunities for family enterprises in adapting to technological change and leveraging innovations to remain competitive.

Topic area 5: The role of government policies in managing labor market transformations and ensuring equitable outcomes.

Media:

Teaching will be in the form of a lecture and seminar presentations.

Reading List:

Acemoglu, Daron and David Autor (2011). "Skills, tasks and technologies: Implications for employment and earnings", In: Handbook of Labor Economics, Vol. 4, Elsevier, pp. 1043–1171.

Angrist, Joshua and Jörn-Steffen Pischke (2008). Mostly Harmless Econometrics (Princeton: Princeton University Press)

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0173: Project and Innovation Management: Design thinking methodology | Project and Innovation Management: Design thinking methodology

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The final grade consists of both written and oral components. Students are required to prepare a seminar paper, which accounts for 70% of the final grade. To support them throughout the writing process, there will be regular discussions to monitor the progress of their work. In addition, students will deliver an oral presentation on their seminar paper, which constitutes 20% of the grade. Furthermore, students will complete two class exercises, each accompanied by a discussion. Each exercise and its corresponding discussion will be assessed and will contribute 5% to the final grade, for a total of 10%. Active participation in the discussions and class exercises measures the student's ability to use the theoretical knowledge of the seminar to discuss and analyze real complex situations. The written examination and the corresponding presentation are means to evaluate the student's ability to work with a specific problem and incorporate the methods discussed in class to organize and present a solution to a problem. Finally, regular discussions with the instructor measure the student's ability to develop an idea, shape it, and elaborate a structured project within a given timeframe.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

This seminar presents the methodology of design thinking to generate solutions and tackle identified problems, particularly in connection with innovation initiatives in family enterprises. During the different sessions, the stages of design thinking will be covered, and we will explore how they work in connection with various types of case studies (related to technological

development, sustainability, and internal processes in enterprises). We will also explore how artificial intelligence can contribute to optimizing and enriching the application of the methodology and the generation of innovative solutions.

Intended Learning Outcomes:

1. Participants will be able to articulate the core principles of Design Thinking, including empathy, ideation, prototyping, and iteration.
2. Participants will learn to clearly define and reframe problems from a user-centered perspective, creating problem statements that guide effective solutions.
3. Participants will gain proficiency in various ideation techniques, such as brainstorming, mind mapping, and rapid sketching, to generate innovative ideas.
4. Participants will apply Design Thinking methodologies to real-world challenges, demonstrating the ability to navigate the entire process from problem definition to solution implementation.

Teaching and Learning Methods:

We will have content and application sessions where the methodology will be introduced and case studies of its application will be presented. Students will also receive suggested materials to delve deeper into the methodology after class. In the last part of the session, students will be asked to solve exercises related to the application of the methodology. Paralelly, they will work on the topics while constructing their seminar papers.

Media:

Presentation of the topics, case studies, exercises, PPT, Whiteboard

Reading List:

1. Razzouk, R., & Shute, V. (2012). What Is Design Thinking and Why Is It Important? Review Of Educational Research, 82(3), 330-348. <https://doi.org/10.3102/0034654312457429>
2. Frisendal, T. (2012). Design Thinking for Business Analysis. En Management for professionals (pp. 15-24). https://doi.org/10.1007/978-3-642-32844-2_3
3. You, X. (2022). Applying design thinking for business model innovation. Journal Of Innovation And Entrepreneurship, 11(1). <https://doi.org/10.1186/s13731-022-00251-24>.
4. Martin, R., & Euchner, J. (2012). Design thinking. Research-Technology Management, 55(3), 10-14. <https://doi.org/10.5437/08956308x5503003>
5. Rösch, N., Tiberius, V., & Kraus, S. (2023). Design thinking for innovation: context factors, process, and outcomes. European Journal Of Innovation Management, 26(7), 160-176. <https://doi.org/10.1108/ejim-03-2022-0164>
6. Carella, G., Melazzini, M., Cautela, C., & Zurlo, F. (2024). Boosting Design Thinking adoption in organisations through a game#based toolkit: A gamified approach in building facilitators to overcome Design Thinking adoption barriers. Creativity And Innovation Management. <https://doi.org/10.1111/caim.12627>
7. Docherty, C., Mazzei, M., & Steiner, A. (2023). Exploring the impact of design thinking on social enterprise mission-aligned innovation. Entrepreneurship And Regional Development, 1-17. <https://doi.org/10.1080/08985626.2023.2286506>

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0014: Corporate Campus Challenge | Corporate Campus Challenge

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students will present a prototype and business case in a final presentation. Furthermore, they will write a report which will be examined. The final grade is averaged as follows:

60 %: Final presentation

40 %: Scientific paper (critical reflection on the prototype development process by using relevant scientific literature)

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

- Customer needs
- Research & empathic exploration
- Value proposition
- Design thinking methodology
- Business model and ecosystem modeling
- Rapid prototyping
- Scientific reasoning and writing

Intended Learning Outcomes:

At the end of the module, students are able to

- apply methods of idea generation and need-finding to stimulate creativity and recognize business opportunities
- create design prototypes in order to demonstrate their proposed solutions and gather feedback

- create business plans, presentations, and video prototypes in order to effectively communicate business value
- evaluate their ideas by involving peers, academics, and industry partners
- develop and document their ideas in a scientific seminar paper

Teaching and Learning Methods:

In this course, students will work in interdisciplinary teams to develop innovative solutions for challenges provided by industry partners. Over the course of 8 weeks, students will receive coaching sessions, individual mentoring, tutorials, and practical courses on the use of machines and equipment (3D printer, laser cutter, sensors, etc.). Students will present their first results through a presentation and a live prototype to gather feedback. At the end of the course, a working prototype and the business case will be presented during a final presentation.

Media:

Slides, scientific papers, handbooks, physical events

Reading List:

Lewrick, M., Link, P., & Leifer, L. (2018). The design thinking playbook: Mindful digital transformation of teams, products, services, businesses and ecosystems. John Wiley & Sons.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0028: Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship | Advanced Seminar Innovation and Entrepreneurship: Topics in Corporate Entrepreneurship

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic related to corporate entrepreneurship. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade, approx. 10 pages) The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade, approx. 10 minutes). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis).

Repeat Examination:

(Recommended) Prerequisites:

Basic knowledge of corporate entrepreneurship and research methodologies is recommended.

Content:

Foundations of Corporate Entrepreneurship (CE): Key concepts, definitions, and the role of CE in modern businesses.

Strategic Thinking in CE: Developing an entrepreneurial mindset to identify and leverage opportunities.

Implementing CE: Steps to integrate entrepreneurial practices into corporate operations.

Building an Entrepreneurial Organization: Creating a culture that fosters innovation and risk-taking.

Sustaining CE: Strategies to maintain entrepreneurial energy and adapt to changing markets.

Leadership and Culture: The impact of leadership styles and organizational culture on entrepreneurship.

Entrepreneurial Traits and Behavior: Key personal characteristics that support entrepreneurial initiatives.

Creativity and Motivation: Harnessing creativity, motivation, and HRM practices to drive innovation.

Incentive Systems and Psychological Ownership: Enhancing engagement through effective incentives and ownership.

Entrepreneurial Intensity: Measuring and increasing the level of entrepreneurial activity within corporations.

Focus Areas of CE: Examining CE at the levels of the phenomenon, organization, individual, and dynamic interactions.

Challenges in CE: Addressing obstacles faced by corporations and strategies to overcome them.

Intended Learning Outcomes:

Understand the current state of research in corporate entrepreneurship.

Develop the ability to critically analyze and synthesize academic literature.

Improve academic writing and presentation skills.

Identify research gaps and formulate theoretical and practical implications.

Teaching and Learning Methods:

In-presence seminar sessions

Literature review and independent research.

Group discussions and collaborative learning (optional group work).

Interactive feedback sessions.

Presentation of results and peer review

Media:

Moodle platform for resources and updates.

Academic databases for literature access.

Presentation tools (e.g., PowerPoint, posters)

Reading List:

Core reading will be provided on Moodle. Students are encouraged to explore additional academic articles and books relevant to their selected topic.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0029: Advanced Seminar Innovation & Entrepreneurship: Managing the Family Enterprise | Advanced Seminar Innovation & Entrepreneurship: Managing the Family Enterprise

Version of module description: Gültig ab summerterm 2020

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work in a group on a specific topic related to managing the family enterprise. The examination consists of two parts:

1) Projekt thesis (75 % of the overall grade) The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to analyze effectively the problem stated in their work and know how to structure and write a project thesis.

2) Presentation of the project thesis (25 % of the overall grade). Students should demonstrate that:

- they have gained the skills to present their project thesis to an academic audience.
- they are able to answer questions related to specific parts of their work.

The final grade will be based on the two parts (75 % project thesis and 25 % presentation and discussion of the seminar thesis).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Good English language skills are required.

Content:

This course deals with the management of family firms. Given that the majority of firms around the world are family businesses (ranging between 75-90 %), it is crucial to understand better the particularities of family businesses. Given that the boundaries between the family and the business

overlap, certain challenges may arise that makes family businesses unique. The objective of this course is to better understand the challenges and opportunities family firms face and how managers, owners, and advisers can best deal with them. By combining various theoretical frameworks with hands-on examples, we will elaborate on how to manage family enterprises successfully. We will cover topics such as:

- strength and weaknesses in family firms,
- succession and governance issues in family enterprises,
- management of key resources in family family firms,
- financial management in family firms and,
- transgenerational entrepreneurship.

Intended Learning Outcomes:

- Understand the family business phenomenon and strengths and weaknesses of family firms.
- Understand the importance of governance in family firms. Evaluate how different types of governance suit the needs of family firms differing in size.
- Reflect on how family firms use the “family” as a strategic asset and how do they manage those resources best. How do different institutional contexts influence the emergence of family firms?
- Reflect on financial management in family firms. Evaluate how does the family background bias financial behaviour, such as investment, financing, and risk taking?
- Know and apply different concepts of transgenerational entrepreneurship and be able to advise family firms how they can sustain in the long run.
- Reflect on governance and succession topics.
- (1) the search, understanding, and analysis of academic literature, (2) methods for data collection and data analysis, (3) the structuring and writing of a paper, and (4) giving presentations including a subsequent discussion

Teaching and Learning Methods:

- Through lectures, supported by a presentation, the instructor will provide the theoretical foundations on managing the family enterprise.
- A series of group discussions and exercises as well as exchanging ideas, questions, and comments will encourage an experiential learning atmosphere.
- Guest speakers will share practical insights and will strengthen the understanding of key concepts and processes, and expose perspectives complementary to the topic of the seminar.
- Within their group work, students identify challenges and develop recommendations. This is supported by feedback sessions with the instructor.
- In a presentation, students present their groupwork.

Media:

Powerpoint slides, Zoom etc.

Reading List:

Amongst others (for detailed reading list, please see Moodle):

- Aldrich, H. E., & Cliff, J. E. (2003). The pervasive effects of family on entrepreneurship: Toward a family embeddedness perspective. *Journal of Business Venturing*, 18(5), 573-596.
- Bird, M. & Zellweger, T. (2018): Relational embeddedness and firm growth: Comparing spousal and sibling entrepreneurs. *Organization Science* 29 (2), 264-283.
- Schulze, W. S., Lubatkin, M. H., Dino, R. N., & Buchholtz, A. K. (2001). Agency relationships in family firms: Theory and evidence. *Organization Science*, 12(2), 99-116.
- Sirmon, D. G., & Hitt, M. A. (2003). Managing resources: Linking unique resources, management, and wealth creation in family firms. *Entrepreneurship Theory and Practice*, 27(4), 339-358.
- Zellweger, T. (2017): *Managing the Family Business*. Edward Elgar Publishing: Massachusetts.

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0029): Managing the Family Enterprise
(Seminar, 4 SWS)

Bird M, Kosberg M

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0037: Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations | Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 146	Contact Hours: 34

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The final grade consists of both written and oral components. Students are required to prepare a seminar paper, which accounts for 70% of the final grade. To support them throughout the writing process, there will be regular discussions to monitor the progress of their work. In addition, students will deliver an oral presentation on their seminar paper, which constitutes 20% of the grade. Furthermore, students will complete two class exercises, each accompanied by a discussion. Each exercise and its corresponding discussion will be assessed and will contribute 5% to the final grade, for a total of 10%. The seminar paper and the corresponding presentation are a means to measure the student's ability to understand a scientific subject, evaluate literature, and develop, conduct, and analyze questionnaires/surveys. By doing a presentation, students show that they can summarize the subject, present it to an audience, and to conduct a discussion about the presented subject. Regular discussions with the instructor measure the student's ability to develop an idea from initial concepts to the complete picture within a given timeframe. Participation in class exercises measures the students ability to apply their theoretical knowledge in practice and to reflect on it afterwards.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

None

Content:

Basic terms of contract negotiations (like BATNA and ZOPA etc.);
Negotiation strategies and concepts (Harvard negotiation concept, win-win, win-lose etc.);
Effects of behavioural economics and negotiation tactics based on it (anchoring effect etc.);

Negotiation tactics (based on deception, pressure and defensive tactics etc.);
Principal agent problem in negotiations;
Communication (question techniques, answer techniques, argumentation techniques);
Emotions (core concerns framework);
Gender aspects in contract negotiations.

Intended Learning Outcomes:

Students learn the economic and psychological basics of contract negotiations as well as important communication tools for negotiations. They know the Harvard negotiation concept, including its weaknesses and the most important strategies in contract negotiations. They are familiar with the essential tactics. Students know how to apply these tactics or, respectively, how to react to these tactics if applied by the negotiation partner. They can distinguish different negotiation styles. Students have experienced the functioning of individual tactics during negotiation simulations and are able to understand the dynamics underlying a specific negotiation.

In the area of interdisciplinary competences, students strengthen their communication and argumentative skills (especially through the negotiation simulations) and their ability to work in a team (especially through group work in the context of presentations and negotiation simulations in a team).

Teaching and Learning Methods:

This module is held as an interactive seminar/lecture. Negotiation simulations and games are integrated into the course so that students learn to implement tactics and strategies. The negotiation games are designed to simulate practice. Students are encouraged to actively participate in the negotiation simulations and to get involved in the subsequent discussions. Video recordings are used to reflect on what has been experienced. Moreover, students will do research on a specific research question and write a seminar paper. In this framework, students will have to perform research of reference materials, design a questionnaire, conduct interviews with negotiators and analyse the answers. In order to support the students in their work individual appointments will be offered. Students will present their work in class.

Media:

Simulations, Exercises, Videos, Self-Tests, PPT, Whiteboard

Reading List:

Jung/Krebs, The Essentials of Contract Negotiation (2019)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Economics & Policy: Business-to-Business Contract Negotiations (WIHN0037)
(Seminar, 4 SWS)
Gelvez Alvarez L, Jung S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0046: Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises | Advanced Seminar Innovation & Entrepreneurship: Sustainability In Entrepreneurship and Family Enterprises

Version of module description: Gültig ab winterterm 2020/21

Module Level: Master	Language: English	Duration: one semester	Frequency:
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each seminar participant will work individually on a specific topic in the field of sustainable entrepreneurship / sustainable management in family enterprises. The examination consists of two parts:

1) Seminar thesis (75 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics dealt within the course
- they show that they are able to write a paper that follows a clear logic and is based on sound literature.
- they are able to detect promising research questions and know how to structure and write a research paper.

2) Presentation of the seminar thesis (25 % of the overall grade). Students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer questions related to specific parts of their work

The final grade will be based on the two parts (75 % seminar thesis and 25 % presentation and discussion of the seminar thesis). Active engagement in the course will be expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English

Content:

This course addresses various aspects of sustainability available to new ventures and established family enterprises. Sustainable entrepreneurship (SE) emphasizes that entrepreneurs may pursue additional goals such as social and ecological goals that may alleviate social and ecological problems. It explores the opportunities that entrepreneurs create, the challenges entrepreneurs encounter, and the ways in which entrepreneurs exploit opportunities to contribute toward enhancing the society while simultaneously starting and sustaining a profit-oriented business. This course also examines the role of sustainability in family enterprises:

Course content will comprise:

- The importance of sustainable entrepreneurship in solving grand societal challenges of the 21st century
- Motives and barriers of sustainable entrepreneurship
- Overview of theories and concepts in sustainable entrepreneurship
- Stakeholder management for entrepreneurs
- Sustainable management in the context of family enterprises
- Sustainability in entrepreneurship and family enterprises in the context of developing countries
- Recognizing and exploiting social and ecological opportunities
- Practical examples of sustainable ventures

Intended Learning Outcomes:

After completing the seminar students, should understand how entrepreneurs and family enterprises can contribute to solving societal challenges. Therefore, students will be able to:

- understand the concept of sustainable entrepreneurship and sustainability in family enterprises
- understand and evaluate how entrepreneurs can enact on sustainable opportunities
- understand the entrepreneurs' motives and barriers to engage in sustainable entrepreneurship
- reflect on the importance of diversity in achieving sustainability in entrepreneurship and family enterprises
- evaluate strategic actions of sustainable enterprises
- evaluate hands-on examples and cases of how sustainable entrepreneurship can be implemented in practice with specific focus on the Heilbronn region: Those examples may relate to various sectors such as the public governance sector, the food and agriculture, environmental and mobility sector.

In addition, students will be able:

- to search, understand, and analyze academic literature by evaluating key findings and explaining complex research studies in a comprehensible and interesting manner,
- to apply methods for data collection and data analysis, - to structure and write a seminar thesis, and
- to present and subsequently discuss findings of their seminar thesis.

Teaching and Learning Methods:

' -Through lectures, supported by Power-Point presentations, the instructor will provide the theoretical foundations of sustainable entrepreneurship

- The content is discussed in the course by exchanging ideas, questions, and comments to encourage an open learning atmosphere
- Every sessions contains exercises, where the students apply their learnings in a group work (e.g. to present one start-up, enterprise or organization that has already successfully implemented the topic/concept presented in the lecture)
- Guest speakers will share practical insights and will strengthen the understanding of key concepts and processes and expose perspectives complementary to the topic of the seminar
- For their seminar thesis, students will investigate a topic within the subject of this course to develop a seminar thesis. Students will receive feedback from the instructor.
- In a final presentation, students present the results of their project thesis

Media:

Powerpoint, Zoom- & Breakout-Sessions, Online Simulations

Reading List:

Basic literature (for detailed reading list, see Moodle):

- Cruz, C.; Larraza-Kintana, M. Garcés-Galdeano, L. Berrone, P. (2014). Are family firms really more socially responsible? *Entrepreneurship Theory and Practice* 38(6), 1295–1316.
- Gast, J., Gundolf, K., & Cesinger, B. (2017). Doing business in a green way: A systematic review of the ecological sustainability entrepreneurship literature and future research directions. *Journal of Cleaner Production*, 147, 44-56.
- Le Breton-Miller, I., & Miller, D. (2016). Family firms and practices of sustainability: A contingency view. *Journal of Family Business Strategy*, 7(1), 26-33.
- Lumpkin, G. T., Bacq, S., & Pidduck, R. J. (2018). Where change happens: community#level phenomena in social entrepreneurship research. *Journal of Small Business Management*, 56(1), 24-50.
- Patzelt H., & Shepherd D.A. (2001) Recognizing opportunities for sustainable development. *Entrepreneurship Theory and Practice*. 35(4), 631-652.
- Sarason, Y., & Dean, T. J. (2019). Lost Battles, Trojan Horses, Open Gates, and Wars Won: How entrepreneurial firms co-create structures to expand and infuse their sustainability missions in the acquisition process. *Academy of Management Perspectives*, 33(4), 469-490.
- Sarasvathy, S. D., & Ramesh, A. (2019). An effectual model of collective action for addressing sustainability challenges. *Academy of Management Perspectives*, 33(4), 405-424.
- Shevchenko, A., Levesque, M. & Pagell, M. (2016). Why firms delay reaching true sustainability. *Journal of Management Studies* 53, 911-35.
- Zahra, S. A., Gedajlovic, E., Neubaum, D. O., & Shulman, J. M. (2009). A typology of social entrepreneurs: Motives, search processes and ethical challenges. *Journal of Business Venturing*, 24(5), 519-532.
- Lumpkin, T. & Katz, J. (2011). *Social and sustainable entrepreneurship*. Emerald Group Publishing: Bingley (UK).
- Weidinger, C., Fischler, F. & Schmidpeter, R. (2014). *Sustainable entrepreneurship: Business success through sustainability*. Springer: Berlin (Germany).

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0046): Sustainability in Entrepreneurship and Family Enterprises (Seminar, 4 SWS)

Bird M, Kaufmann L

For further information in this module, please click campus.tum.de or [here](#).

Catalogue of Elective Modules: Digital Enterprises | Wahlkatalog: Digital Enterprises

Module Description

MGTHN0074: Contemporary Strategies in the Automotive Industry | Contemporary Strategies in the Automotive Industry

Version of module description: Gültig ab winterterm 2021/22

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The performance of students for this course will be determined by:

- 40% of a comprehensive industry and company analysis in form of PowerPoint presentation
- 40% of continuous monitoring of real-time event, market movement, company announcement
- 20% of processing and consolidating all theory from the seminar and information in self-study into a final project presentation, delivered in PowerPoint deck and verbally

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English, fundamentals in business administration and/or strategic management

Content:

This seminar aims to deliver an overview of recent development of discussion in the automotive industry, and to offer students the chance to consolidate the knowledge from strategic and competitive management to analyse case studies independently.

Automotive is one of the most important pillars of Germany's economy. It is now facing the challenges of transformation: rising consciousness for environmental protection from the general public and legislator, pressing need for alternative powertrain technologies, deeper involvement and engagement from the tech powerhouses, ever complicating partnership and competition landscape.

This seminar examines the issues above and the standpoint of major market participants, both established OEMs and new entrants - their reactions, strategic positions, and options. Furthermore, the interpretation of strategy related topics, e.g. core company character, corporate governance, decision-making process, business communication will also be important components to complete the scope of this seminar.

Intended Learning Outcomes:

After the course students will be able to:

- Understand the fundamental theories of automotive management and industry-specific topics
- Apply the knowledge from strategic & international management, as well as competitive strategy in practical oriented course work
- Gain experience in building strategic framework for a market participant and evaluating the impact of influential factors
- Enhance the ability of academic reading, information gathering and research, data analysis
- Further develop communication skills through verbal business presentation and group discussion
- Build up critical thinking and interpretation skills

Teaching and Learning Methods:

This seminar will be aided by PowerPoint slide deck to provide introduction of fundamental theories. Students will be required to self-reading and closely following the real-time market movement.

"War gaming" methods to simulate real-time strategy deployment: each student to prepare keynote to present the strategy analysis of one market participant of their choice - so the students would exercise their ability of information gathering, data processing and business presentation.

Group discussion to ignite the exchange of ideas and provide each other feedbacks.

Industry experts will be invited as guest speakers to provide empirical insights.

Media:

PowerPoint, flipchart, case study materials etc.

Reading List:

Uwe Winkelhake (2022), The Digital Transformation of the Automotive Industry

Jens Diehlmann, Joachim Häcker (2013), Automotive Management. Navigating the next decade of auto industry transformation

Léa Platel (2010), Rethinking communication strategies in the automotive industry

Andreas Blum (2009), Co-opetition in the Automotive Industry

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0084: Advanced International Excursion: Global Strategy | Advanced International Excursion: Global Strategy

Version of module description: Gültig ab summerterm 2022

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 9	Total Hours: 270	Self-study Hours: 180	Contact Hours: 90

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each student will work on a seminar thesis and participate in the discussion with international firms and universities. The examination consists of two parts:

1) Seminar thesis (70 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics discussed in the course
- they have applied the frameworks introduced in the course
- they show that they are able to write a paper that follows scientific standards
- they are able to structure and write a seminar thesis

2) Participation in discussion during excursion (30 % of the overall grade).

The students should demonstrate that:

- they have gained the skills to contribute to discussions with firms and universities
- Active engagement in the course is expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

English oral and written fluency

Content:

Multinational firms need to succeed globally to thrive in today's business environment. Thus, these firms need managers who are well-versed in international business environments. This course aims to provide students an immersive and intense international experience through a 9-day excursion abroad.

For 2022, the destination will be Paris.

Intended Learning Outcomes:

After completing this course students should be able to:

-Feel comfortable and gain proficiency in communicating and doing business abroad

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via in-class discussions
- Build up critical thinking, ethnorelative perspectives and interpretation skills

Teaching and Learning Methods:

This module includes the following teaching and learning methods:

- Immersive learning of strategy, localization, and value chain activities in a different country through an excursion
- Key concepts and empirical findings from the field of strategic management will be provided through discussions with multinational firms.
- Students will acquire additional knowledge from practical literature through their own work on their seminar theses.

Media:

PowerPoint, research papers, case scenarios

Reading List:**Responsible for Module:**

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced International Excursion: Global Strategy (MGTHN0084) (Seminar, 6 SWS)

Li C, Schilder J

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0095: Social Media Marketing | Social Media Marketing

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Grading based on a written multiple-choice examination of 90 minutes (100%) with 35 multiple-choice assignments, designed to assess students' ability to outline and explain core concepts of social media marketing in a structured manner. The exam includes both knowledge-based questions and applied tasks that require students to demonstrate their understanding of electronic word-of-mouth marketing, influencer marketing, and digital consumer feedback. Additionally, case studies or practical scenarios will be used to evaluate students' ability to critically reflect on emerging topics such as meme marketing or social media firestorms.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Completed course on fundamentals in Marketing (mandatory)

Content:

Social media have gained strong importance in both practice and academia. Corresponding digital technologies changed the way how consumers communicate, how they gather information, and how they spend large portions of their life. In this spirit, social media also opened up new opportunities for marketers to target and reach potential customers online. The lecture will cover the basics of social media (marketing), (electronic) word-of-mouth marketing, endorser marketing, cancelling, meme marketing, decoding digital consumer feedback, and new social media formats.

Intended Learning Outcomes:

Upon successful participation, students will be able to outline core concepts of social media and social media marketing. Students will understand and be able to explain the importance of electronic word of mouth marketing. Moreover, students will be able to outline differences between endorser and influencer marketing. They will be capable to explain key mechanism of influencer

marketing and persuasion processes. Students will have an understanding of novel topics in social media marketing as well, these include meme marketing, the threat of cancelling or social media firestorms. Students will have an overview of approaches to analyze social media data for marketing purposes. Students will be aware of new directions in social media and will be able to discuss what this implies for stakeholders. Finally, students will be capable to read and understand current research in social media marketing and to discuss their findings.

Teaching and Learning Methods:

The lecture will consist of input by the lecturer with interactive class discussions. Lectures will include critical discussions of scientific papers.

Media:

PowerPoint, videos, AI image generation software, research papers, virtual reality devices, exercise sheets

Reading List:

Ismagilova, Elvira; Dwivedi, Yogesh K.; Slade, Emma; Williams, Michael (2017): Electronic Word of Mouth (eWOM) in the Marketing Context. A State of the Art Analysis and Future Directions: Springer.

Appel, G., Grewal, L., Hadi, R., & Stephen, A. T. (2020). The future of social media in marketing. Journal of the Academy of Marketing Science, 48(1), 79-95.

Further reading material will be provided in the course.

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0099: Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy | Advanced Seminar Operations & Technology: Supply Chain Tactics & Strategy

Version of module description: Gültig ab winterterm 2022/23

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The seminar prepares students to make tactical choices supporting organizational performance and strategy. The seminar enables students to understand the relationships between operations and supply chains and other areas of the organizations and to design reliable, fast, and efficient operations and supply chains.

Students will be guided to learn critically analyzing value chain reports and to identify problems and areas for potential improvements. Later, students will demonstrate their ability to apply these skills in a simulated value chain setting. After completing this seminar, students can identify value chain challenges, identify potential improvements, coordinate critical decisions across functional areas, and forecast results from the solutions implemented. As such, this seminar will prepare students for applying academic knowledge to industry practice.

Central to the seminar is a simulation consisting of several rounds. Students will be assigned to groups of four participants and have to formulate a strategy for their company. During each round, they will be required to make tactical and strategic choices to maximize value and prepare a four-page summary justifying the choices made and the expected results. This summary is part of the overall assessment (75% of the final grade, team grade). This assessment method helps students reason about the options available in each round, make informed decisions, develop predictive capabilities, work together in a team, and condense necessary information formally. Furthermore, students must submit a four-page report at the end of the course, including a critical reflection about teamwork and individual takeaways (25% of the final grade, individual grade). This assessment method allows students to reflect on their capabilities to work in a team and strengthen their personal learnings from the seminar.

At the beginning of the course, students will be advised on the report structure and submission deadlines.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Desirable but not mandatory: Production and Logistics, Management Science

Content:

Subject-specific content

- Strategic Alignment
- Value Chain
- Supply Chain Tactical Decisions
- Supply Chain Management
- Sales and Operations Planning

Intended Learning Outcomes:

Upon completion of the module, students are able to:

- understand the need to construct operations and supply chains that align with corporate-level strategies and identify appropriate actions
- analyze the relationships between operations and supply chain management and other areas of the organization (e.g., marketing, finance, accounting, engineering, R&D, etc.)
- evaluate the feasibility of functional-level actions with corporate-level strategies and assess the impact of their choices on the organization's performance
- coordinate actions and address competing interests across functional areas
- evaluate alternative tactical decisions and convince with compelling reasoning for their choice of action
- critically reflect upon their ability to work in a team and provide advanced improvement recommendations for effective teamwork

Teaching and Learning Methods:

Experiential learning

The module is designed to enable students to acquire knowledge on current (behavioral) operations and supply chain management topics through hands-on experience and reflection. To this aim, a web-based business simulation game will be employed. The business simulation game will engage participants in turning around a manufacturer of fruit juices by making tactical and strategic decisions on four critical areas of the organization: Purchasing, Operations, Supply Chain, and Sales. Students are required to analyze key performance indicators, define a strategy for turning around the organization, identify potential improvements, and make tactical decisions in each of the organization's four areas.

Seminar sessions take place physically, and the methods combine mini-lectures, hands-on experience, group reflection, and questions and feedback sessions. Each week, there will be two

90-minutes sessions that will allow for feedback, sparring sessions, and the introduction of new concepts.

Students will get an overview of the course in the first two weeks. They will be informed about the examination and expectations. Furthermore, students will be introduced to highly recognized supply chain strategy and integration papers. Students will also see examples and learn to interpret data from sales, operations, supply chain, and purchasing reports.

After setting the theoretical scene, students will form small groups of size 4 (groups of smaller size might also be formed depending on the total number of participants), and they will start working on the business simulation game. The simulation advances over several rounds and students must present their analysis, evaluation, decisions, and expected results. Further details will be shared during the seminar.

Media:

Research papers and further material to be shared via Moodle, in-class discussions

Reading List:

Readings for general preparation:

Myerson, P. (2012). Lean supply chain and logistics management. McGraw-Hill.

Croson, R., Donohue, K. (2006). Behavioral causes of the bullwhip effect and the observed value of inventory information, *Management Science*, 52(3), pp. 323-336. DOI: 10.1287/mnsc.1050.0436

Flynn, B.B., Huo, B. and Zhao, X. (2010), The impact of supply chain integration on performance: A contingency and configuration approach. *Journal of Operations Management*, 28: 58-71. DOI: 10.1016/j.jom.2009.06.001

Frohlich, M.T., Westbrook, R. (2001). Arcs of integration: an international study of supply chain strategies. *Journal of Operations Management*, 19, pp. 185-200. DOI: 10.1016/S0272-6963(00)00055-3

Further materials will be discussed in the session.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0111: Cases in industry 4.0 | Cases in industry 4.0

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The idea of this course is to train students in managerial decision-making to strengthen their positive impact on their next job. Accordingly, the coursework and examinations are structured.

Each week during this semester, there will be sessions of 2 times 90 minutes where we will discuss teaching case studies revolving around management issues in Industry 4.0. Students must prepare each case and answer preparatory questions (60% of the final grade, individual grade). Class participation and active contributions are graded as 40% of the final grade (individual grade). This course will allow students to learn firsthand about management issues in Industry 4.0. During those sessions, we will cover a variety of relevant technologies and their interplay.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Production and Logistics, Strategic Management

Content:

Subject-specific content

- Industry 4.0
- Management issues/conflicts of interests
- Implementation of digital technologies
- Blockchain, IoT, Cloud, AI, AR, VR, etc.

Methodological content

- Quantitative assessment
- Modelling

Intended Learning Outcomes:

Upon completion of the module, students can

- understand and analyze state-of-the-art approaches to industry 4.0,
- evaluate the advantages and disadvantages of the most recent technologies
- critically analyze firms and supply chains within their context and make superior recommendations,
- and criticize innovative approaches by assessing potential shortcomings.

Teaching and Learning Methods:

Lecture

Each week, students will be given one case study (approx. 10-20 pages), read those, answer a few graded case-related questions (approx. 2-4), and attend the classroom session. In those sessions, students will actively contribute and engage in the discussion. While correctness and plausibility are essential, this class also seeks diverse views and appreciates reasonably different perspectives.

Media:

Case studies (to be shared via Moodle)

Reading List:

Readings for general preparation:

Babich, V., & Hilary, G. (2020). OM Forum—Distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management*, 22(2), 223-240.

Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113-122.

Wuttke, D., Upadhyay, A., Siemsen, E., & Wuttke-Linnemann, A. (2022). Seeing the Bigger Picture? Ramping up Production with the Use of Augmented Reality. *Manufacturing & Service Operations Management*.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0113: How to launch a platform start-up | How to launch a platform start-up

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

1) A final written exam determines 60% of the grade. The final written exam (60 minutes) is a means to assess students' understanding of the basic and advanced theories of the economics and management of digital platforms. Students have to show that they understand the core economic concepts underlying platform business models. They have to prove that they are familiar with the strategies and tactics to start a digital platform business. They have to show that they can critically evaluate the different options for designing a platform business model. Students may use a non-programmable calculator and a non-electronic dictionary for the exam.

2) Case study group work with presentation determines the remaining 40% of the grade. Students will be assigned into groups to work on a case study about a platform company. Students have to show that they can analyze the case based on the knowledge acquired in the course and suggest recommendations for the platform company. The grade for the case study group work is based on an in-class presentation by the group.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

none

Content:

In the 21st century, we see a number of companies like Uber or Airbnb that provide value to their customers by operating a platform and not by selling products. This rather new business model allows users and producers to interact with each other while the platform provider sets up the infrastructure for doing so.

This course gives an introduction to digital platforms to understand the underlying concepts of those business models and highlights the importance of the critical mass and network effects. Important principles for launching a digital platform are explained and different challenges that can appear in the process are outlined. By the end of the course, students will be able to understand digital platforms as a business model in general and be able to develop promising strategies in the launching phase. For a deeper understanding of the theoretical concepts, participants will apply the learnings on a case study which provides them with the necessary skills for analyzing cases and using previously acquired knowledge to assess them.

Intended Learning Outcomes:

After the course students will be able to:

- understand the concepts of digital platforms as a business model.
- develop strategies for launching a platform and understand the importance of the critical mass and network effects.
- apply instruments that allow to overcome the challenges in the launch process in a market with and without competition.

Teaching and Learning Methods:

Lecture: the lecture serves to build a thorough theoretical understanding of the concepts and methods introduced in the course.

Case studies: students will be assigned into groups to work on a case study of a real-world platform start-up with the goal to identify solutions for their challenges.

Media:

Slides

Case studies

Reading List:

Förderer, Jens: Erfolgsmodell Digitale Plattformen: Geschäftsmodelle – Netzwerkeffekte – Community-Management. 2022. Schäffer-Poeschel Verlag, Stuttgart.

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0115: Advanced Seminar: Digital Technologies in Operations Management | Advanced Seminar: Digital Technologies in Operations Management

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The key idea of this seminar is to lead students towards academic work and help prepare them for their master's thesis. In addition, they will learn about recent developments in digital technologies. Accordingly, the coursework and examinations are structured.

During the semesters, students will prepare two short presentations on research methods and skills. They will work in small teams. Each of the two presentations (ca. 45 minutes) accounts for 10% of the final grade and is a team grade.

Upon completion of this module, students will demonstrate their ability to cope with advanced research studies, comprehend and assess study outcomes, compare contributions of several studies, and transfer theoretical concepts to practice through a seminar paper (50% of the final grade; length of 15 pages including references). This method requires students to assess and replicate a study themselves.

In addition, they will prove their ability to communicate even complicated relationships and methods to their peer students through a presentation in a comprehensible fashion. They will further guide and moderate an ensuing discussion throughout which they will demonstrate their ability to criticize and assess innovative approaches and their potential shortcomings. The presentation, including moderation of ensuing discussion, accounts for 30% of the final grade (presentation duration 30 minutes + 30 minutes discussion). Complementing the written seminar paper, the presentation is targeted toward students who have not studied the same paper; this poses the challenge of presenting theoretical work interestingly while breaking down complex relationships into understandable information without losing rigor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Management Science, Production and Logistics

Content:

Subject-specific content

- Digital technologies potentially including but not limited to AR, VR, AI, Blockchain, and IoT.
- Management focus
- Incentive conflicts that arise when introducing new technologies in operations management
- Behavioral operations management

Methodological content (depending on assigned papers and limited to an introductory level)

- Research Design
- Analytical Modelling
- Game Theory
- Event Study Method
- Econometrics

Intended Learning Outcomes:

Upon completion of the module, students can

- understand and analyze state-of-the-art approaches to digital technologies in operations management,
- understand selected research papers and evaluate their key findings,
- present and explain complex research studies in a comprehensible and engaging manner,
- evaluate suggested management solutions in the digital technologies in an operations management context and assess their strengths and weaknesses,
- create management recommendations based on recent academic studies.

Teaching and Learning Methods:

Seminar

The seminar consists of two parts. During the semester, we will meet regularly to discuss select papers on digital technologies in operations management. Students will also explore and present information on select research methods (e.g., econometrics, game theory, and behavioral experiments). This part is complemented by a few interactive lectures.

At the end of the semester, we several sessions take placed blocked where students will present their assigned research studies (final assessment).

Media:

Research papers (to be shared via Moodle), student presentations, further material to be shared via Moodle, online discussions

Reading List:

Readings for general preparation (i.e., what is research? What marks a contribution?):

Wacker, J. G. (1998). A definition of theory: research guidelines for different theory-building research methods in operations management. *Journal of operations management*, 16(4), 361-385.

Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of management review*, 14(4), 490-495.

Examples for typical papers that we will discuss (this list is not meant to be comprehensive but gives some indication of topics to be covered):

Babich, V., & Hilary, G. (2020). OM Forum—Distributed ledgers and operations: What operations management researchers should know about blockchain technology. *Manufacturing & Service Operations Management*, 22(2), 223-240.

Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113-122.

Further papers will be provided at the beginning of the semester.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0122: Campus Challenge: Successful Investing with AI and Large Language Models | Campus Challenge: Successful Investing with AI and Large Language Models

Version of module description: Gültig ab summerterm 2023

Module Level:	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises group-based scientific coursework (approx. 20-30 pages) about the group's investment strategy.

The investment strategy should incorporate language models systematically and creatively.

Students will receive more information on the available tools during the kick-off meeting.

The coursework should comprehensively document the investment strategy (investment idea, tools used, prompt engineering, data preparation, data analysis, performance evaluation and interpretation, and suggestions for further improvement). In addition to the written scientific report, candidates must present their strategy within an approximately 20-minute presentation and answer questions from their audience.

To maintain the seminar atmosphere, the number of participating groups is generally limited. While Heilbronn students are prioritized, students from all TUM locations may also participate if the course is offered in a hybrid/online format. The targeted maximum number of participants is 24. More information will be provided in the introductory session.

The written assignment will be weighted 80%, and the formal presentation will make up 20% of the final grade. The group whose investment strategy performs best may receive an additional grade improvement in the live-test. The assessment is generally based on group work. If there are significant differences across the quality of the report or presentation, grades may be adapted individually. The students should therefore highlight their contributions to the report.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;
Basic knowledge in scientific writing;
Interest in investing in financial markets and the application of AI tools

Content:

The students will motivate, create and implement an investment strategy for the equity market using AI and/or large language models and/or sophisticated chatbots. The investment strategy will be developed based on group-work.

Candidates are expected to motivate their investment strategies with respective literature to ensure a systematic and structured manner. Candidates are expected to learn how to work with sophisticated chatbots, language models, and Application Programming Interfaces (APIs).

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- motivate, create, and implement an investment strategy;
- work with language models / sophisticated chatbots;
- conduct an empirical project in a team;
- deliver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise regular meetings with the team members about the current status of the stock portfolio;

Methods of teaching comprise group-based coursework and the formal presentation of obtained results;

The learning methods of the students primarily comprise the following activities:

- Collaborative Implementation of the investment strategy (theory, tools used, prompt engineering, data preparation, data analysis, results, and interpretation, room for improvements);
- The theory may be developed based on an independent literature review (usage of scientific articles published in top international journals);
- Collaborative writing of a scientific report;
- Exercise a deductive, logical, and consistent argumentation to derive an investment strategy;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction/teaching methods are adequate to foster/extend the student's ability to conduct independent academic work and elaborate thematically complex contents on their own. It is considered a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

The students will receive a list of relevant literature during the kick-off meeting at the beginning of the semester.

De Prado, Marcos Lopez. Advances in financial machine learning. John Wiley & Sons, 2018.

Lopez-Lira, Alejandro and Tang, Yuehua, Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models, 2023

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Campus Challenge: Successful Investing with AI and Large Language Models (MGTHN0122)
(Seminar, 4 SWS)

Breitung C, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0123: Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals | Advanced Seminar Finance & Accounting: Artificial Intelligence and the Sustainable Development Goals

Version of module description: Gültig ab summerterm 2023

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

To successfully pass the module, the participants have to demonstrate methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations. The participants proof these competences through several presentations with written reports and active participation in discussions.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

No prerequisites required.

Content:

Artificial intelligence (AI) is one of the most profound ongoing technological developments which has the potential to strongly impact various fields of business and society. This module critically analyzes the effects of AI-applications on the sustainable development goals (SDGs) of the United Nations. Units of the module: Political and conceptual background of the SDGs; methods to assess impacts on the micro-, meso-, and macro-level; discussion of supportive and contradictory impacts of specific AI-applications on SDGs.

Intended Learning Outcomes:

After successful completion of the module, the participants have methodological and conceptual competences to critically reflect about general and specific impacts of artificial intelligence on the sustainable development goals (SDGs) of the United Nations.

Teaching and Learning Methods:

The module consists of an interactive introductory lecture and presentations of the participants with topic-specific written reports.

Media:

Textbook, presentation slides, and e-learning (Moodle).

Reading List:

Dodds, Donoghue und Leiva Roesch. 2016. Negotiating the sustainable development goals: A transformational agenda for an insecure world.

Kanie und Biermann. 2017. Governing through goals: Sustainable development goals as governance innovation.

Saetra. 2022. AI for the sustainable development goals.

Sobti und Sharma. 2018. An introduction to sustainable development goals.

Walker, Pekmezovic und Walker. 2019. Sustainable development goals: Harnessing business to achieve the SDGs through finance, technology, and law reform.

Responsible for Module:

Stich, Michael; Prof. Dr. rer. oec.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0130: CEO Strategy Series | CEO Strategy Series

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The assessment of this course is based on short quizzes administered at the end of each session. These quizzes both test listening comprehension and encourage students to critically reflect on the content in light of the theoretical foundations introduced at the beginning of the semester. To successfully complete the course, students must submit at least 70% of the quizzes. Each quiz contains 7 questions. The final grade is calculated from the aggregated quiz results.

In addition, students may earn a grade bonus of up to 30% through active oral participation. This requires asking relevant and well-prepared questions during the sessions, which should be linked to the visiting CEOs and their companies as well as to current developments, global trends, or issues related to digital strategy.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization.

Content:

This module is the flagship event of TUM C.HN and will take place on Tuesdays or Fridays at 3 pm (each date will feature one guest). It covers practical insights on firm strategies, whereby lessons can be learned directly from CEOs themselves. The covered strategy topics may include internationalization, the analysis of the internal and external environment, M&A and other types of strategic alliances, as well as digital transformation, the role of digital enterprises, and the strategic challenges arising from new technologies, online platforms, and data-driven business models. This course therefore combines classical perspectives of strategic management with current

developments in the digital economy, providing students with valuable insights for their future work as strategizers in a corporate environment.

Please note: Registration is mandatory. There will be a participants list at the entrance and police presence for each event.

Intended Learning Outcomes:

After attending this course, students will have learned about different perspectives on developing and executing firm strategies in practice. They will be able to analyze companies and CEOs in their function as strategizers, and apply the knowledge gained in an in-depth study of relevant strategy literature by interacting with CEOs. With this approach, students will gain theoretical and practical knowledge, strengthen their understanding of digital enterprises and the role of digital transformation in strategy, and are additionally able to improve their discussion- and soft skills.

Teaching and Learning Methods:

In this seminar, there will be both Q&As and podium discussions with CEOs. Students will meet several CEOs and managers of real-life companies, gain insights into their strategies and related concepts, and interact with them in Q&A sessions. To prepare, students should develop questions that build on the concepts introduced in the first session and reflect the relevance of digital enterprises and digital transformation in corporate strategy. Active participation in all sessions is expected.

Media:

Powerpoint

Reading List:

- 1) Hanson, D., Hitt, M. A., Ireland, R. D., & Hoskisson, R. E. (2016). Strategic management: Competitiveness and globalisation. Cengage AU.
- 2) Barney, J. B. (2002). Strategic management: From informed conversation to academic discipline. *Academy of Management Perspectives*, 16(2), 53-57.
- 3) Nag, R., Hambrick, D. C., & Chen, M. J. (2007). What is strategic management, really? Inductive derivation of a consensus definition of the field. *Strategic management journal*, 28(9), 935-955
- 4) Smith, K. G., & Hitt, M. A. (Eds.). (2005). *Great minds in management: The process of theory development*. OUP Oxford.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

CEO Strategy Series (MGTHN0130) (Seminar, 4 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0165: Sustainable Management - Bridging Economy and Ecology through Technology | Sustainable Management - Bridging Economy and Ecology through Technology

Version of module description: Gültig ab winterterm 2024/25

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 160	Contact Hours: 20

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of seminar papers (60%) and presentation (40%): The seminar begins with block lectures, after which students independently write a seminar paper, present their findings, and actively engage in discussions about their work. Possible seminar paper topics include the development of energy-efficient building technologies, advancements in CO2 capture and storage, the implementation and potential of smart grids, modern heat recovery systems, innovations in sustainable waste management, the decarbonization of industrial processes, and urban planning.

Repeat Examination:

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management and organizations.

Content:

This course introduces the fundamentals of sustainability, covering ecological principles, the significance of ecosystems, and the science behind climate change, including its global impacts. It examines historical and modern responses to climate-related disasters, focusing on technological solutions like clean air regulations and sulfur dioxide controls, and highlights the importance of a circular economy.

Intended Learning Outcomes:

After attending this course, students will have gained a comprehensive understanding of sustainability, including ecological principles, climate change, and its impacts. They will be able to explain, analyze, and critically reflect on historical and current climate-related challenges and

solutions driven by technological advancements. Students will have built an understanding of political, legal, and economic frameworks related to sustainability, with practical insights into corporate sustainability practices, exemplified by Knauf. Moreover, they will acquire knowledge on energy efficiency, sustainable energy sources, and technological innovations. They will prepare a written assignment, hold a short presentation and lead a discussion on a sustainability related topic, which enhances their ability to apply these concepts practically and improves their discussion skills. Furthermore, students will have the option to discuss actual activities at the multinational company Knauf, where they can observe and learn firsthand how a leading company integrates sustainability into its operations and practices.

Teaching and Learning Methods:

Students will write a seminar paper on various sustainability topics and present their findings to the class. After each presentation, there will be discussions to elaborate on the topics and reflect on the concepts. Additionally, students can see how sustainability practices are implemented in a real-world setting at Knauf. These methods aim to deepen the understanding, enhance analytical and presentation skills, and apply theoretical knowledge practically.

Media:

Powerpoint

Reading List:**Responsible for Module:**

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Sustainable Management - Bridging Economy and Ecology through Technology (MGTHN0165)
(Seminar, 4 SWS)

Li C, Saling R

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0181: Advanced Seminar Operations & Technology: Analytics in Manufacturing Management | Advanced Seminar Operations & Technology: Analytics in Manufacturing Management

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours:	Contact Hours:

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

This seminar aims to guide students toward academic research and prepare them for their master's thesis. The topics focus on advanced analytics in manufacturing management, covering AI methods, modeling techniques, optimization algorithms, and rigorous analysis of algorithmic properties, such as performance guarantees. While students are not expected to master all these techniques, they should be willing to develop deep expertise in one and understand its relevance for manufacturing decision-making. Additionally, they should be able to present their findings clearly and didactically to their peers.

Students work in small teams and complete two presentations and a term paper (~15 pages). The first presentation (30 minutes, 20% of the grade) provides a structured overview of a recent research article and formulates a research question for the seminar thesis. The second presentation (30 minutes, 30% of the grade) is the final presentation, showcasing their research findings. The term paper accounts for 50% of the grade.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Foundations in statistics and management science

Content:

This course focuses on advanced analytics in manufacturing, emphasizing algorithms, modeling, simulations, and theoretical analysis. It is open to all advanced analytics methodologies, provided they meaningfully support manufacturing decision-making.

Additionally, the course integrates practical experience by incorporating potential talks from industry practitioners and potential excursions to manufacturing facilities.

Intended Learning Outcomes:

Students will develop a comprehensive understanding of the scientific process, further develop critical thinking skills, and the ability to solve business problems efficiently. They will gain firsthand experience in manufacturing decision-making through practitioner interactions and engagement with cutting-edge research on data-driven manufacturing. Additionally, they will acquire hands-on expertise in advanced analytics methods applied to manufacturing decision-making.

Teaching and Learning Methods:

Teaching methods include presentations, interactive learning, e-learning, group discussions, and (if feasible) discussions and presentations by practitioners. Additionally, if feasible, we aim to include excursions to manufacturing and related facilities, as well as interactions with leading scientists in the field.

Media:

Zoom, literature databases (Ebsco, ScienceDirect etc.), standard solvers (Gurobi, Hexaly etc.), databases and case studies, programming environments and software development platforms, simulation environments etc.

Reading List:

We focus on high-profile and influential ideas in advanced analytics for manufacturing, typically published in leading journals such as Operations Research, Management Science, Manufacturing & Service Operations Management, INFORMS Journal on Optimization, European Journal of Operational Research, and INFORMS Journal on Computing. Below are examples of relevant articles:

[AI] Senoner, J., Netland, T., & Feuerriegel, S. (2022). Using explainable artificial intelligence to improve process quality: Evidence from semiconductor manufacturing. *Management Science*, 68, 5704-5723.

[mathematical analysis] Drozdowski, M., Jaehn, F., & Paszkowski, R. (2017). Scheduling position-dependent maintenance operations. *Operations Research*, 65, 1657-1677.

[modeling] Möhring, R., Schulz, A. S., Stork, F., & Uetz, M. (2003). Solving project scheduling problems by minimum cut computations. *Management Science*, 49, 330-350.

Da Col, G., & Teppan, E. C. (2022). Industrial-size job-shop scheduling with constraint programming. *Operations Research Perspectives*, 9, 100249.

[algorithmics] Koca, Y., Martagan, T. G., Adan, I. J. B. F., Maillart, L., & van Ravenstein, B. (2023). Increasing biomanufacturing yield with bleed-feed: Optimal policies and insights. *Manufacturing & Service Operations Management*, 25, 108-125.

Barnhart, C., Jacquillat, A., & Schmid, A. (2024). Robotic warehousing operations : A learn-then-optimize approach to large-scale neighborhood search. *INFORMS Journal on Optimization*, 1-24.

Responsible for Module:

Otto, Alena; Prof. Dr. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0012: Digital Finance | Digital Finance [DF]

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The module examination consists of a 90-minute written exam, in which the candidates should reproduce the knowledge imparted in the lectures. The students are also required to provide transfer knowledge. It is tested whether students have understood how Digital Finance relates to Traditional Finance and other financial research disciplines; which interdependencies exist among the research disciplines, etc. Dangers and Opportunities of FinTech applications should be understood in detail. Further, students are required to conduct and assess mathematical calculations. The written exam also comprises open questions as well as multiple-choice-tasks.

Students may have the opportunity to improve their final grade by up to 0.3 points by voluntarily participating in group work and presentations on selected topics in Digital Finance. This opportunity depends on factors such as the number of participants and available capacity, and is decided on a semester-by-semester basis by the instructor.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge of corporate finance/financial mathematics is required (e.g. material covered in the course Corporate Finance); joint election with the module "Advanced Seminar in Finance/Accounting: Current Research Topics in Digital Finance" highly recommended

Content:

The aim of the course is to give a comprehensive overview on Digital Finance, not covered elsewhere in courses of Corporate Finance within the Master in Management. It provides a coverage of the most recent relevant topics in the field of Digital Finance. Basic knowledge in the field of financial mathematics is a pre-requisite.

The module comprises the most important and most recent developments in Digital Finance. Specifically, it comprises (but is not limited to) the following areas:

- History of Digital Finance and how it relates to other financial research disciplines;
- Robo-Advisors;
- Crowd Funding;
- Peer-2-Peer-Lending;
- Social Media in Finance;
- Digital Payments;
- Cryptocurrencies;
- Digital Technologies (Big Data and Machine Learning; Textual Analysis; Blockchain)
- Cybersecurity;
- Digital Finance and the Market Efficiency Debate;
- The Future of Digital Finance;
- Regulation;
- etc.

Intended Learning Outcomes:

The students acquire detailed knowledge on how technological innovations are changing existing financial services and products, their implications and possible applications within the financial industry. Specifically, upon completion of the module, students will be able to identify changes and opportunities of Digital Finance and its implications for the overall macroeconomic stability.

Also, students will be able to fully understand limitations of the digital transformation within the financial industry and its underlying causes. Finally, the students will be able to assess how and why digital finance applications influence the market (in-)efficiency of existing financial markets. In particular, the students are able to analyze complex financial market relations while taking into consideration various influencing factors (as for instance irrational investor behavior).

Teaching and Learning Methods:

Types of instruction comprise lecture courses as well as practical courses;

Methods of teaching include lectures, presentations as well as guest lectures;

the learning methods of the students primarily comprise the following activities:

- Follow-up of course contents;
- Exercise and execution of financial calculations;
- Preparation and execution of presentations;
- Ability to answer advanced thematic issues;
- etc.

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to fully understand and elaborate the in-depth the thematic content.

Media:

Lecture slides; whiteboard; exercise sheets; exercise portfolio; flipchart; powerpoint; films

Reading List:

Due to the topicality of this lecture, the usage of reference books is only possible in a limited manner. Still, the below-listed references are considered as a solid starting point for the contents of this lecture.

Goldstein, I., Jiang, W., & Karolyi, G. A. (2019). To FinTech and Beyond. The Review of Financial Studies, 32(5), 1647-1661.

Chuen, D. L. K., & Deng, R. H. (2017). Handbook of blockchain, digital finance, and inclusion: Cryptocurrency, FinTech, InsurTech, regulation, ChinaTech, mobile security, and distributed ledger. Academic Press.

Fatas, A. (Ed.; 2019). The Economics of Fintech and Digital Currencies. CEPR Press.

If applicable, further literature will be announced at the first session.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Digital Finance (WIHN0012) (Vorlesung mit integrierten Übungen, 4 SWS)

Müller S, Osterrieder D

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0013: Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance | Advanced Seminar Finance & Accounting: Current research Topics in Digital Finance

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises a scientific coursework (approx. 25-30 pages), which is supposed to address the research questions posed within each given seminar topic. The task is to be conducted as a group work (i.e. 2-3 students per topic). Additional to the written scientific report, candidates are required to showcase main findings within a 30-minute presentation and answer further questions following their presentation.

In doing so, the candidates are required to prove that they have dealt with the given topic in a comprehensive manner, that their selection and evaluation of literature is adequate and that they have in-depth analyzed the topic while simultaneously putting it in a higher context of financial research. Also, it can be assessed whether the students are able to answer further questions in a correct and comprehensive manner.

The written assignment will be weighted 2/3 and the formal presentation will make 1/3 of the final grade. Each student will be assessed separately.

To maintain the seminar atmosphere, the number of participants is generally limited. If the course is offered in a hybrid format, Munich students may also participate in the course. The targeted maximum number of participants is 12. More information will be provided in the introductory session.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;

basic knowledge in scientific writing;
combination with the module "Digital Finance" is recommended

Content:

Candidates are supposed to build groups of 2-3 students and choose one seminar topic out of a set of pre-defined scientific issues. The pre-selection omprises, but is not limited to topics of the following areas:

- Asset Pricing of Cryptocurrencies: Market (In-)Efficiency, Behavioral Finance & Bubbles;
- Financial Innovations: Foundations and Opportunities & Threats to Financial Market Stability;
- Wisdom of the Crowd? Information Asymmetries, Adverse Selection, and Moral Hazard in Peer-to-Peer Lending Markets Around the World;
- Robo-Advisors: Overview of Services, Benefits, and Limitations;
- Social Investing: Overview, Benefits, and Limitations;
- Machine Learning: An Overview of ML Applications for Finance;
- etc.

Candidates are expected to conduct a comprehensive literature review of the most important scientific articles capturing the chosen seminar topic in order to address the given research questions in a systematic and structured manner.

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- describe the effects of current FinTech developments for existing financial products and processes;
- exhibit a comprehensive overview on current research topics in the field of Digital Finance;
- identify chances and opportunities of these developments (especially with regard to the overall macroeconomic stability);
- conduct an independent literature analysis at the highest possible, international level;
- analyze and evaluate references in a systematic manner;
- combine topics in Digital Finance with theories/methods/models of traditional finance in an independent manner;
- deriver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise an initial lecture (overview course) as well as the corresponding main seminar;

Methods of teaching include a group work and the formal presentation of obtained results;

the learning methods of the students primarily comprise the following activities:

- independent literature research (usage of scientific articles published in international top journals);
- Collaborative writing of a scientific report;
- Exercise of a deductive, logic and consistent argumentation to specifically address and answer the posed research questions;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to conduct independent academic work and to elaborate thematically complex contents on their own. It is considered to be a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

Machi, L. A., & McEvoy, B. T. (2012). The literature review: Six steps to success (2nd ed.). Corwin Press.

A separate reading list for each topic will be provided in the introductory lecture.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Finance & Accounting (WIHN0013): Current research Topics in Digital Finance (Seminar, 4 SWS)

Gong Z, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0017: Advanced Seminar Innovation & Entrepreneurship: Digital Innovation | Advanced Seminar Innovation & Entrepreneurship: Digital Innovation

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Seminar paper with presentation: The students independently and individually prepare a scientific seminar paper and present it during the course.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Very good English language skills are required. Students should also have an understanding of basic principles of management, innovation, and organization as well as empirical research methods. Attendance of the course Empirical Research Methods in Management and Economics recommended, but not mandatory.

Content:

The course prepares students for their Master's Thesis by introducing them to (1) the search, understanding, and analysis of academic literature, (2) methods for data collection and data analysis, (3) the structuring and writing of a paper, and (4) giving presentations to an academic audience including a subsequent discussion. The course exposes students to digital platforms and open forms of organizing innovation.

Intended Learning Outcomes:

After the course students will be able to:

- Define, explain and apply selected key concepts in the field of digital innovation
- Read and understand academic literature
- Write and present an academic paper

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via presentations and group discussions
- Build up critical thinking and interpretation skills

Learning Objectives

At the end of this course, students will be able to demonstrate understanding, critical assessment and application of the following:

- Learn how to evaluate academic literature
- Interact with an academic debate
- Prepare, elaborate and defend an academic argument
- Specialist knowledge on their selected topic of interest

Teaching and Learning Methods:

Introductory session will provide introductions into the core topics and scientific writing in lecture style. All other sessions will be focused on the individual student presentations, in which each student presents their seminar paper to the class and all other students are encouraged to discuss the papers.

Media:

Powerpoint

Reading List:

Literature is semester-specific and will be distributed in class

Responsible for Module:

Förderer, Jens; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0019: Inventory Management | Inventory Management

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In order to check the learning outcomes an oral exam (20 minutes) is performed.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

- the content of the course Production and Logistics
- Basics in probability calculus
- Programming skills in Python

Content:

The following topics are discussed

Demand Data for Inventory Control,

3) Performance measures for inventory control
control policies

approaches

7) Joint replenishments

safety stock

10) Multi echelon inventory systems

11) Inventory Management in the digital Age

1) Analysis of

2) ABC Analysis, XYZ Analysis

4) Inventory

5) Service and cost oriented

6) Optimization of cycle stock

8) Optimization of

9) Managing Spare Parts Inventories

Intended Learning Outcomes:

The students

- know fundamental problems and trade-offs while managing inventories in supply chains,
- are able to model inventory problems,
- know common inventory policies and are able to analyse them,

- can determine optimal policy parameters for common inventory policies,
- are able to use stochastic inventory models to optimize safety stock.

Teaching and Learning Methods:

Case studies, presentation, discussions, computer work

Media:

Reading List:

- Kiesmüller, G.P. (2024) Inventory Management. Books on demand
- Tempelmeier, H. (2006): Inventory Management in Supply Networks. 2nd edition. Books on demand
- Axsäter, S. (2004): Inventory Control. Kluwer's International Series.
- Montgomery D.C.; Runger, G.C. (2003) Applied Statistics and Probability for Engineers. 3rd edition. Wiley

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

Inventory Management (WIHN0019) (Vorlesung mit integrierten Übungen, 4 SWS)

Kiesmüller G, Sahin K

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0022: Advanced Seminar Operations & Supply Chain Management: Operations Management | Advanced Seminar Operations & Supply Chain Management: Operations Management

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination consists of a written Seminar paper (75%) to check the learning outcomes 1-5, as well as an oral presentation (25%) in order to check learning outcome 6. The seminar paper should cover 15 - 20 pages and is written in the style of current publications of peer-reviewed journal articles. At the end of the module students present their work in a 20 minutes presentation + 15 minutes discussion and have to initialize and moderate the discussion on a selected paper of their fellow participants.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

The courses Management Science and Production and Logistics

Content:

The advanced seminar in logistics and supply chain management focuses on recent research progress on varying topics, e.g. after sales service, procurement.

- In a seminar on After Sales Service the following topics can be discussed: Forecasting Spare parts Demand, Inventory Management of Spare Parts, Reliability Analysis, Maintenance Strategies, Level of Repair Analysis, Repair Kit Problem, Design of Service Supply Chains, Additive Manufacturing of Spare Parts

-In a seminar on procurement the following topics can be discussed: outsourcing, global sourcing, multiple sourcing, order splitting, e-procurement, auctions, procurement on online spot markets,

Intended Learning Outcomes:

The objective of the module is to equip the participants with the necessary skills and tools for a successful master thesis project. Specifically, the aim is to be able to:

- 1 Read and understand recent research contributions
- 2 Pursue interesting research questions
- 3 Conduct a literature study and/or numerical study and/or implementation
- 4 Structure and organize research methods and results
- 5 Write a seminar paper
- 6 Present research finding and defend them in a discussion

Teaching and Learning Methods:

In an introductory session, the current theme of the module is explained by the lecturer and the various available seminar topics are elaborated in detail. Also information on relevant literature for the problem settings is introduced, which forms the basis of the students' seminar paper. After the introductory session, students will work out the topic on their own, by using their abilities of conducting literature research, mathematical modelling, programming and analyses. Throughout the whole time, they receive guidance from a supervisor of the chair. Different milestones are to be achieved at specific dates, such as a preliminary outline of the seminar paper, first research results and the final paper. Following the submission of the final paper, presentations and discussions of all students' seminar papers are conducted, usually spanning one or several days, where amongst others also presentation, moderation and discussion skills are trained.

Media:

Presentation, Various form of literature (Journal Articles, Books, Report, Conference Proceedings, etc.)

Reading List:

'Dependet on seminar focus, e.g.:

E. vanWingerden, R.J.I.Basten,

R.Dekker, W.D.Rustenburg (2014)

More grip on inventory control through improved forecasting:

A comparative study at three companies. IJPE 157, 2020-237

Joeri Poppe, Rob J.I.

Basten, Robert N. Boute, Marc R. Lambrecht. (2017) Numerical study of inventory management under various maintenance policies. Reliability Engineering and System Safety 168, 262-273.

Bram Westerweel, Rob J.I. Basten, Geert-Jan van Houtum (2018). Traditional or Additive Manufacturing? Assessing component design options through lifecycle cost analysis. EJOR 270, 570-585

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Supply Chain Management: Operations Management
(WIHN0022) (Seminar, 4 SWS)

Kiesmüller G, Proschmann D

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0031: Advanced Seminar Operations & Supply Chain Management: Digital Operations | Advanced Seminar Operations & Supply Chain Management: Digital Operations

Version of module description: Gültig ab summerterm 2020

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination consists of a written Seminar paper (75%) to check the learning outcomes 1-5, as well as an oral presentation (25%) in order to check learning outcome 6. The seminar paper should cover 15 - 20 pages and is written in the style of current publications of peer-reviewed journal articles. At the end of the module students present their work in a 20 minutes presentation + 15 minutes discussion and have to initialize and moderate the discussion on a selected paper of their fellow participants.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

The courses Management Science and Production and Logistics

Content:

The advanced seminar in logistics and supply chain management focuses on recent research progress on varying topics, e.g. after sales service, procurement.

- In a seminar on After Sales Service the following topics can be discussed: Forecasting Spare parts Demand, Inventory Management of Spare Parts, Reliability Analysis, Maintenance Strategies, Level of Repair Analysis, Repair Kit Problem, Design of Service Supply Chains, Additive Manufacturing of Spare Parts

-In a seminar on procurement the following topics can be discussed: outsourcing, global sourcing, multiple sourcing, order splitting, e-procurement, auctions, procurement on online spot markets,

Intended Learning Outcomes:

The objective of the module is to equip the participants with the necessary skills and tools for a successful master thesis project. Specifically, the aim is to be able to:

- 1 Read and understand recent research contributions
- 2 Pursue interesting research questions
- 3 Conduct a literature study and/or numerical study and/or implementation
- 4 Structure and organize research methods and results
- 5 Write a seminar paper
- 6 Present research finding and defend them in a discussion

Teaching and Learning Methods:

In an introductory session, the current theme of the module is explained by the lecturer and the various available seminar topics are elaborated in detail. Also information on relevant literature for the problem settings is introduced, which forms the basis of the students' seminar paper. After the introductory session, students will work out the topic on their own, by using their abilities of conducting literature research, mathematical modelling, programming and analyses. Throughout the whole time, they receive guidance from a supervisor of the chair. Different milestones are to be achieved at specific dates, such as a preliminary outline of the seminar paper, first research results and the final paper. Following the submission of the final paper, presentations and discussions of all students' seminar papers are conducted, usually spanning one or several days, where amongst others also presentation, moderation and discussion skills are trained.

Media:

Presentation, Various form of literature (Journal Articles, Books, Report, Conference Proceedings, etc.)

Reading List:

'Dependet on seminar focus, e.g.:

E. vanWingerden, R.J.I.Basten,

R.Dekker, W.D.Rustenburg (2014)

More grip on inventory control through improved forecasting:

A comparative study at three companies. IJPE 157, 2020-237

Joeri Poppe, Rob J.I.

Basten, Robert N. Boute, Marc R. Lambrecht. (2017) Numerical study of inventory management under various maintenance policies. Reliability Engineering and System Safety 168, 262-273.

Bram Westerweel, Rob J.I. Basten, Geert-Jan van Houtum (2018). Traditional or Additive Manufacturing? Assessing component design options through lifecycle cost analysis. EJOR 270, 570-585

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0032: Business Forecasting | Business Forecasting

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In order to check whether students can calculate forecasts using the Python/nixtla, they receive data that needs to be analysed. They have to select a suitable method and determine the forecasts. Thus, the first part of the exam is a practice performance (programming task, 30%). In the second part of the exam, students must present the results of a small project that focuses on forecasting in a presentation (20 minutes, 70%) and be able to defend their approach in the subsequent discussion.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

- Basics in statistics
- Basics in probability calculus
- Programming skills in Python
- Basics in Forecasting

Content:

The following topics are discussed
analysis

1) Basics of time series

2) Arima

3) Hierarchical Forecasting

4) Global models

5) Random Forrest

6) Multivariate Forecasting

7) LGBM, XGBoost

8) Neural Networks

Intended Learning Outcomes:

The students

- understand the importance of data analysis for business forecasting
- know different forecasting techniques
- are able to select the right forecasting method and apply it
- can use Python and the forecasting libraries from nixtla to compute forecasts
- are able to present forecasts and explain their derivation

Teaching and Learning Methods:

Group work, Programming with Python, Presentations, Exercises

Media:

PowerPoint Slides, Screencasts, Video conferences, Exercises

Reading List:

- Hyndman, R.J., Athanasopoulos, G. (2012) Forecasting: principles and practice. Otexts
- Ord, K., Fildes, R. (2013) Principles of Business Forecasting. Cengage Learning.
- Joseph, J., Tackes, J., (2024) Modern Time Series Forecasting with Python. 2nd edition.

Responsible for Module:

Kiesmüller, Gudrun; Prof. Dr. rer. nat. Dr. rer. pol. h. c.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Other Electives in Management | Sonstige wirtschaftswissenschaftliche Wahlmodule

Module Description

MGTHN0071: Advanced Seminar Management & Marketing: Strategic Management | Advanced Seminar Management & Marketing: Strategic Management

Version of module description: Gültig ab winterterm 2021/22

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of a seminar paper and presentation: In this course, students are to independently and individually prepare a scientific seminar paper and give a presentation on it.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, organization, and empirical research methods. Recommended (but not mandatory) is the attendance of the course Empirical Research Methods in Management and Economics.

Content:

The students will be prepared for their Master's Thesis by teaching them how to search, understand and analyse academic literature and methods für data collection and data analysis. Moreover, they learn how to structure and write a paper and give presentations to an academic audience, including a discussion on it.

Intended Learning Outcomes:

After attending the course the students can define, explain and apply selected key concepts in the field of strategic management. They will be able to read and understand academic literature and write and present an academic paper. The students will improve their writing and written communication skills, strengthen their verbal skills with presentations and group discussions and

gain skills in critical thinking and interpretation. After the course the students will be able to show their understanding, critical assessment and application of how to evaluate academic literature, interact within an academic debate, put together, elaborate and defend an academic argument as well as specialist knowledge on the topic of interest, they selected.

Teaching and Learning Methods:

The first session will be lecture-like, providing introductions to the core topics and scientific writing. The other sessions will have a focus on the individual student presentations, in which each student presents their seminar paper to the class. After the presentations all students are encouraged to discuss the papers.

Media:

Powerpoint

Reading List:

The literature is selected specifically for each semester and will be distributed in class.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Management & Marketing: Strategic Management (MGTHN0071) (Seminar, 4 SWS)

Li C, Saling R, Schilder J

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0073: Doing Business in Asia | Doing Business in Asia

Version of module description: Gültig ab winterterm 2021/22

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Each student will work on a seminar thesis and present their results to determine the final grade. The examination consists of two parts:

1) Seminar thesis (70 % of the overall grade)

The students should demonstrate that:

- they have gained a deeper knowledge of the topics discussed in class
- they have applied the frameworks introduced in class
- they show that they are able to write a paper that follows scientific standards
- they are able to detect promising research questions and know how to structure and write a research paper

2) Presentation of the seminar thesis (30 % of the overall grade).

The students should demonstrate that:

- they have gained the skills to present their seminar thesis to an academic audience
- they are able to answer specific questions related to their seminar thesis

Active engagement in the course will be expected.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

English oral and written fluency, as well as knowledge of the basics of scientific writing and empirical research.

Content:

The economic importance and potential of Asian markets has increased considerably in recent decades. In this module students will discuss different business practices in Asia with a particular focus on the major Asian economies such as China, Japan, India and South Korea etc. Students will look at differences in the institutional environment (1). Examine common business practices in

Asia (2). Furthermore, students will apply intercultural frameworks and concepts from the literature (3) and equipped with this knowledge engage in ethnorelative discussions on cultural variations and their implication for strategic management decision (4).

Intended Learning Outcomes:

After completing this course students should be able to:

- Define, explain and apply selected key concepts from the fields of cultural studies and international business
- Read, understand and evaluate academic literature
- Write and present their academic papers
- Gain specialist knowledge on their selected topic of interest

Skills Objectives

- Improve writing and written communication skills
- Enhance verbal skills via presentations and in-class discussions
- Build up critical thinking, ethnorelative perspectives and interpretation skills

Teaching and Learning Methods:

This module includes the following teaching and learning methods:

- Key concepts and empirical findings from the fields of International Business and Cultural Studies will be provided through lectures with PowerPoint slides and regularly recurring discussions in class.
- Specific frameworks and concept will be applied in group discussions and case scenarios.
- Students are encouraged to shift their perspectives and adopt an ethnorelative perspective by questioning and debating specific concepts and ethnocentric approaches
- Students will acquire additional knowledge from academic literature through their own research projects.

Media:

PowerPoint, research papers, case scenarios

Reading List:

Davies, R. J., & Ikeno, O. (2002). The Japanese mind: Understanding contemporary Japanese culture / edited by Roger J. Davies & Osamu Ikeno. Tuttle Pub.

Sorge, A., Noorderhaven, N. G., & Koen, C. (2015). Comparative international management (Second edition). RoutledgeTaylor & Francis Group.

Suder, G., Tsai, T., & Varma, S. (2020). Doing business in Asia (1st). SAGE.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0079: Advanced Seminar Management & Marketing: Digital Marketing | Advanced Seminar Management & Marketing: Digital Marketing

Version of module description: Gültig ab summerterm 2022

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Seminar participants work in groups investigating a topic related to privacy in virtual reality. Groups write a seminar paper (80%, 20 pages) and present their results to their peers (20%).

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

none

Content:

In an introductory block, seminar participants will be familiarized with key topics in the area of Virtual Reality Privacy. By critically dissecting extant scientific articles and adapting their knowledge gained in the primer sessions, participants train to identify research gaps in the domain of privacy research. Seminar participants learn both to precisely formulate their scientific work and to comprehensibly convey their ideas to the community by formally writing down their research proposal and presenting/discussing it in front of their peers.

Intended Learning Outcomes:

Learn to work with scientific literature / Learn to systematically review the scientific literature / Learn how to present research results to others and defend own work / Learn how to structure and write a seminar report

Teaching and Learning Methods:

The seminar comprises introductory lectures, group work and presentations in front of all class members. The introductory lectures are selected to familiarize students with the subject in an

efficient manner. Group work is selected as it allows students to have a joint learning experience. Students will have the opportunity to reflect on their work by frequently discussing their work with the supervisors. Presenting their results to others, encourages students to reflect on their learning experience.

Media:

Reading List:

- Buck, L., & McDonnell, R. (2022). Security and Privacy in the Metaverse: The Threat of the Digital Human. https://wenjietseng.com/assets/pdf/SSPXR22_submissions/SSPXR22_paper_2.pdf
- Cowan, K., Javornik, A., & Jiang, P. (2021). Privacy concerns when using augmented reality face filters? Explaining why and when use avoidance occurs. *Psychology & Marketing*, 38(10), 1799-1813.
- Giaretta, A. (2022). Security and Privacy in Virtual Reality--A Literature Survey. <https://doi.org/10.48550/arXiv.2205.00208>
- Hilken, T., Keeling, D. I., Chylinski, M., de Ruyter, K., Golf Papez, M., Heller, J., ... & Alimamy, S. (2022). Disrupting marketing realities: A research agenda for investigating the psychological mechanisms of next#generation experiences with reality#enhancing technologies. *Psychology & Marketing*, 39, 1660-1671.
- Scarpi, D., Pizzi, G., & Matta, S. (2022). Digital technologies and privacy: State of the art and research directions. *Psychology & Marketing*, forthcoming.

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0080: Advanced Seminar Economics & Econometrics: Economics of Education | Advanced Seminar Economics & Econometrics: Economics of Education

Version of module description: Gültig ab winterterm 2023/24

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The coursework involves reading of scientific papers and further academic elaboration and research on a specific topic related to the seminar's overall theme.

The students will be asked to present their finding in class in a 20 minutes presentation.

The examination will consist of the oral in-class presentation. Active class room participation is expected during the entire course of the seminar.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Principles of Economics (for Master in Management)

Content:

This seminar will cover key topics within the Economics of Education such as:

Topic area 1: The role of human capital for individual success on the labor market and beyond

Topic area 2: The role of human capital for countries' economic growth

Topic area 3: Learning from a life-cycle perspective: How early education determine educational trajectories

Topic area 4: School policy: How do school reforms affect children's skills?

Topic area 5: Higher education: How do students chose whether and what to study?

Topic area 6: Life long learning: Does continued education pay off on the labor market?

Topic area 7: Educational inequality: Determinants, consequences, and policies to foster equality

Intended Learning Outcomes:

Students will learn the key concepts in the research field of Education Economics. The goal is twofold: First, to understand how education, skills and knowledge affects individual (economic) success, innovation and the wealth of nations. Second, to understand what individual and institutional factors determine the level and distribution of skills and knowledge in society. The course builds on original research on that topic. The course will cover basic theoretical consideration, but will have a clear focus on empirical studies with real-world examples and applications.

Teaching and Learning Methods:

Students will self-study selected articles related to the assigned topic and will be coached throughout the semester by an instructor. A key learning objective is the ability to read, understand and reflect upon scientific articles on the seminar topic. Students will perform their own research of reference materials and identify potential gaps in the academic literature and public understanding of the focal topic, and learn to derive policy suggestions based on the scientific evidence.

A further learning objective is the communication of the key insights to the seminar group, i.e. to other students. Students are encouraged to form groups and work in groups.

Media:

Teaching will be in the form of a lecture and seminar presentations.

Reading List:

Core textbook: Lovenheim and Turner (2018). Economics of Education. New York: Worth Publishers.

Responsible for Module:

Lergetporer, Philipp; Prof. Ph.D.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0086: Advanced Strategic Management: Theory and Practice | Advanced Strategic Management: Theory and Practice

Version of module description: Gültig ab summerterm 2022

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of a handout and presentation: In this course, students are to independently and individually read scientific papers and give a presentation on them, apply a chosen theory to a real phenomenon as well as lead a discussion. Additionally, they are supposed to create a handout for the assigned papers.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization, and empirical research methods.

Content:

The course will be taught using a seminar or discussion style. On any given day, students will be “discussion leaders” for the material. Some sessions will feature book reviews or discussions of classic papers. Other readings will be recent papers that should be critiqued in-depth and used to identify future research directions and empirical challenges. This also includes applying a chosen theory to a real phenomenon. The course is intended to be cumulative, so as we go, students should think across the theories and perspectives underlying strategic management research.

Intended Learning Outcomes:

The students should have an awareness of the key research challenges and opportunities presented by the fundamental issues in strategy. They should understand the core behavioral assumptions employed in strategy and management research and how they relate to research in the disciplines. The students will develop an awareness of classic theoretical works that can be used in strategy and management. Furthermore, they will develop the skills to evaluate the

theoretical build-up of hypotheses in research in strategy and management. Each student will be able to compare and contrast each of the major theories used in strategic management. They will gain the ability to discuss and elaborate on scientific literature in the field of strategic management.

Teaching and Learning Methods:

Everyone should thoroughly prepare all of the readings assigned for each session. For each reading, one or more of the participants will be designated to be a discussion leader and will provide a summary and developmental review of the paper, as appropriate. This also includes the application of a chosen theory to a real phenomenon. They should prepare a written handout for the participants of the course (2-3 pages). These will be useful to facilitate the discussion. The review, as well as the class discussion that follows, should go well beyond offering a mere summary of the paper since it is expected that everyone will have read the material before class. In particular, the discussion leader should relate a reading to other research, underscore implications for the field, critique it carefully and offer suggestions for improvement, and identify research opportunities that would be worthwhile pursuing. The discussions should focus on debating the merits of different theories and these items above to enhance the value of the course for everyone. The goal of doing these reviews is to increase the understanding of the material and improve the students' ability to evaluate and carry out research. The discussion may be carried out in cooperation with the Bachelor's Course "Strategic Management: Theory and Practice".

Media:

Powerpoint

Reading List:

The literature is selected specifically for each semester and will be distributed in class.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0090: Master Thesis Seminar: Strategic Management | Master Thesis Seminar: Strategic Management

Version of module description: Gültig ab summerterm 2022

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Preparation of a thesis proposal and presentation: In this course, students are to independently and individually prepare a thesis proposal and give a presentation on it. If students have already accomplished the Advanced Seminar Strategic Management, where they developed a topic for their thesis, they should present their progress in this course. If students did not attend the Advanced Seminar Strategic Management, a proposal for a possible thesis topic should be developed and presented in this course.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Required are very good English language skills, as well as an understanding of basic principles of management, innovation and organization and empirical research methods. Recommended (but not mandatory) is the attendance of the course Advanced Seminar Strategic Management.

Content:

The students will be prepared for their Master's Thesis by teaching them how to search, understand and analyze academic literature and methods for data collection and data analysis. Moreover, they learn how to structure and write a thesis and give presentations to an academic audience, including a discussion on it.

The seminar is closely linked to the course Advanced Seminar: Strategic Management. More specifically students who already created a proposal in the Advanced Seminar and decided to conduct their thesis at the professorship for Strategic Management will only have to present their progress. Students who did not attend the Advanced Seminar will create and present their thesis

proposals in this course. Having a joint discussion with students from both courses provides them with increased feedback on their research ideas.

Intended Learning Outcomes:

After attending the course the students can define, explain and apply selected key concepts in the field of strategic management. They will be able to read and understand academic literature and write and present an academic paper. The students will improve their writing and written communication skills, strengthen their verbal skills with presentations and group discussions, and gain skills in critical thinking and interpretation. After the course, the students will be able to show their understanding, critical assessment, and application of how to evaluate academic literature, interact within an academic debate, put together, elaborate and defend an academic argument as well as specialist knowledge on the topic of interest, they selected. The course provides specific preparation for writing a thesis at the Professorship for Strategic Management.

Teaching and Learning Methods:

Some sessions will be lecture-like, providing introductions to the core topics and scientific writing. The other sessions will have a focus on individual student presentations, in which each student presents their progress to the class. After the presentations, all students are encouraged to discuss the presented results.

Media:

Powerpoint

Reading List:

The literature is selected specifically for each semester and will be distributed in class.

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Master Thesis Seminar: Strategic Management (MGTHN0090) (Seminar, 4 SWS)

Elmendorf M, Meng Y

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0119: Advanced Project Studies | Advanced Project Studies

Version of module description: Gültig ab winterterm 2016/17

Module Level: Master	Language: English	Duration: one semester	Frequency:
Credits:* 12	Total Hours: 360	Self-study Hours: 330	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Grading is based on a project work. The project work consists of a written project report (20-30 pages, 50% of grade) and a final presentation (30 minutes, 50% of grade). A student team of at least 2 students works on a specific problem set within a company or any other similar institution. The team runs through several project stages: problem definition, division of work/tasks, decision-making processes, and realization. In that, the students show that they can develop appropriate strategies to cope with the set of problems. They show that they are able to compose the state of research. In addition, they demonstrate their ability to develop their own specific approach to a solution based on scientific knowledge as well as methodical skills. Students demonstrate their ability within a team to manage resources, and deadlines through timely submission of the enumerated tasks. Students demonstrate that they are able to complete the tasks of their project in a team environment. By presenting their project report, students show their ability to summarize and communicate the evolvement of the project in a clear, well-structured, and convincing manner to the supervisors from the company and the university. Additionally, they show that they are able to respond competently to questions and discussions related to the suggested solutions brought by the audience.

Grading will mainly take into account the overall working outcome of the project with respect to the initial problem set, the selection and application of the chosen methodology as well as the analyses and discussion of the main findings. The project work is set up in a way that enables the identification and evaluation of each student's individual contribution to the project's success.

Repeat Examination:

Next semester / End of Semester

(Recommended) Prerequisites:

Basic knowledge in Business Administration

Content:

The project study consists of a specific problem statement or challenge that a company or any similar institution faces. This challenge may have a research-related or practical character. The project study and its findings regarding the outlined problem set are based on students' academic knowledge gained through their study programs.

Examples of topics covered in the context of a project study include (non-exhaustive list):

- Analyzing potential sales volumes of a new market
- Identifying potential optimization actions regarding a supply chain
- Creating a financing concept for a company
- explaining problems of the logistic sector and developing appropriate optimization solutions
- Developing specific use cases for new electronic payment procedures and deriving appropriate product specifications
- Capturing and processing key performance indicators (KPIs) in controlling and the development of recommended actions
- Developing and conceptualizing a marketing strategy and deriving recommendations for implementation in the given market- or company environment

Intended Learning Outcomes:

After successful participation in the module, students are able to work on a project in a systematic and academic manner. They can contribute their own part to a team's work output. Students are able to exchange in a professional and academic manner within a team. They show that they are able to integrate involved persons into the various tasks considering the group situation. Furthermore, the students conduct solution processes through their constructive and conceptual acting in a team. They can make this contribution in a time-limited environment. The students can capture and identify problem sets. Furthermore, they can analyze appropriate methodologies for problem-solving. They are able to infer the appropriate methodologies and to adapt these. On this basis, they can develop analytical solutions. Finally, they can evaluate the developed solutions regarding the problem set. Finally, students are able to summarize and clearly and convincingly communicate the evolution of their project and the developed solutions to an audience of professionals and academics.

Teaching and Learning Methods:

The team-based development (at least 2 students) of the project solution encourages the students to deal soundly with an academic or practical subject based on their previously acquired theoretical knowledge. Teamwork is particularly suitable for tackling problem sets and writing a report, for developing constructive critiques for others, and for implementing appropriate solutions to these critiques. The project may happen at the premises of the respective company/institution or from a remote location. They are able to communicate the evolvement of the project by composing a project report and preparing a presentation of their solutions to the supervisors from the company as well as the university. The project is supervised jointly by mentors from the respective company/institution and the professor of the TUM School of Management. With regard to content, the project study takes an approximate time of three months.

Media:

literature, presentations

Reading List:

Project Management Institute (2013): A Guide to the Project Management Body of Knowledge (PMBOK® Guide) - Fifth Edition

Further specific literature based on the topic

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0121: Transformative Impacts of Generative AI in Marketing | Transformative Impacts of Generative AI in Marketing

Version of module description: Gültig ab summerterm 2023

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students will form groups (of 2 or 3 students) to conduct a research project exploring the impact of conversational AI in a selected marketing domain. The progress and quality of the project will be monitored throughout the semester. The coursework consists of two parts, a group presentation in class (30%, 45 minutes) and a seminar report (70%, 30-40 slides).

Presentation: Each group will present their research findings to the class at the end of the semester. The presentations will be assessed based on clarity, depth of research, critical analysis, and the effectiveness of communication.

Seminar Report: Each group will prepare a seminar report, in the form of an expanded PowerPoint presentation, summarizing the principal discoveries from their research. The report should reflect a comprehensive understanding of the selected marketing domain and the impact of conversational AI on it.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

It is required that participants have successfully completed the course 'Marketing and Innovation Management' before taking this course.

Content:

The advent of advanced AI models, such as ChatGPT, has sparked an unprecedented surge in the application of conversational artificial intelligence (AI). Its increasing availability and sophistication are predicted to catalyze fundamental shifts in learning and working. Businesses must determine strategies to effectively harness these innovative tools to facilitate employee education, devise AI-driven business strategies, and enhance process efficiencies.

This course seeks to explore and illuminate the potential consequences of conversational AI on core marketing processes. To initiate this exploration, an initial set of sessions will acquaint students with advancements in AI applications for both research and practical applications. Students will be required to form groups, selecting a specific marketing domain of interest such as advertising, product development, or branding. The primary course project will involve the preparation of a comprehensive report detailing the prospective influence of AI on the chosen marketing domain and associated processes.

Throughout the semester, these student groups will engage in intensive project work, presenting their findings and participating as critical discussants in peer presentations. The culmination of the project will be a slide deck summarizing the group's principal discoveries. The objective of these slide decks is to foster a legacy of knowledge that will be utilized as educational material in future semesters, enabling an evolving understanding of this rapidly advancing field among succeeding cohorts of students.

Intended Learning Outcomes:

By the end of this course, students will be able to:

Understand the Fundamentals of Conversational AI: Grasp the key concepts and recent developments in the field of conversational AI, such as ChatGPT. Students will learn about its rise and the ongoing transformations in learning, working, and business strategy caused by its deployment.

Connect AI Applications to Marketing Domains: Develop the ability to understand and articulate the implications of AI in different aspects of marketing, such as advertising, product development, and branding.

Engage in Collaborative Research: Formulate strategies to conduct team-based research.

Students will be expected to actively collaborate, investigate, and critically analyze the impact of conversational AI on their chosen marketing domain.

Develop Critical Analysis and Presentation Skills: Cultivate skills in critical analysis, discussion, and public speaking. This will be achieved through group presentations, peer review, and open dialogue about the work of others.

Synthesize Research into a Comprehensive Report: Generate a thorough report on the potential influence of AI in the selected marketing domain. This will enhance their skills in written communication, information synthesis, and understanding of the topic.

Teaching and Learning Methods:

This seminar starts with a series of foundational lectures providing an overview of conversational AI. Following this introduction, students will be divided into groups and start their project work under the guidance of a supervisor. As they progress, students will engage in collaborative discussions concerning their research advancements, both with the supervisor as well as other teams. This promotes a critical evaluation of their work as well as that of their peers, fostering a rich exchange of ideas and insights. Upon completion of their projects, all groups will be tasked with presenting their findings in a classroom setting. The culmination of their efforts will be documented in a comprehensive seminar report, which will take the form of an expanded PowerPoint presentation.

Media:

Reading List:

Huang, M. H., & Rust, R. T. (2021). A strategic framework for artificial intelligence in marketing. Journal of the Academy of Marketing Science, 49, 30-50.

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0122: Campus Challenge: Successful Investing with AI and Large Language Models | Campus Challenge: Successful Investing with AI and Large Language Models

Version of module description: Gültig ab summerterm 2023

Module Level:	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises group-based scientific coursework (approx. 20-30 pages) about the group's investment strategy.

The investment strategy should incorporate language models systematically and creatively.

Students will receive more information on the available tools during the kick-off meeting.

The coursework should comprehensively document the investment strategy (investment idea, tools used, prompt engineering, data preparation, data analysis, performance evaluation and interpretation, and suggestions for further improvement). In addition to the written scientific report, candidates must present their strategy within an approximately 20-minute presentation and answer questions from their audience.

To maintain the seminar atmosphere, the number of participating groups is generally limited. While Heilbronn students are prioritized, students from all TUM locations may also participate if the course is offered in a hybrid/online format. The targeted maximum number of participants is 24. More information will be provided in the introductory session.

The written assignment will be weighted 80%, and the formal presentation will make up 20% of the final grade. The group whose investment strategy performs best may receive an additional grade improvement in the live-test. The assessment is generally based on group work. If there are significant differences across the quality of the report or presentation, grades may be adapted individually. The students should therefore highlight their contributions to the report.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;
Basic knowledge in scientific writing;
Interest in investing in financial markets and the application of AI tools

Content:

The students will motivate, create and implement an investment strategy for the equity market using AI and/or large language models and/or sophisticated chatbots. The investment strategy will be developed based on group-work.

Candidates are expected to motivate their investment strategies with respective literature to ensure a systematic and structured manner. Candidates are expected to learn how to work with sophisticated chatbots, language models, and Application Programming Interfaces (APIs).

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- motivate, create, and implement an investment strategy;
- work with language models / sophisticated chatbots;
- conduct an empirical project in a team;
- deliver answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise regular meetings with the team members about the current status of the stock portfolio;

Methods of teaching comprise group-based coursework and the formal presentation of obtained results;

The learning methods of the students primarily comprise the following activities:

- Collaborative Implementation of the investment strategy (theory, tools used, prompt engineering, data preparation, data analysis, results, and interpretation, room for improvements);
- The theory may be developed based on an independent literature review (usage of scientific articles published in top international journals);
- Collaborative writing of a scientific report;
- Exercise a deductive, logical, and consistent argumentation to derive an investment strategy;
- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction/teaching methods are adequate to foster/extend the student's ability to conduct independent academic work and elaborate thematically complex contents on their own. It is considered a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

The students will receive a list of relevant literature during the kick-off meeting at the beginning of the semester.

De Prado, Marcos Lopez. Advances in financial machine learning. John Wiley & Sons, 2018.

Lopez-Lira, Alejandro and Tang, Yuehua, Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models, 2023

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Campus Challenge: Successful Investing with AI and Large Language Models (MGTHN0122)
(Seminar, 4 SWS)

Breitung C, Müller S

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0124: Advanced Seminar Operations & Technology: Research on the Finance and Operations Interface | Advanced Seminar Operations & Technology: Research on the Finance and Operations Interface

Version of module description: Gültig ab summerterm 2023

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

This module enables students to understand the financial and operational constraints faced by firms and how these two areas overlap in salient ways. Students with an operations background will be introduced to financing aspects in supply chains and operations, whereas students with a finance background will learn more about the operations implications of financing decisions. Students will have to deeply analyze several critical topics from traditional topics such as moral hazard or information asymmetry problems to novel research directions such as supply chain finance. Upon completion of this module, the students will be able to identify potential challenges by the firms or financial institutions. The students will also develop capabilities to establish the core problems and formulate the necessary objectives and constraints for setting an academic paper/thesis on the related topic. Moreover, by preparing the discussion for the module, the students will be able to enhance their skills to communicate complicated concepts and dynamics of the research in a comprehensive way.

The presentation on demonstrating one of the problems from the reading material accounts for 20% of the final grade (group grade). A seminar paper on deriving the solution to problems related to those learned throughout the seminar accounts for 80% of the final grade (individual grade). A detailed guide on content and deadline will be given at the beginning of the semester.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Desirable but not mandatory: Production and Logistics, Management Science

Content:

Subject-specific content

- Supply chain management
- Finance and Operations Interface

Methodological content (depending on assigned papers and limited to an introductory level)

- Game Theory
- Econometrics

Intended Learning Outcomes:

Upon completion of the module, students are able to

- understand the financial and operational constraints faced by firms and how these two areas overlap in salient ways,
- derive and prove given solutions in research studies,
- develop critical perspectives on operational problems and objectives setting with the understanding of how finance matters,
- generate practical insights from research studies,
- and be prepared for writing a master's thesis in the relevant field.

Teaching and Learning Methods:

Seminar

This module will be held hybrid to enable the participation of all students who are interested in this topic regardless of their physical location (i.e., there is no need to be present in Heilbronn). Further details will be shared after the final enrollment is confirmed.

Students will be provided with one main text on various topics related to the finance and operation interface. Additional research papers will be given for more specific cases of the theoretic concepts. Each week students will have to be prepared with a set of questions on the topic that would require a comprehensive and thorough understanding of the topic, for example, the dynamics of a model and deriving or proving the propositions. The lecturer will give a short review and facilitate the session while the main discussions will be done by the students to derive the solutions to the given set of questions.

Media:

Research papers and further material to be shared via Moodle, hybrid platform

Reading List:

Reading for main topics:

Babich, V., & Birge, J. R. (2020). Foundations and Trends at the Interface of Finance, Operations, and Risk Management. SSRN Electronic Journal. <https://doi.org/10.2139/ssrn.3693440>

Examples of papers that we will discuss (this list is not meant to be comprehensive):

Babich, V., & Hilary, G. (2020). Distributed Ledgers and Operations: What Operations Management Researchers Should Know about Blockchain Technology. *Manufacturing & Service Operations Management*, 22, 223-240.

Babich, V., Marinesi, S., & Tsoukalas, G. (2020). Does crowdfunding benefit entrepreneurs and venture capital investors? *Manufacturing & Service Operations Management*, 23(2), 508-524.

Yang, S. A., Birge, J. R., & Parker, R. P. (2015). The supply chain effects of bankruptcy. *Management Science*, 61(10), 2320-2338.

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Operations & Technology (MGTHN0124): Research on the Finance and Operations Interface (Seminar, 4 SWS)

Sriraman S, Wuttke D

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0180: Empirical Consumer Research for Marketing Decisions | Empirical Consumer Research for Marketing Decisions

Version of module description: Gültig ab summerterm 2025

Module Level: Master	Language:	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In this seminar, students will actively engage in developing their own research ideas. Throughout the course, they will work in small groups to conceptualize, design, and refine a research project related to the topics covered in the module. Individual feedback sessions will support the development of these projects.

The final examination consists of three components: (1) A scientific paper (15–30 pages) written and submitted as a group project (70% of the final grade), (2) Each student's individual contribution to the research process, assessed through peer evaluation (10%), (3) A written summary and critical evaluation of a scientific paper (approximately 3 pages) (20%). To pass the module, students must receive a grade of 4.0 or higher in each of the three components.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Students should have a solid foundational understanding of marketing principles and should be able to apply basic statistical concepts such as standard deviation, correlation, and linear regression.

Content:

This seminar focuses on a selected psychological construct relevant to marketing decision making. Students will explore its theoretical foundations and measurable impact on consumer behavior, as well as the implications for marketing decision-making. Drawing on current scientific literature, students will identify research gaps related to the construct and formulate research questions with managerial relevance. Lecture sessions will cover topics such as methods of data collection and analysis in marketing research, interpretation of results, academic writing, and the translation of psychological insights into actionable marketing strategies.

Intended Learning Outcomes:

In this seminar, students will learn to:

1. Explain and critically assess key psychological constructs and their relevance to marketing decisions.
2. Identify research gaps related to consumer psychology in marketing decision-making.
3. Design appropriate data collection strategies to gather relevant consumer data.
4. Apply statistical techniques to analyze consumer data and extract actionable insights.
5. Synthesize and interpret findings based on both primary and secondary data to draw evidence-based conclusions and implications for marketing decisions.
6. Critically assess scientific papers, with particular emphasis on their practical implications for marketing and the ethical considerations associated with those implications.

Teaching and Learning Methods:

To support the learning outcomes, the seminar integrates theoretical foundations with practical application, ensuring both academic depth and real-world relevance. Students will actively engage in lectures by analyzing and discussing scientific papers, with particular attention to their managerial implications and ethical dimensions. Working in teams, students will design and carry out empirical research projects addressing real-world consumer behavior challenges. These projects will involve collecting primary data, integrating secondary data where relevant, and applying psychological and statistical methods to derive managerial insights. Through continuous feedback and iterative refinement, students will enhance their research design, analytical skills, and ability to draw practical conclusions from their findings.

Media:

Moodle

Reading List:

Lange, D., & Pfarrer, M. D. (2017). EDITORS' COMMENTS: SENSE AND STRUCTURE—THE CORE BUILDING BLOCKS OF AN "AMR" ARTICLE. *The Academy of Management Review*, 42(3), 407–416. <http://www.jstor.org/stable/26395005>

Responsible for Module:

Eckert, Christine; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Empirical Consumer Research for Marketing Decisions (MGTHN0180) (Seminar, 4 SWS)

Eckert C, Ristau I

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0183: Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research | Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Students work in groups of 3-4 to design and implement a VR-based marketing research experiment. Following experimental design, students will create VR scenarios using industry-standard software and hardware. Final deliverables include: (1) an in-class presentation demonstrating the functional VR experiment and discussing findings (30%), and (2) a comprehensive written report (15-20 pages) documenting the research question, methodology, technical implementation with code, results, data analysis, and reflection on the research process (70%).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic programming experience, preferably C# or similar object-oriented language (highly-recommended)

Content:

In an introductory block, participants will acquire key methodological skills in experimental design, virtual reality implementation, and programming for marketing research in immersive environments. Students will critically reflect on how to use VR for marketing research based on relevant scientific literature. Based on literature review and supervisor consultations, students will conceptualize and develop original VR implementations.

The seminar will conclude with in-class project presentations and written documentation of all VR implementation code and assets.

Intended Learning Outcomes:

Students will acquire skills in the following areas:

- Software development: Using a game engine (Unity) and programming (C#) to implement virtual reality studies in a marketing context
- Experimental design: Conceptualizing and planning VR study implementations
- Literature analysis: Reviewing, evaluating, and synthesizing relevant scientific literature
- Academic communication: Presenting research results and defending methodological choices
- Technical documentation: Preparing a comprehensive seminar report that includes software architecture, code documentation, and programming implementation

Teaching and Learning Methods:

The seminar starts with introductory lectures providing examples of virtual reality as a research environment for investigating marketing research questions. Lectures introduce the Extended Reality Lab equipment and guide the implementation of VR experimental designs. Subsequently, student groups develop their experimental implementations while receiving ongoing supervision. Students are also expected to pilot-test their VR implementations and collect data with actual participants. Finally, all groups present their results in class and document their work in a seminar report.

Media:

PowerPoint, videos, AI image generation software, research papers, virtual reality devices, exercise sheets

Reading List:

Will be published at the start of the semester

Responsible for Module:

Meißner, Martin; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Management & Marketing: Virtual Reality Applications in Marketing Research (MGTHN0183) (Seminar, 4 SWS)

Meißner M, Siqueira Rodrigues L

For further information in this module, please click campus.tum.de or [here](#).

Module Description

MGTHN0188: Customer Relationship Management for Long-Term Profitability | Customer Relationship Management for Long-Term Profitability

Version of module description: Gültig ab winterterm 2025/26

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

In this seminar, students will actively engage in developing their own research ideas. Throughout the course, they will work in small groups to evaluate current trends in customer relationship management (CRM), and conceptualize, design, and refine a research proposal. Individual feedback sessions will support the development of these proposals.

The final examination consists of three components: (1) A scientific paper proposal (10 pages) written and submitted as a group project (60% of the final grade), (2) Each student's individual contribution to the research process, assessed through peer evaluation (10%), (3) A presentation & discussion of a scientific paper (45min) (30%). To pass the module, students must receive a grade of 4.0 or higher in each of the three components.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Students should have a solid foundational understanding of marketing principles and statistics.

Content:

This seminar explores how organizations can strategically manage customer relationships to drive long-term profitability. Through a series of interactive lectures and case discussions, students will examine key concepts including the fundamentals of CRM, strategic implementation, GDPR compliance, technology platforms, CRM metrics, permission marketing, and loyalty programs. A central component of the seminar is the analysis and application of customer lifetime value to support data-driven marketing decisions. Students will critically engage with current academic literature through presentations. They will develop original research proposals that are grounded in

academic theory, demonstrate managerial relevance, and be informed by in-depth interviews with CRM professionals.

Intended Learning Outcomes:

In this seminar, students will learn to:

1. Understand and apply key theories and strategic frameworks of CRM to real-world business contexts.
2. Evaluate CRM technologies, metrics, and loyalty programs for their effectiveness in driving customer engagement and long-term profitability.
3. Analyze the impact of GDPR and permission marketing on CRM practices and strategic decision-making.
4. Calculate and interpret Customer Lifetime Value (CLV) to inform marketing strategies and resource allocation.
5. Design an academically grounded and managerially relevant CRM research proposal.

Teaching and Learning Methods:

The seminar combines interactive lectures, case-based discussions, and hands-on exercises to foster both theoretical understanding and practical application of CRM concepts. Students will engage in group work, article presentations, and guided discussions to deepen their critical thinking and communication skills. Students will work in teams to design research proposals based on interviews with CRM managers, supported by structured feedback sessions to refine their academic and practical contributions.

Media:

Moodle

Reading List:

Kumar, V., & Reinartz, W. (2018). Customer relationship management: Concept, strategy, and tools (3rd ed.). Springer.

Responsible for Module:

Eckert, Christine; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

Customer Relationship Management for Long-Term Profitability (MGTHN0188) (Seminar, 4 SWS)
Eckert C

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0011: Advanced Seminar Operations & Supply Chain Management: Supply Chain Finance & Supply Chain Resilience | Advanced Seminar Operations & Supply Chain Management: Supply Chain Finance & Supply Chain Resilience [SCF & SCRM]

Version of module description: Gültig ab summerterm 2019

Module Level: Master	Language: English	Duration: one semester	Frequency: winter semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Upon completion of this module, students will demonstrate their ability to cope with advanced research studies, to comprehend and assess study outcomes, to compare contributions of several studies, and to transfer theoretical concepts to practice through a seminar paper (70% of final grade; length of 15 pages including references). This method requires students to formulate an academic paper themselves; it can be seen as an exercise towards writing a master thesis.

In addition, they will prove their ability to communicate even complicated relationships and methods to their peer students through a presentation in a comprehensible fashion. They will further guide and moderate an ensuing discussion throughout which they will demonstrate their ability to criticize and assess innovative approaches and their potential shortcomings. The presentation including moderation of ensuing discussion accounts for 30% of the final grade (presentation duration 30 minutes + 30 minutes discussion). Complementing the written seminar paper, the presentation is targeted towards students who have not read the set of same papers; this poses the challenge to present theoretic work in an interesting fashion while breaking down complex relationships into understandable information without losing rigor.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

Management Science, Production and Logistics

Please note: all classes will be 100% online. There will be classes during the semester and at the end. Students do not have to be in HN.

Content:

Subject-specific content

- Supply Chain Finance
- Reverse Factoring
- Supply Chain Risk Management
- Finance and Operations Interface

Methodological content (depending on assigned papers and limited to an introductory level)

- Research Design
- Analytical Modelling
- Game Theory
- Event Study Method
- Econometrics

Intended Learning Outcomes:

Upon completion of the module, students are able to

- understand and analyze state-of-the-art approaches to supply chain finance and supply chain risk management,
- understand selected research papers and evaluate their key findings,
- present and explain complex research studies in a comprehensible and interesting manner,
- evaluate suggested management solutions in the supply chain finance and supply chain risk management context and assess their strengths and weaknesses,
- create management recommendations based on recent academic studies,
- and criticize innovative approaches by assessing potential shortcomings.

Teaching and Learning Methods:

Seminar

Each students will be provided with three research papers and asked to study them intensively. Each student in this seminar receives a different set of papers. As one form of guidance, students will be provided with a detailed set of questions that they need to answer. Students are asked to prepare a seminar paper, that is, an academic essay in which they critically reflect upon the questions they are provided with. This seminar paper will be written in a scientific style. At a later stage in the semester, the actual seminar will take place, where students present their findings and moderate an ensuing discussion. Throughout the semester, Moodle will be leveraged to provide ongoing feedback and incentives to start working early on on the assigned questions. At the beginning of the semester, specific milestones with deadlines will also be provided.

Media:

Research papers (to be shared via Moodle), student presentations, further material to be shared via Moodle, online discussions

Reading List:

Readings for general preparation (i.e., what is research? What marks a contribution?):

Wacker, J. G. (1998). A definition of theory: research guidelines for different theory-building research methods in operations management. *Journal of operations management*, 16(4), 361-385.

Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of management review*, 14(4), 490-495.

Examples for typical papers that we will discuss (this list is not meant to be comprehensive, but gives some indication of topics to be covered):

Hendricks, K. B., & Singhal, V. R. (2003). The effect of supply chain glitches on shareholder wealth. *Journal of operations Management*, 21(5), 501-522.

Peura, H., Yang, S. A., & Lai, G. (2017). Trade credit in competition: a horizontal benefit. *Manufacturing & Service Operations Management*, 19(2), 263-289.

Wuttke, D. A., Rosenzweig, E. D., & Heese, H. S. (2019). An empirical analysis of supply chain finance adoption. *Journal of Operations Management*, 65(3), 242-261.

Yang, S. A., Birge, J. R., & Parker, R. P. (2015). The supply chain effects of bankruptcy. *Management Science*, 61(10), 2320-2338.

Further papers will be provided at the beginning of the semester

Responsible for Module:

Wuttke, David; Prof. Dr. rer. pol. habil.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0020: Empirical Research Project in Finance | Empirical Research Project in Finance

Version of module description: Gültig ab summerterm 2024

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 150	Contact Hours: 30

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises an individual or group-based scientific coursework (approx. 30 pages) about a research project, which is supposed to be conducted together with and under the guidance of the project supervisors.

The research projects will be fixed by the project supervisors in advance and will be in accordance with their research interests.

The coursework should comprehensively document the research project (research question, literature contribution, data elicitation and data preparation, data analysis, results and interpretation, further research questions). Additional to the written scientific report, candidates are required to showcase main findings within a 30-minute presentation and answer further questions following their presentation. In doing so, it can be assessed to what extent the students have been able to successfully conduct the research project.

To maintain the seminar atmosphere, the number of participating groups is generally limited. While Heilbronn students are prioritized, students from all TUM locations may also participate if the course is offered in a hybrid/online format. The targeted maximum number of participants is 8 to 12. More information will be provided in the introductory session.

The written assignment will be weighted 80% and the formal presentation will make 20% of the final grade. The assessment is based on the groupwork (no individual assessment).

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;

basic knowledge in scientific writing;
motivation letter

Content:

The research project is in the field of empirical capital market research and will be announced at the beginning of each semester. The project will be developed and implemented in joint meetings and on the basis of independent (group-)work.

Candidates are expected to conduct a comprehensive literature review of the most important scientific articles related to the research topic in order to address the given research questions in a systematic and structured manner. It is further expected that candidates learn how to work with capital market databases and mathematical-statistical software packages and learn how to apply appropriate analysis methods.

This module requires an intensive supervision by and collaboration with the lecturers. The maximum number of course participants is therefore set to six. Candidates are chosen based on various submitted documents (transcript of records, motivation letter and CV).

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- conduct an independent literature analysis at the highest possible, international level;
- work with different capital market databases;
- work with different mathematical-statistical software packages and other tools;
- conduct an empirical research project alone and in a research team;
- derive answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise regular meetings with the lecturer about the current status of the research project and further steps as well as the corresponding main seminar;

Methods of teaching comprise a group-based coursework and the formal presentation of obtained results;

the learning methods of the students primarily comprise the following activities:

- Independent literature research (usage of scientific articles published in international top journals);
- Collaborative Implementation of the research project (research question, literature contribution, data elicitation and data preparation, data analysis, results and interpretation, further research questions);
- Collaborative writing of a scientific report;
- Exercise of a deductive, logic and consistent argumentation to specifically address and answer the posed research questions;

- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to conduct independent academic work and to elaborate thematically complex contents on their own. It is considered to be a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

A separate reading list for the research topic will be provided at the beginning of the semester. To familiarize with the basic software packages and econometric methods the following literature might be useful:

Kohler, U., Kreuter, F. (2012). Data Analysis Using Stata. Stata Press.

Müller, A., Guido, S. (2016). Introduction to Machine Learning with Python: A Guide for Data Scientists. O'Reilly.

Wooldridge, J. M. (2016). Introductory Econometrics. A Modern Approach (6th ed.). Cengage Learning.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0024: Advanced Seminar Finance & Accounting: Current Research Topics in Empirical Capital Market Research | Advanced Seminar Finance & Accounting: Current Research Topics in Empirical Capital Market Research

Version of module description: Gültig ab winterterm 2019/20

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 165	Contact Hours: 15

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination comprises a scientific coursework (approx. 20 pages), which is supposed to address the research questions posed within each given seminar topic. The task is to be conducted alone or as a group work (i.e. 2-3 students per topic). It will be communicated in the introductory session if individual work or group work is required (depending on class size). Additional to the written scientific report, candidates are required to showcase main findings within a 30-minute presentation and answer further questions following their presentation.

To maintain the seminar atmosphere, the number of participants is generally limited. If the course is offered in a hybrid format, Munich students may also participate in the course. The targeted maximum number of participants is 12. More information will be provided in the introductory session.

In doing so, the candidates are required to prove that they have dealt with the given topic in a comprehensive manner, that their selection and evaluation of literature is adequate and that they have in-depth analyzed the topic while simultaneously putting it in a higher context of financial research. Also, it can be assessed whether the students are able to answer further questions in a correct and comprehensive manner.

The written assignment will be weighted 2/3 and the formal presentation will make 1/3 of the final grade. Each student will be assessed separately.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Basic knowledge in corporate finance;
basic knowledge in scientific writing

Content:

Candidates are supposed to choose one seminar topic out of a set of pre-defined scientific issues. Those topics will be published on Moodle in advance and will be presented in detail during the introduction session.

Candidates are expected to conduct a comprehensive literature review of the most important scientific articles capturing the chosen seminar topic in order to address the given research questions in a systematic and structured manner.

For certain topics, students may be asked to reconstruct empirical results of a materially relevant research paper.

Intended Learning Outcomes:

Upon completion of the module, students will be able to ...

- exhibit a comprehensive overview on current research topics in the field of Empirical Capital Market Research;
- identify chances and opportunities of these developments (especially with regard to the overall macroeconomic stability);
- conduct an independent literature analysis at the highest possible, international level;
- analyze and evaluate references in a systematic manner;
- combine current topics in Empirical Capital Market Research with theories/methods/models of traditional finance in an independent manner;
- derive answers to posed research questions in a systematic and structured manner;
- create/draft a scientific report independently

Teaching and Learning Methods:

Types of instruction comprise an initial lecture (overview course) as well as the corresponding main seminar;

Methods of teaching include an individual work or group work and the formal presentation of obtained results;

the learning methods of the students primarily comprise the following activities:

- independent literature research (usage of scientific articles published in international top journals);
- Collaborative writing of a scientific report;
- Exercise of a deductive, logic and consistent argumentation to specifically address and answer the posed research questions;

- Preparation and execution of a final presentation;
- Ability to answer advanced thematic issues

The chosen types of instruction / methods of teaching are considered adequate to foster/extend the students' ability to conduct independent academic work and to elaborate thematically complex contents on their own. It is considered to be a good preparation for the students' master theses.

Media:

Exercise sheets, PowerPoint

Reading List:

Berk/DeMarzo (2023), Corporate Finance, 6th. Edition, Pearson.

Machi, L. A., & McEvoy, B. T. (2012). The literature review: Six steps to success (2nd ed.). Corwin Press.

A separate reading list for each topic will be provided in the introductory lecture.

Responsible for Module:

Müller, Sebastian; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0044: Advanced Seminar Innovation & Entrepreneurship: From Idea to Venture | Advanced Seminar Innovation & Entrepreneurship: From Idea to Venture

Version of module description: Gültig ab winterterm 2020/21

Module Level: Master	Language: English	Duration: one semester	Frequency: summer semester
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

Seminar participants will work in groups during the entire semester to develop a business idea. The examination will be based on the final presentation, i.e. a pitch of the developed business idea (60 %) and the final report (40 %). The students will defend their business idea in front of a jury (lecturers, fellow students and other experts). Active engagement in the course will be expected.

The final presentation and report will be evaluated based on the following criteria:

- Product/Service: Clear problem statement - Why do customers need this product and how is it different to existing products/services. Clear description of value proposition.
- Structure of the presentation: depth and breadth of the presented solution. Clear and systematic logic and argumentation. Quality of presented data.
- Business Model: How well is the business model described and how feasible it is.
- Engagement/Quality of presentation: Creativity of the presented solution. Clarity and formal quality.

Repeat Examination:

Next semester

(Recommended) Prerequisites:

Fluency in spoken and written English

Content:

This course will support you in developing, and evaluating business ideas for creating technology-based ventures. The course will also introduce you to product innovation and product development methods that will help you to turn your business idea into a viable venture.

Theory related content will comprise:

- Understanding basic concepts related to an entrepreneurial mindset and
- Further concepts related to new venture creation

Methodological content will comprise:

- Design Thinking
- Lean Start-up approach
- SCRUM
- Stage-gate model
- Business Model Canvas

Intended Learning Outcomes:

After completion of the seminar students will:

- understand the basic characteristics of an entrepreneurial mindset,
- evaluate the suitability of potential co-founders/team-members as well as strategic partners for the successful implementation of a business idea,
- know how to develop a business idea into a product/service, and
- be able to design a business model and pitch a business idea.

In addition, students will be able to:

- understand and apply the principles of innovation and product development using concepts such as the lean start-up approach, design thinking, SCRUM and the Stage-gate model

Teaching and Learning Methods:

- Through lectures, supported by power-point presentations, the instructor will provide the theoretical foundations and approaches on how to generate and evaluate viable business ideas.
- The content is discussed in the course by openly exchanging ideas. Questions, and comments will encourage an experiential learning atmosphere.
- Students will participate in a design thinking workshop.
- Guest speakers will share practical insights and will strengthen the students' understanding of the entrepreneurial mindset and the process of starting a business.
- Simulation of a real-life pitch event.

Media:

Powerpoint, Zoom- & Breakout-Sessions, Flipchart

Reading List:

Basis Literatur (bitte Moodle beachten für detaillierte Leseliste):

- Camuffo, A., Cordova, A., Gambardella, A., & Spina, C. (2020). A scientific approach to entrepreneurial decision making: Evidence from a randomized control trial. *Management Science*, 66(2), 564-586.
- Christensen, C. M. (1997). *The Innovator's Dilemma: When new technologies cause great firms to fail*. Harvard Business Review Press: Boston (US).

- Cooper, R. G.; Edgett, Scott J. (2009). Product Innovation and Technology Strategy. BookSurge Publishing: Charleston (US).
- Cross, N. (2011). Design thinking: Understanding how designers think and work. Berg: New York (US).
- Livingston, J. (2007). Founders at work: Stories of startup's early days. Apress: New York (US).
- Martin, R. (2009). The Design of Business. Harvard Business Press: Boston (US).
- Moore, G. A. (2014). Crossing the Chasm, 3rd Edition: Marketing and selling disruptive products to Mainstream. Harper Business: New York (US).
- Paulynice, J. P. (2019). From Idea to Reality: An entrepreneur's guide to meaningful business growth. Paulynice Consulting Group: Worcester (US).
- Ries, E. (2011). The Lean Startup: How today's entrepreneurs use continuous innovation to create radically successful businesses. Crown Business: New York (US).
- Thiel, P. (2015). Zero to One: Notes on Start-ups, or How to Build the Future. Crown Business: New York (US).
- Todaro, D. (2019). The Epic Guide to Agile: More Business Value on a Predictable Schedule with SCRUM. R9 Publishing LLC: New Hampshire (US).
- Osterwalder, A., & Pigneur, Y. (2010). Business model generation: a handbook for visionaries, game changers, and challengers. John Wiley & Sons: Hoboken (US).

Responsible for Module:

Bird, Miriam; Prof. Dr.

Courses (Type of course, Weekly hours per semester), Instructor:

Advanced Seminar Innovation & Entrepreneurship (WIHN0044): From Idea to Venture (Seminar, 4 SWS)

Bird M, Scheidt B

For further information in this module, please click campus.tum.de or [here](#).

Module Description

WIHN0049: Economic Analysis of Contracts, Competition and Companies | Economic Analysis of Contracts, Competition and Companies

Version of module description: Gültig ab winterterm 2020/21

Module Level: Master	Language: English	Duration: one semester	Frequency: irregularly
Credits:* 6	Total Hours: 180	Self-study Hours: 120	Contact Hours: 60

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The examination includes an oral report where the students will present and discuss their work for the seminar paper (30% of the grade). Moreover, students will prepare a seminar paper. In order to support students in writing process, there will be regular discussions about the progression of the project (seminar paper and regular discussions = 50% of the grade). On top of that, students' participation in the classes' discussions of cases will be evaluated (20% of the grade). The active participation in the case discussions measures the students' ability to use the theoretical knowledge of the seminar to analyze real complex situations. The written examination and the corresponding presentation are means to evaluate the student's ability to work with a case, incorporate legal and economic concepts and present a sound study that incorporates the seminar's topics. In the oral report, students also show that they can summarize, academically present, and discuss the subject. Finally, regular discussions with the instructor measure the student's ability to develop an idea, shape it and elaborate a structured project within a given timeframe.

Repeat Examination:

End of Semester

(Recommended) Prerequisites:

None

Content:

Market structures (perfect competition, monopoly, oligopoly, monopolistic competition)
Anti-competitive practices: horizontal and vertical restraints (dumping, price fixing, cartels, etc.)
European regulation and cases.
Economic criteria to analyze situations related to these topics in business environments.

Intended Learning Outcomes:

Students get to know different market structures and their regulation. They acquire elements to analyze and discuss how those structures and the legal mechanisms affect consumers and firms. Participants will gain a basic understanding of contractual structures that are relevant for the economic analysis of the legal framework. Finally, students will learn how to combine their knowledge about topics with the discussion of cases, integrating praxis elements into their analysis. Thus, they can evaluate whether practices in this area are legal and/or economically sensible according to context information.

Argumentative and analytical competencies are strengthened since cases are always open for debate. Their ability to propose creative solutions and work in a team will also be exercised continuously in their work cases.

Teaching and Learning Methods:

This seminar presents concepts and tools to analyze competition practices and market structures from a legal and an economic point of view. The seminar integrates a case study with the topics, so students have the opportunity to apply their learnings and implement them in their analysis. Students will study a specific case, expose their understanding, and analyze the case using legal and regulatory elements and economic criteria. Finally, students will present their work in class.

Media:

Case studies, exercises, PPT, Whiteboard

Reading List:

Mankiw, Principles of Economics (2014); Kovac/Vandenberghe, Economic evidence in EU Competition Law (2018); Posner, Economic Analysis of Law (2014)

Responsible for Module:

Jung, Stefanie; Prof. Dr. jur.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

Master's Thesis | Master's Thesis

Module Description

MGTHN9277: Master's Thesis | Master's Thesis

Version of module description: Gültig ab winterterm 2016/17

Module Level: Master	Language: English	Duration: one semester	Frequency: winter/summer semester
Credits:* 30	Total Hours: 900	Self-study Hours: 900	Contact Hours:

Number of credits may vary according to degree program. Please see Transcript of Records.

Description of Examination Method:

The Master's Thesis is the final paper focusing on a specific topic in business administration or economics. The thesis usually requires six months of work. Students describe and analyze the state of research on a particular topic. Based on the scientific knowledge and methodical skills acquired during their studies, students autonomously find an answer to their research question or provide a solution to a specific problem. The Master's Thesis is supervised by a TUM School of Management professor or a professor who teaches in the Master in Management & Digital Technology program.

Repeat Examination:

Next semester / End of Semester

(Recommended) Prerequisites:

Students can start their thesis after completing at least 48 credits, of which at least 18 credits are from the technology specialization and 6 credits are from methods.

Content:

The Master's Thesis focuses on a research topic in business administration or economics, often with a particular focus on business analytics and digital technology. The thesis is supervised by a TUM School of Management professor or a professor who teaches in the Master in Management & Digital Technology program. It can be done in collaboration with a company or a research institution. The thesis must be completed within six months.

Intended Learning Outcomes:

At the end of the module Master's Thesis, students can independently and systematically complete a scientific project. Students deploy their scientific knowledge and methodical skills to the specific

subject. They describe the state-of-the-art knowledge in the particular field, conduct the research, evaluate the findings, and classify them within the scientific and or practical discussion. Students are able to independently address new and complex research questions and develop their own solutions and recommendations.

Teaching and Learning Methods:

The thesis should familiarize students with scientific work and give them deep insights into a specific topic. Therefore, students apply the knowledge and methodical skills they acquired during their studies and create a scientific manuscript within the set time frame.

Media:

literature, presentations

Reading List:

specific literature based on the topic

Responsible for Module:

Li, Chengguang; Prof. Dr. rer. pol.

Courses (Type of course, Weekly hours per semester), Instructor:

For further information in this module, please click campus.tum.de or [here](#).

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